ELMFIRE Validation Guide

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ELMFIRE version: not declared
Compiled: October 22, 2026

About This Guide

This guide compares ELMFIRE predictions with observations for the active validation cases. Each report describes the event, input data, modeling choices, comparison methods, and uncertainties. The opening summary presents the calculated decisions and the computational information available during report preparation. A completed simulation does not by itself establish validation.

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1 How to Read This Guide

Start with the validation summary for the recorded computational environment, declared parallel allocation, and scientific decision table. The case identifiers in the summary link to the corresponding case-report entries. The common landscape-validation evidence then explains the metrics and limitations shared by the current cases, followed by each complete standalone case report with its event-specific provenance, configuration, figures, results, and limitations.

Interpret PASS and FAIL only where a case defines a justified acceptance decision. CHARACTERIZED means that descriptive metrics exist without a justified pass threshold. Missing, unrun, incomplete, or otherwise non-decisional evidence is shown as NOT EVALUABLE. A completed narrative alone does not establish validation evidence. The environment recorded during report preparation is not necessarily the execution environment of every included case.

2 Validation summary

2.1 Computational environment recorded during report preparation

The following information describes the environment available during preparation of this guide. It does not establish the environment used for earlier simulations. An available compiler is not necessarily the compiler used to build the model. ELMFIRE was not executed to obtain these observations; an undeclared model release remains unknown.

Configuration	Recorded value
Recorded at (Coordinated Universal Time)	2026-10-23T01:45:07+00:00
Declared ELMFIRE release	not declared
Operating system	macOS-15.7.7-arm64-arm-64bit
Processor architecture	arm64
Message Passing Interface (MPI) implementation	not available
Available Fortran compiler	GNU Fortran (Homebrew GCC 14.2.0_1) 14.2.0
Python version	CPython 3.12.9
Scientific analysis libraries	NumPy 1.26.4; Matplotlib 3.9.2; PyYAML 6.0.1; Rasterio 1.5.0; GDAL not installed; Pandas 2.2.2; GeoPandas not installed; Shapely not installed

2.2 Comparison type and declared parallel allocation

The comparison type distinguishes isolated and coupled processes or the spatial scale of validation. The process count is the allocation declared for each case, not a measurement of resources used in an earlier simulation. Missing declarations are not inferred.

Case	Comparison type	MPI processes
antelope_2021	Landscape-scale validation	1
august_complex_2020	Landscape-scale validation	1
beckwourth_complex_2021	Landscape-scale validation	1
bobcat_2020	Landscape-scale validation	1
caldor_2021	Landscape-scale validation	1
camp_fire	Landscape-scale validation	50
creek_2020	Landscape-scale validation	1
czu_lightning_complex_2020	Landscape-scale validation	1
dixie_2021	Landscape-scale validation	1
dolan_2020	Landscape-scale validation	1
glass_2020	Landscape-scale validation	1
july_complex_2020	Landscape-scale validation	1
kincade_2019	Landscape-scale validation	1
knp_complex_2021	Landscape-scale validation	1
lnu_lightning_complex_2020	Landscape-scale validation	1
mccash_2021	Landscape-scale validation	1
mcfarland_2021	Landscape-scale validation	1
monument_2021	Landscape-scale validation	1
north_complex_2020	Landscape-scale validation	1

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Case	Comparison type	MPI processes
red_salmon_complex_2020	Landscape-scale validation	1
river_complex_2021	Landscape-scale validation	1
scu_lightning_complex_2020	Landscape-scale validation	1
slater_and_devil_2020	Landscape-scale validation	1
sqf_complex_2020	Landscape-scale validation	1
tamarack_2021	Landscape-scale validation	1
thomas_fire	Landscape-scale validation	50
tubbs_fire	Landscape-scale validation	50
w_5_cold_springs_2020	Landscape-scale validation	1
walker_2019	Landscape-scale validation	1
windy_2021	Landscape-scale validation	1
zogg_2020	Landscape-scale validation	1

2.3 Scientific decision summary

The decisions below follow the established calculated metrics and case-specific acceptance criteria. Completing a simulation or preparing a report does not establish PASS. Missing or insufficient scientific evidence is reported as NOT EVALUABLE.

Total cases: 31 **PASS:** 0 **FAIL:** 0 **CHARACTERIZED:** 0 **NOT EVALUABLE:** 31

Case	Category	Title	Decision
antelope_2021	Landscape-scale validation	2021 Antelope fire reconstruction against GOFER observations	**NOT EVALUABLE**
august_complex_2020	Landscape-scale validation	2020 August Complex fire reconstruction against GOFER observations	**NOT EVALUABLE**
beckwourth_complex_2021	Landscape-scale validation	2021 Beckwourth Complex fire reconstruction against GOFER observations	**NOT EVALUABLE**
bobcat_2020	Landscape-scale validation	2020 Bobcat fire reconstruction against GOFER observations	**NOT EVALUABLE**
caldor_2021	Landscape-scale validation	2021 Caldor Fire reconstruction against GOFER hourly perimeters	**NOT EVALUABLE**
camp_fire	Landscape-scale validation	2018 Camp Fire landscape reconstruction	**NOT EVALUABLE**
creek_2020	Landscape-scale validation	2020 Creek fire reconstruction against GOFER observations	**NOT EVALUABLE**
czu_lightning_complex_2020	Landscape-scale validation	2020 CZU Lightning Complex reconstruction against GOFER hourly perimeters	**NOT EVALUABLE**
dixie_2021	Landscape-scale validation	2021 Dixie fire reconstruction against GOFER observations	**NOT EVALUABLE**
dolan_2020	Landscape-scale validation	2020 Dolan fire reconstruction against GOFER observations	**NOT EVALUABLE**
glass_2020	Landscape-scale validation	2020 Glass Fire reconstruction against GOFER hourly perimeters	**NOT EVALUABLE**
july_complex_2020	Landscape-scale validation	2020 July Complex fire reconstruction against GOFER observations	**NOT EVALUABLE**
kincade_2019	Landscape-scale validation	2019 Kincade Fire reconstruction against GOFER hourly perimeters	**NOT EVALUABLE**

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Case	Category	Title	Decision
knp_complex_2021	Landscape-scale validation	2021 KNP Complex fire reconstruction against GOFER observations	NOT EVALUABLE
lnu_lightning_complex_2020	Landscape-scale validation	2020 LNU Lightning Complex fire reconstruction against GOFER observations	NOT EVALUABLE
mccash_2021	Landscape-scale validation	2021 McCash fire reconstruction against GOFER observations	NOT EVALUABLE
mcfarland_2021	Landscape-scale validation	2021 McFarland fire reconstruction against GOFER observations	NOT EVALUABLE
monument_2021	Landscape-scale validation	2021 Monument fire reconstruction against GOFER observations	NOT EVALUABLE
north_complex_2020	Landscape-scale validation	2020 North Complex fire reconstruction against GOFER observations	NOT EVALUABLE
red_salmon_complex_2020	Landscape-scale validation	2020 Red Salmon Complex fire reconstruction against GOFER observations	NOT EVALUABLE
river_complex_2021	Landscape-scale validation	2021 River Complex fire reconstruction against GOFER observations	NOT EVALUABLE
scu_lightning_complex_2020	Landscape-scale validation	2020 SCU Lightning Complex fire reconstruction against GOFER observations	NOT EVALUABLE
slater_and_devil_2020	Landscape-scale validation	2020 Slater and Devil fire reconstruction against GOFER observations	NOT EVALUABLE
sqf_complex_2020	Landscape-scale validation	2020 SQF Complex fire reconstruction against GOFER observations	NOT EVALUABLE
tamarack_2021	Landscape-scale validation	2021 Tamarack fire reconstruction against GOFER observations	NOT EVALUABLE
thomas_fire	Landscape-scale validation	2017 Thomas Fire landscape reconstruction	NOT EVALUABLE
tubbs_fire	Landscape-scale validation	2017 Tubbs Fire landscape reconstruction	NOT EVALUABLE
w_5_cold_springs_2020	Landscape-scale validation	2020 W-5 Cold Springs fire reconstruction against GOFER observations	NOT EVALUABLE
walker_2019	Landscape-scale validation	2019 Walker fire reconstruction against GOFER observations	NOT EVALUABLE
windy_2021	Landscape-scale validation	2021 Windy fire reconstruction against GOFER observations	NOT EVALUABLE
zogg_2020	Landscape-scale validation	2020 Zogg fire reconstruction against GOFER observations	NOT EVALUABLE

Decision vocabulary. PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

3 Common landscape-validation evidence

The current Camp, Thomas, and Tubbs landscape cases were reconstructed from the supplied historical-fire datasets. Each is designed to compare a 50-member ELMFIRE ensemble with cumulative Visible Infrared Imaging Radiometer Suite (VIIRS) active-fire detections. The case reports retain the event-specific input provenance, namelist rationale, evidence status, and unresolved limitations.

Their common spatial metrics are Jaccard overlap, precision, recall, F_1, predicted-to-observed area ratio, and Brier score. Together these distinguish spatial mismatch, overgrowth, undergrowth, scale

bias, and probabilistic error. Because the recovered archives define no acceptance thresholds and do not quantify VIIRS-hull uncertainty, completed runs are reported as CHARACTERIZED rather than PASS or FAIL unless justified thresholds are declared before inspecting results. Unresolved model-input provenance, VIIRS collection or granule provenance, and event-specific calibration history are stated explicitly in the standalone reports.

2021 Antelope fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Antelope (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 1 August 2021 at 17:00 UTC to 10 September 2021 at 02:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 942 hourly perimeters, a final area of 599.09 km^2 and an official size of 145,632 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

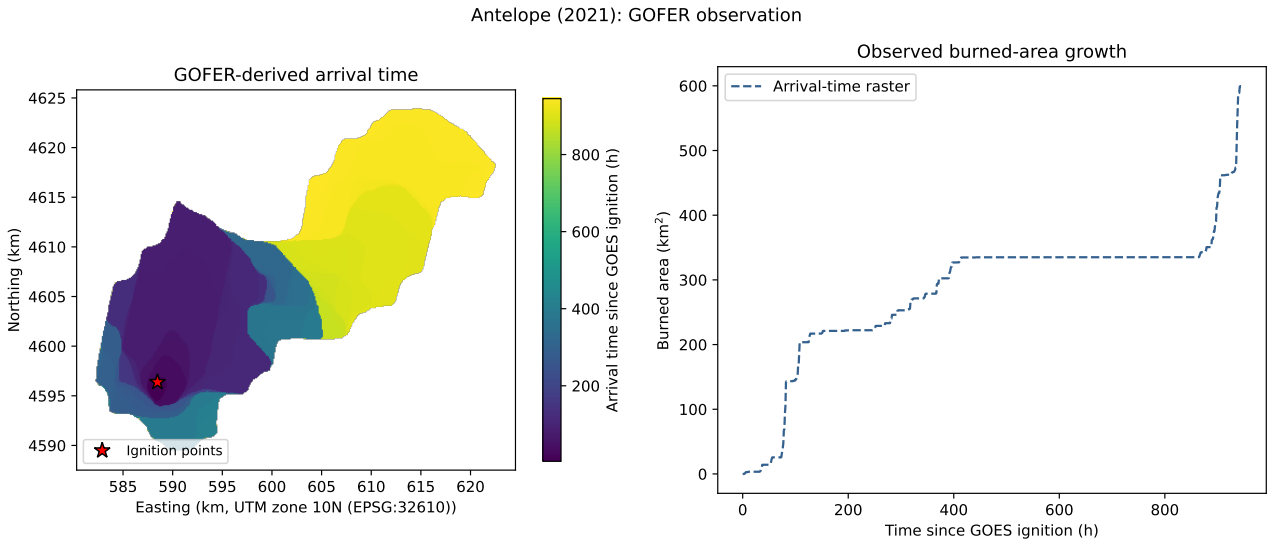


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,067 by 1,867 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 3,473 km^2. Elevation ranges from 1,175 to 2,618 m; median slope is 3°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 35%, and canopy base height has median 0.5 m.

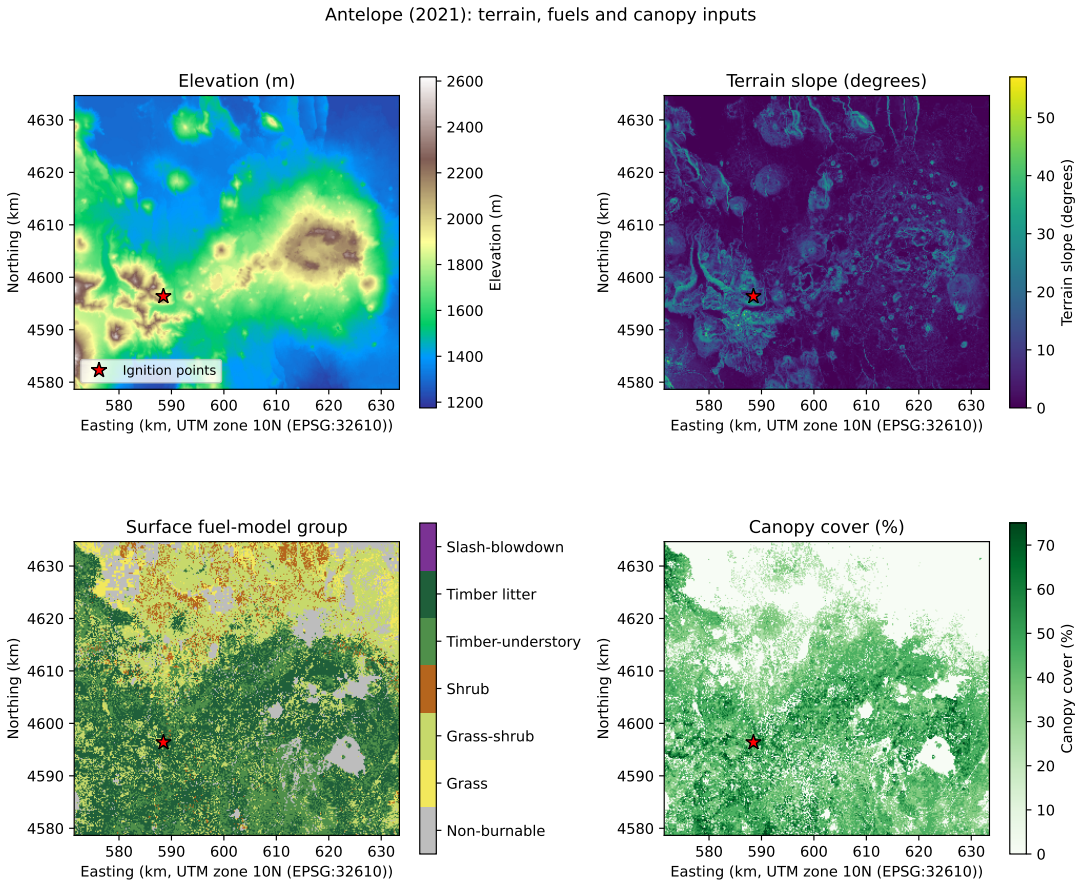


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

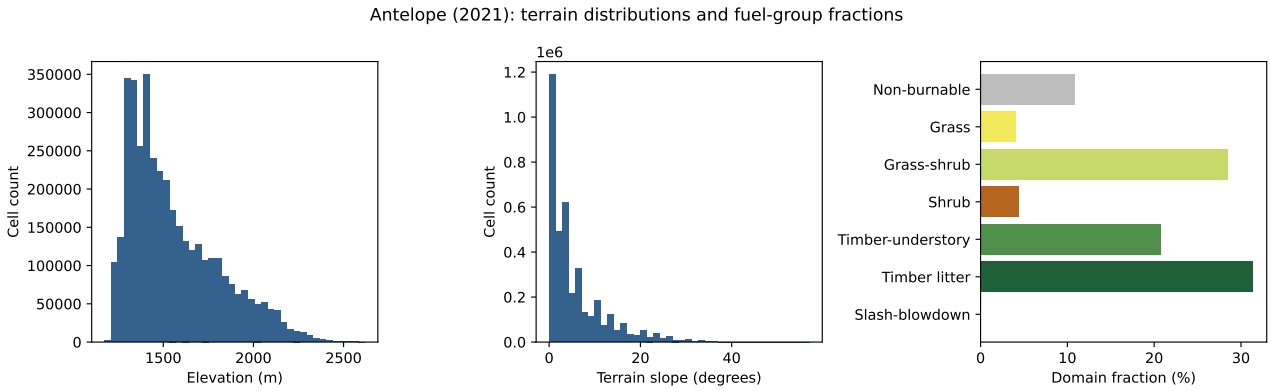


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 943 hourly fields. Median wind speed is 4.49 mph and its 95th percentile is 10.6 mph. Direction is summarized circularly: the mean direction is 293° and mean resultant length is 0.228. Median 1-h dead-fuel moisture is 5.92%. Spatial and temporal variability remain important when these domain summaries are interpreted.

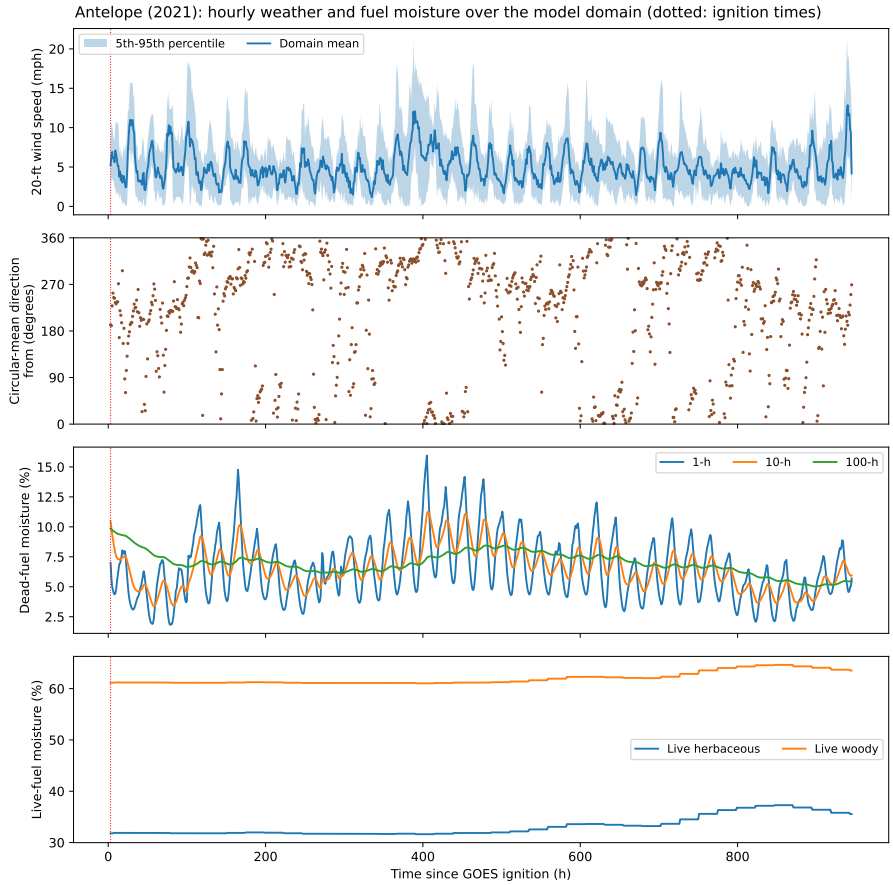


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

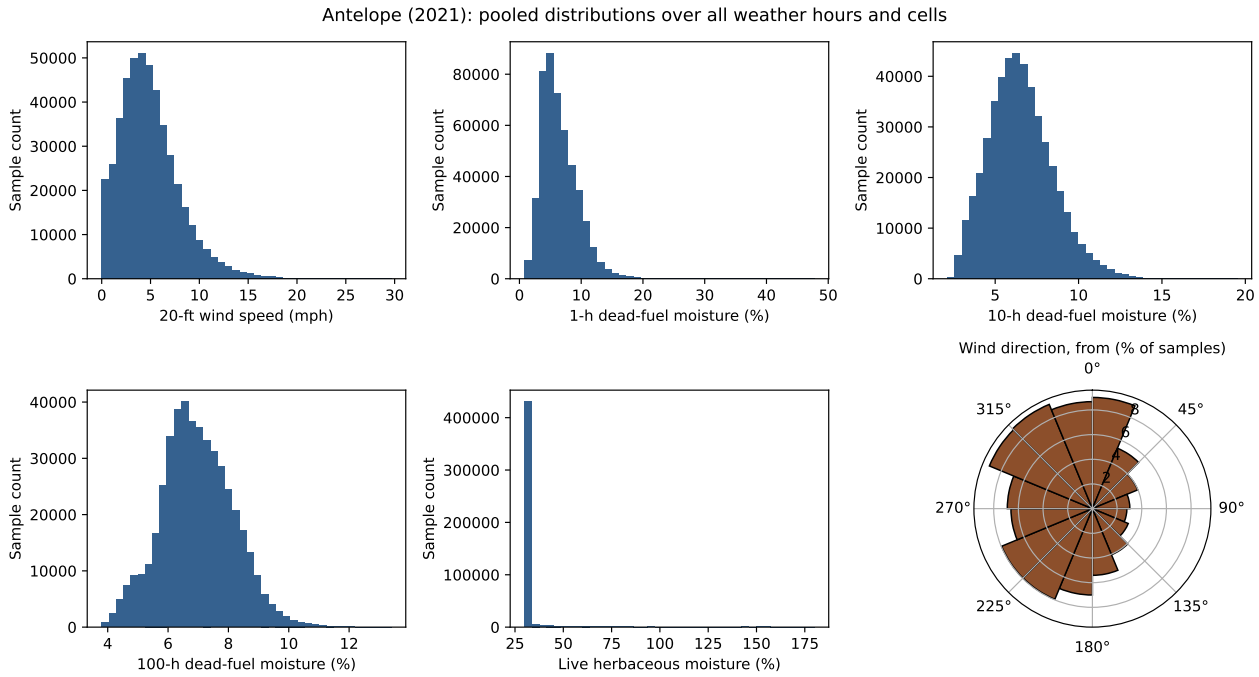


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 942 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 943 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 3 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

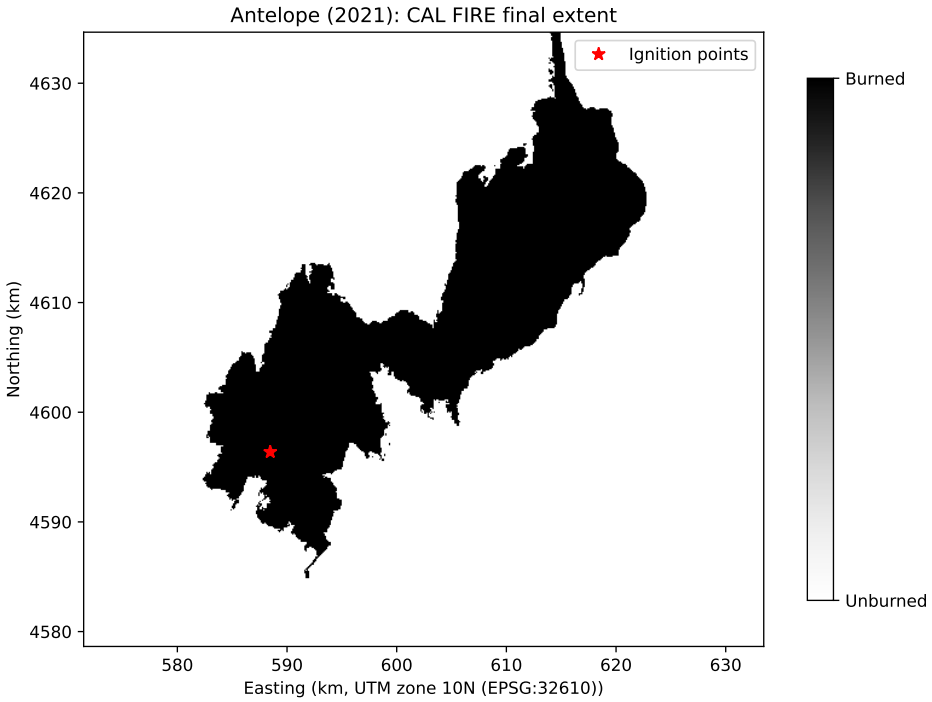


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 587 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Antelope reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 August Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the August Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 16 August 2020 at 21:00 UTC to 16 October 2020 at 22:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1465 hourly perimeters, a final area of 4342.79 km^2 and an official size of 1,032,648 ac. The configuration uses 12 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

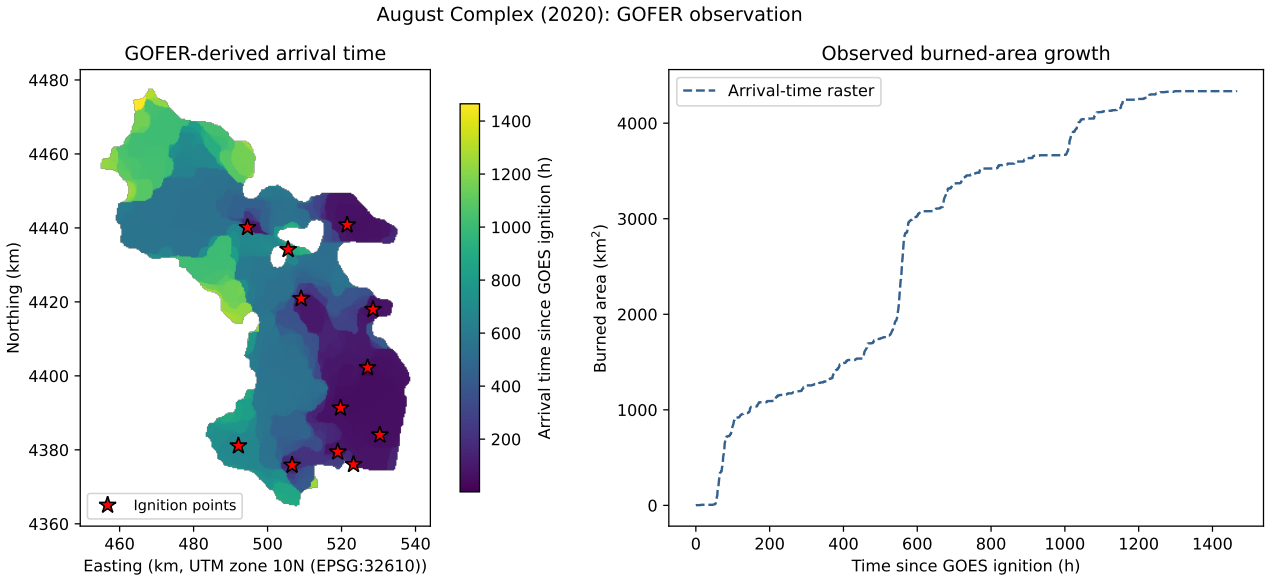


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 3,533 by 4,467 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 14,204 km^2. Elevation ranges from 34 to 2,460 m; median slope is 17°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 45%, and canopy base height has median 0.9 m.

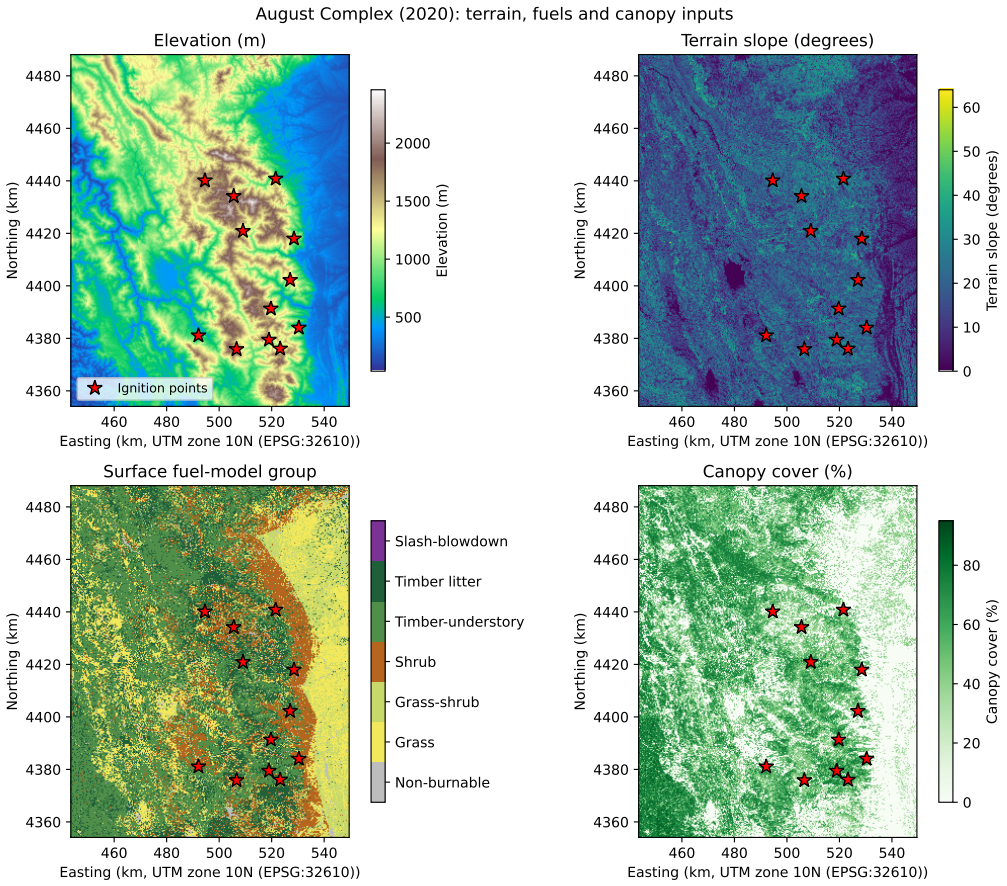


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

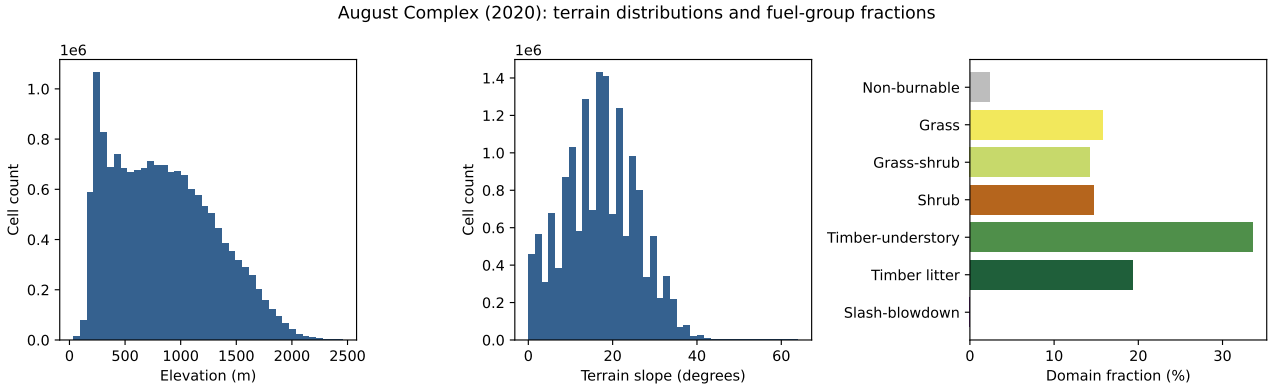


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,466 hourly fields. Median wind speed is 3.81 mph and its 95th percentile is 9.92 mph. Direction is summarized circularly: the mean direction is 302° and mean resultant length is 0.134. Median 1-h dead-fuel moisture is 7.05%. Spatial and temporal variability remain important when these domain summaries are interpreted.

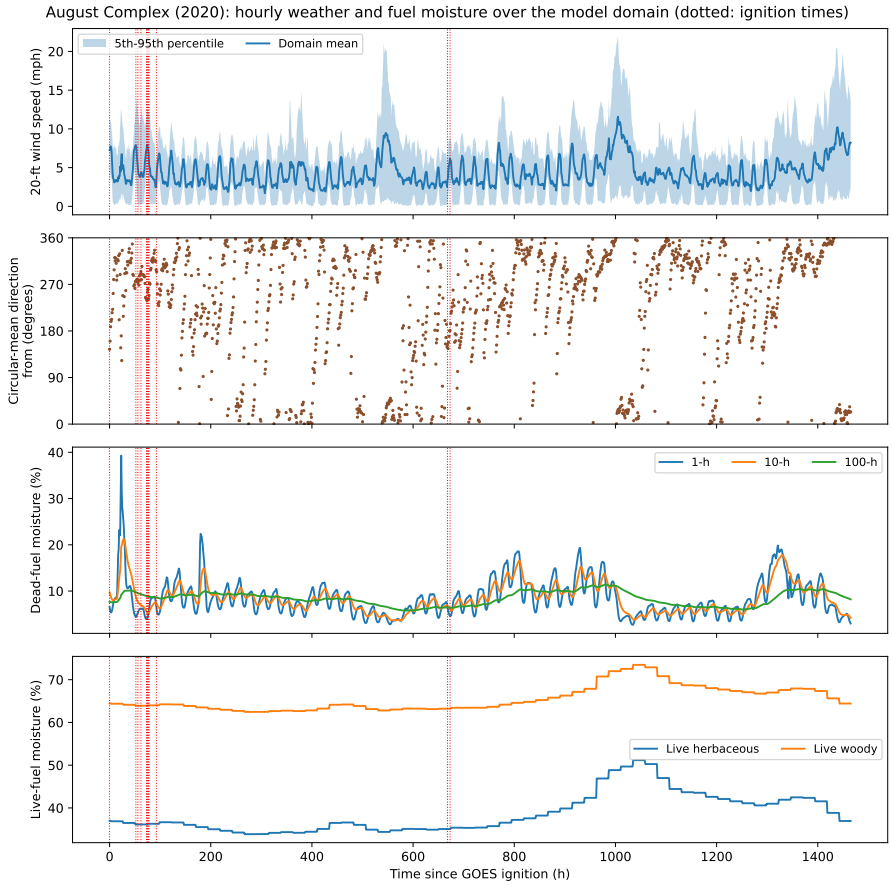


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

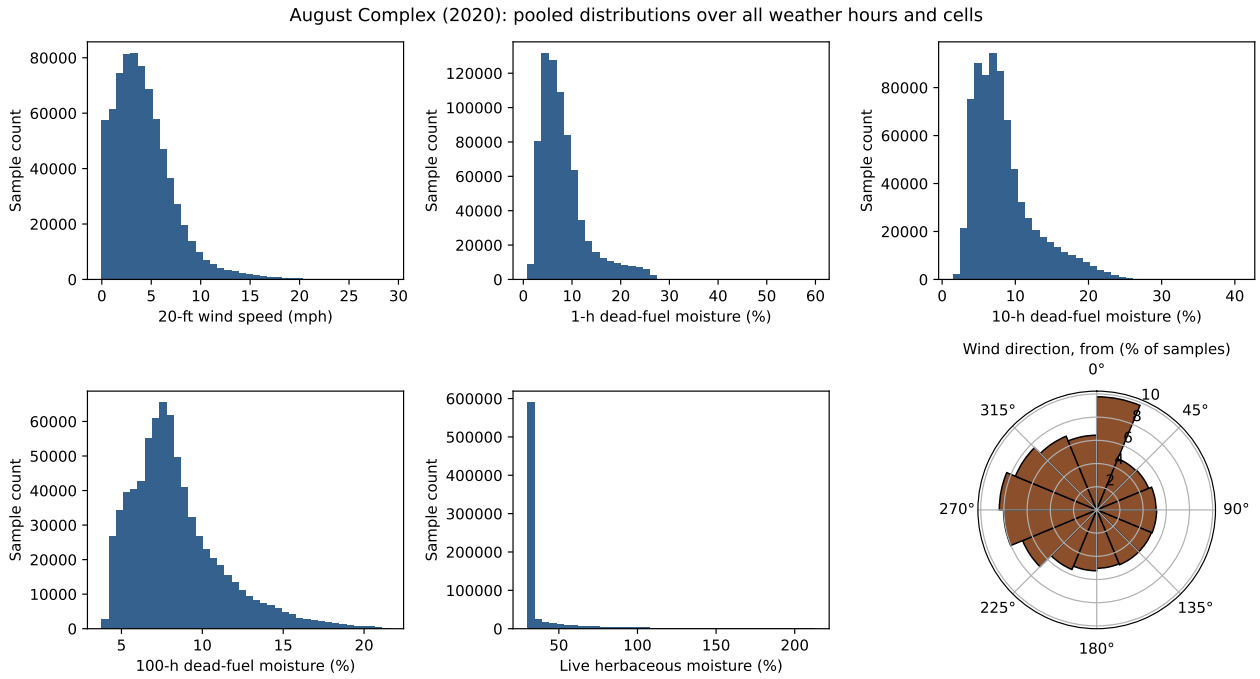


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1465 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1466 hourly weather fields and 12 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 0 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

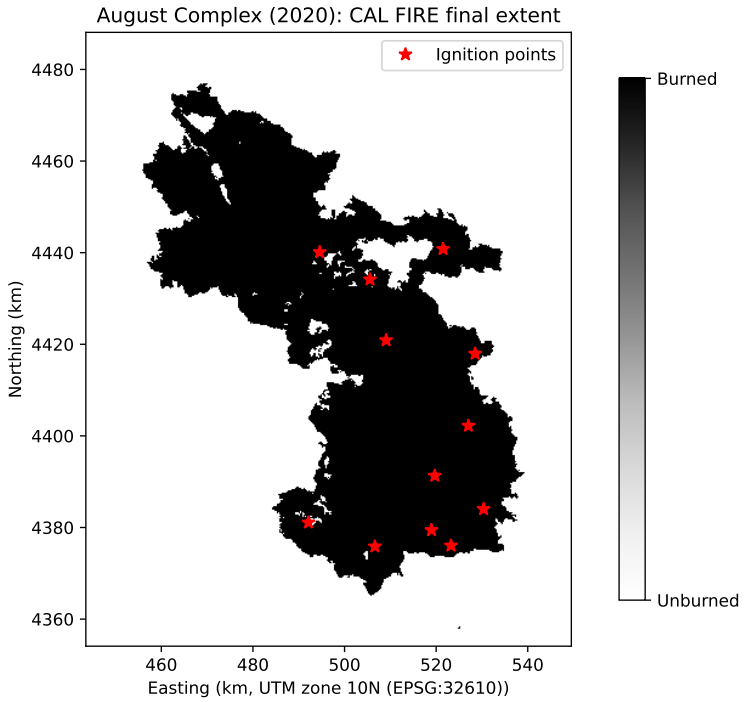


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 4,176 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the August Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 Beckwourth Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Beckwourth Complex (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 30 June 2021 at 23:00 UTC to 5 August 2021 at 03:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 844 hourly perimeters, a final area of 558.307 km^2 and an official size of 105,670 ac. The configuration uses 2 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

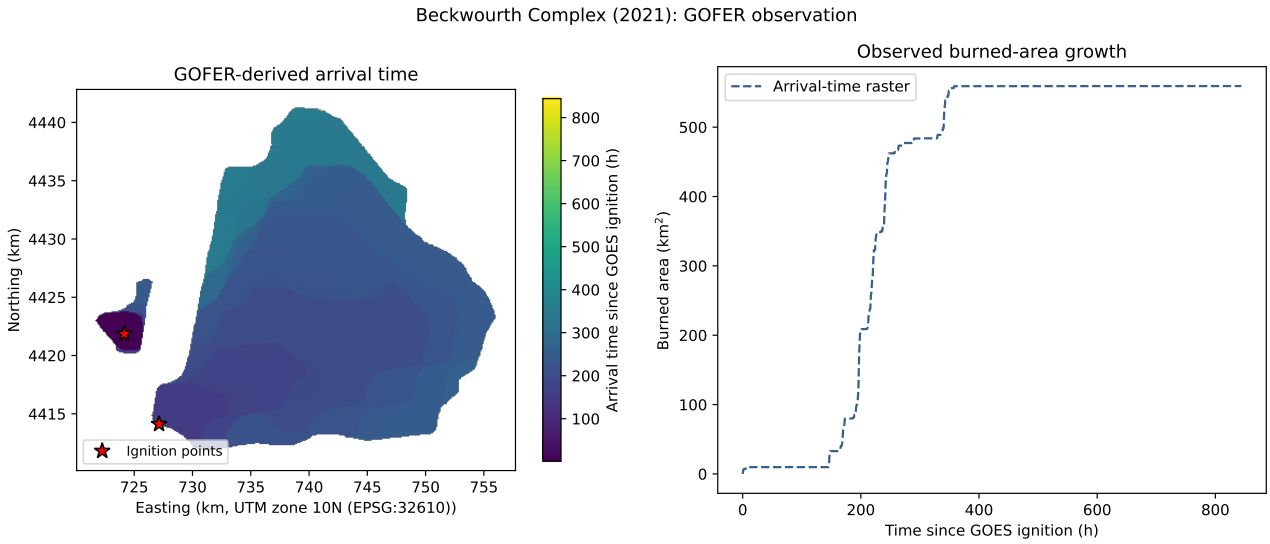


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,867 by 1,733 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 2,912 km^2. Elevation ranges from 1,209 to 2,526 m; median slope is 6°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

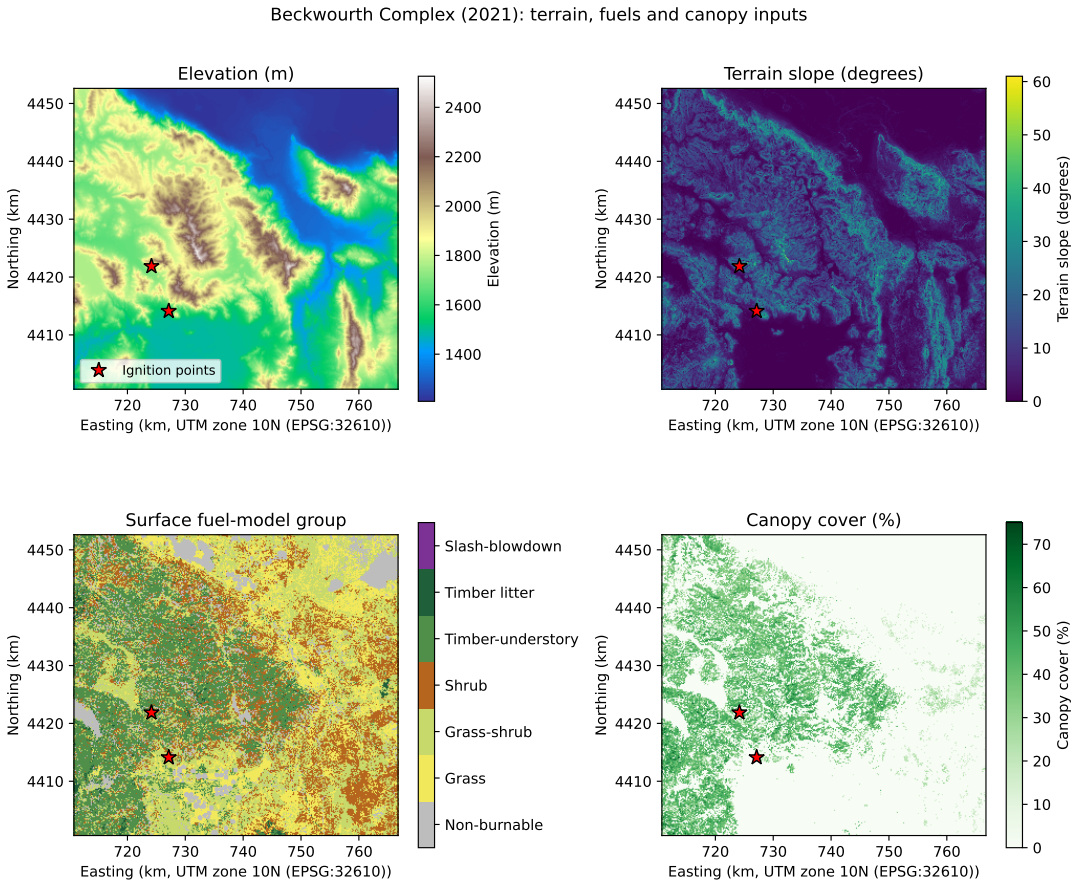


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

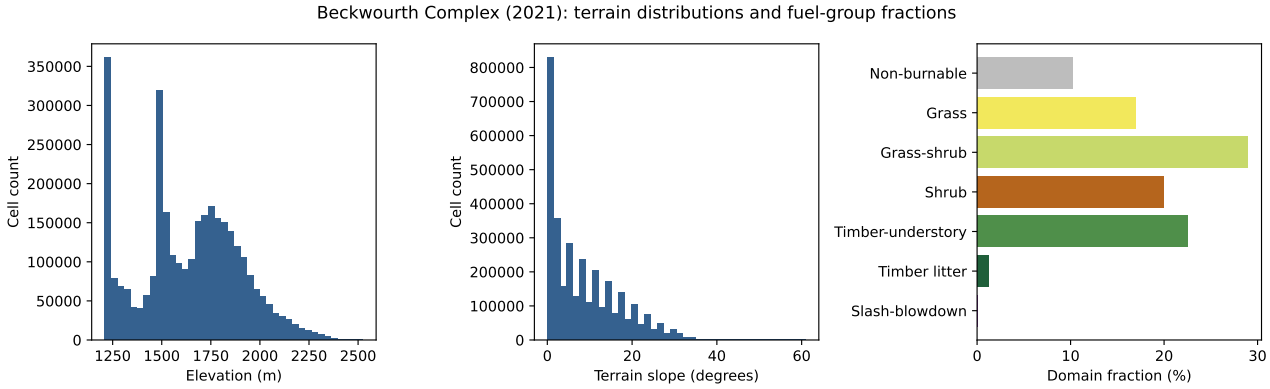


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 845 hourly fields. Median wind speed is 4.52 mph and its 95th percentile is 12.8 mph. Direction is summarized circularly: the mean direction is 263° and mean resultant length is 0.36. Median 1-h dead-fuel moisture is 5.26%. Spatial and temporal variability remain important when these domain summaries are interpreted.

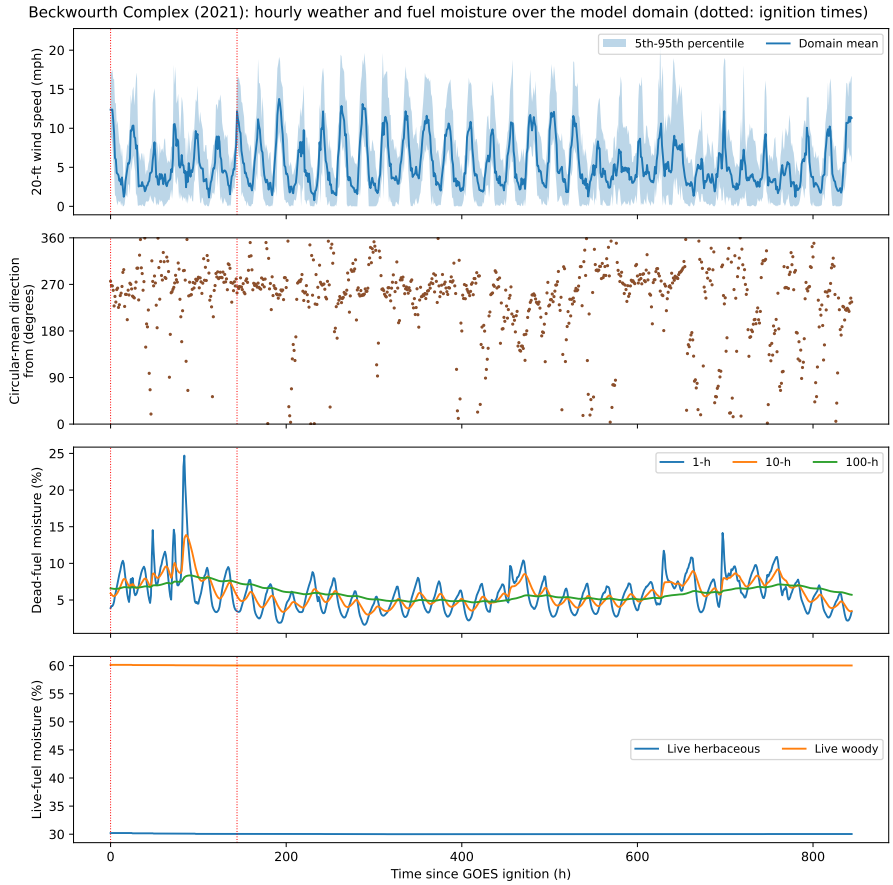


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

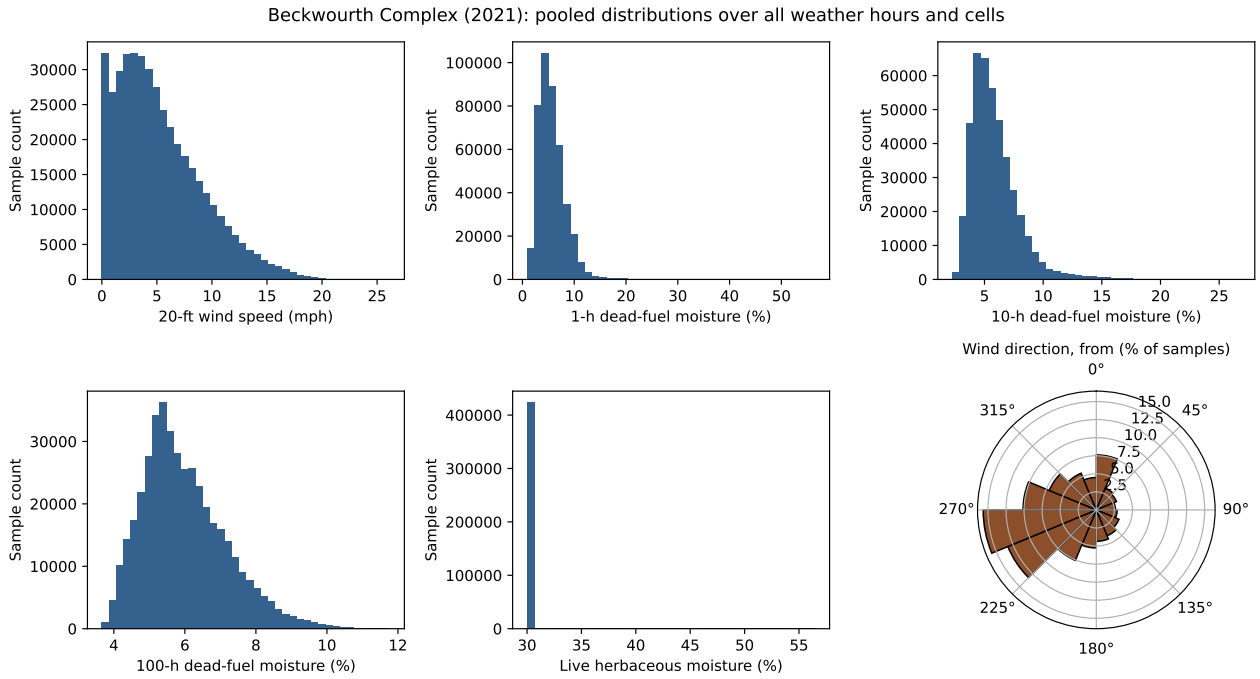


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 844 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 845 hourly weather fields and 2 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 0 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

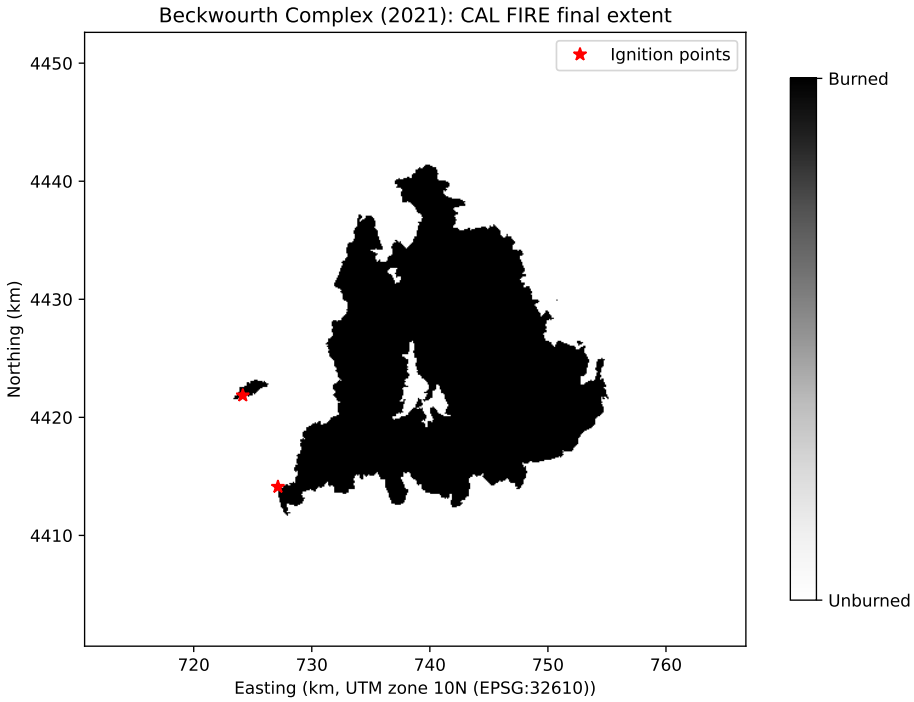


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 427 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Beckwourth Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 Bobcat fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Bobcat (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 6 September 2020 at 19:00 UTC to 24 September 2020 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 433 hourly perimeters, a final area of 583.864 km^2 and an official size of 115,997 ac. The configuration uses 3 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

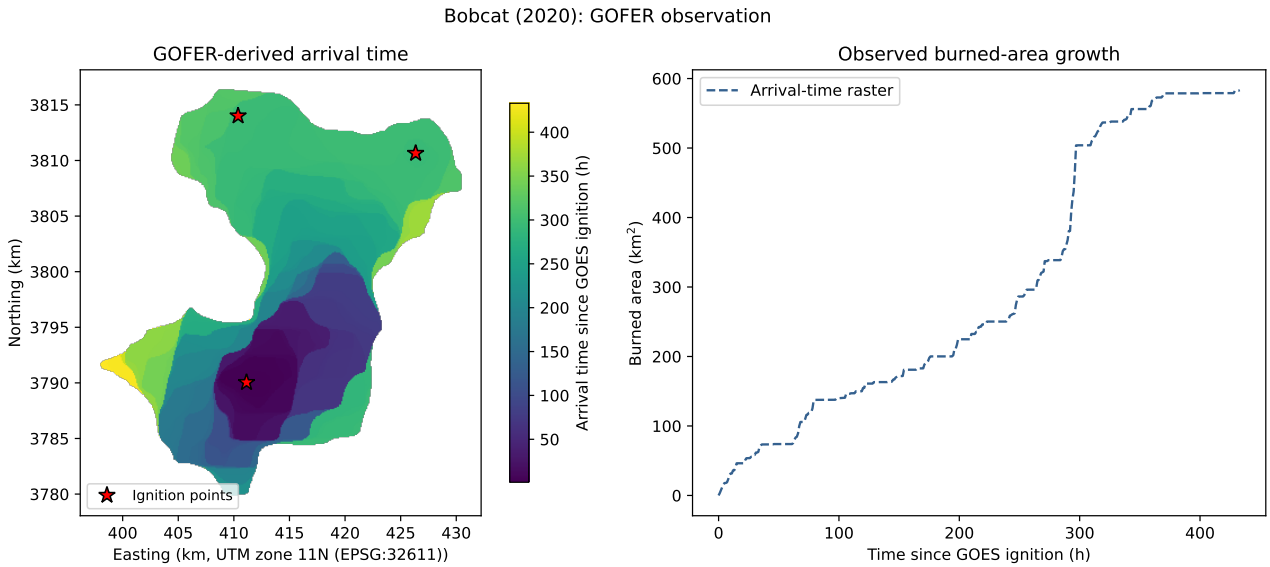


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,800 by 1,933 cells at 30 m spacing in UTM zone 11N (EPSG:32611), covering 3,131 km^2. Elevation ranges from 54 to 3,066 m; median slope is 14°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

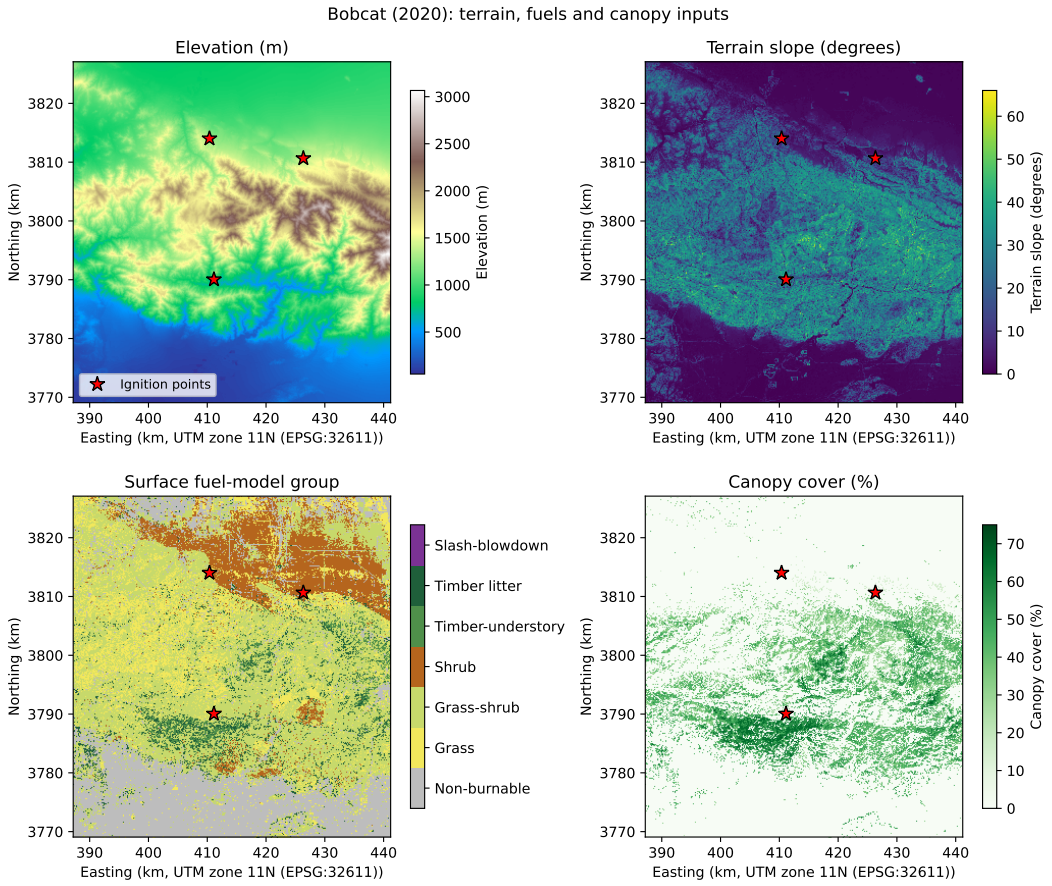


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 11N (EPSG:32611).

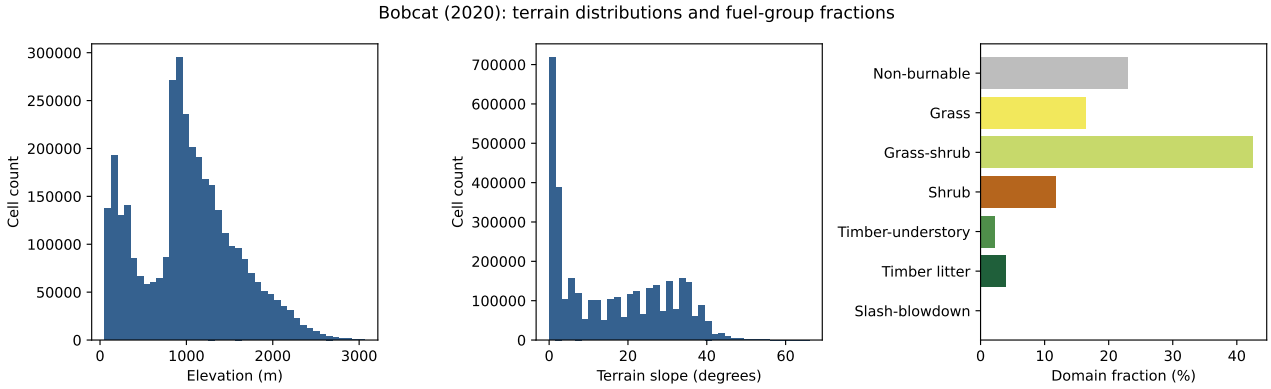


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 434 hourly fields. Median wind speed is 4.63 mph and its 95th percentile is 13.5 mph. Direction is summarized circularly: the mean direction is 247° and mean resultant length is 0.183. Median 1-h dead-fuel moisture is 4.83%. Spatial and temporal variability remain important when these domain summaries are interpreted.

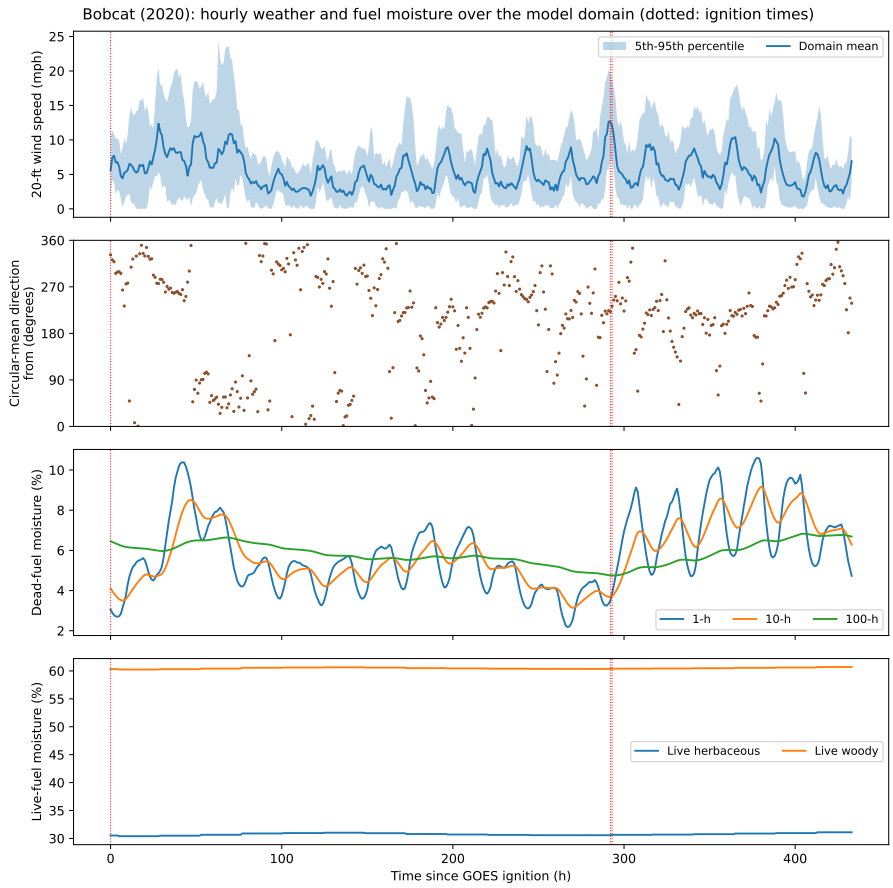


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

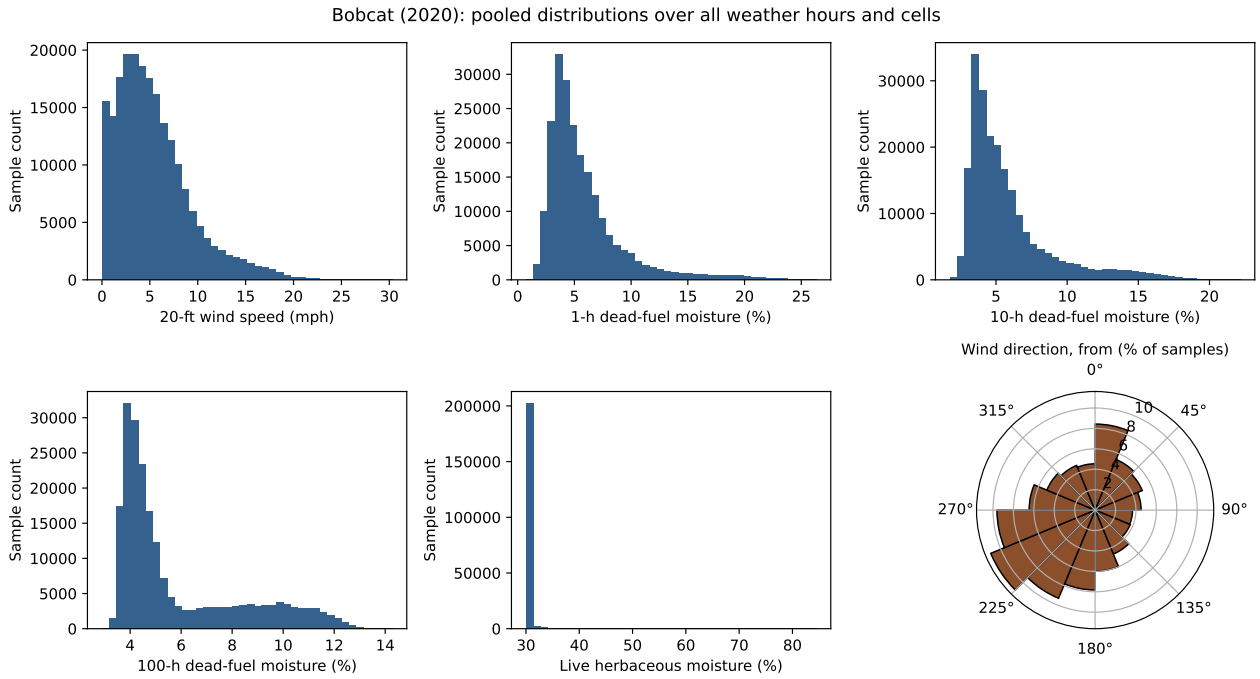


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 433 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 434 hourly weather fields and 3 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 0 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

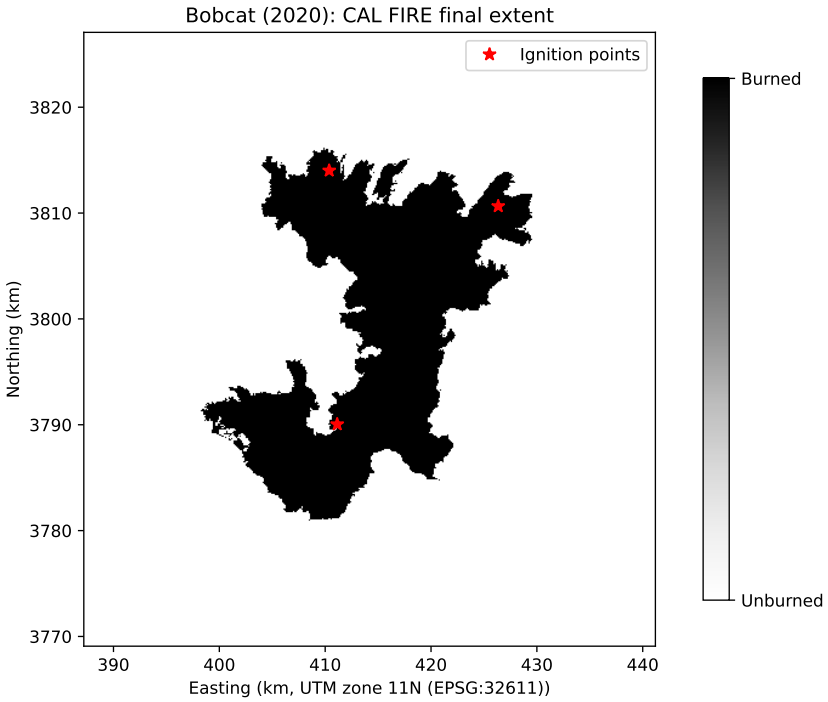


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 469 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Bobcat reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 Caldor Fire Reconstruction against GOFER Hourly Perimeters

1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Caldor Fire (2021) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 626 h, from the first GOES fire detection at 15 August 2021 at 01:00 UTC to the last GOFER perimeter at 10 September 2021 at 03:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. No model parameter was tuned against the GOFER observation of this fire; the spotting configuration was revised after comparisons for the Kincade and CZU Lightning Complex fires.

2 Reference event, experiment, and observations

GOFER records the Caldor Fire from 15 August 2021 at 01:00 UTC to 10 September 2021 at 03:00 UTC with 620 hourly perimeters, a final perimeter area of 994 km^2 and an official size of 221,835 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of 0.75 ± 0.21 km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 620 hourly perimeters of this fire on a 51.3-m grid in UTM zone 10N (EPSG:32610). The raster covers 995 km^2, consistent with the final GOFER perimeter area of 994 km^2; arrival times range from 7 to 626 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

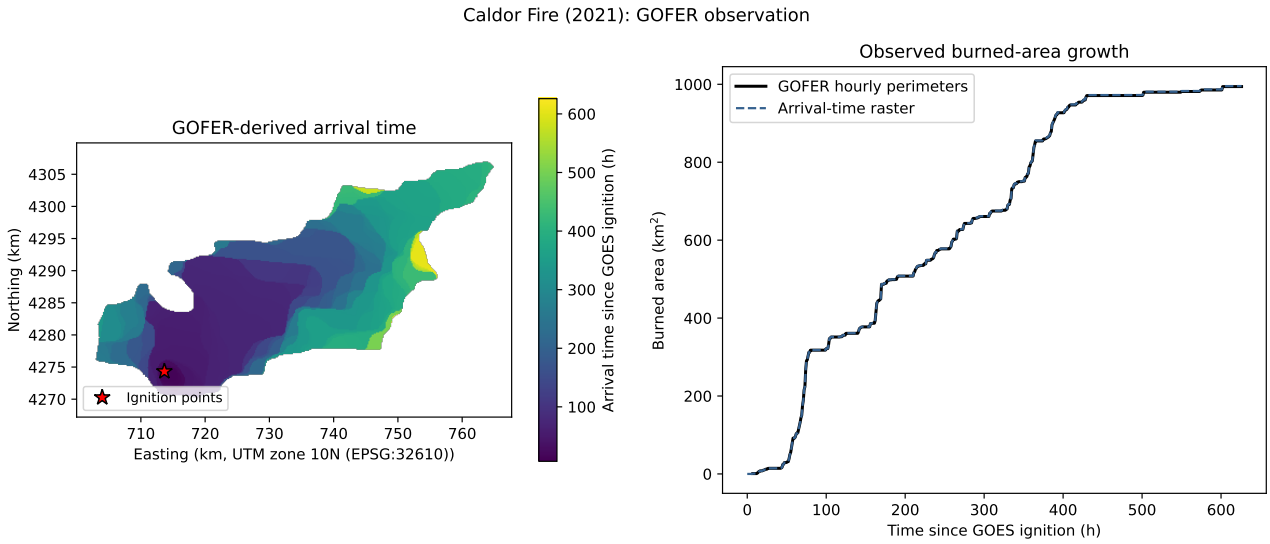


Figure 1: GOFER observation of the Caldor Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

3 Input data provenance and characterization

Input	Source and version	Resolution, units and processing
Terrain	LANDFIRE elevation, slope and aspect [3]	30 m; m and degrees; UTM zone 10N (EPSG:32610)
Surface fuels	LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.2.0	30 m categorical
Canopy	LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.2.0	30 m; percent, m, m and kg m^{-3}
Weather	Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6]	600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees
Fuel moisture	Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather	600 m hourly; percent; derivation method not documented in the available case materials
Ignition	GOFER ignition points [2]	Points with hourly time stamps
Adjustment and initial state	Spread-rate adjustment factor and initial level set	Uniform value of one (no adjustment, nothing burned at the start)

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 2,800 by 1,933 cells of 30 m (4,871 km^2) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 302 to 3,312 m (median 1,722 m) and slope has a median of 13° and a 95th percentile of 32°; the circular-mean aspect is 247°. By area the domain is 3.22% grass, 12.6% grass-shrub, 9.19% shrub, 49.2%

timber-understory, 14.4% timber litter, 0.0394% slash-blowdown and 11.4% non-burnable fuel models. Canopy cover has a median of 45% and canopy base height a median of 0.9 m.

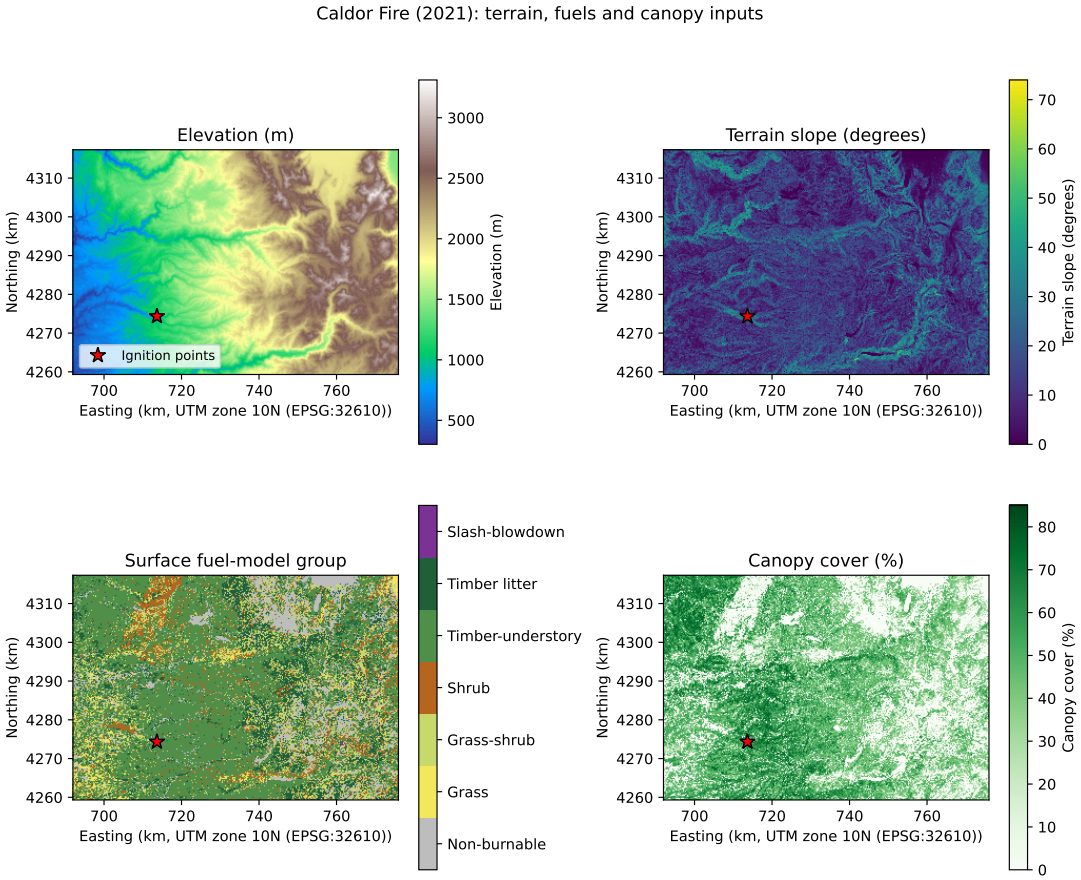


Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

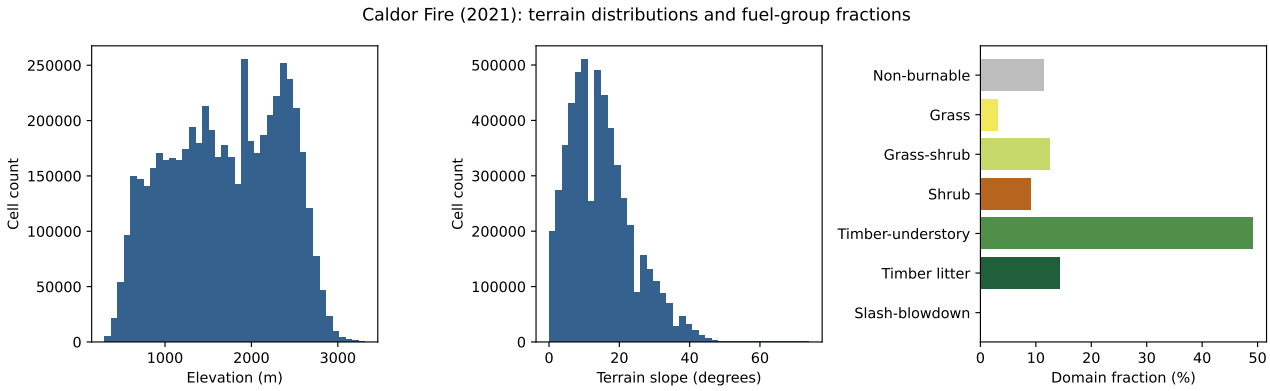


Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 621 hourly fields. Over all hours and cells the 20-ft wind speed has a median

of 3.89 mph and a 95th percentile of 11.4 mph; the circular-mean direction the wind blows from is 206° with a mean resultant length of 0.193 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 5.74% (5th percentile 3.2%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

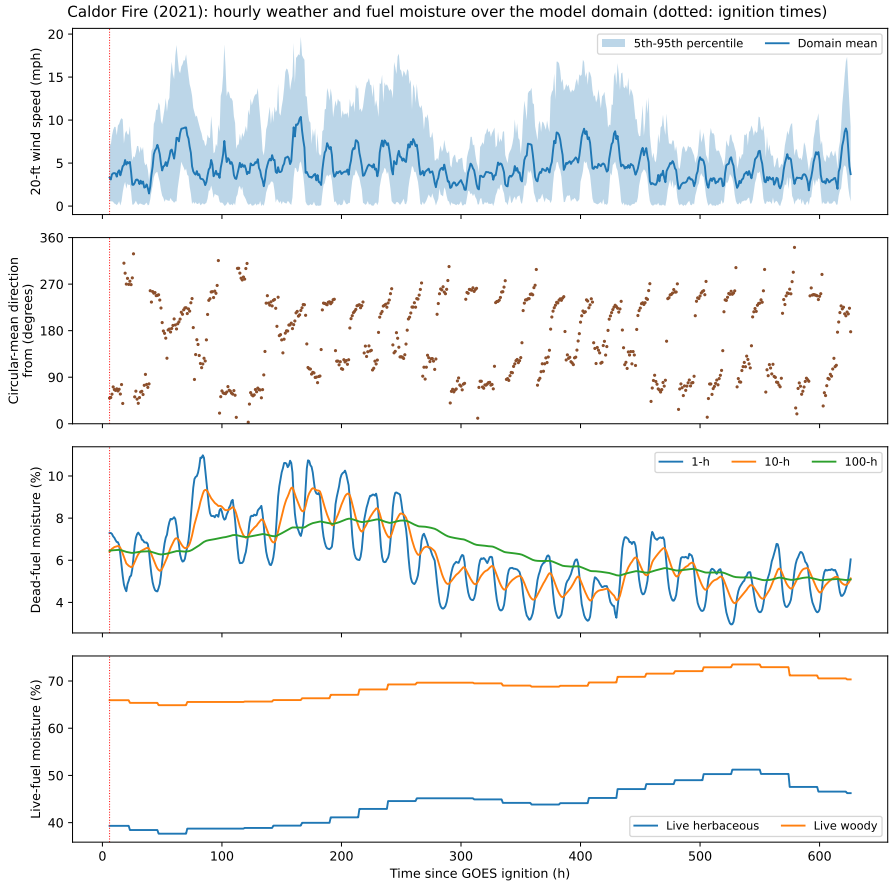


Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

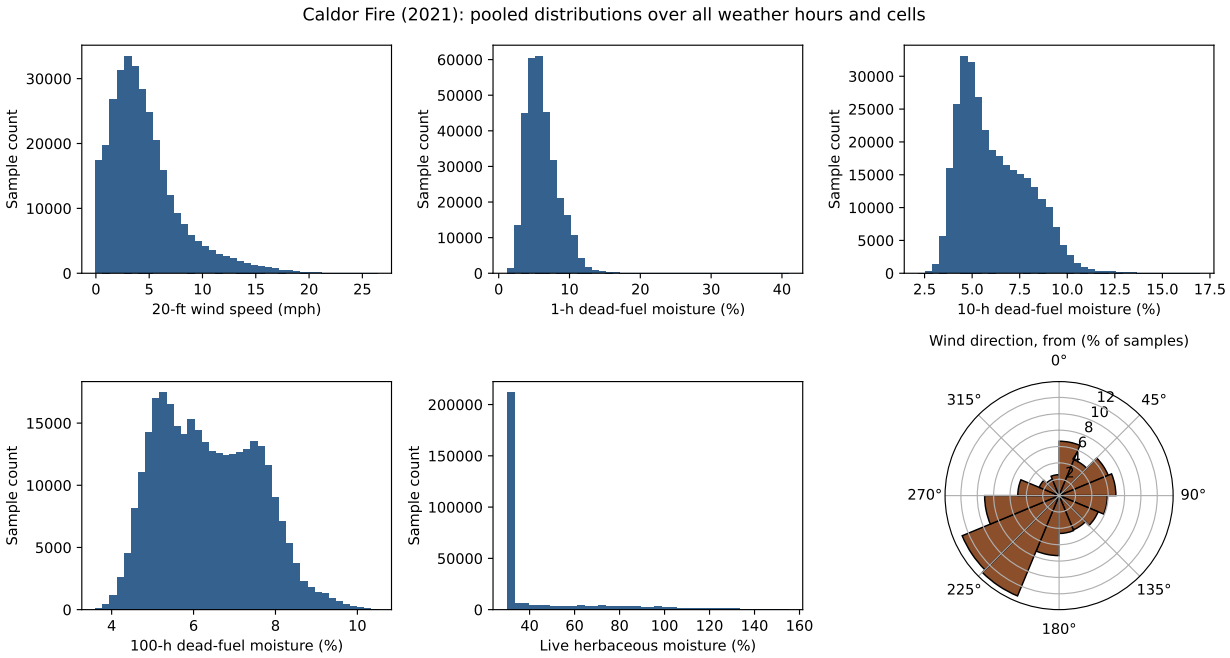


Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 620 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft min^{-1}, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

Choice	Basis	Evidence and limitation
Ignition points and times	documented	GOFER ignition record; no point required moving
Simulation start and stop	documented	First GOFER ignition point and last GOFER perimeter
Fuel vintage	derived	Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire
Weather and moisture	user-specified	RTMA analysis through CloudFire, as stated by the case owner
Crown fire	documented	ELMFIRE default crown-fire treatment, set explicitly
Firebrand spotting	user-specified	Enabled with ELMFIRE default spotting settings until calibration
Domain margin	user-specified	10 km beyond the final GOFER perimeter
Single deterministic member	user-specified	No ensemble; stochastic spotting uses a fixed random seed

Table 2: Basis of consequential configuration choices.

5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 6 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. No simulation results are retained in this dataset. New results must record the executable identity and configuration used.

6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time t the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap J_t summarizes footprint agreement; precision P_t diagnoses commission (simulated burning outside the observation), recall R_t omission (observed burning the simulation misses), F_t balances the two, and the area ratio A_t exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

7 Simulation results and field-level comparison

The current status of this case is **NOT EVALUABLE**. No simulation result is available for this case.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Caldor Fire. Inputs and the observation are characterized in full; the current scientific status is **NOT EVALUABLE** with 0 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

References

[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

2018 Camp Fire Landscape Reconstruction

1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2018 Camp Fire Landscape Reconstruction through 36 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

2 Reference event, experiment, and observations

The configured event start, expressed in Pacific standard time (PST), is 8 November 2018 at 14:00 PST. The comparison instant is 10 November 2018 at 02:00 PST (10:00 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 2260 point records in the geographic coordinate system defined by EPSG:4326; 2017 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

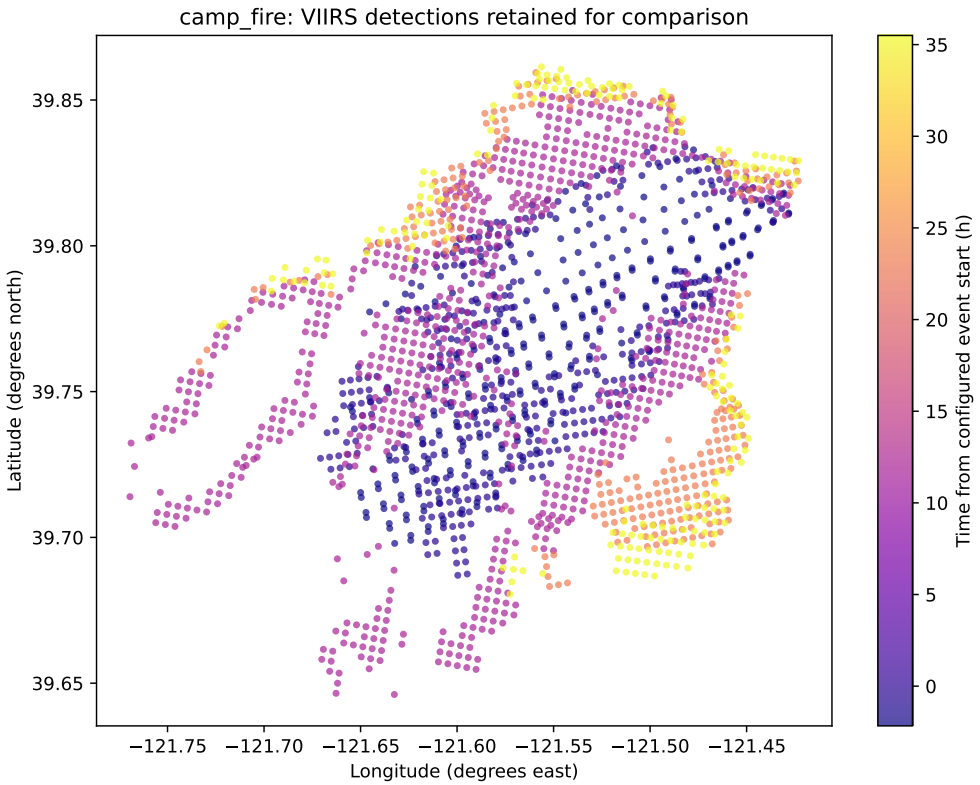


Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `0b0e8f7ce3bbea4665269ff7f5b88dd373f3d9210b3fa4ddc7916b610f724033`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

Field	Minimum	Mean	Standard deviation	Maximum
Elevation	30	470.9	365.7	1620
Slope	0	11.74	10.32	63
Wind speed	0	8.313	6.306	33.35
1-h moisture	6	6	0	6
10-h moisture	7	7	0	7
100-h moisture	8	8	0	8

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

No additional active-range anomaly was identified by the structural and statistical checks; upstream plausibility remains unverified because product metadata are absent.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 329.6° and 2.253°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most 512×512 cells per band, and are labeled accordingly.

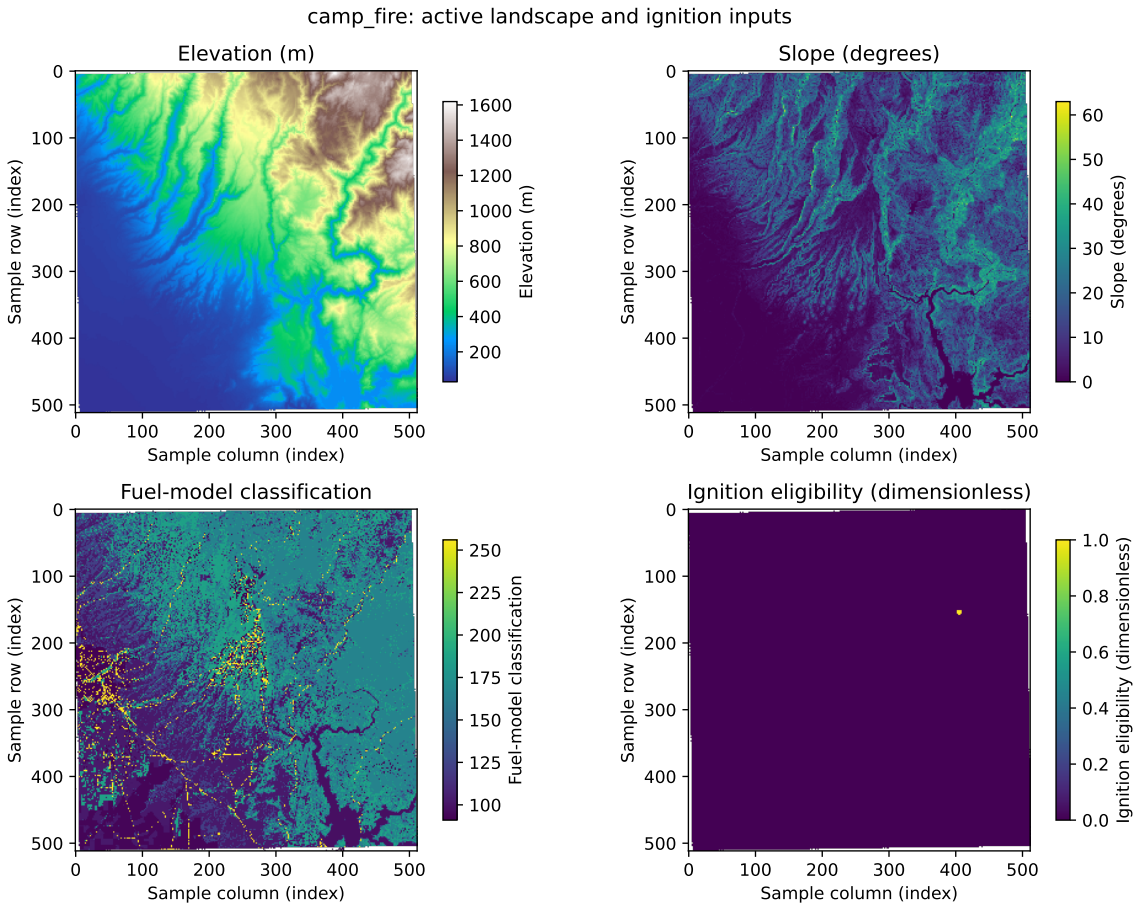


Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

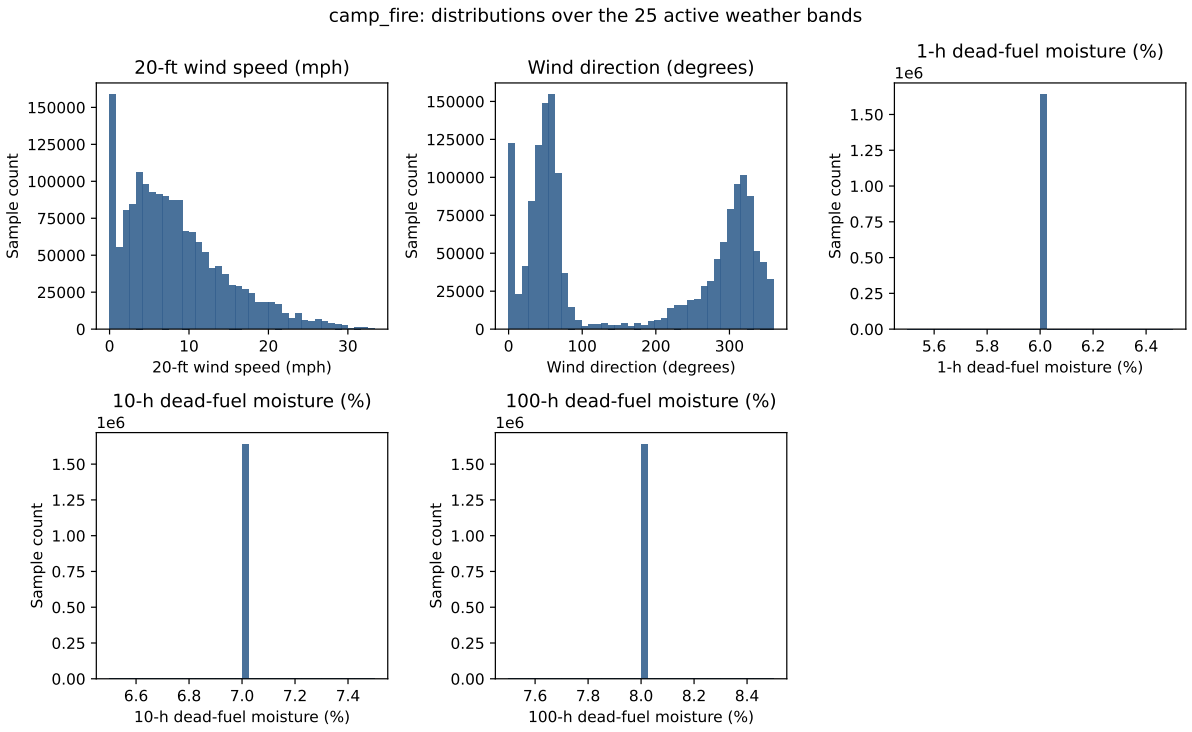


Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

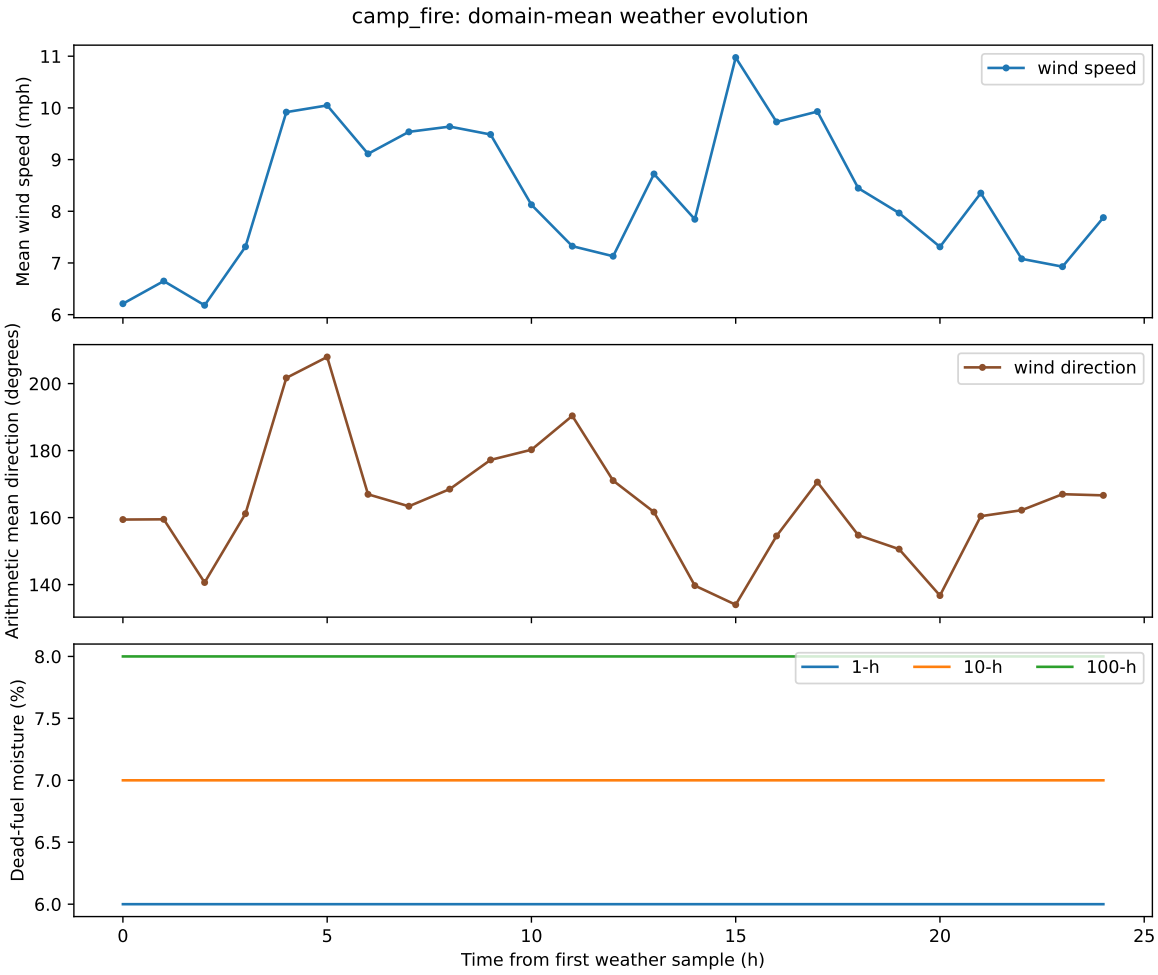


Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

4 Model configuration and rationale

The active domain is 1457 by 1476 cells at 30 m in EPSG:26910. Weather is 37-band weather rasters; bands 1–25 are selected at one-hour intervals, and the ignition treatment is random ignition sampling from the archived ignition mask. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

Choice	Evidence class	Basis and limitation
Landscape, weather, ignition, Monte Carlo, spotting, and WUI values	documented	Retained from the received archive; upstream derivation and event-specific tuning history are unknown.
Model-setting compatibility	derived	Required by the choices supported by the inspected model formulation; no physical values were tuned.
Comparison time	documented	Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent.
$p \geq 0.50$ footprint	selected for this comparison	Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use.
VIIRS hull ratio 0.2	selected for this comparison	Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter.

Table 2: Provenance of consequential configuration and comparison choices.

5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

6 Validation metrics and comparison method

For member i, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and t_c the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c] \,, \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support O, intersection $I = |P \cap O|$, and valid-cell count M, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and F_1 balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

7 Simulation results and field-level comparison

The current scientific status is **NOT EVALUABLE**. No current time_of_arrival GeoTIFFs were found for this case. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported and must not be inferred from WUI input rasters alone.

8 Integrated statistics and calculated metrics

Metric	Current value
Jaccard overlap	**MISSING**
Precision	**MISSING**
Recall	**MISSING**
F_1 score	**MISSING**
Brier score	**MISSING**
Predicted/observed area ratio	**MISSING**
Predicted area (km^2)	**MISSING**
Observation-support area (km^2)	**MISSING**
Ensemble members found	0

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **NOT EVALUABLE**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

2020 Creek fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Creek (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 5 September 2020 at 01:00 UTC to 9 November 2020 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1574 hourly perimeters, a final area of 1614.99 km^2 and an official size of 379,895 ac. The configuration uses 3 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

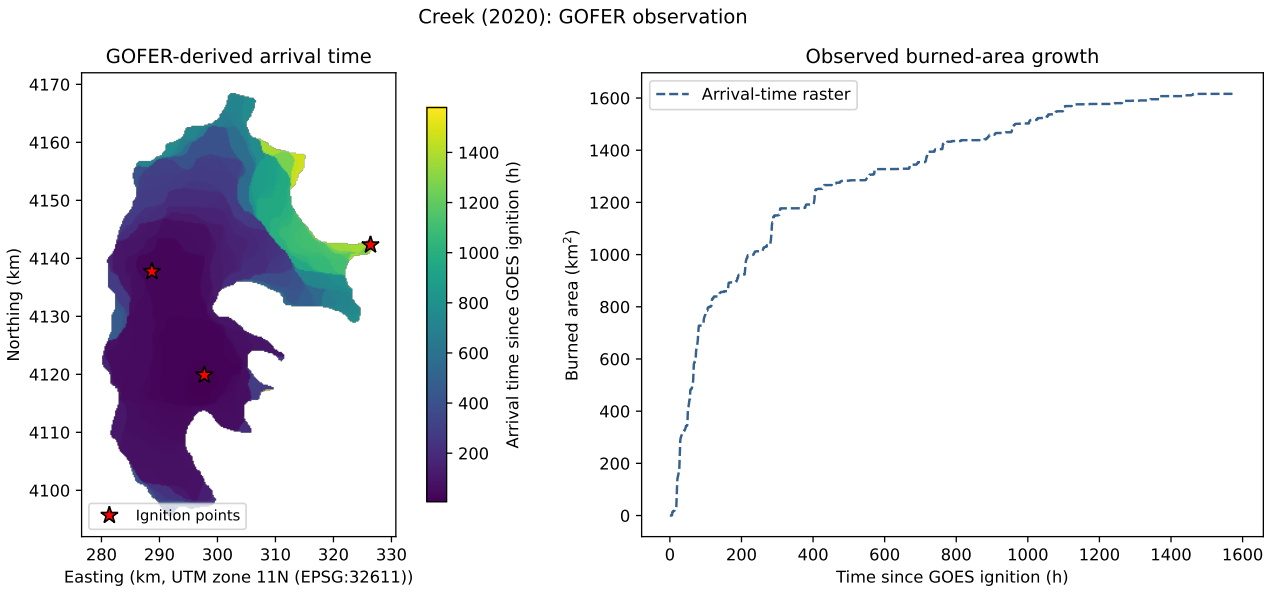


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,333 by 3,200 cells at 30 m spacing in UTM zone 11N (EPSG:32611), covering 6,719 km^2. Elevation ranges from 139 to 4,078 m; median slope is 14°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 35%, and canopy base height has median 0.3 m.

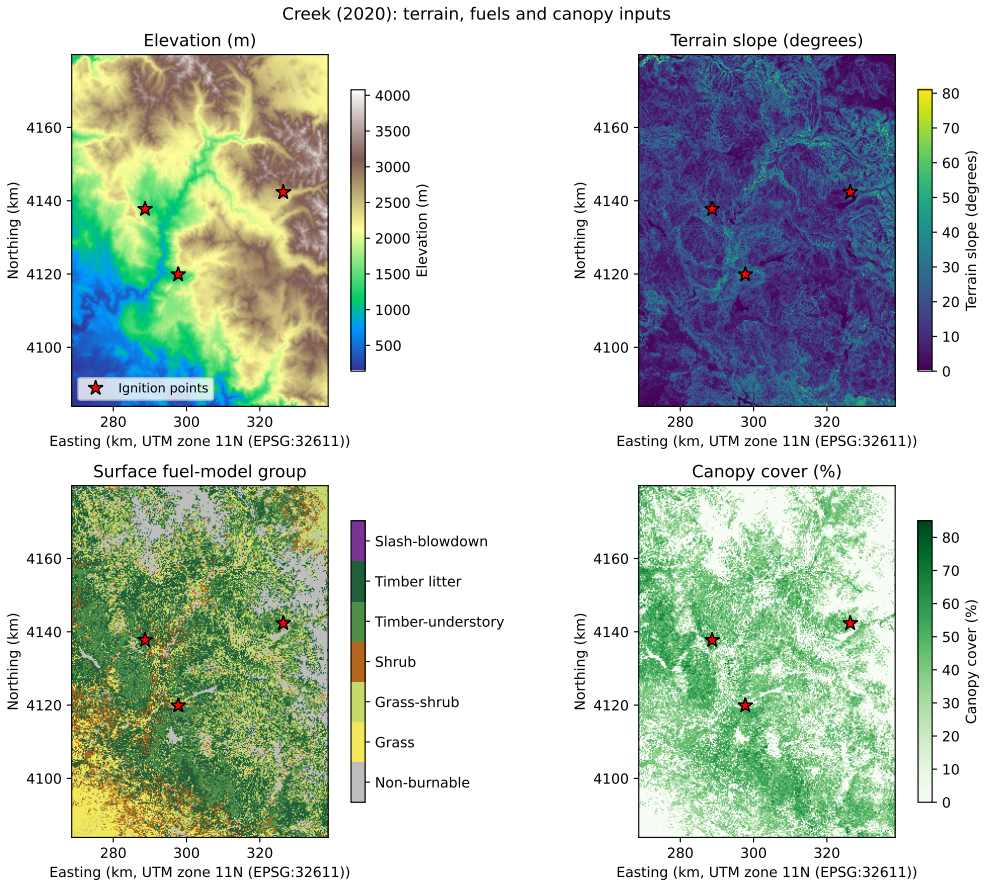


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 11N (EPSG:32611).

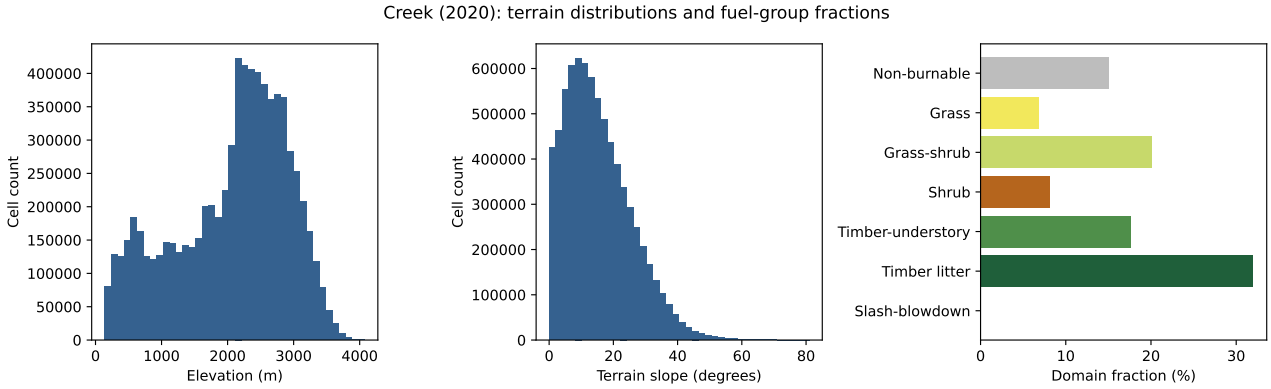


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,575 hourly fields. Median wind speed is 4.07 mph and its 95th percentile is 11.1 mph. Direction is summarized circularly: the mean direction is 298° and mean resultant length is 0.112. Median 1-h dead-fuel moisture is 6.18%. Spatial and temporal variability remain important when these domain summaries are interpreted.

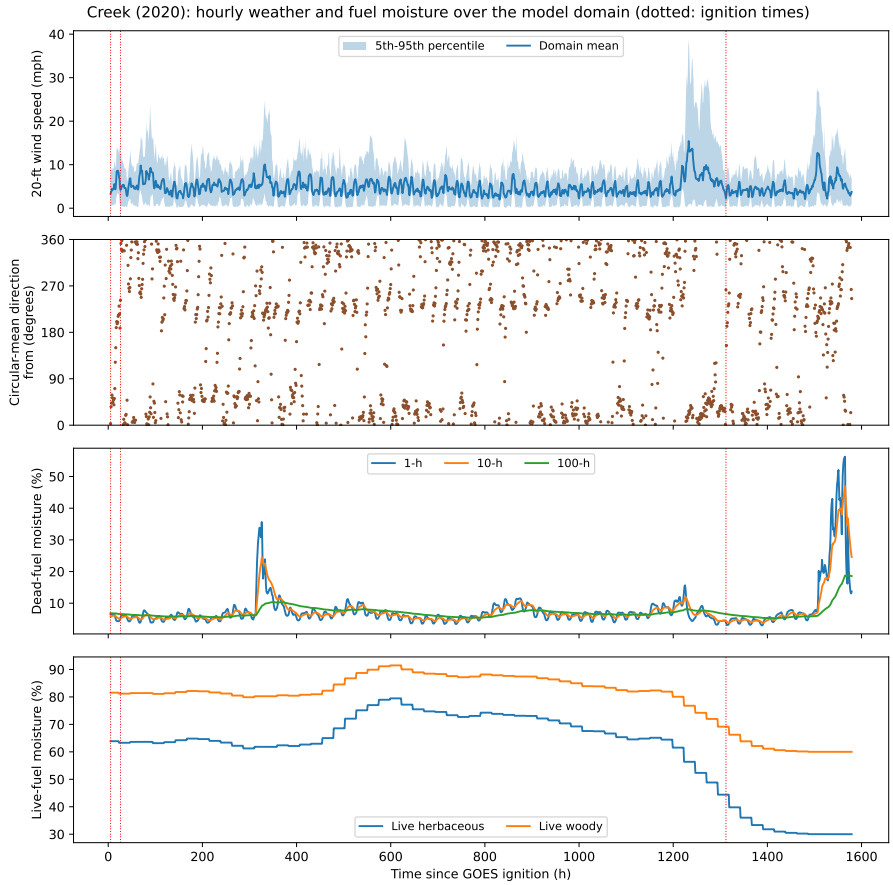


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

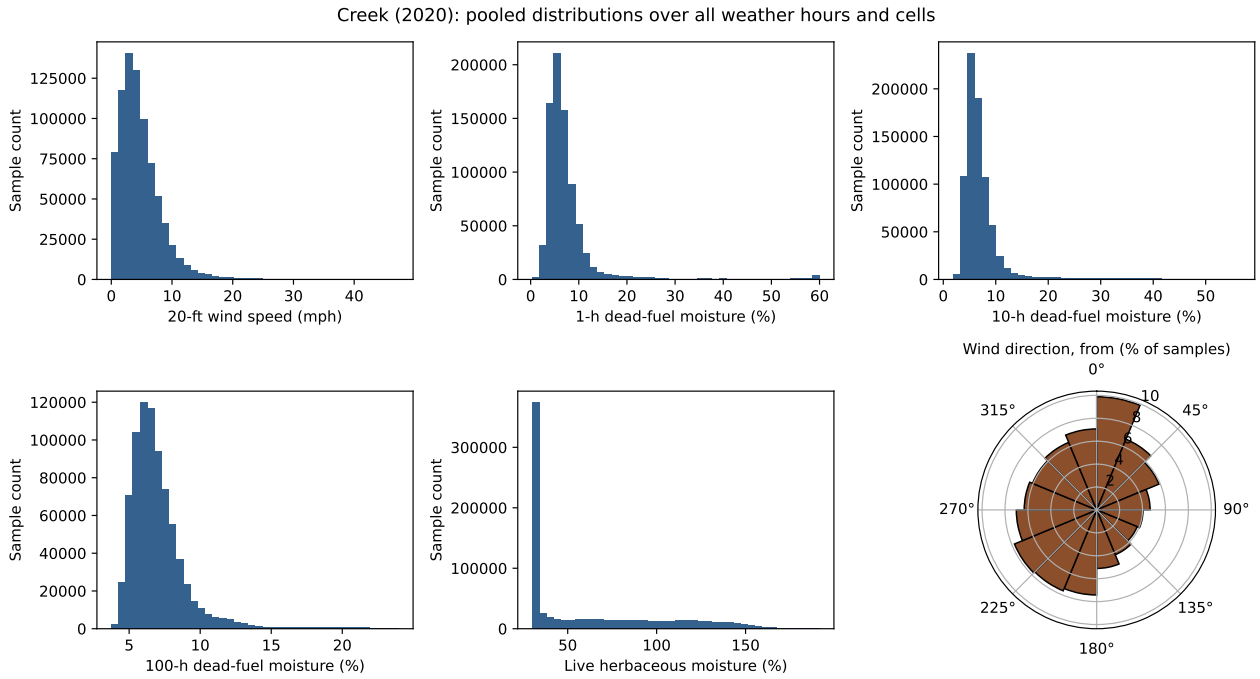


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1574 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1575 hourly weather fields and 3 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 5 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

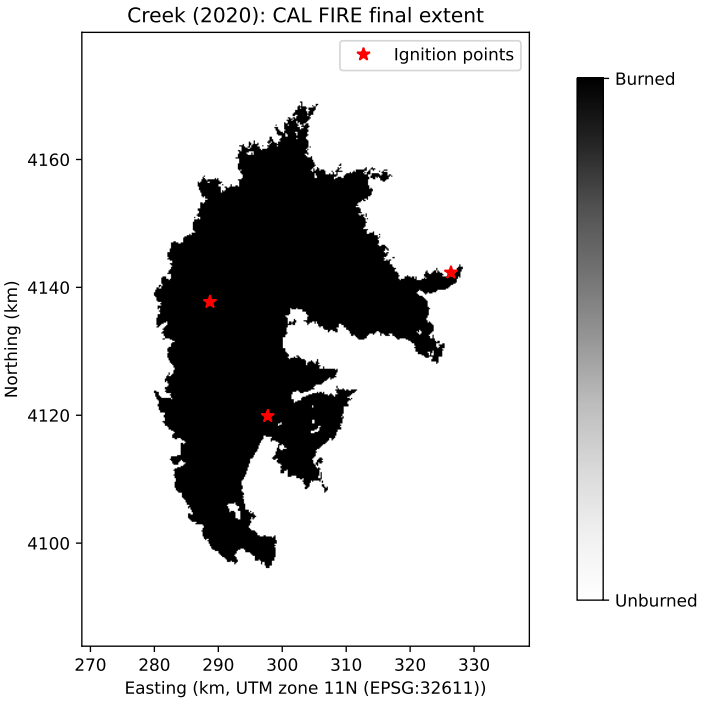


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 1,538 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Creek reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 CZU Lightning Complex Reconstruction against GOFER Hourly Perimeters

1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the CZU Lightning Complex (2020) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 134 h, from the first GOES fire detection at 16 August 2020 at 15:00 UTC to the last GOFER perimeter at 22 August 2020 at 05:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. The firebrand-spotting configuration was changed after the case owner inspected comparisons of this fire with GOFER, so results for this fire are not independent of that choice.

2 Reference event, experiment, and observations

GOFER records the CZU Lightning Complex from 16 August 2020 at 15:00 UTC to 22 August 2020 at 05:00 UTC with 125 hourly perimeters, a final perimeter area of 283 km^2 and an official size of 86,509 ac, and 3 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of 0.75 ± 0.21 km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 125 hourly perimeters of this fire on a 50-m grid in UTM zone 10N (EPSG:32610). The raster covers 283 km^2, consistent with the final GOFER perimeter area of 283 km^2; arrival times range from 10 to 134 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

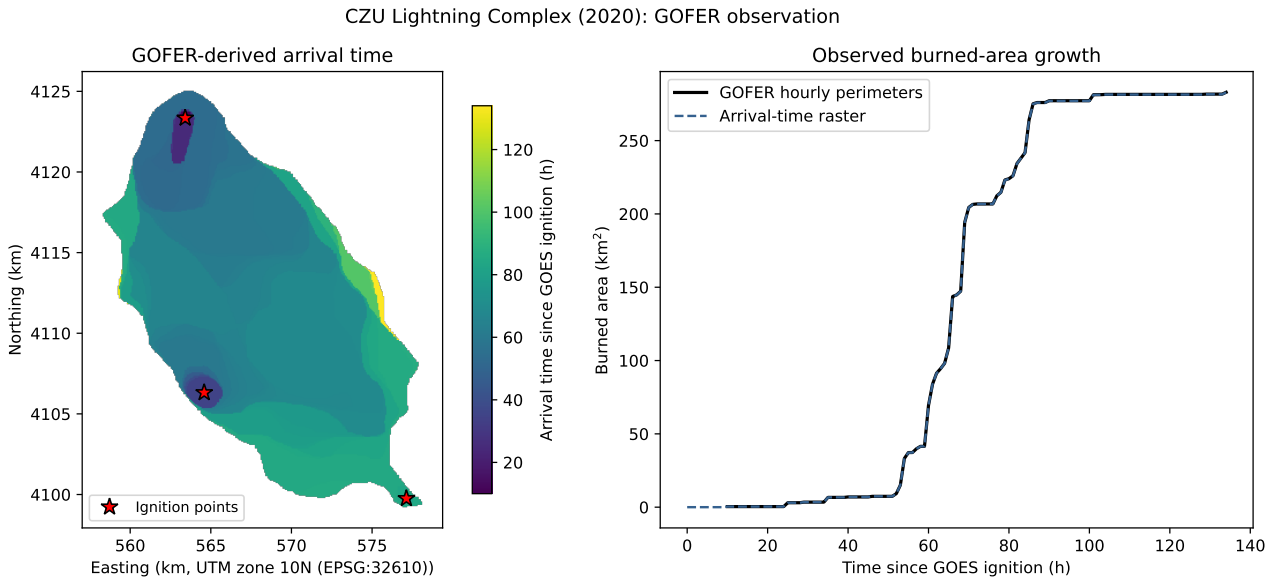


Figure 1: GOFER observation of the CZU Lightning Complex: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

3 Input data provenance and characterization

Input	Source and version	Resolution, units and processing
Terrain	LANDFIRE elevation, slope and aspect [3]	30 m; m and degrees; UTM zone 10N (EPSG:32610)
Surface fuels	LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.1.0	30 m categorical
Canopy	LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.1.0	30 m; percent, m, m and kg m^{-3}
Weather	Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6]	600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees
Fuel moisture	Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather	600 m hourly; percent; derivation method not documented in the available case materials
Ignition	GOFER ignition points [2]	Points with hourly time stamps
Adjustment and initial state	Spread-rate adjustment factor and initial level set	Uniform value of one (no adjustment, nothing burned at the start)

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 1,333 by 1,533 cells of 30 m (1,839 km^2) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from -1 to 985 m (median 148 m) and slope has a median of 9° and a 95th percentile of 31°; the circular-mean aspect

is 223°. By area the domain is 5.31% grass, 7.08% grass-shrub, 1.54% shrub, 9.06% timber-understory, 37.6% timber litter, 0.482% slash-blowdown and 38.9% non-burnable fuel models. Canopy cover has a median of 0% and canopy base height a median of 0 m. Offshore, 283,496 cells outside LANDFIRE coverage were assigned LANDFIRE open-water values; they are non-burnable.

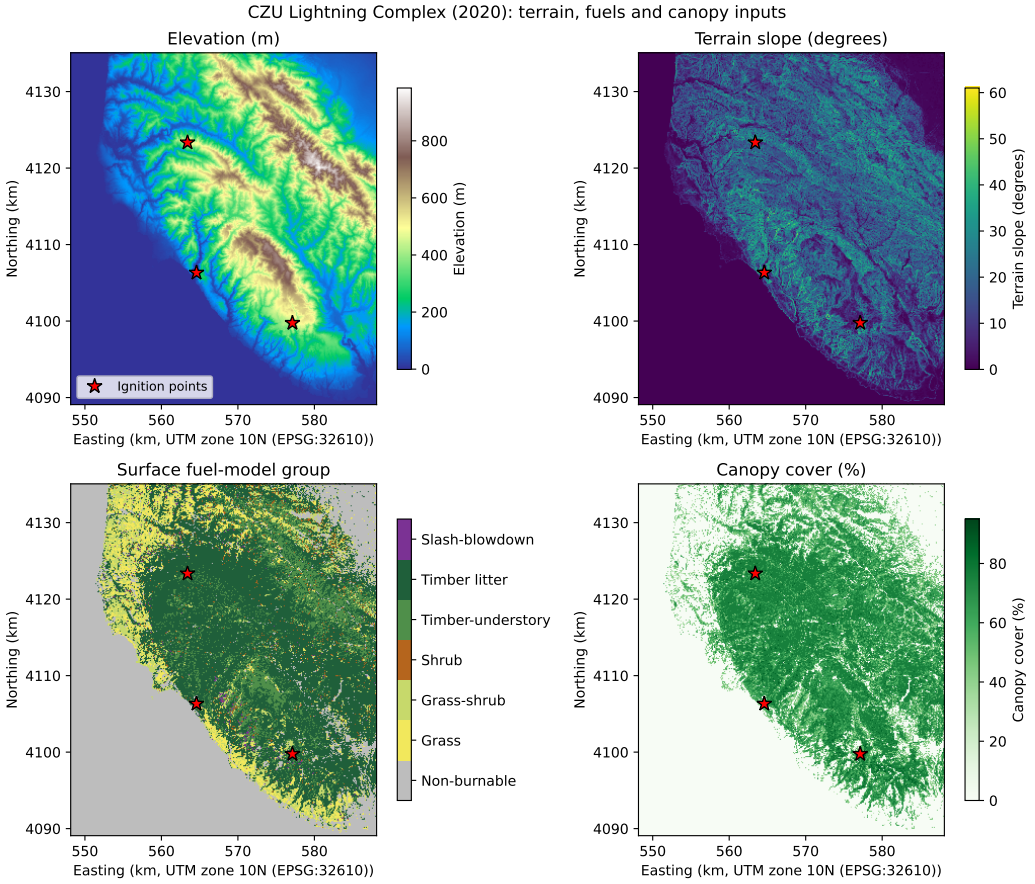


Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

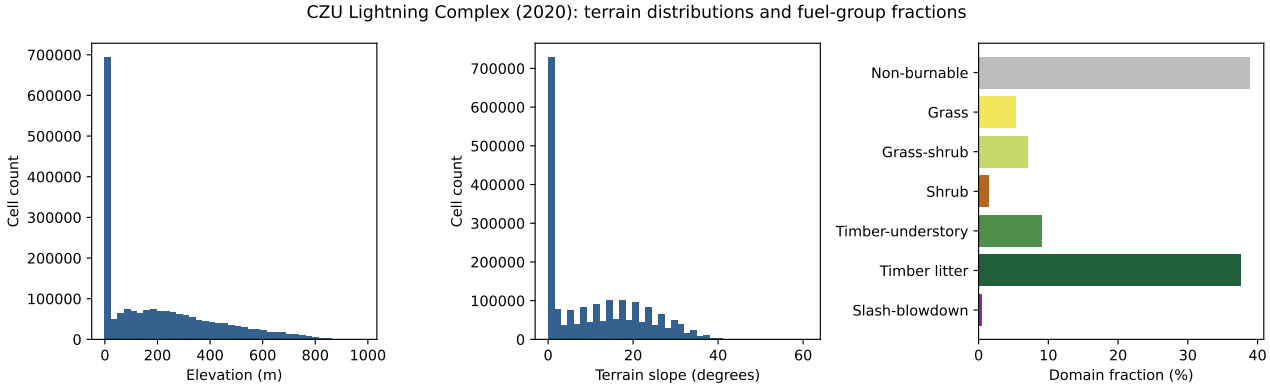


Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 126 hourly fields. Over all hours and cells the 20-ft wind speed has a median of 5.44 mph and a 95th percentile of 16.1 mph; the circular-mean direction the wind blows from is 328° with a mean resultant length of 0.732 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 15.3% (5th percentile 5.01%), and live herbaceous and woody moisture have medians of 160% and 143%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

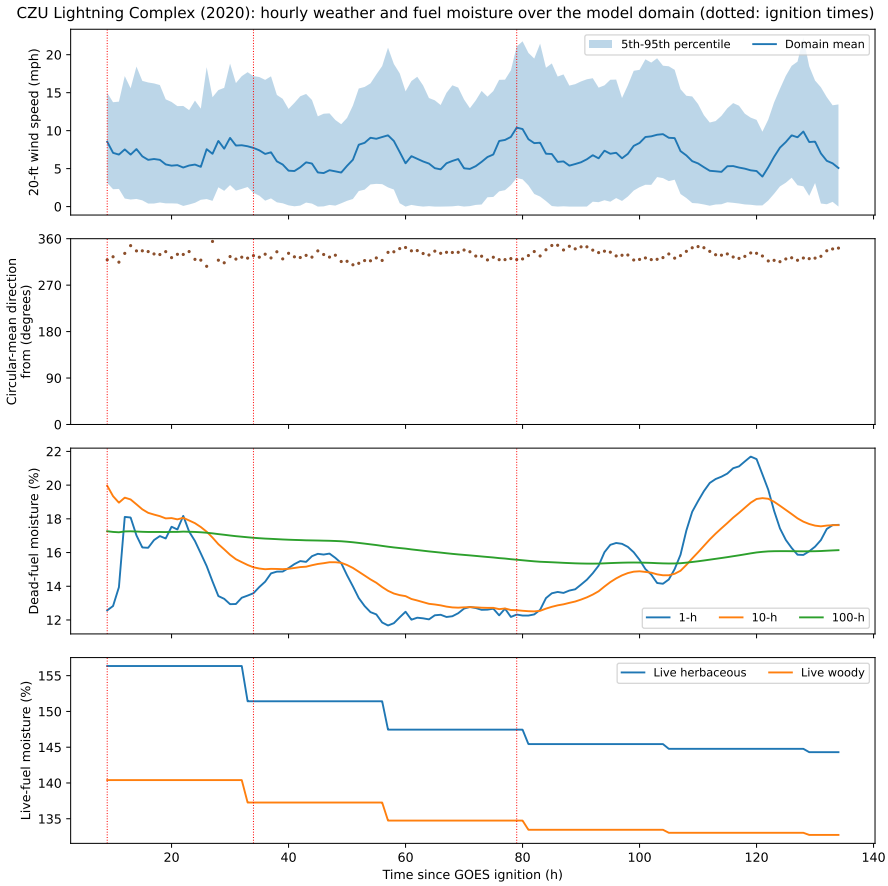


Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

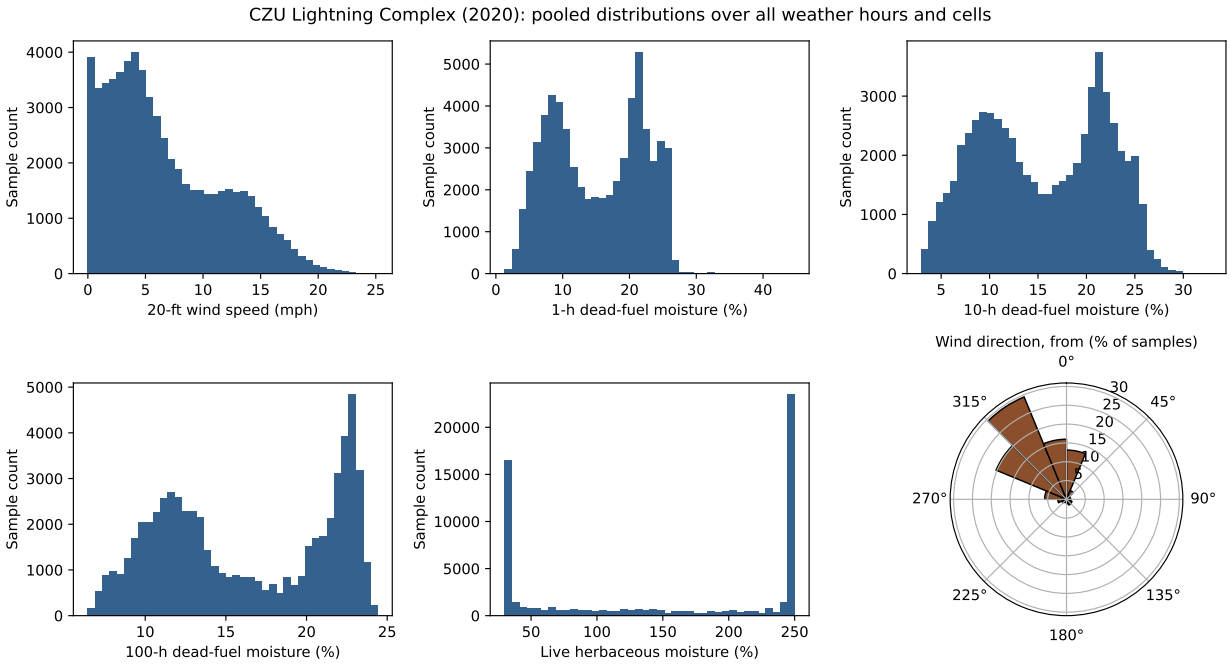


Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 125 h, to the last GOFER perimeter, with 3 point ignitions, each applied at its observed GOFER time. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft min^{-1}, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

Choice	Basis	Evidence and limitation
Ignition points and times	documented	GOFER ignition record; no point required moving
Simulation start and stop	documented	First GOFER ignition point and last GOFER perimeter
Fuel vintage	derived	Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire
Weather and moisture	user-specified	RTMA analysis through CloudFire, as stated by the case owner
Crown fire	documented	ELMFIRE default crown-fire treatment, set explicitly
Firebrand spotting	user-specified	Enabled with ELMFIRE default spotting settings until calibration
Domain margin	user-specified	10 km beyond the final GOFER perimeter
Single deterministic member	user-specified	No ensemble; stochastic spotting uses a fixed random seed

Table 2: Basis of consequential configuration choices.

5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 9 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. No simulation results are retained in this dataset. New results must record the executable identity and configuration used.

6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time t the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap J_t summarizes footprint agreement; precision P_t diagnoses commission (simulated burning outside the observation), recall R_t omission (observed burning the simulation misses), F_t balances the two, and the area ratio A_t exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

7 Simulation results and field-level comparison

The current status of this case is **NOT EVALUABLE**. No simulation result is available for this case.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics. The fire burned near the coast, where the marine layer and fog influenced late growth; the 2.5-km weather analysis may not resolve these effects.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the CZU Lightning Complex. Inputs and the observation are characterized in full; the current scientific status is **NOT EVALUABLE** with 0 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

References

[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

2021 Dixie fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Dixie (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 14 July 2021 at 00:00 UTC to 10 September 2021 at 03:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1390 hourly perimeters, a final area of 4389.17 km^2 and an official size of 963,309 ac. The configuration uses 7 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

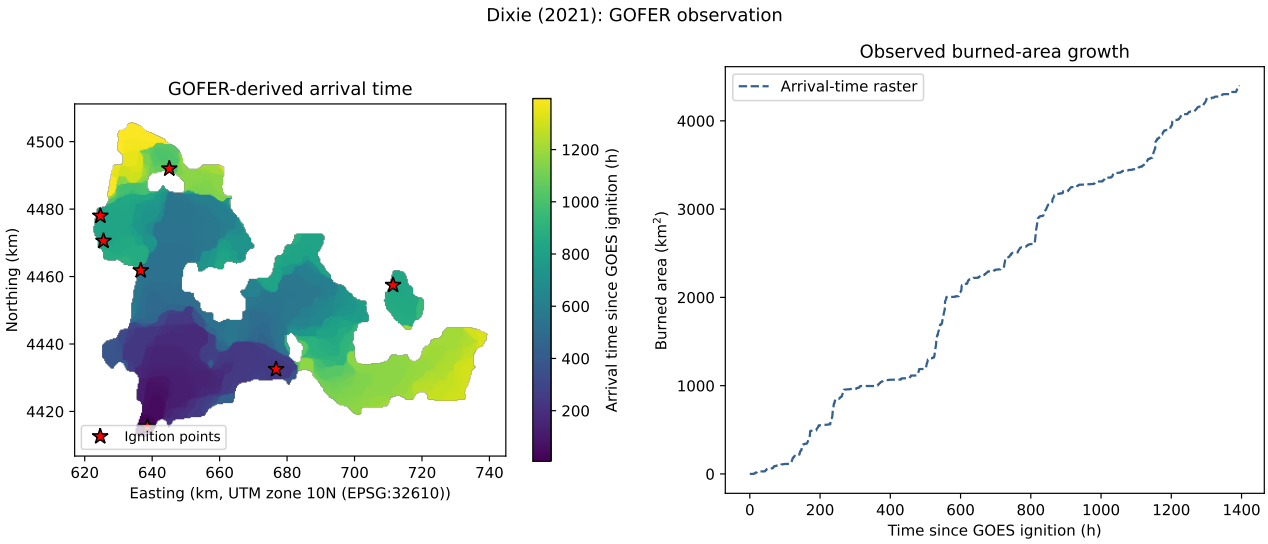


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 4,667 by 3,867 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 16,243 km^2. Elevation ranges from 127 to 3,175 m; median slope is 8°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 35%, and canopy base height has median 0.9 m.

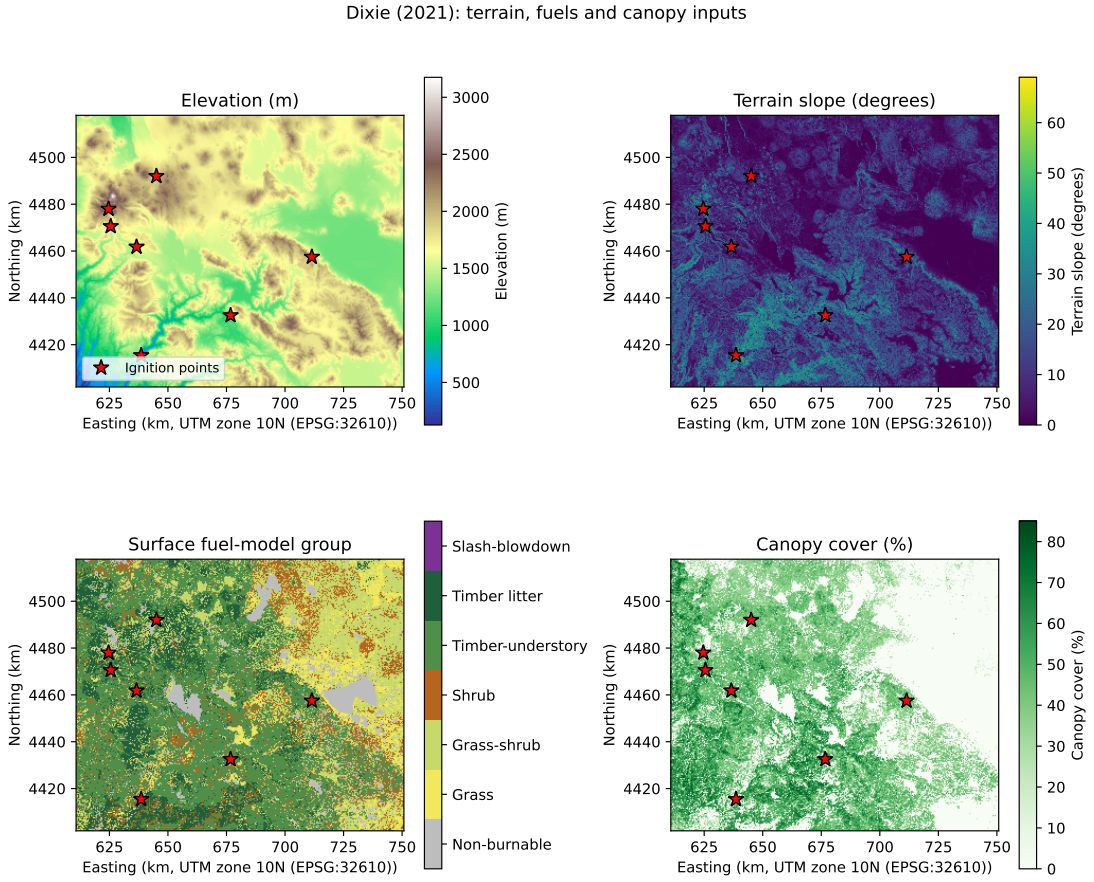


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

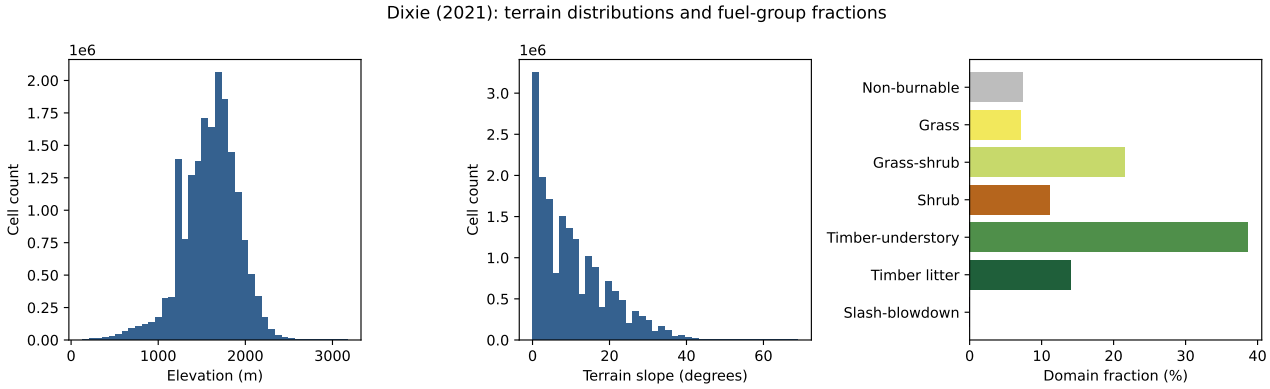


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,391 hourly fields. Median wind speed is 3.78 mph and its 95th percentile is 10.3 mph. Direction is summarized circularly: the mean direction is 262° and mean resultant length is 0.206. Median 1-h dead-fuel moisture is 5.43%. Spatial and temporal variability remain important when these domain summaries are interpreted.

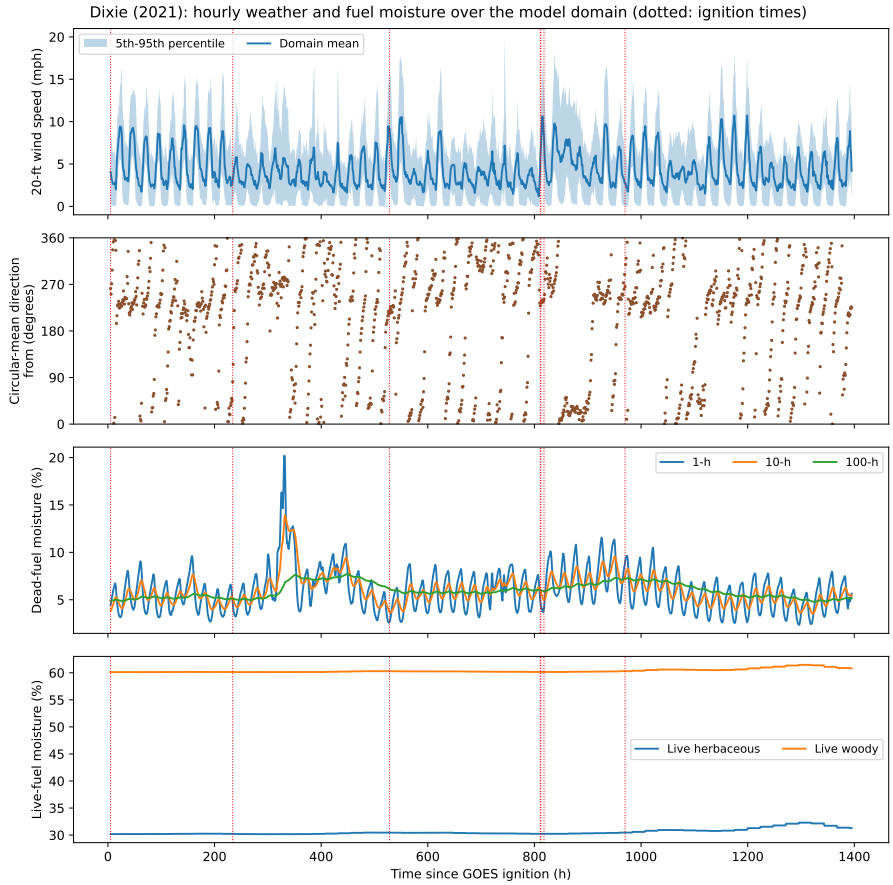


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

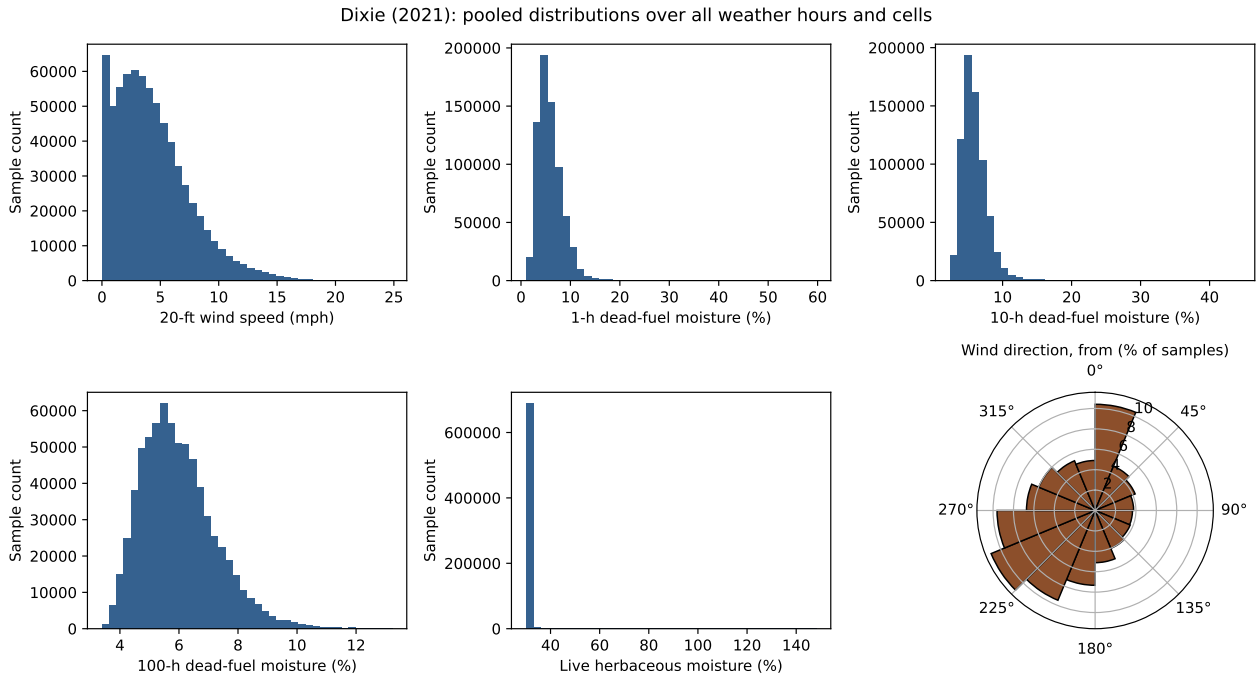


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1390 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1391 hourly weather fields and 7 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 5 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

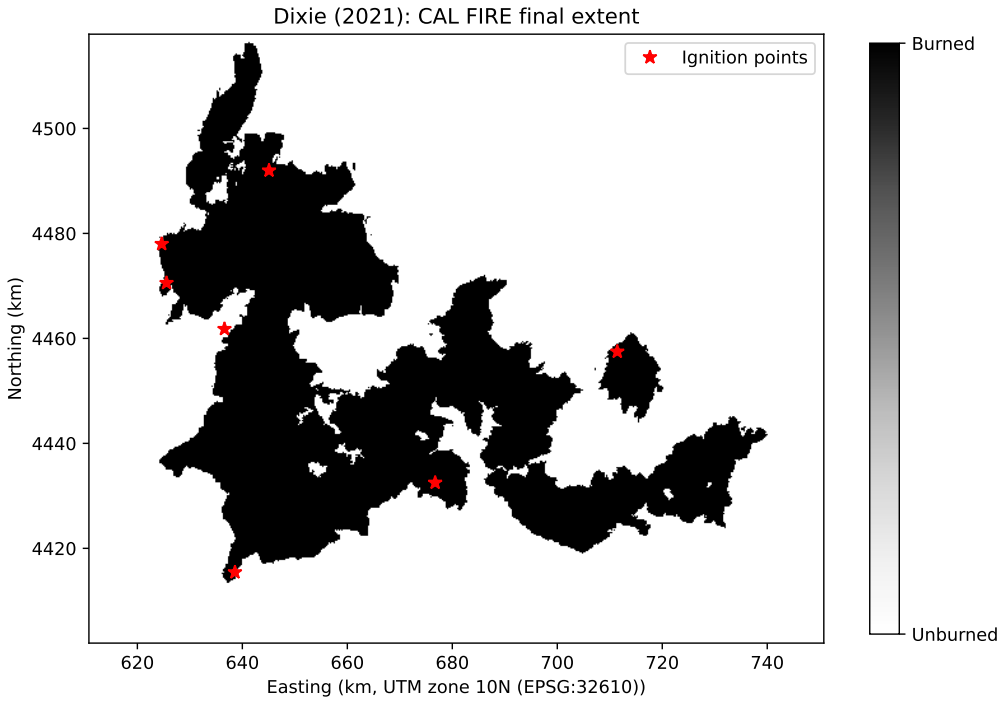


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 3,898 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Dixie reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 Dolan fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Dolan (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 18 August 2020 at 18:00 UTC to 17 September 2020 at 02:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 696 hourly perimeters, a final area of 501.46 km^2 and an official size of 124,924 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

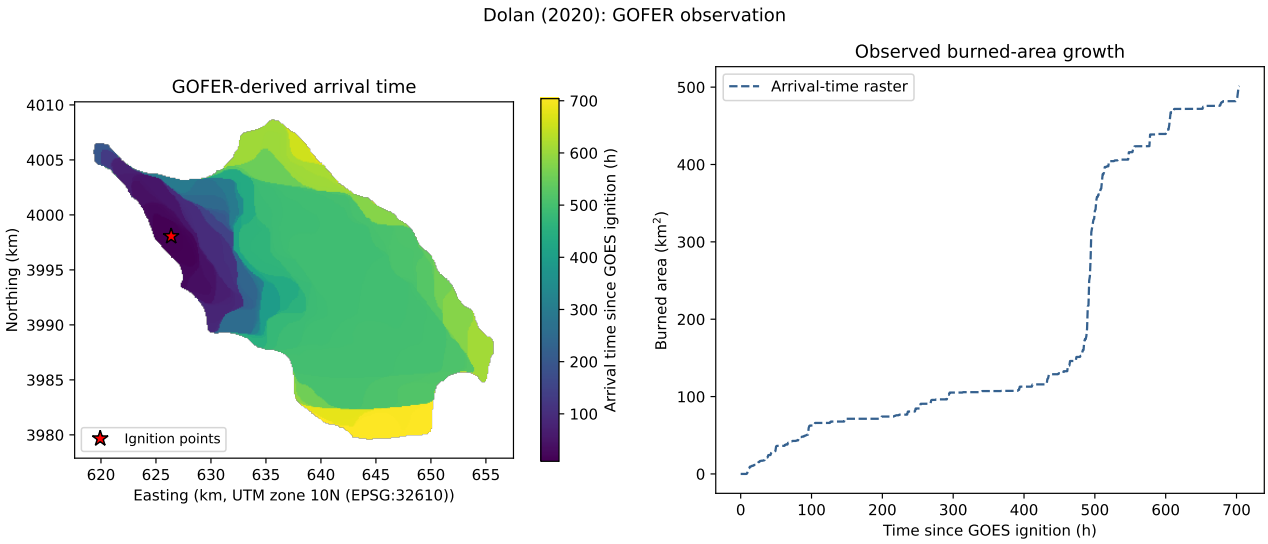


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,933 by 1,667 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 2,900 km^2. Elevation ranges from -1 to 1,780 m; median slope is 15°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

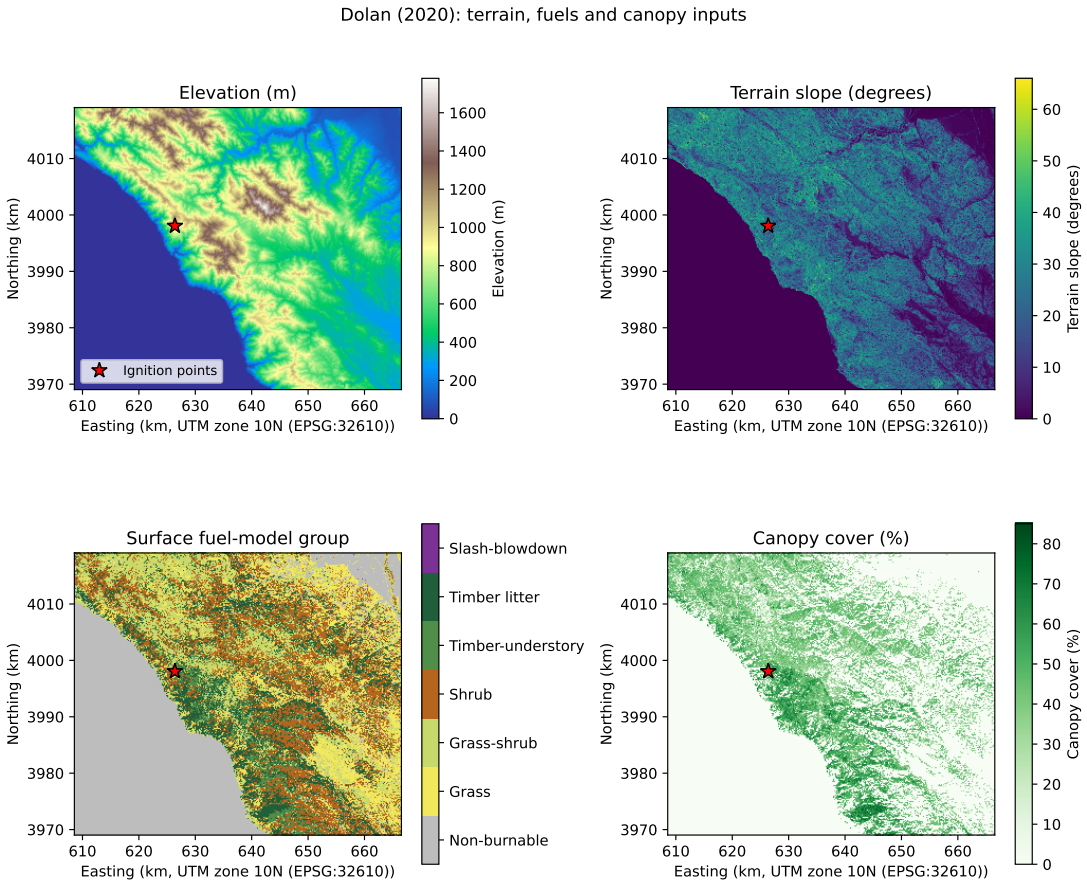


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

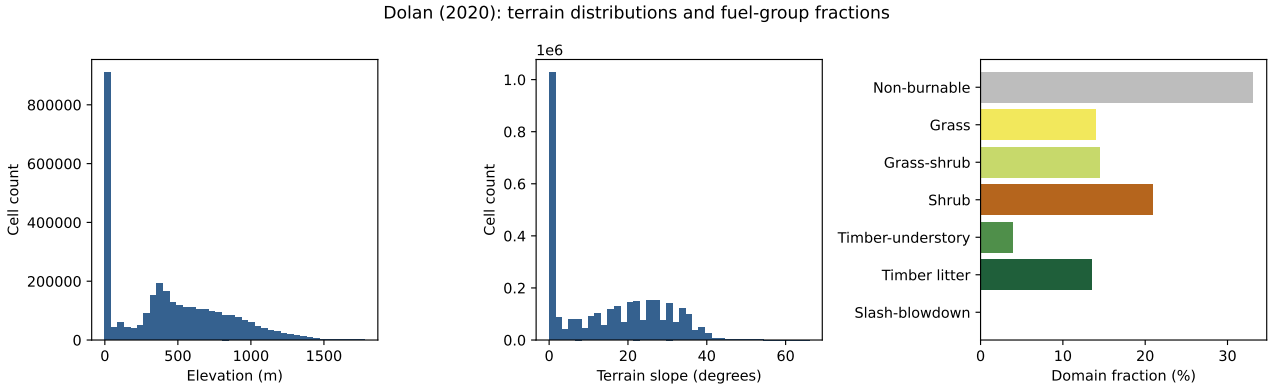


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 697 hourly fields. Median wind speed is 4.21 mph and its 95th percentile is 14.8 mph. Direction is summarized circularly: the mean direction is 331° and mean resultant length is 0.177. Median 1-h dead-fuel moisture is 10.7%. Spatial and temporal variability remain important when these domain summaries are interpreted.

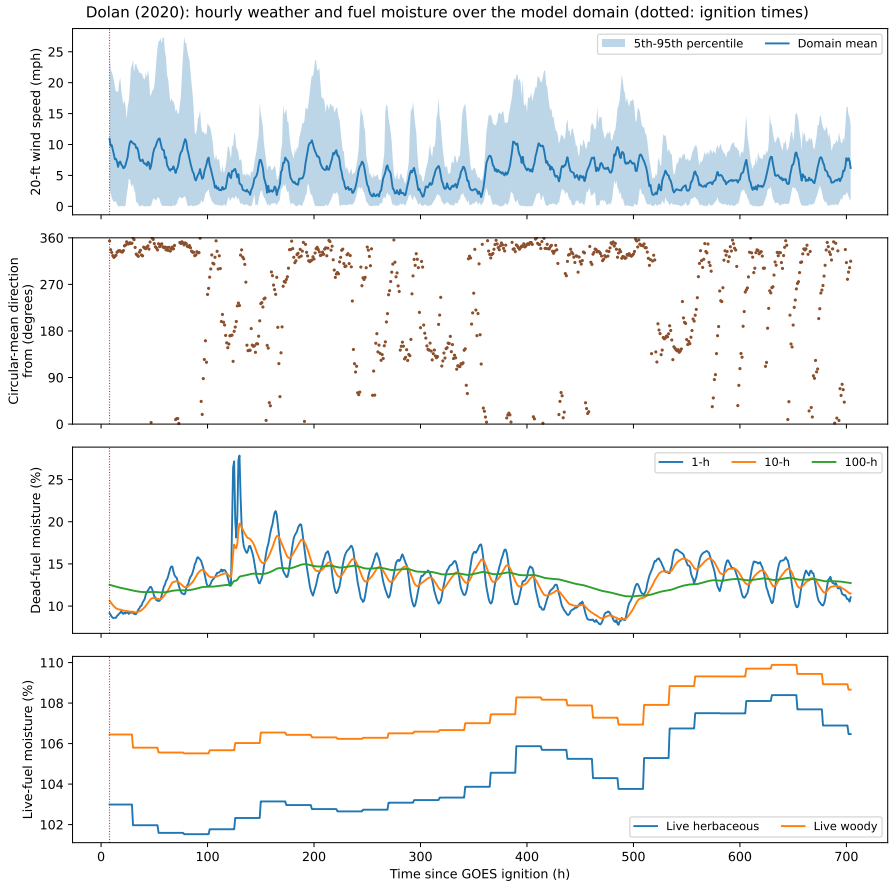


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

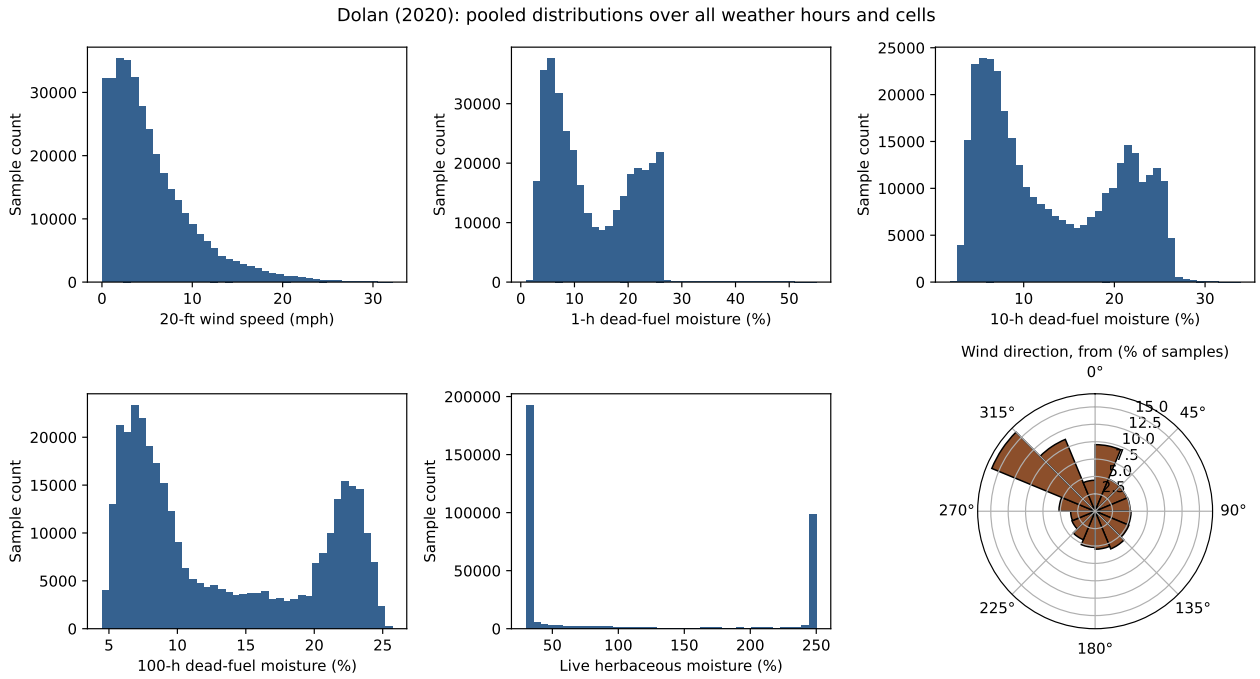


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 696 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 697 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 8 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

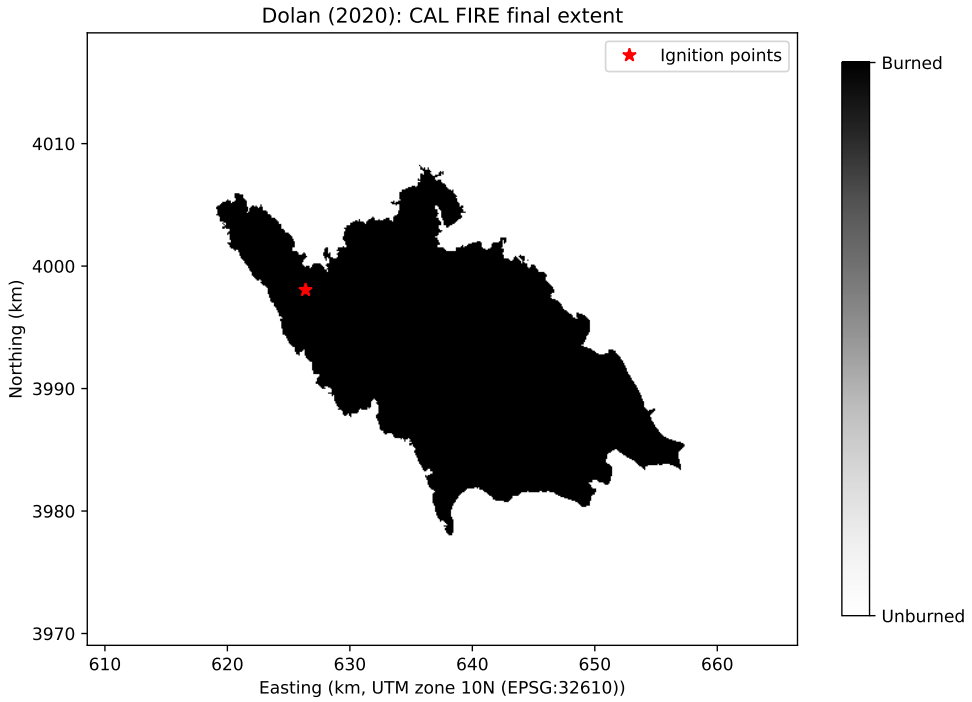


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 504 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Dolan reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 Glass Fire Reconstruction against GOFER Hourly Perimeters

1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Glass Fire (2020) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 189 h, from the first GOES fire detection at 27 September 2020 at 10:00 UTC to the last GOFER perimeter at 5 October 2020 at 07:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. No model parameter was tuned against the GOFER observation of this fire; the spotting configuration was revised after comparisons for the Kincade and CZU Lightning Complex fires.

2 Reference event, experiment, and observations

GOFER records the Glass Fire from 27 September 2020 at 10:00 UTC to 5 October 2020 at 07:00 UTC with 189 hourly perimeters, a final perimeter area of 353 km^2 and an official size of 67,484 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of 0.75 ± 0.21 km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 189 hourly perimeters of this fire on a 50-m grid in UTM zone 10N (EPSG:32610). The raster covers 352 km^2, consistent with the final GOFER perimeter area of 353 km^2; arrival times range from 1 to 189 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

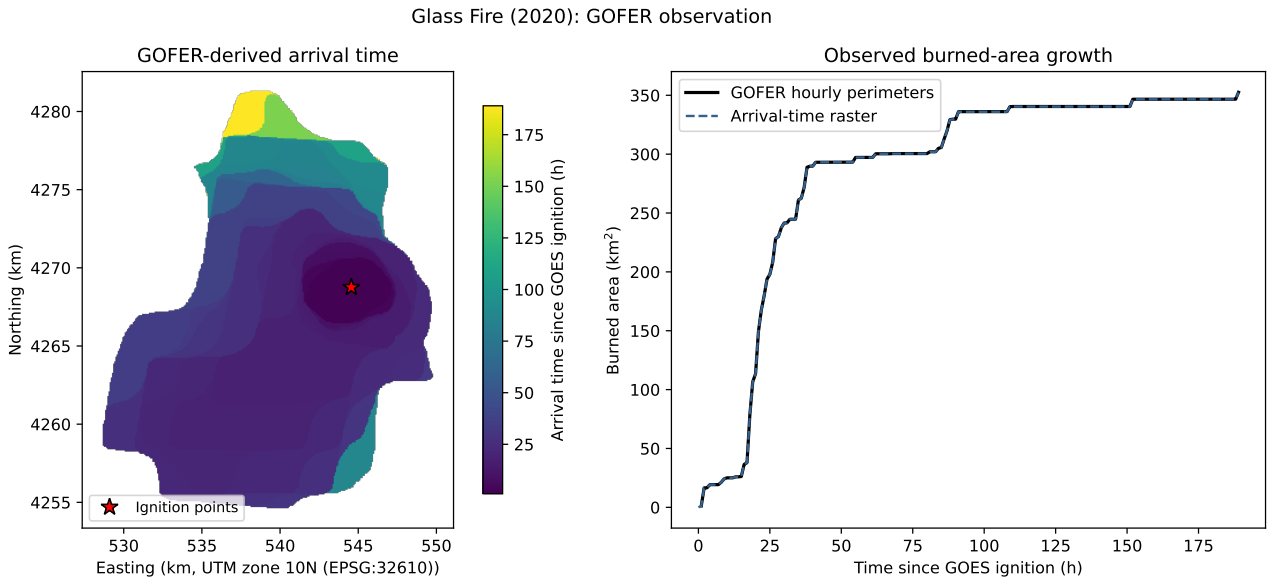


Figure 1: GOFER observation of the Glass Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

3 Input data provenance and characterization

Input	Source and version	Resolution, units and processing
Terrain	LANDFIRE elevation, slope and aspect [3]	30 m; m and degrees; UTM zone 10N (EPSG:32610)
Surface fuels	LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.1.0	30 m categorical
Canopy	LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.1.0	30 m; percent, m, m and kg m^{-3}
Weather	Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6]	600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees
Fuel moisture	Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather	600 m hourly; percent; derivation method not documented in the available case materials
Ignition	GOFER ignition points [2]	Points with hourly time stamps
Adjustment and initial state	Spread-rate adjustment factor and initial level set	Uniform value of one (no adjustment, nothing burned at the start)

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 1,400 by 1,600 cells of 30 m (2,016 km^2) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 14 to 1,318 m (median 275 m) and slope has a median of 12° and a 95th percentile of 29°; the circular-mean

aspect is 214°. By area the domain is 14.1% grass, 11.4% grass-shrub, 17.4% shrub, 16.9% timber-understory, 22% timber litter, 0.0231% slash-blowdown and 18.2% non-burnable fuel models. Canopy cover has a median of 0% and canopy base height a median of 0 m.

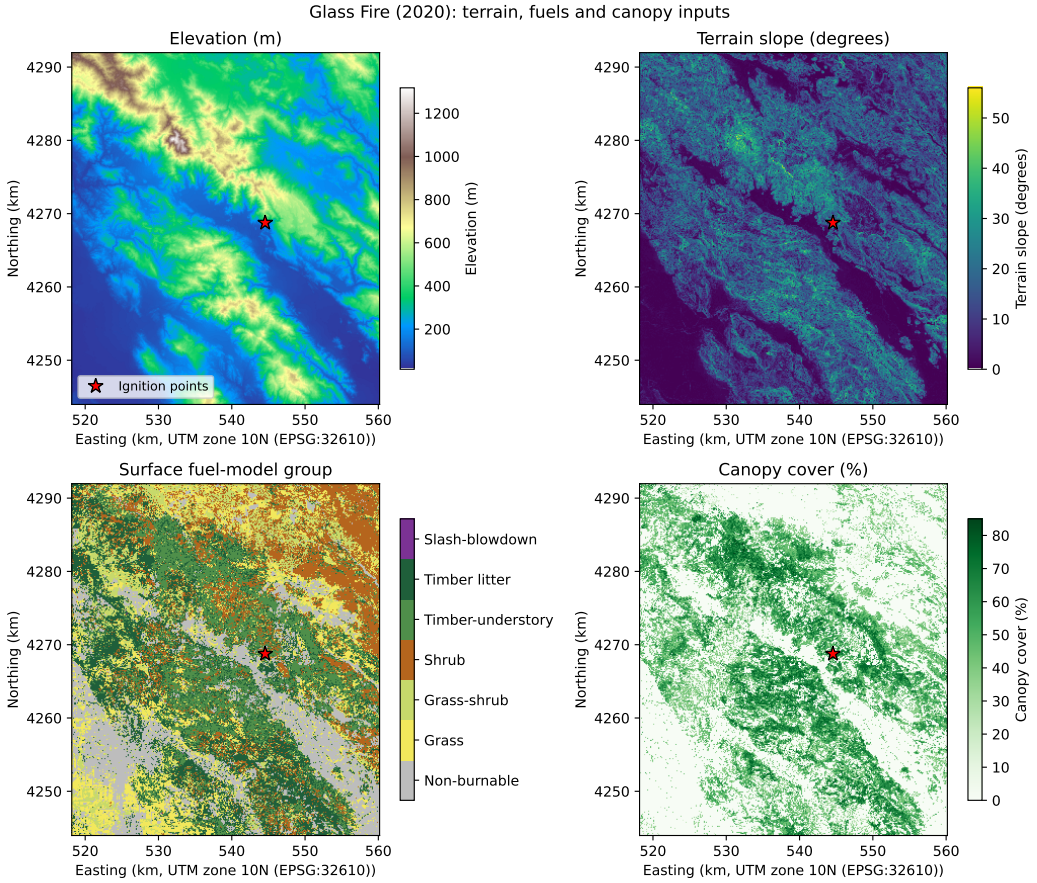


Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

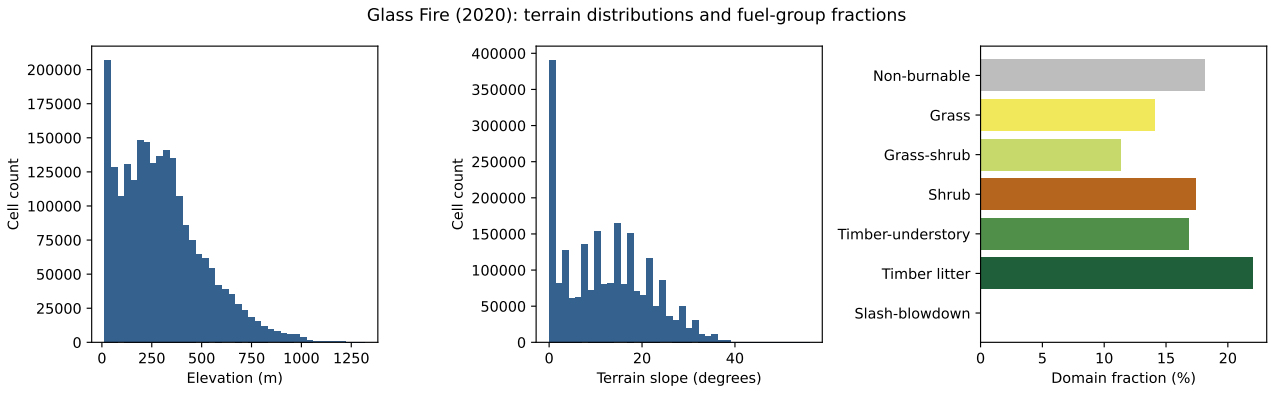


Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 190 hourly fields. Over all hours and cells the 20-ft wind speed has a median of 2.96 mph and a 95th percentile of 10.6 mph; the circular-mean direction the wind blows from is 322° with a mean resultant length of 0.194 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 5.61% (5th percentile 2.87%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

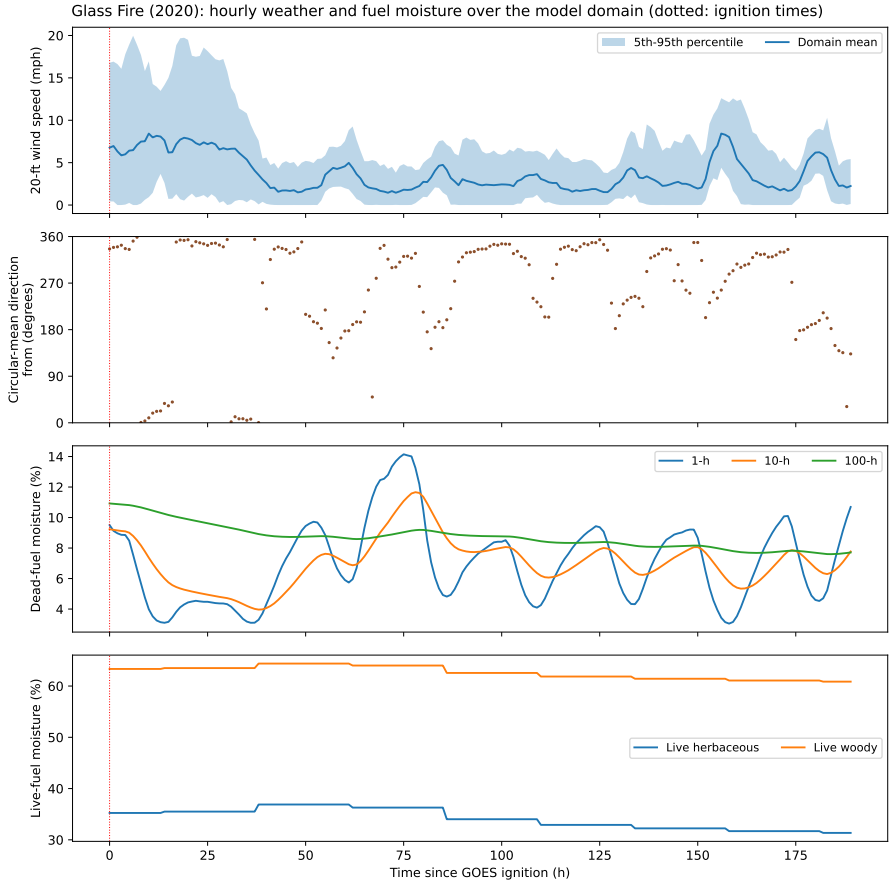


Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

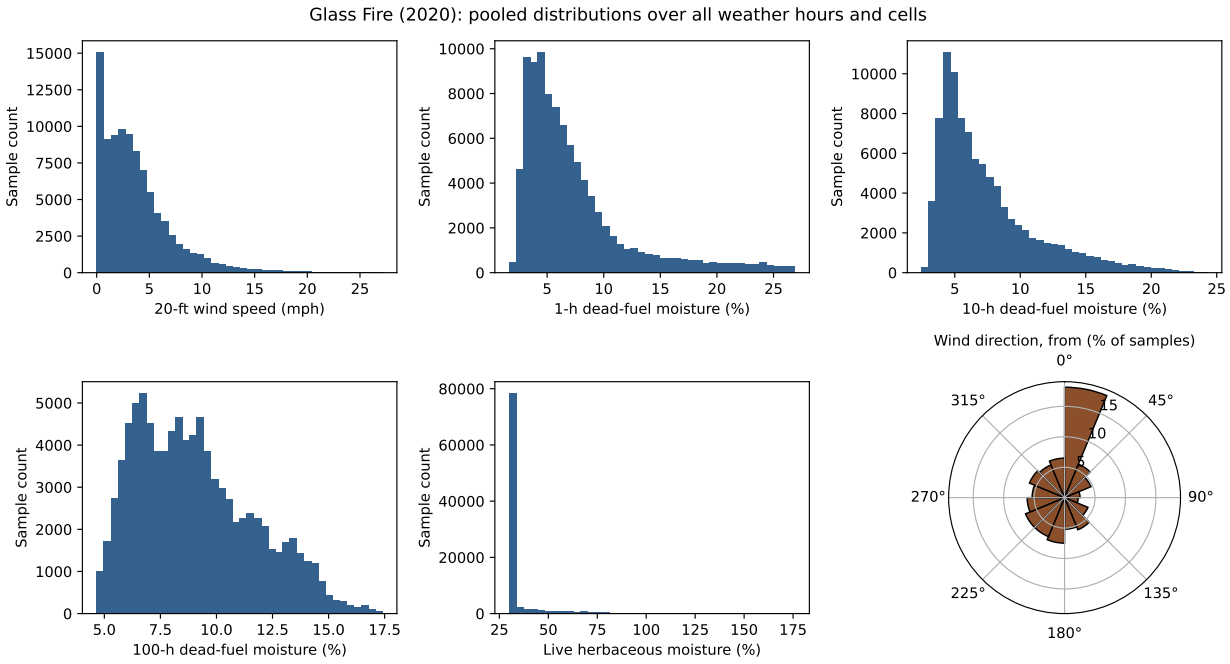


Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 189 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft min^{-1}, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

Choice	Basis	Evidence and limitation
Ignition points and times	documented	GOFER ignition record; no point required moving
Simulation start and stop	documented	First GOFER ignition point and last GOFER perimeter
Fuel vintage	derived	Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire
Weather and moisture	user-specified	RTMA analysis through CloudFire, as stated by the case owner
Crown fire	documented	ELMFIRE default crown-fire treatment, set explicitly
Firebrand spotting	user-specified	Enabled with ELMFIRE default spotting settings until calibration
Domain margin	user-specified	10 km beyond the final GOFER perimeter
Single deterministic member	user-specified	No ensemble; stochastic spotting uses a fixed random seed

Table 2: Basis of consequential configuration choices.

5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 0 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. No simulation results are retained in this dataset. New results must record the executable identity and configuration used.

6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time t the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap J_t summarizes footprint agreement; precision P_t diagnoses commission (simulated burning outside the observation), recall R_t omission (observed burning the simulation misses), F_t balances the two, and the area ratio A_t exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

7 Simulation results and field-level comparison

The current status of this case is **NOT EVALUABLE**. No simulation result is available for this case.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Glass Fire. Inputs and the observation are characterized in full; the current scientific status is **NOT EVALUABLE** with 0 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

References

[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

2020 July Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the July Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 22 July 2020 at 17:00 UTC to 31 July 2020 at 03:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 198 hourly perimeters, a final area of 173.948 km^2 and an official size of 83,261 ac. The configuration uses 3 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

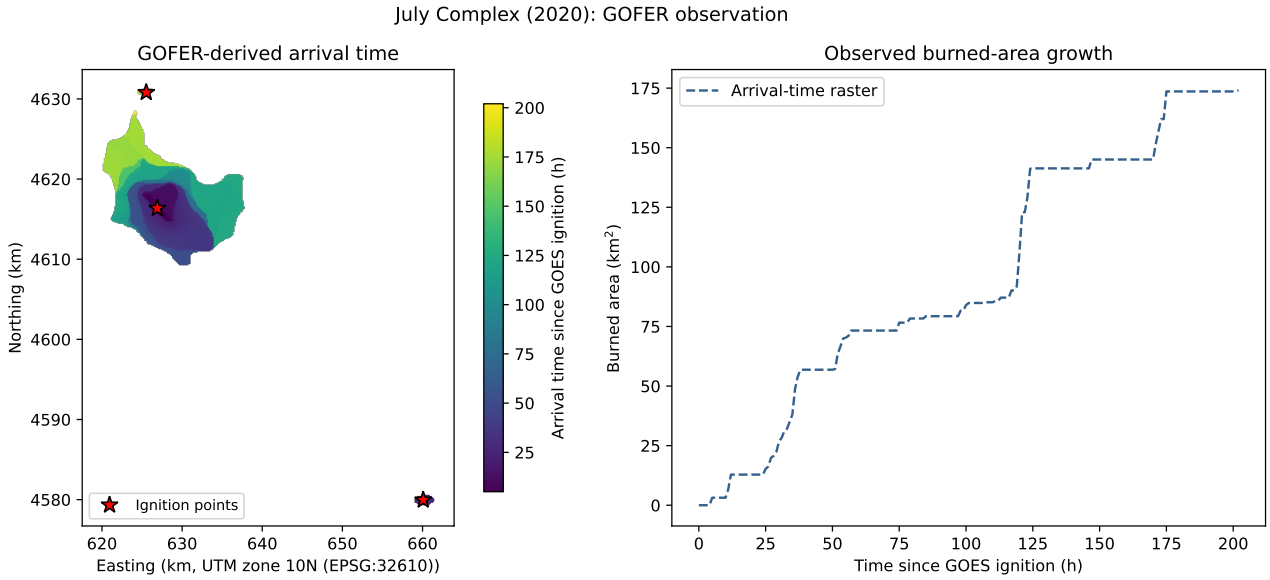


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,067 by 2,467 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 4,589 km^2. Elevation ranges from 1,127 to 2,407 m; median slope is 1°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

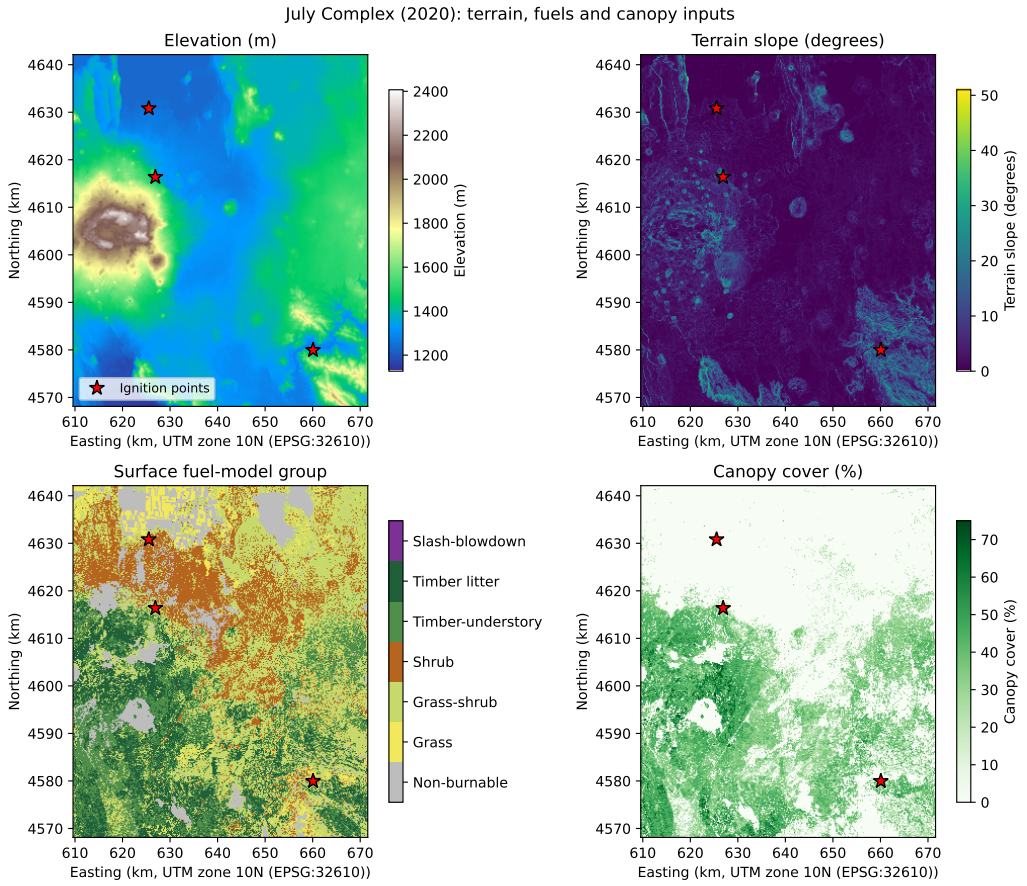


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

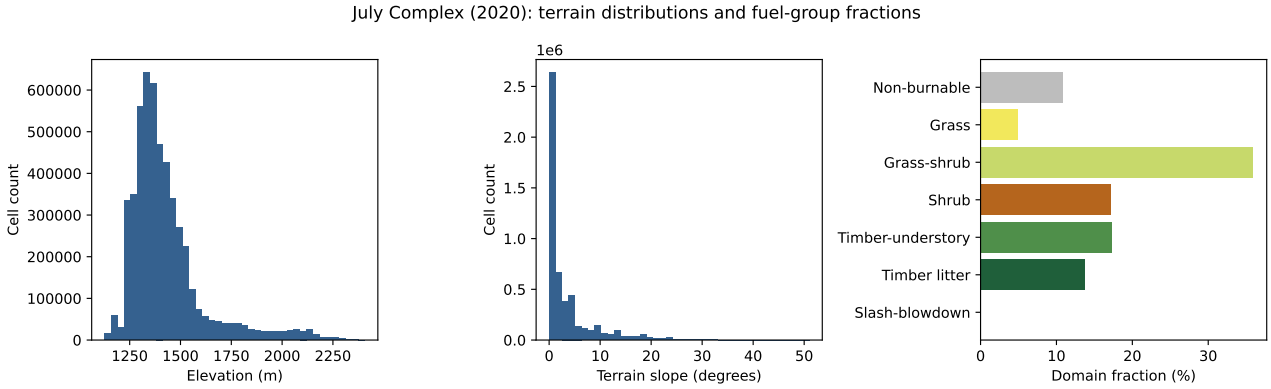


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 199 hourly fields. Median wind speed is 3.9 mph and its 95th percentile is 10.5 mph. Direction is summarized circularly: the mean direction is 268° and mean resultant length is 0.261. Median 1-h dead-fuel moisture is 7.25%. Spatial and temporal variability remain important when these domain summaries are interpreted.

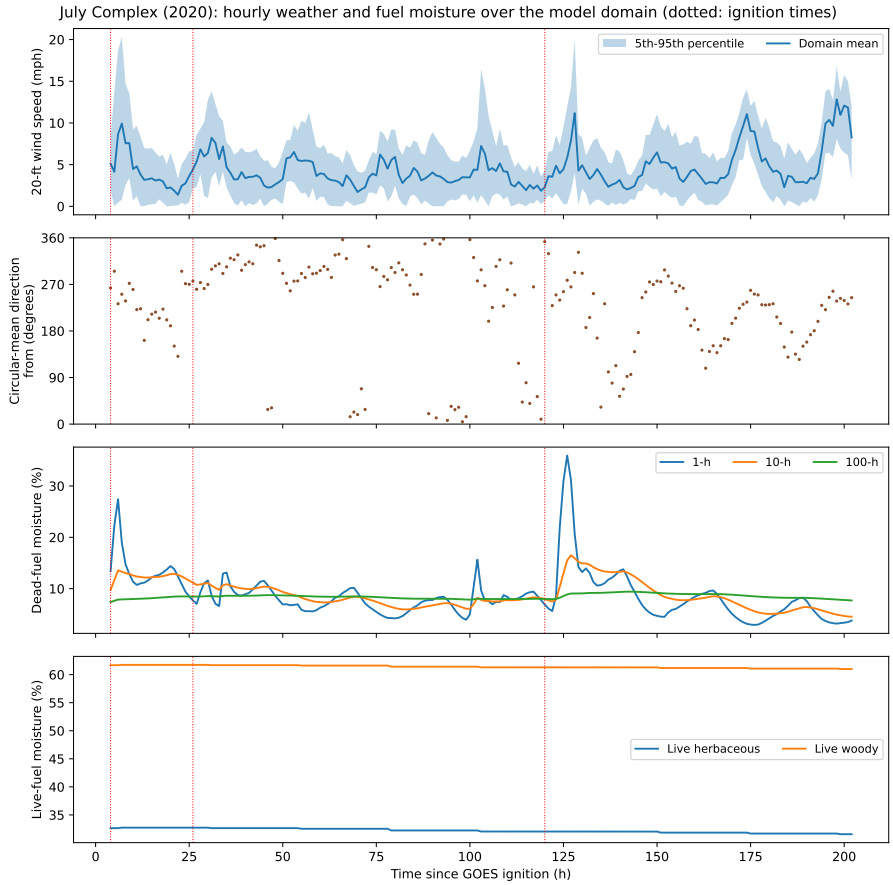


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

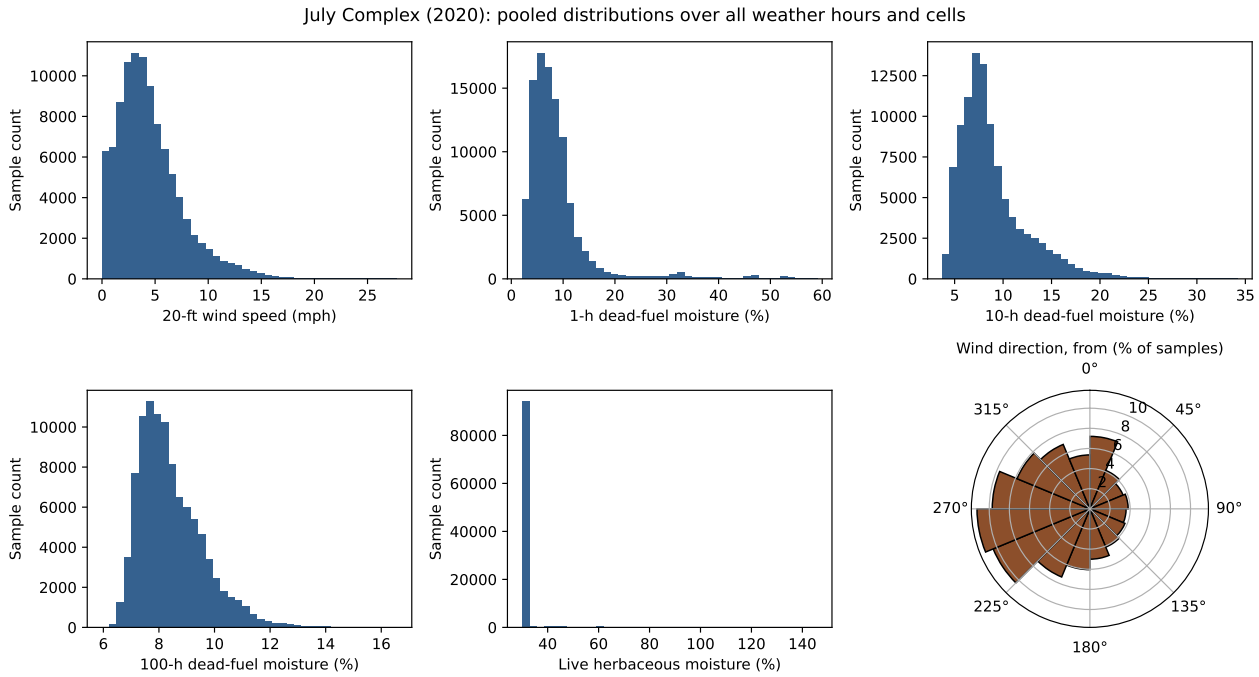


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 198 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 199 hourly weather fields and 3 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 4 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

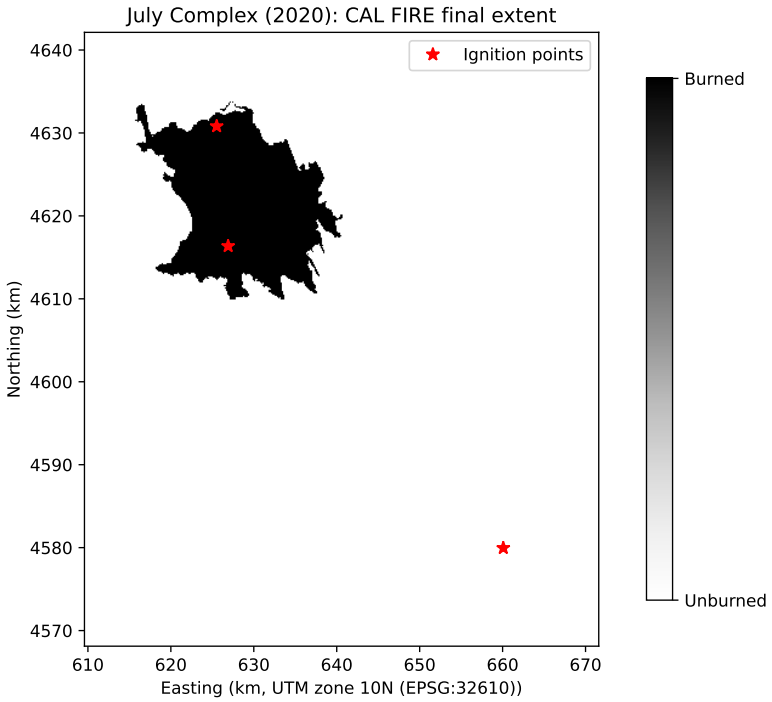


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 329 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the July Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2019 Kincade Fire Reconstruction against GOFER Hourly Perimeters

1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Kincade Fire (2019) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 134 h, from the first GOES fire detection at 24 October 2019 at 04:00 UTC to the last GOFER perimeter at 29 October 2019 at 18:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. The firebrand-spotting configuration was changed after the case owner inspected comparisons of this fire with GOFER, so results for this fire are not independent of that choice.

2 Reference event, experiment, and observations

GOFER records the Kincade Fire from 24 October 2019 at 04:00 UTC to 29 October 2019 at 18:00 UTC with 134 hourly perimeters, a final perimeter area of 347 km^2 and an official size of 77,758 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of 0.75 ± 0.21 km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 134 hourly perimeters of this fire on a 50-m grid in UTM zone 10N (EPSG:32610). The raster covers 347 km^2, consistent with the final GOFER perimeter area of 347 km^2; arrival times range from 1 to 134 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

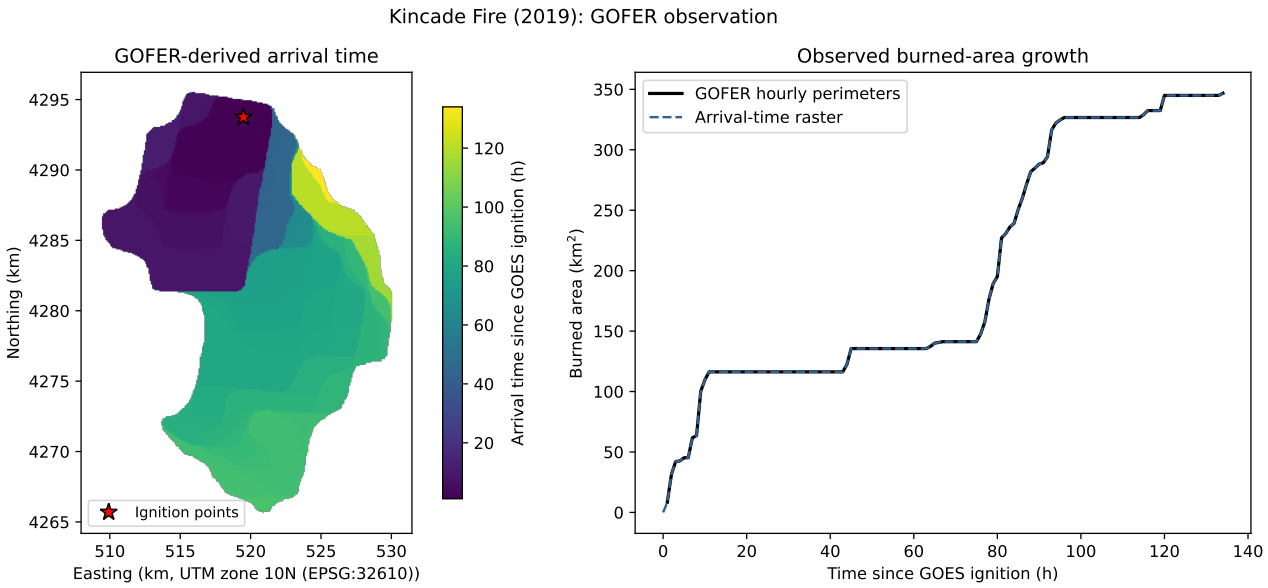


Figure 1: GOFER observation of the Kincade Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

3 Input data provenance and characterization

Input	Source and version	Resolution, units and processing
Terrain	LANDFIRE elevation, slope and aspect [3]	30 m; m and degrees; UTM zone 10N (EPSG:32610)
Surface fuels	LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.0.0_2019	30 m categorical
Canopy	LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.0.0_2019	30 m; percent, m, m and kg m^{-3}
Weather	Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6]	600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees
Fuel moisture	Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather	600 m hourly; percent; derivation method not documented in the available case materials
Ignition	GOFER ignition points [2]	Points with hourly time stamps
Adjustment and initial state	Spread-rate adjustment factor and initial level set	Uniform value of one (no adjustment, nothing burned at the start)

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 1,400 by 1,667 cells of 30 m (2,100 km^2) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 2 to 1,442 m (median 314 m) and slope has a median of 14° and a 95th percentile of 30°; the circular-mean

aspect is 201°. By area the domain is 16.3% grass, 7.73% grass-shrub, 12% shrub, 29.5% timber-understory, 18% timber litter, 0.137% slash-blowdown and 16.4% non-burnable fuel models. Canopy cover has a median of 35% and canopy base height a median of 0.3 m.

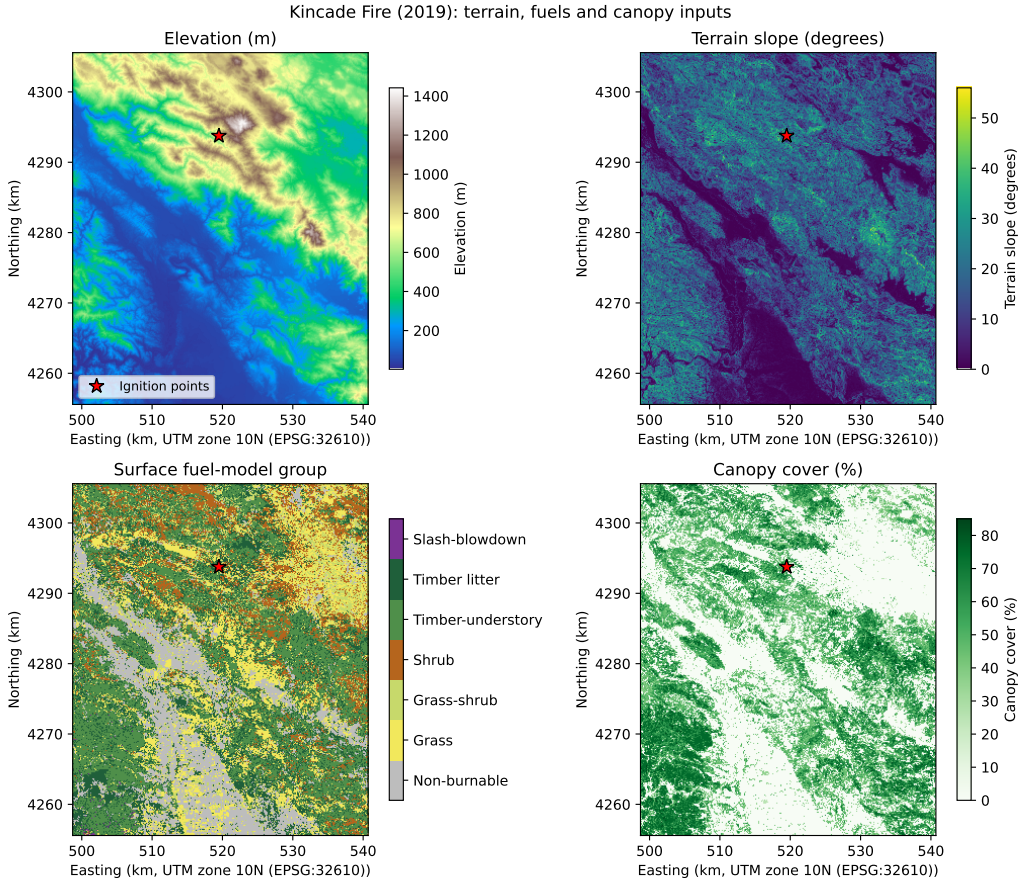


Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

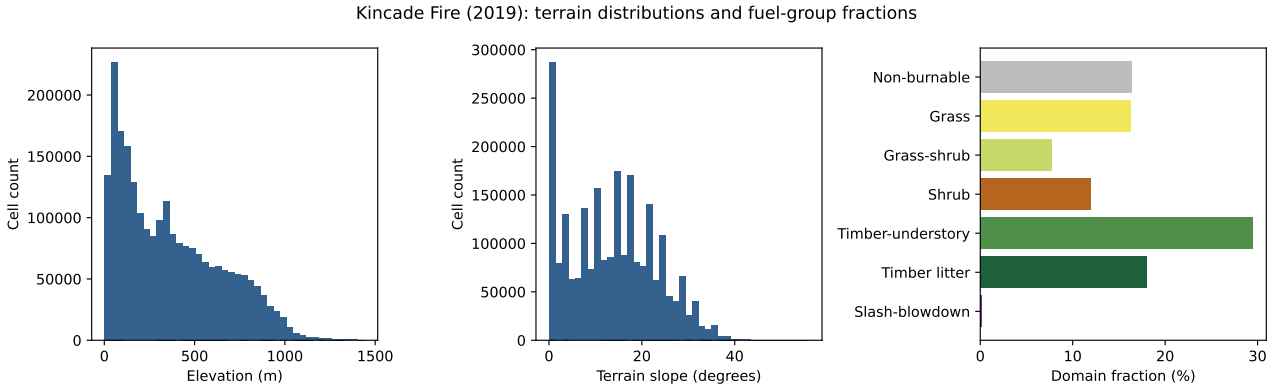


Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 135 hourly fields. Over all hours and cells the 20-ft wind speed has a median of 4.1 mph and a 95th percentile of 19.2 mph; the circular-mean direction the wind blows from is 1.42° with a mean resultant length of 0.309 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 4.57% (5th percentile 2.75%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

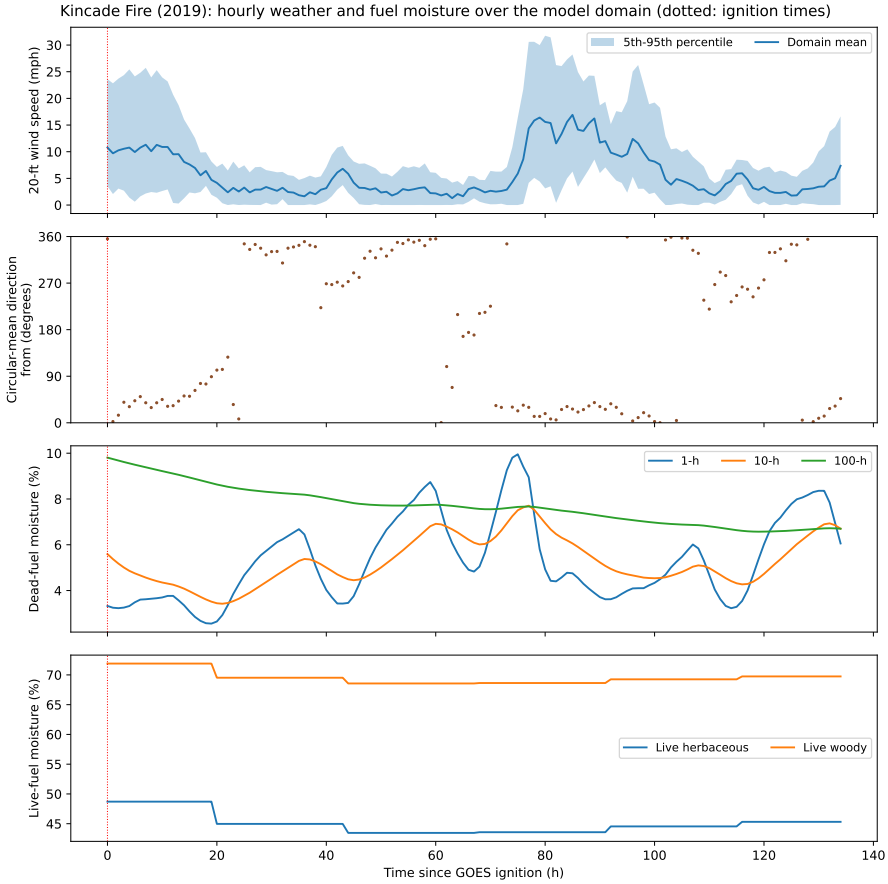


Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

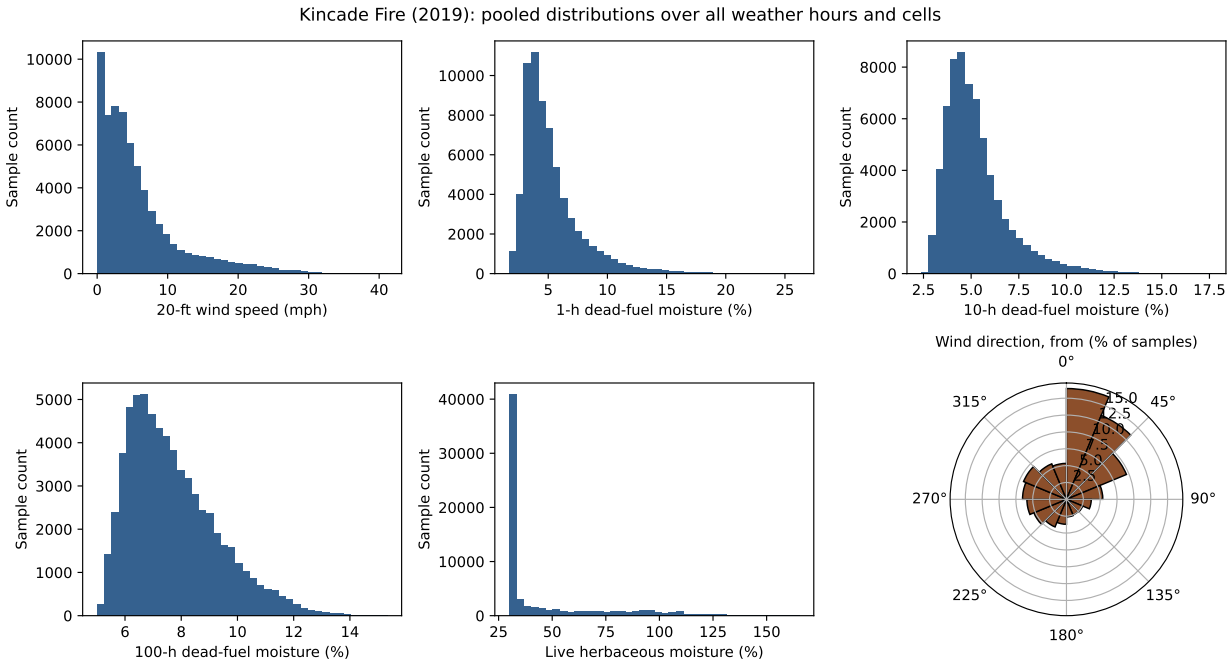


Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 134 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft min^{-1}, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

Choice	Basis	Evidence and limitation
Ignition points and times	documented	GOFER ignition record; no point required moving
Simulation start and stop	documented	First GOFER ignition point and last GOFER perimeter
Fuel vintage	derived	Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire
Weather and moisture	user-specified	RTMA analysis through CloudFire, as stated by the case owner
Crown fire	documented	ELMFIRE default crown-fire treatment, set explicitly
Firebrand spotting	user-specified	Enabled with ELMFIRE default spotting settings until calibration
Domain margin	user-specified	10 km beyond the final GOFER perimeter
Single deterministic member	user-specified	No ensemble; stochastic spotting uses a fixed random seed

Table 2: Basis of consequential configuration choices.

5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 0 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. No simulation results are retained in this dataset. New results must record the executable identity and configuration used.

6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time t the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap J_t summarizes footprint agreement; precision P_t diagnoses commission (simulated burning outside the observation), recall R_t omission (observed burning the simulation misses), F_t balances the two, and the area ratio A_t exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

7 Simulation results and field-level comparison

The current status of this case is **NOT EVALUABLE**. No simulation result is available for this case.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result.

No simulation. No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Kincade Fire. Inputs and the observation are characterized in full; the current scientific status is **NOT EVALUABLE** with 0 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

References

[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

2021 KNP Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the KNP Complex (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 10 September 2021 at 14:00 UTC to 7 October 2021 at 02:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 633 hourly perimeters, a final area of 388.647 km^2 and an official size of 88,307 ac. The configuration uses 3 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

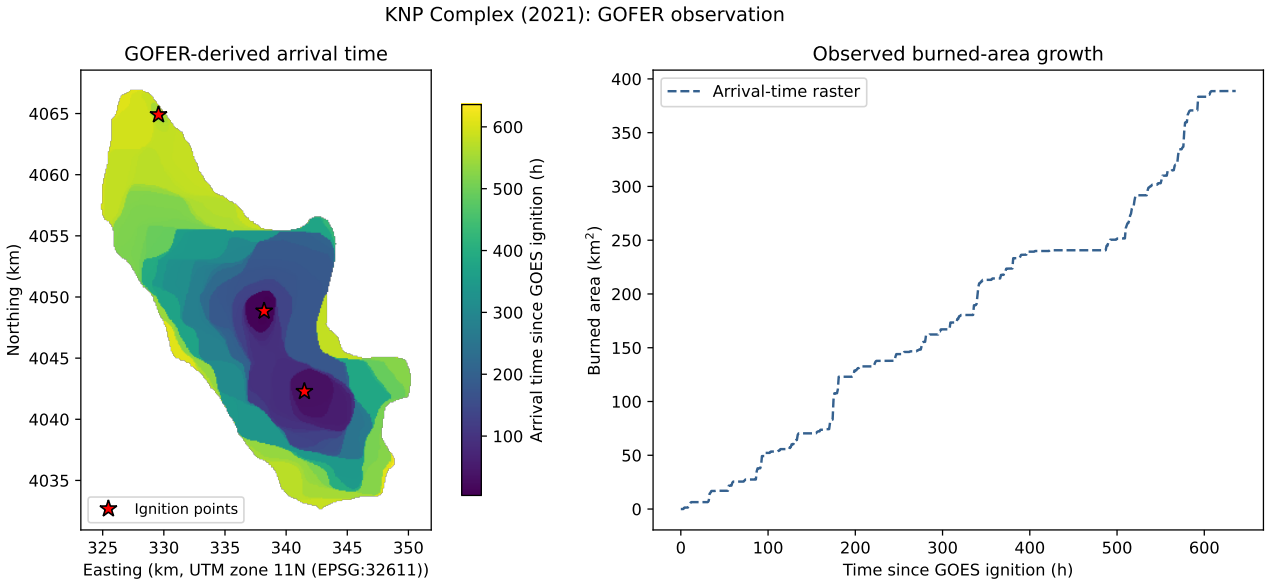


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,600 by 1,867 cells at 30 m spacing in UTM zone 11N (EPSG:32611), covering 2,688 km^2. Elevation ranges from 130 to 3,789 m; median slope is 20°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 35%, and canopy base height has median 0.9 m.

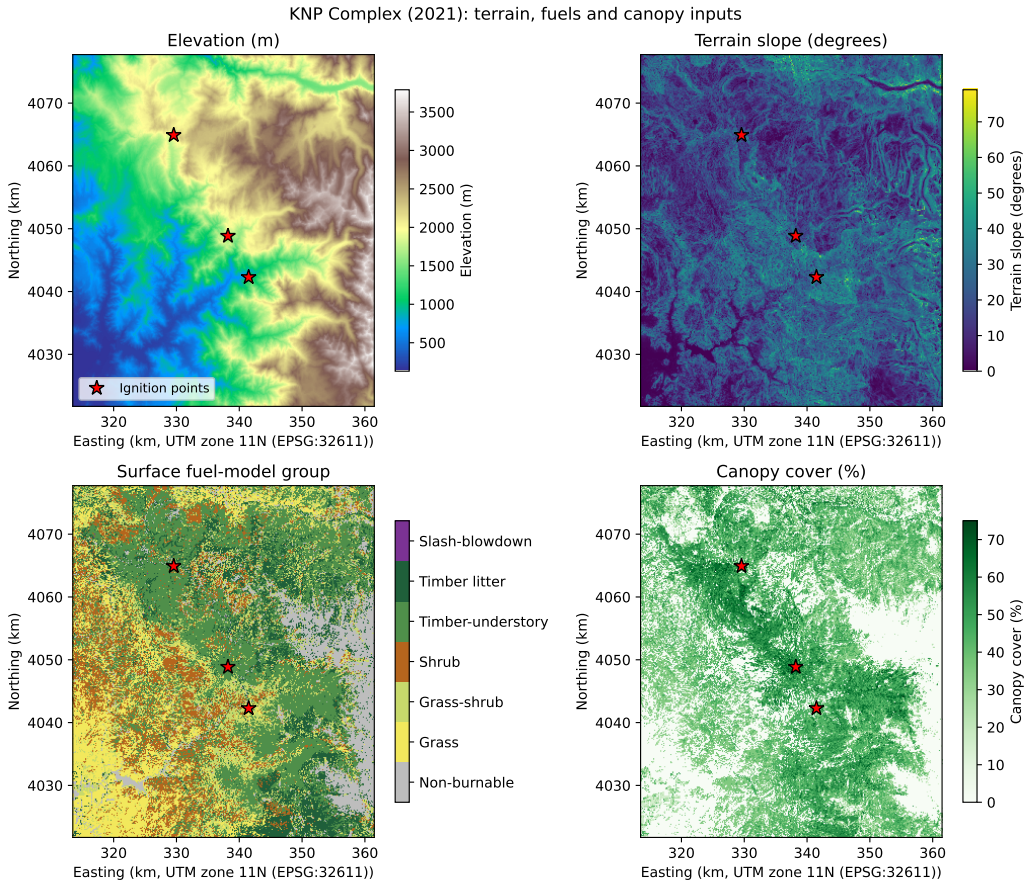


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 11N (EPSG:32611).

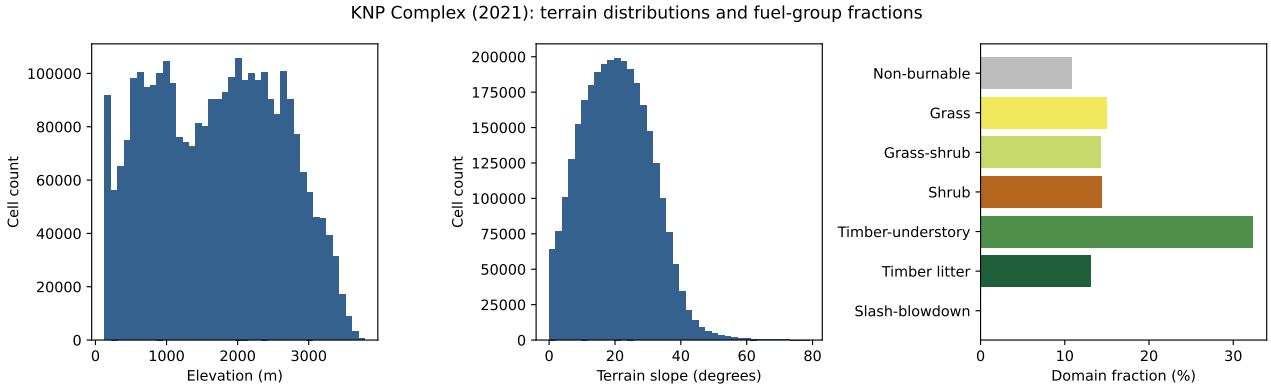


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 634 hourly fields. Median wind speed is 3.43 mph and its 95th percentile is 8.84 mph. Direction is summarized circularly: the mean direction is 252° and mean resultant length is 0.129. Median 1-h dead-fuel moisture is 6.66%. Spatial and temporal variability remain important when these domain summaries are interpreted.

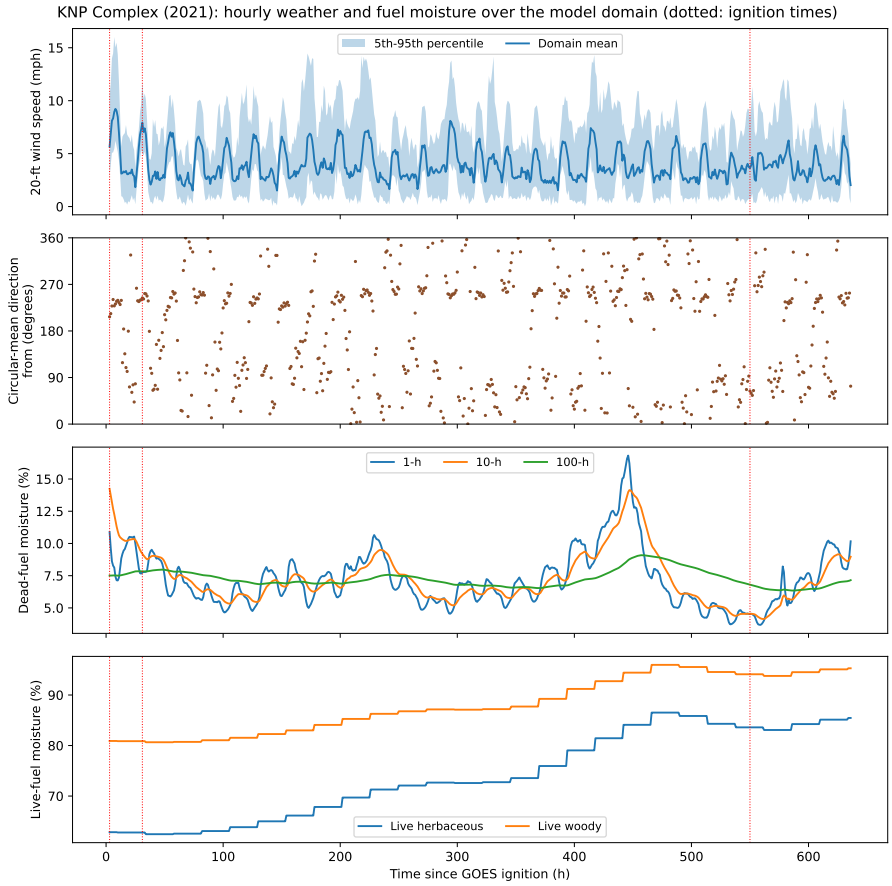


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

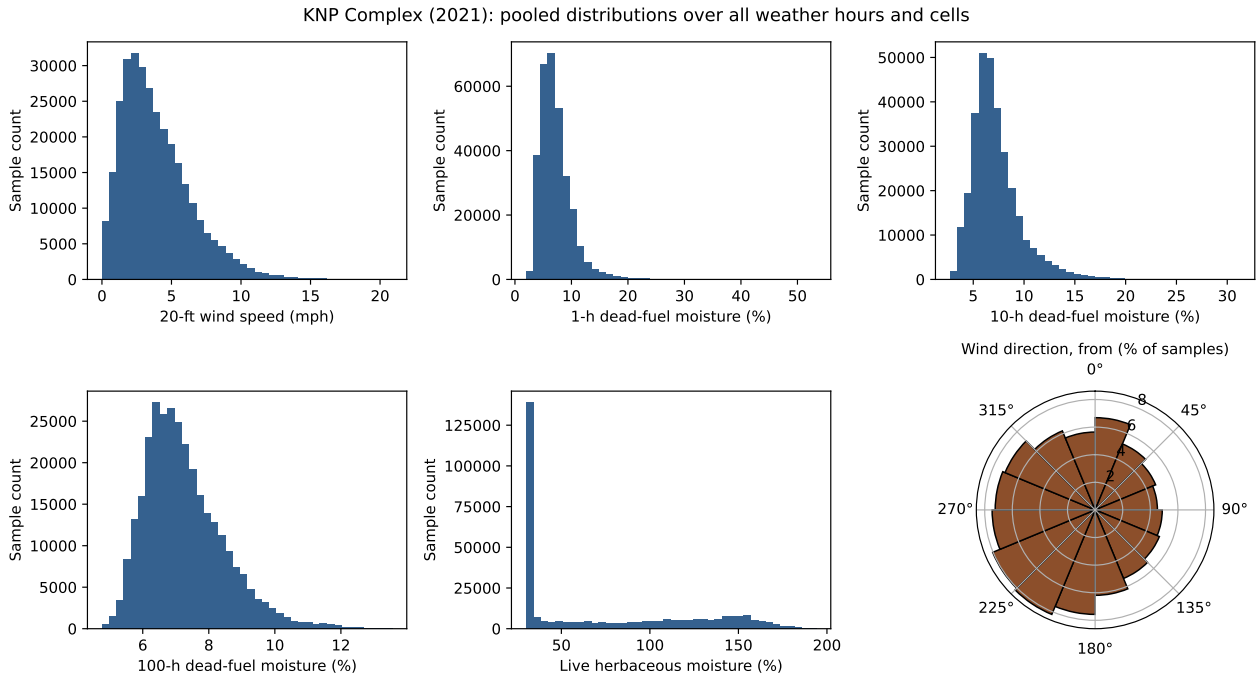


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 633 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 634 hourly weather fields and 3 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 3 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

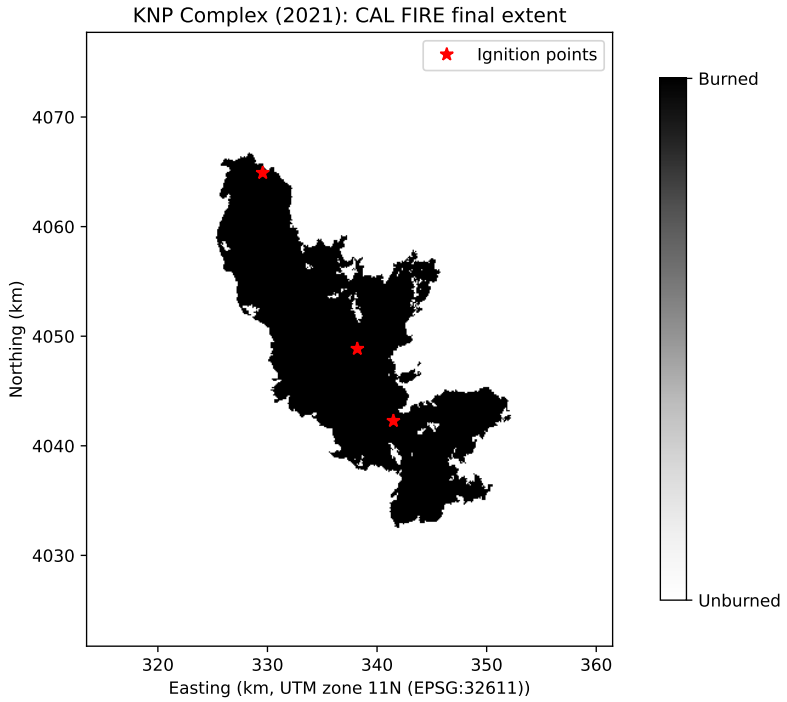


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 357 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the KNP Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 LNU Lightning Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the LNU Lightning Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 17 August 2020 at 13:00 UTC to 28 August 2020 at 21:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 270 hourly perimeters, a final area of 1538.75 km^2 and an official size of 363,220 ac. The configuration uses 10 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

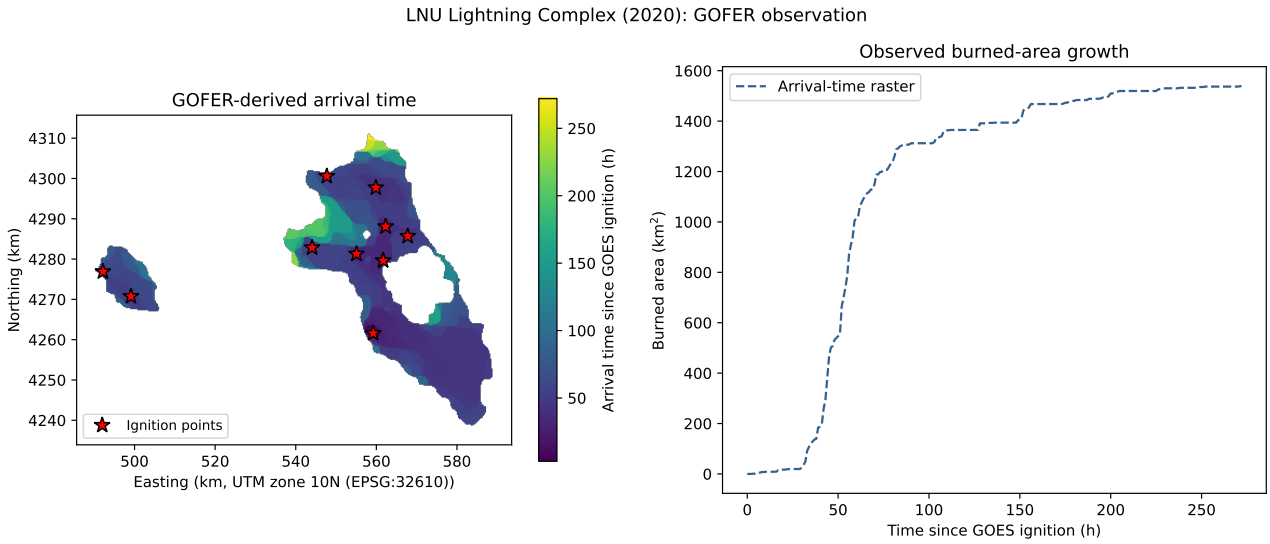


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 4,000 by 3,133 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 11,279 km^2. Elevation ranges from -4 to 1,442 m; median slope is 9°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

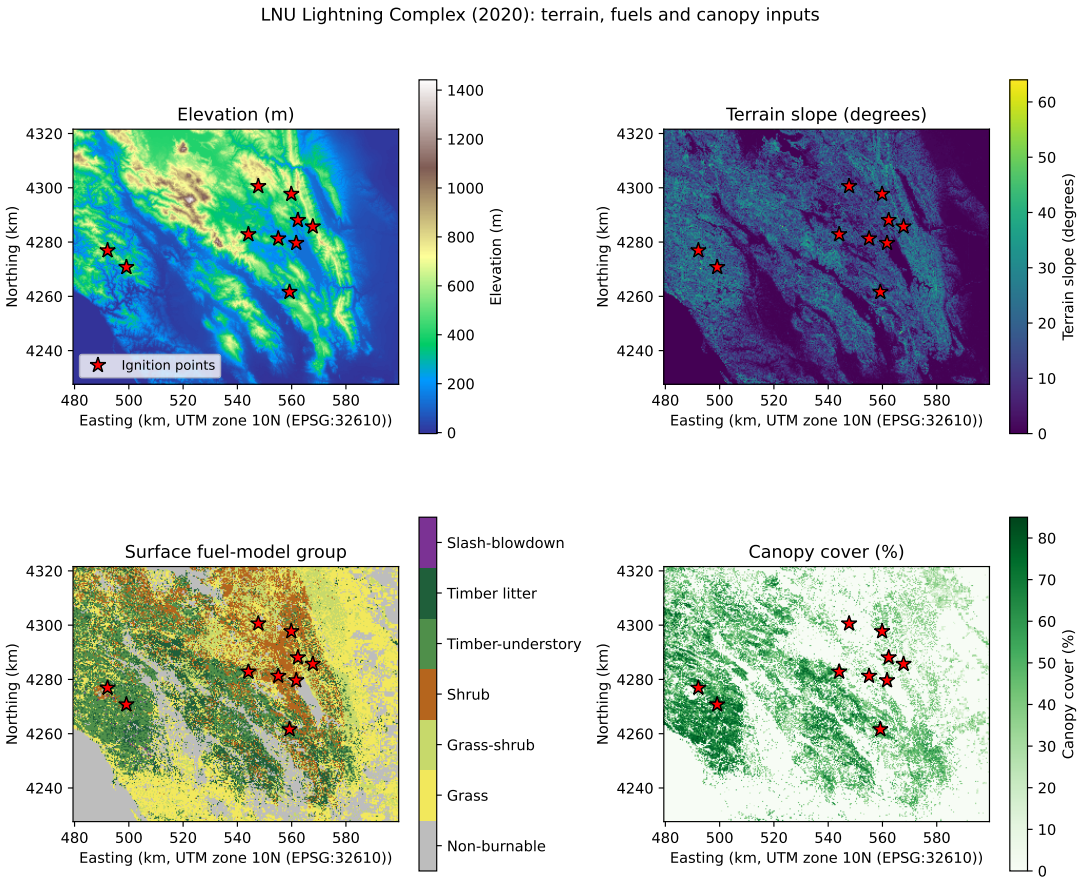


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

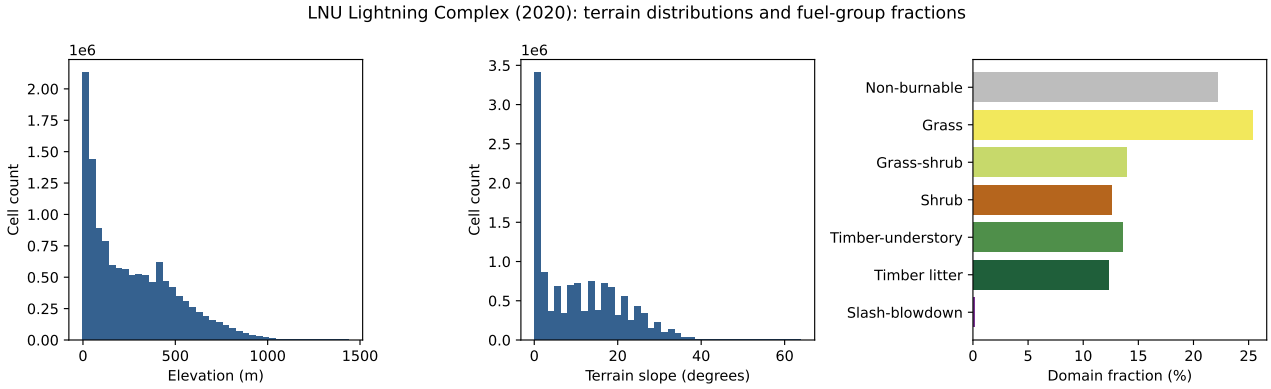


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 271 hourly fields. Median wind speed is 4.39 mph and its 95th percentile is 12.1 mph. Direction is summarized circularly: the mean direction is 259° and mean resultant length is 0.263. Median 1-h dead-fuel moisture is 9.39%. Spatial and temporal variability remain important when these domain summaries are interpreted.

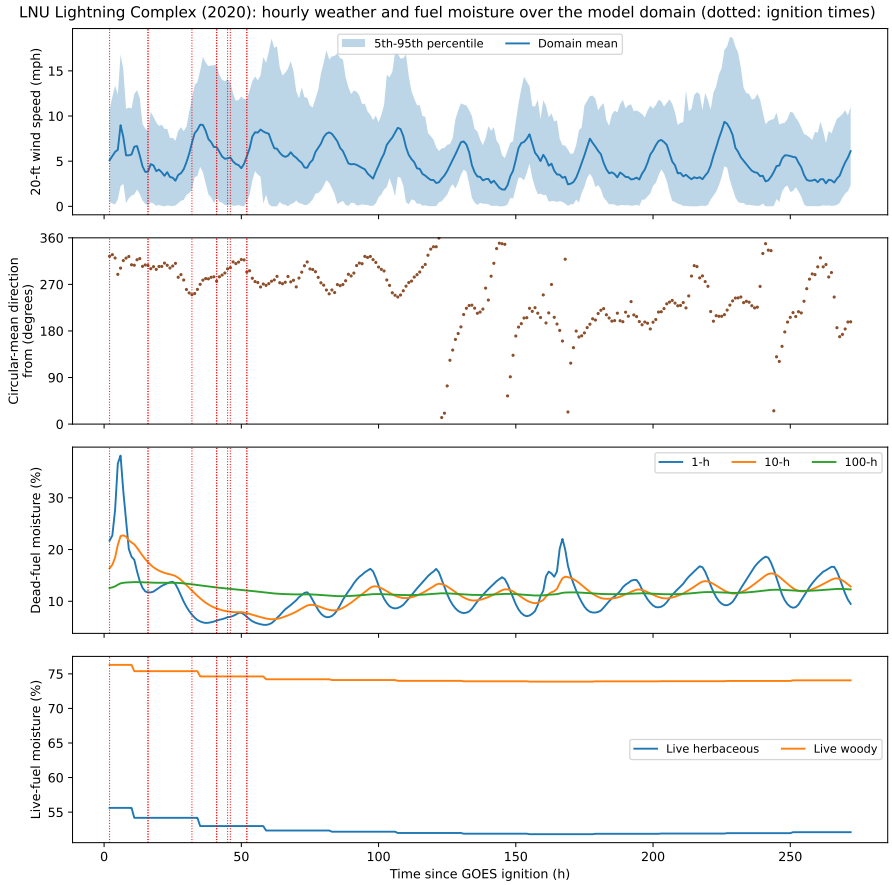


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

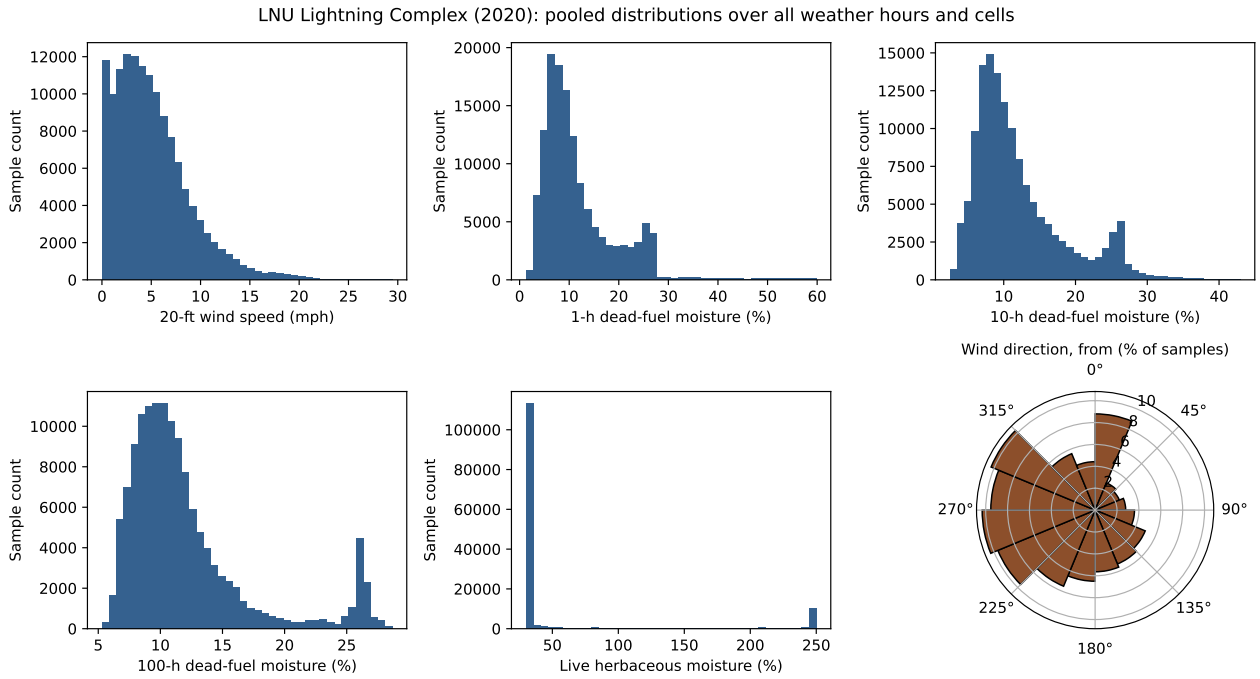


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 270 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 271 hourly weather fields and 10 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 2 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

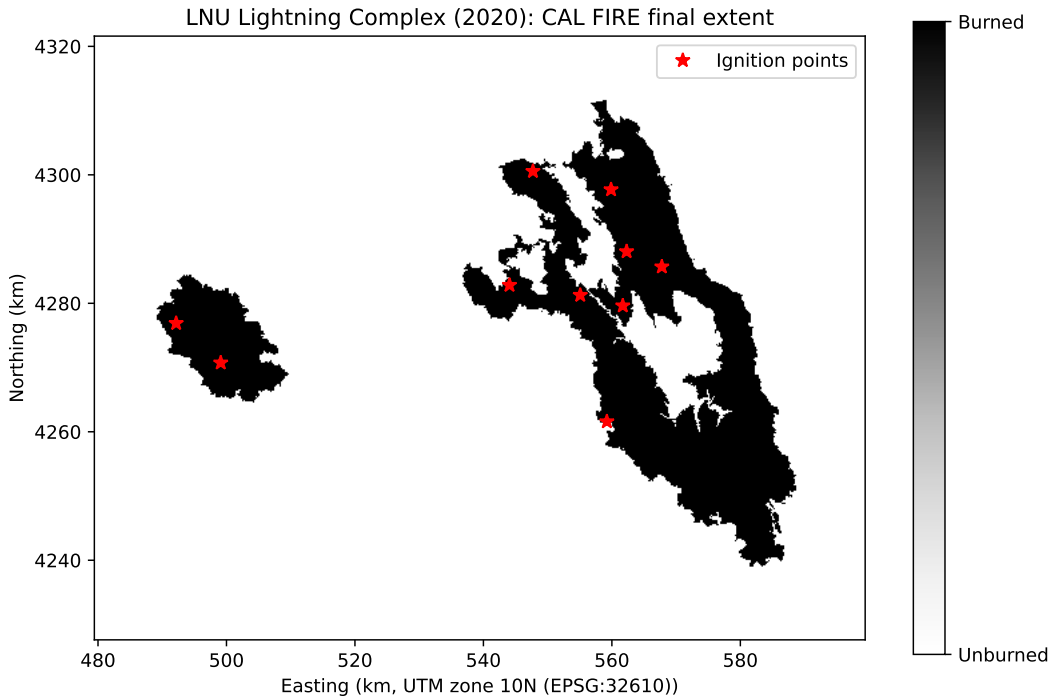


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 1,458 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the LNU Lightning Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 McCash fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the McCash (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 1 August 2021 at 02:00 UTC to 24 September 2021 at 08:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1297 hourly perimeters, a final area of 405.783 km^2 and an official size of 94,962 ac. The configuration uses 3 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

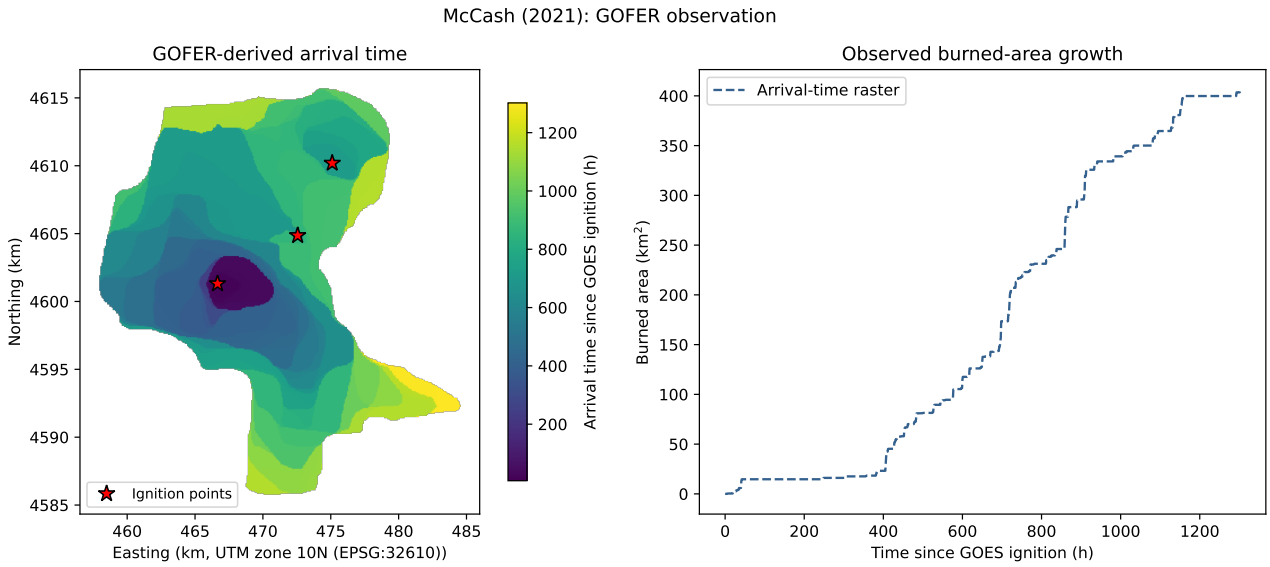


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,600 by 1,667 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 2,400 km^2. Elevation ranges from 118 to 2,519 m; median slope is 27°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 45%, and canopy base height has median 1.3 m.

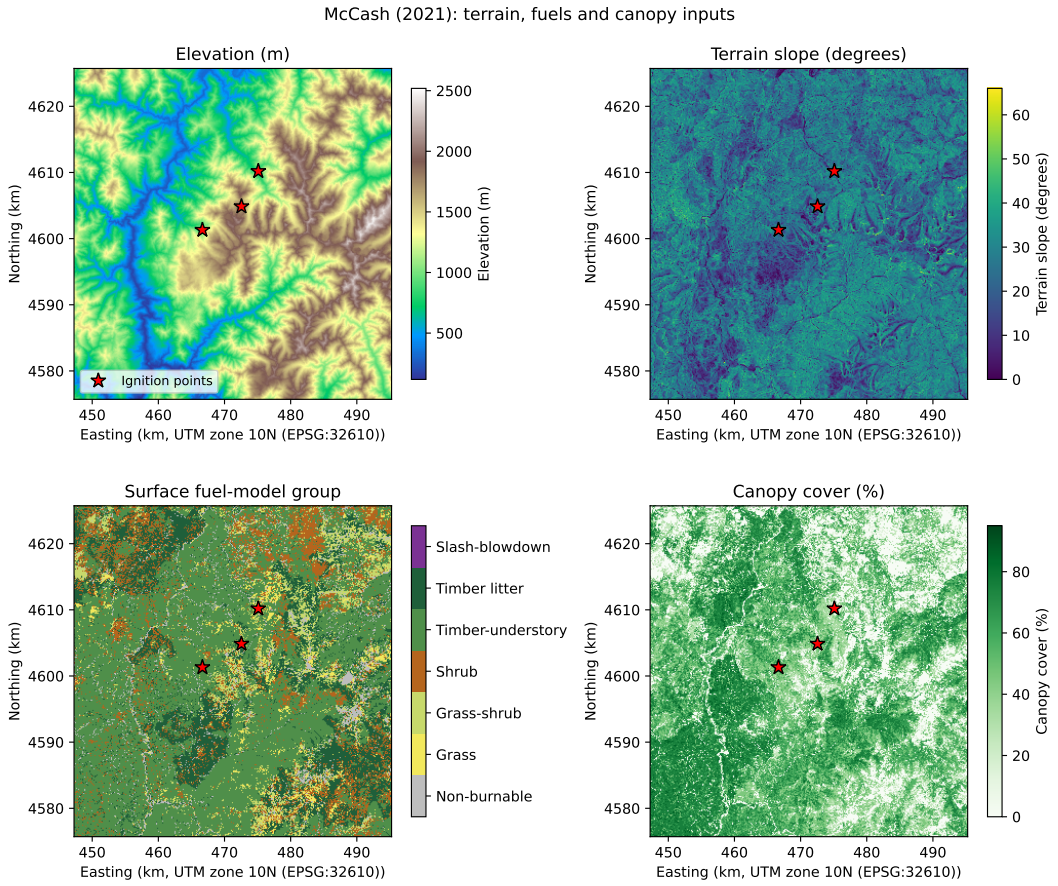


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

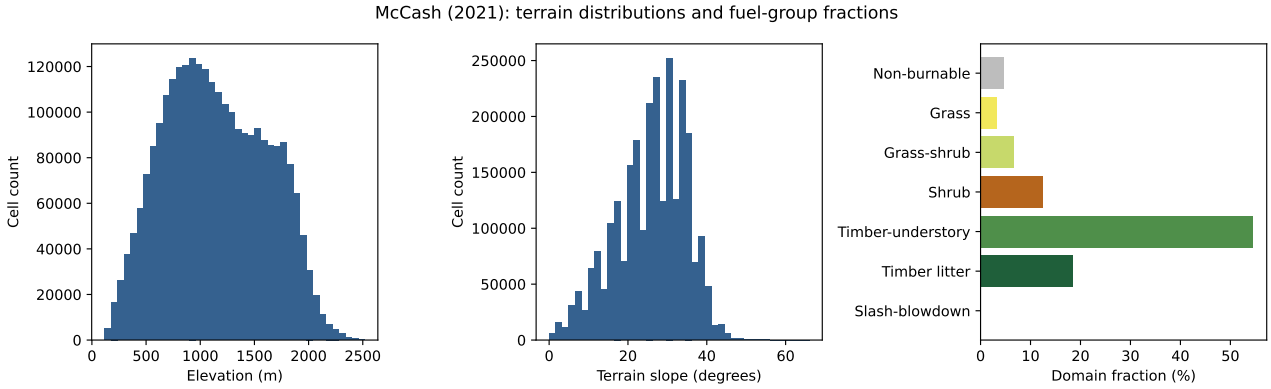


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,298 hourly fields. Median wind speed is 4.28 mph and its 95th percentile is 10 mph. Direction is summarized circularly: the mean direction is 315° and mean resultant length is 0.1. Median 1-h dead-fuel moisture is 6.52%. Spatial and temporal variability remain important when these domain summaries are interpreted.

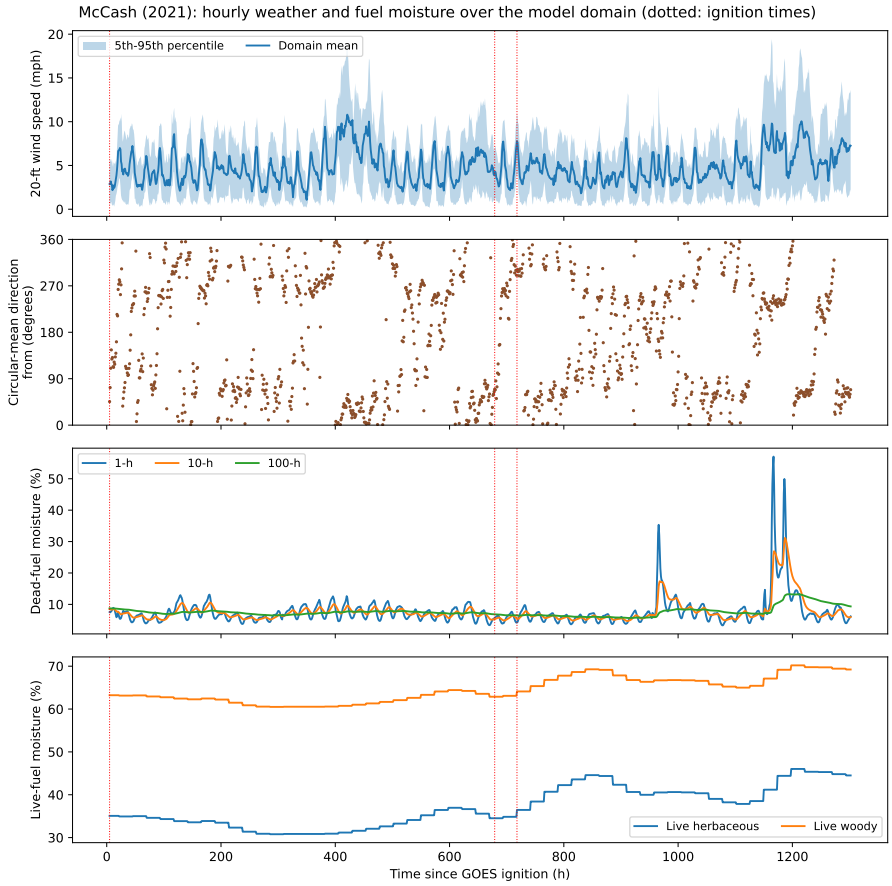


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

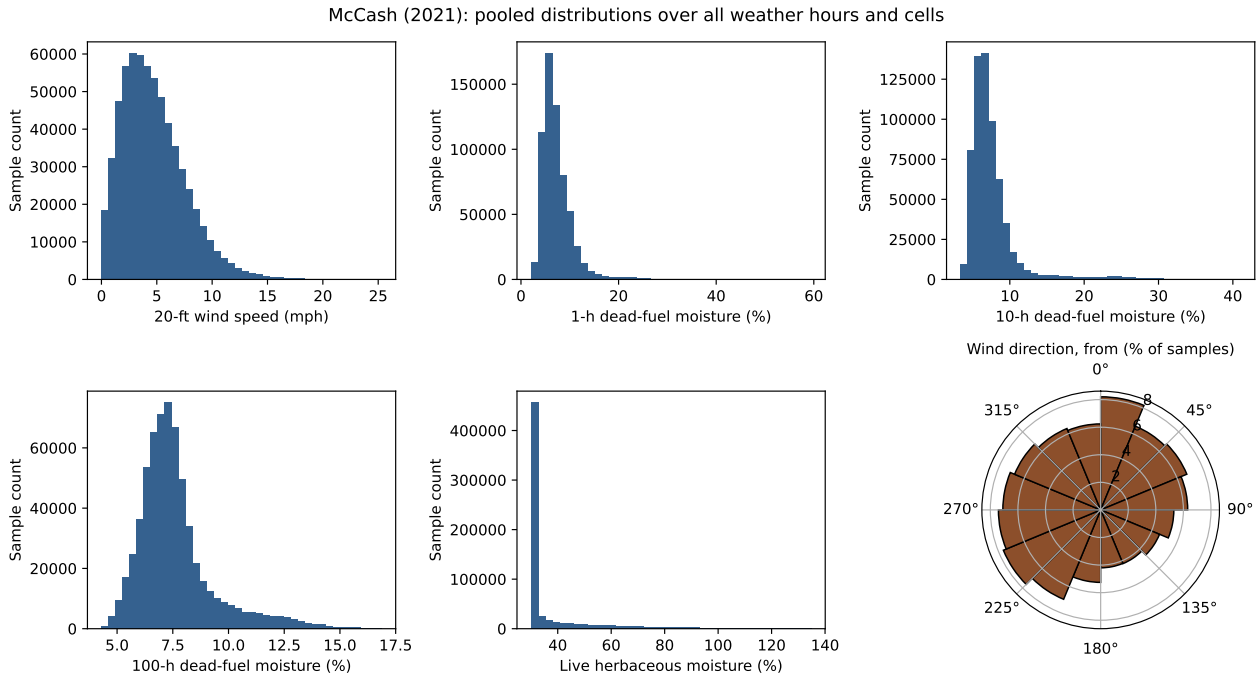


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1297 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1298 hourly weather fields and 3 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 5 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

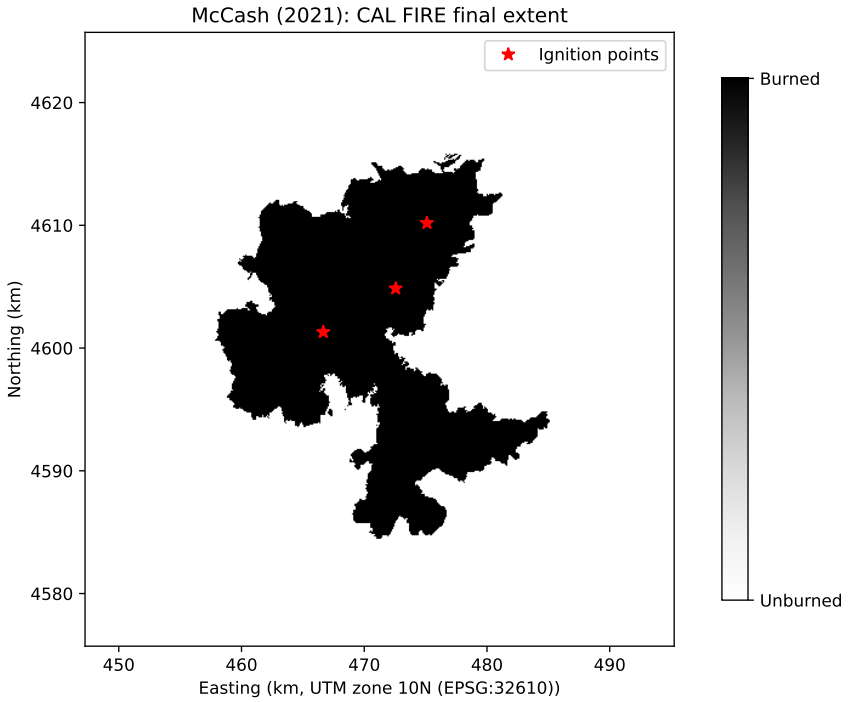


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 384 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the McCash reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 McFarland fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the McFarland (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 30 July 2021 at 01:00 UTC to 25 August 2021 at 22:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 637 hourly perimeters, a final area of 566.952 km^2 and an official size of 122,653 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

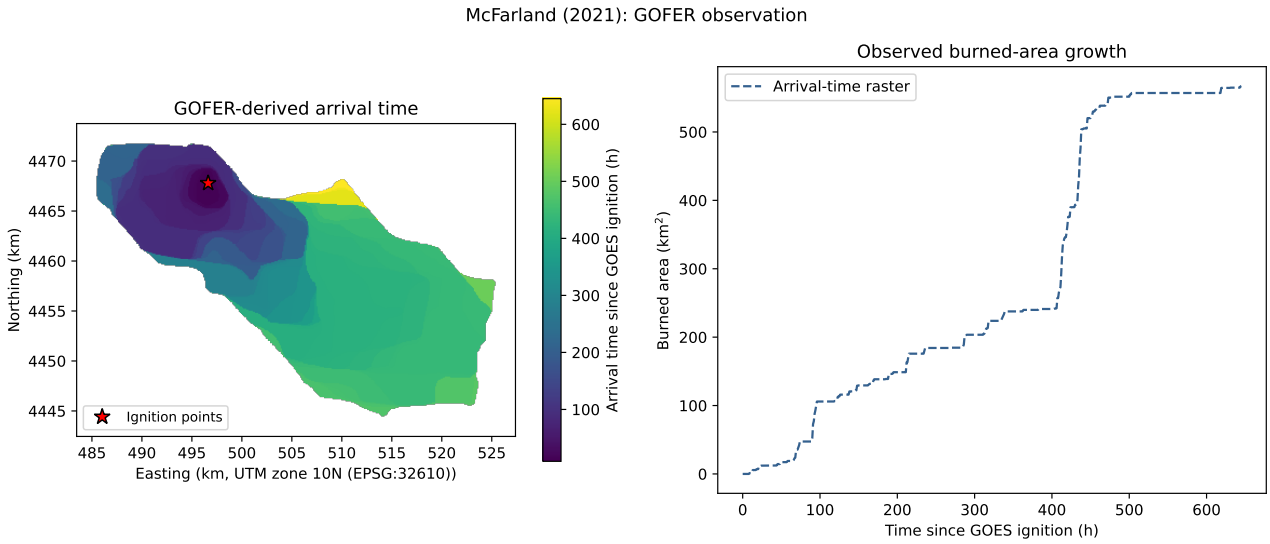


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,000 by 1,600 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 2,880 km^2. Elevation ranges from 194 to 2,386 m; median slope is 19°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 25%, and canopy base height has median 0.3 m.

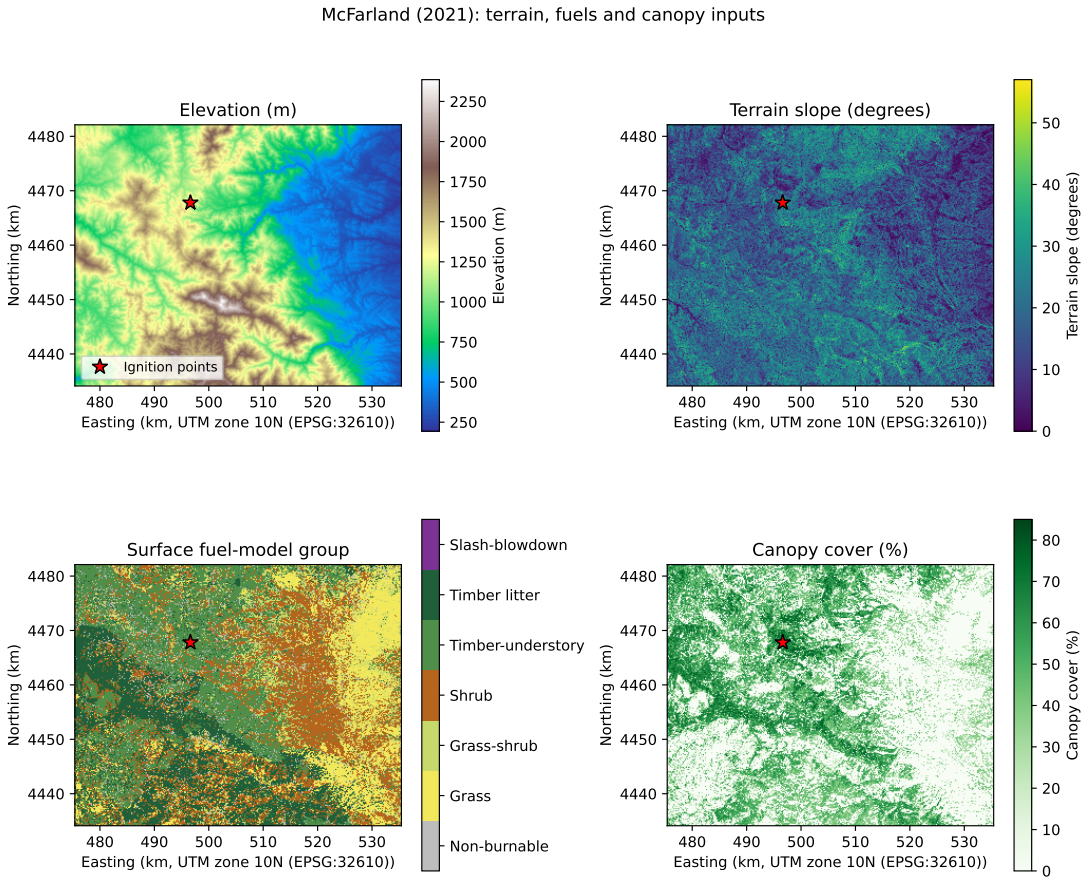


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

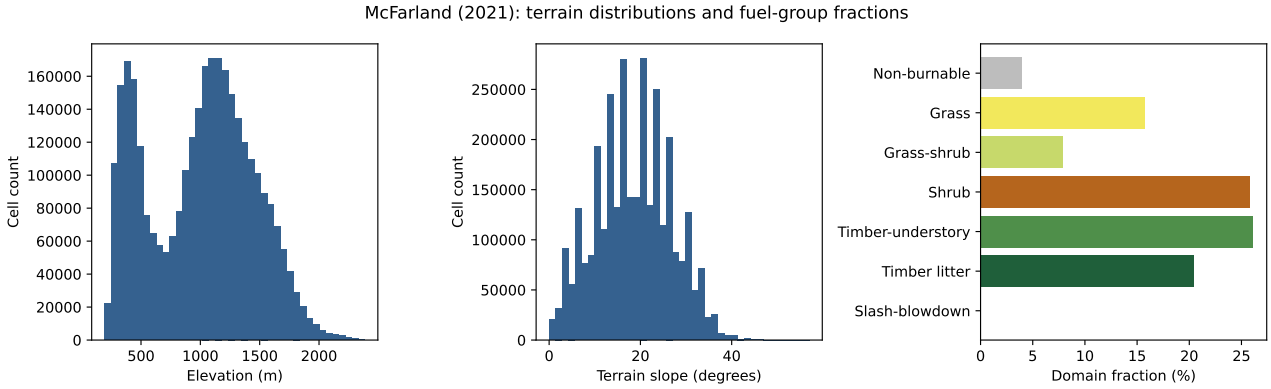


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 638 hourly fields. Median wind speed is 3.95 mph and its 95th percentile is 9.14 mph. Direction is summarized circularly: the mean direction is 41.3° and mean resultant length is 0.0629. Median 1-h dead-fuel moisture is 5.95%. Spatial and temporal variability remain important when these domain summaries are interpreted.

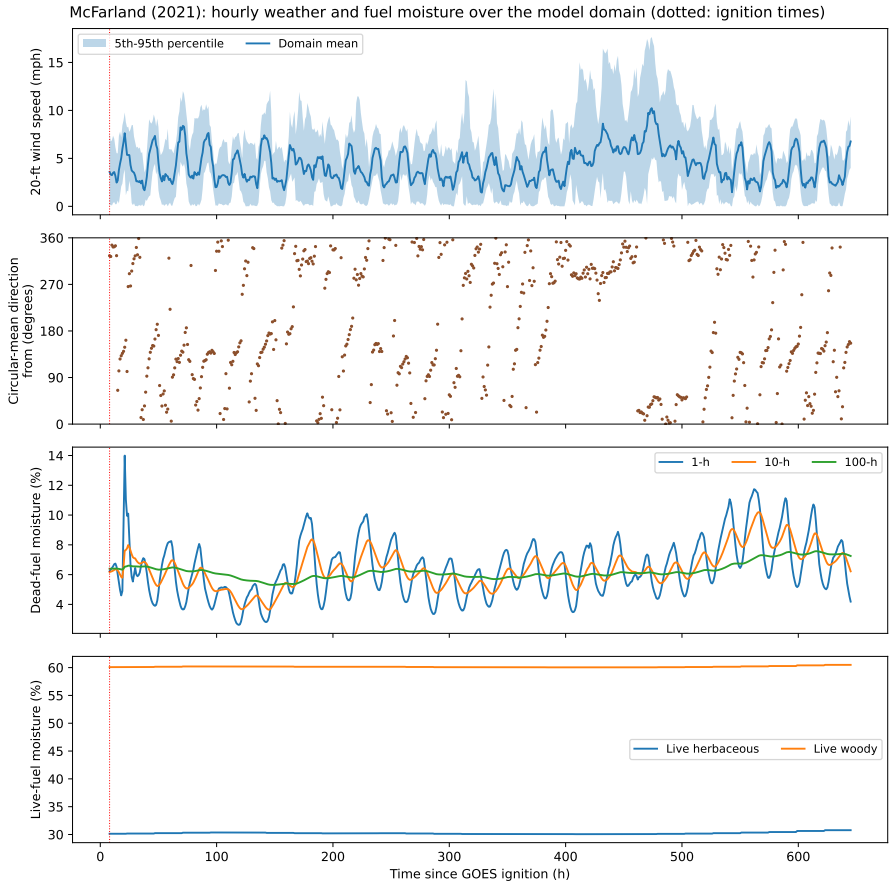


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

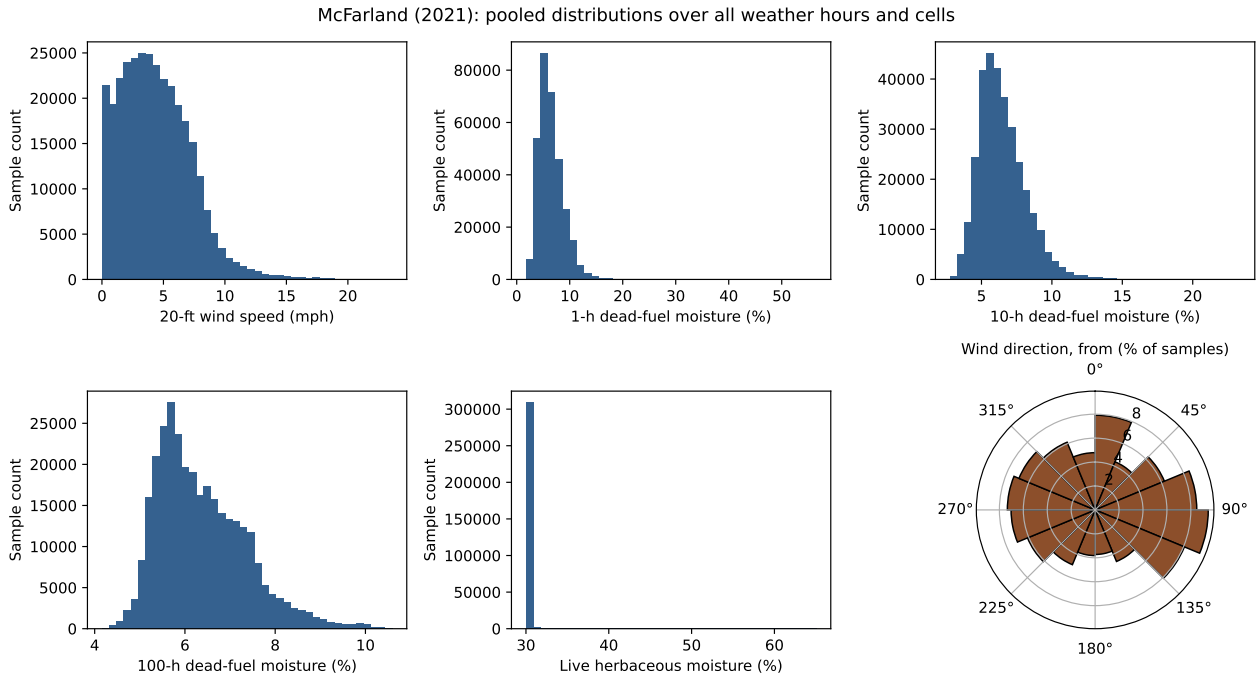


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 637 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 638 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 8 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

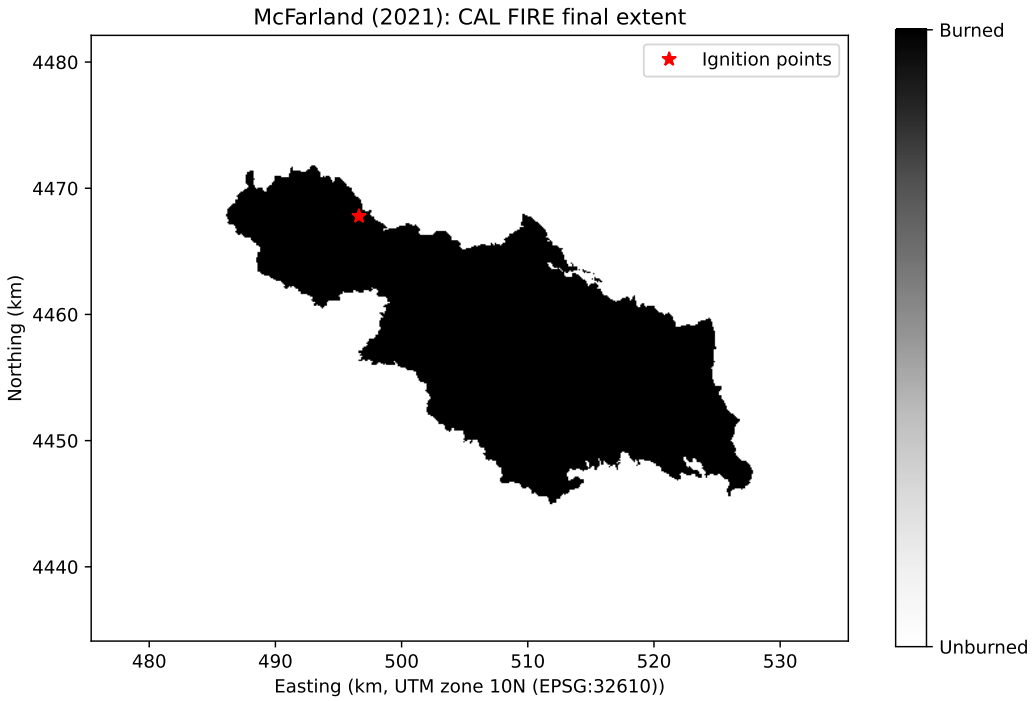


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 495 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the McFarland reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 Monument fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Monument (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 31 July 2021 at 01:00 UTC to 17 September 2021 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1168 hourly perimeters, a final area of 922.504 km^2 and an official size of 223,124 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

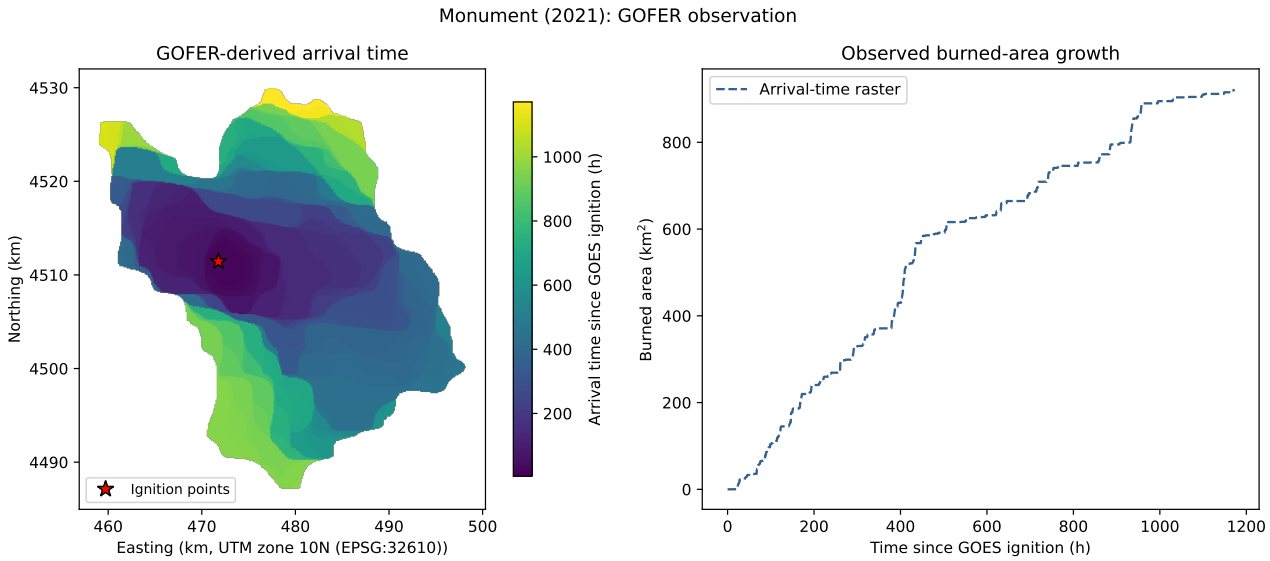


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,000 by 2,133 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 3,839 km^2. Elevation ranges from 129 to 2,726 m; median slope is 24°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 45%, and canopy base height has median 1.2 m.

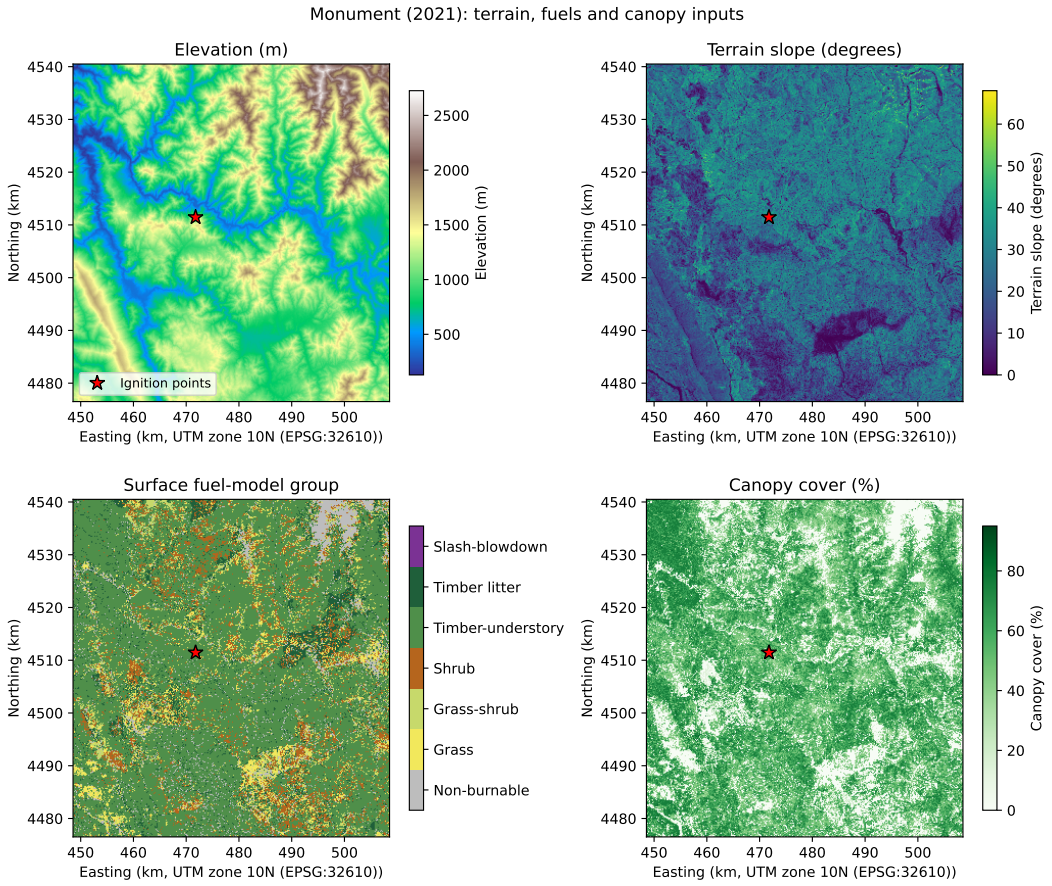


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

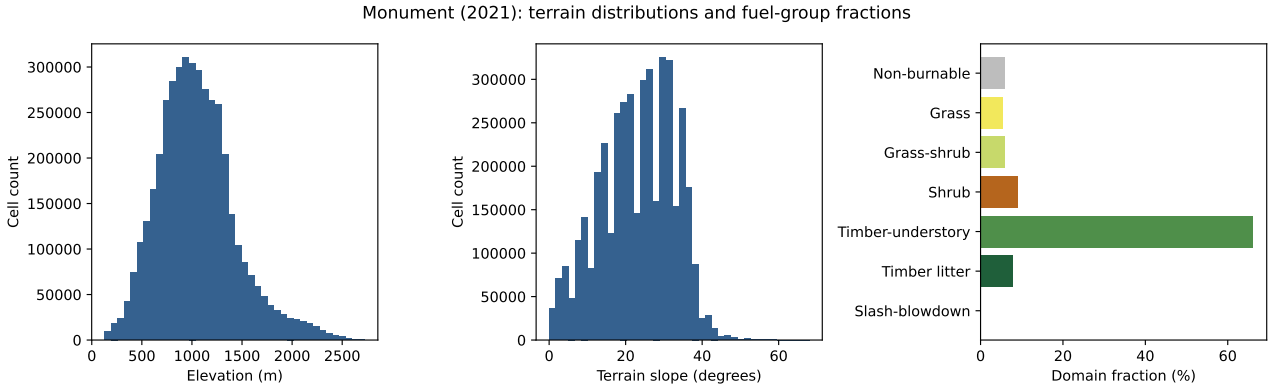


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,169 hourly fields. Median wind speed is 3.66 mph and its 95th percentile is 9.04 mph. Direction is summarized circularly: the mean direction is 304° and mean resultant length is 0.121. Median 1-h dead-fuel moisture is 6.42%. Spatial and temporal variability remain important when these domain summaries are interpreted.

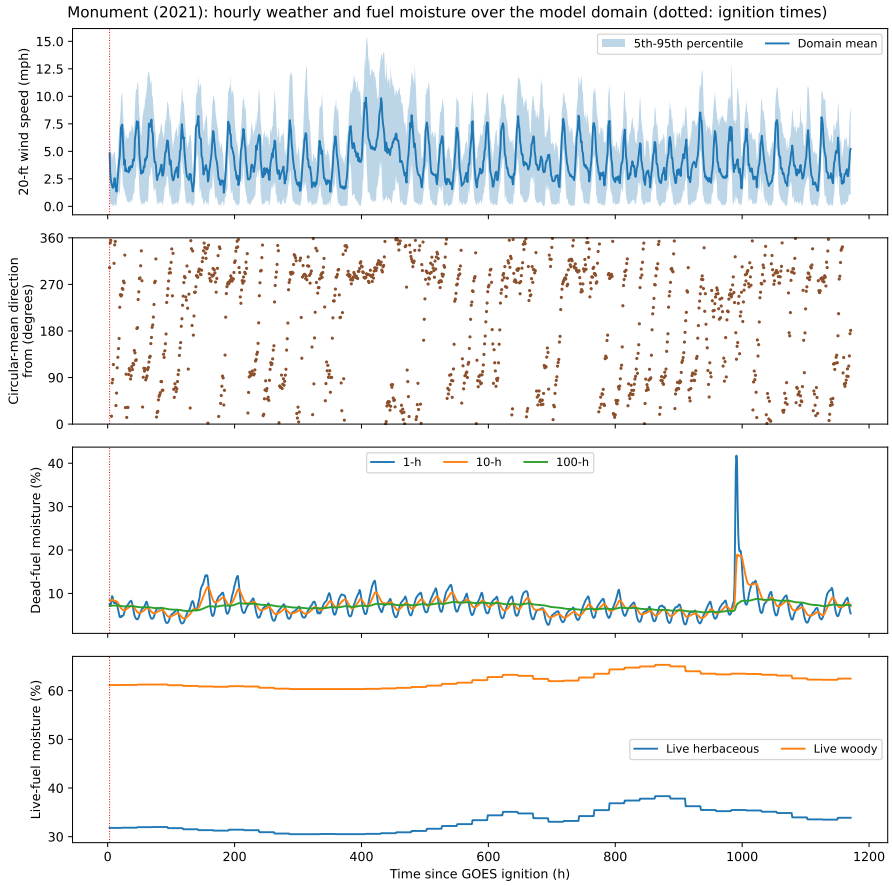


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

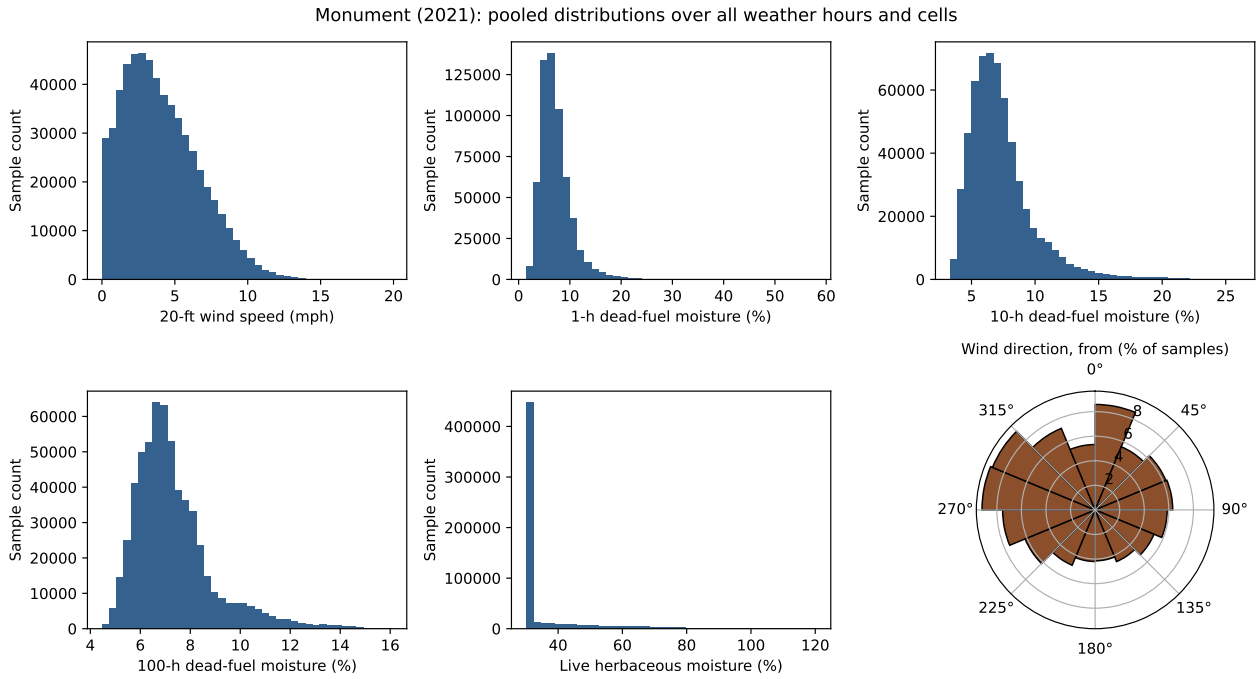


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1168 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1169 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 3 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

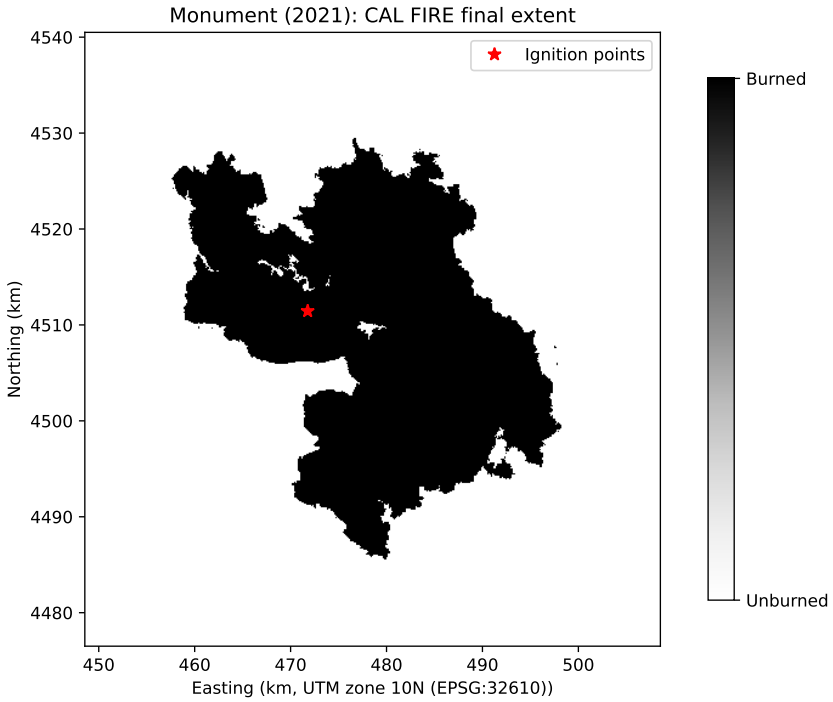


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 902 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Monument reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 North Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the North Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 17 August 2020 at 16:00 UTC to 1 October 2020 at 16:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1075 hourly perimeters, a final area of 1344.5 km^2 and an official size of 318,935 ac. The configuration uses 6 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

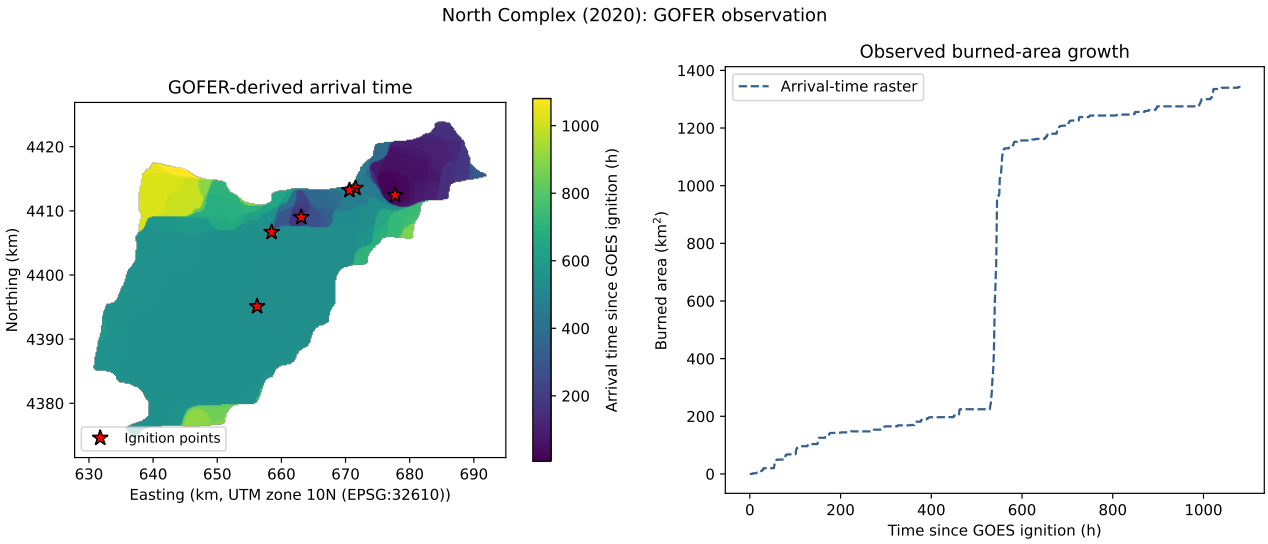


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,800 by 2,333 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 5,879 km^2. Elevation ranges from 30 to 2,587 m; median slope is 15°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 45%, and canopy base height has median 1.3 m.

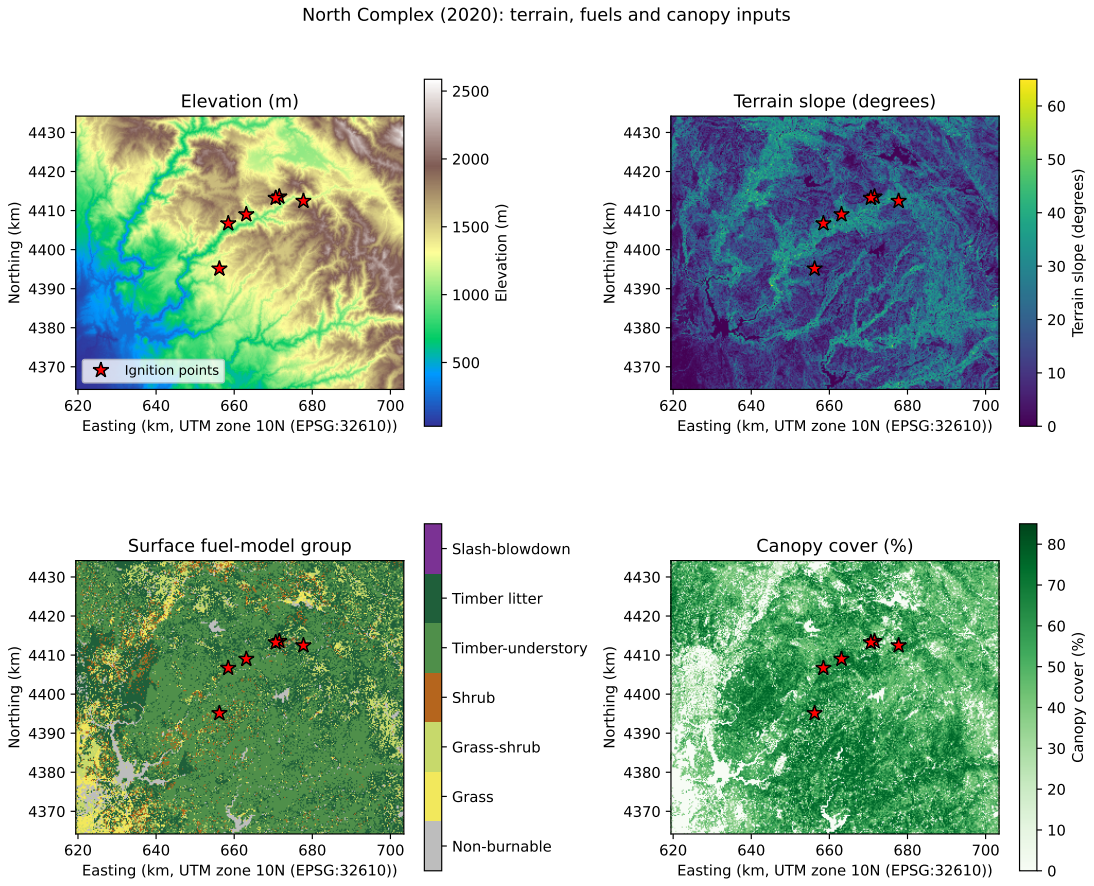


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

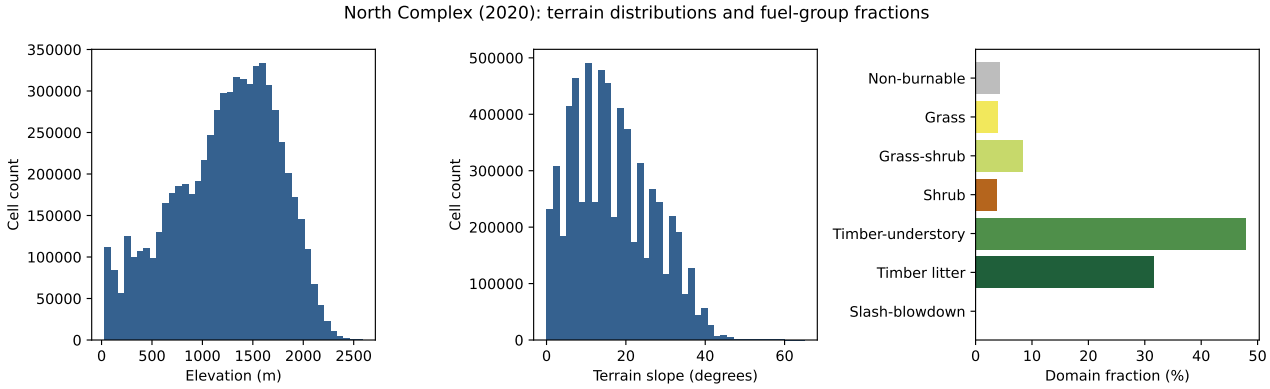


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,076 hourly fields. Median wind speed is 3.64 mph and its 95th percentile is 10.2 mph. Direction is summarized circularly: the mean direction is 256° and mean resultant length is 0.032. Median 1-h dead-fuel moisture is 6.7%. Spatial and temporal variability remain important when these domain summaries are interpreted.

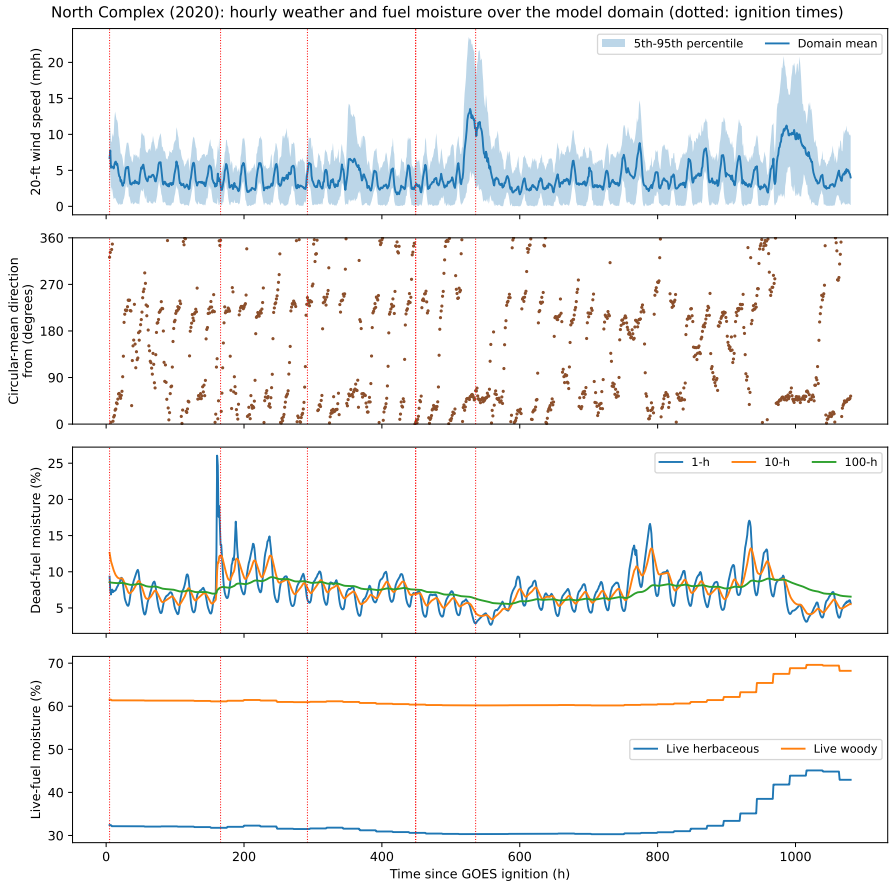


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

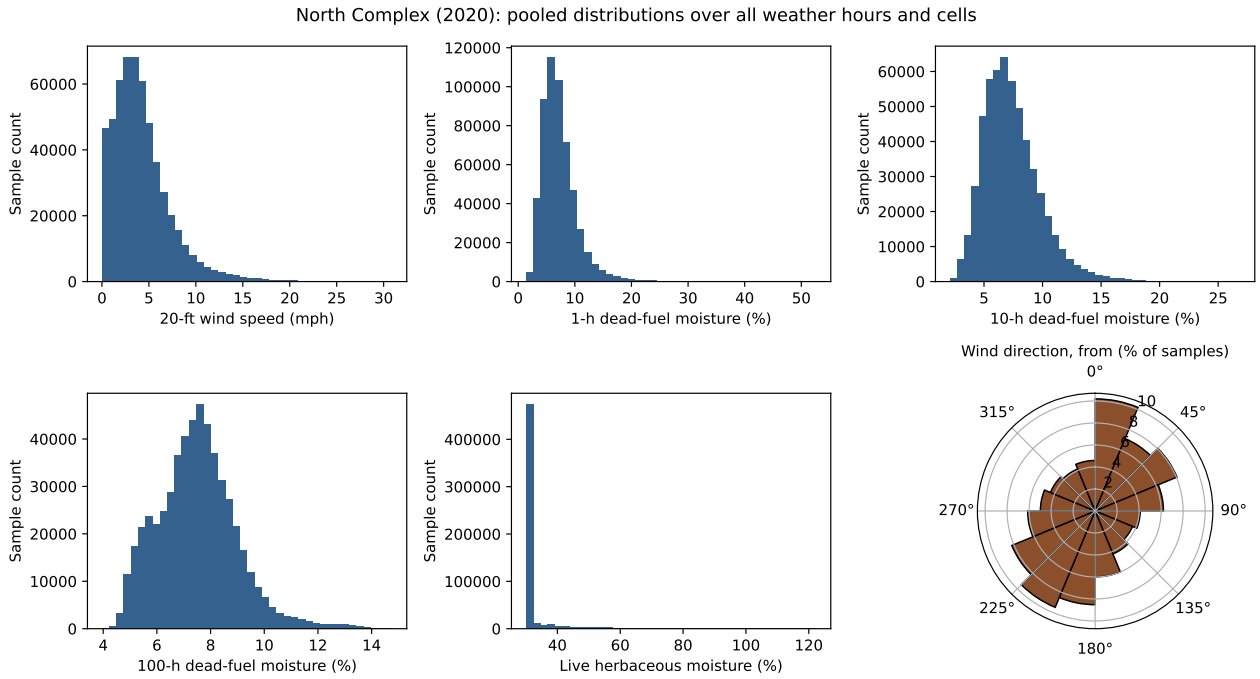


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1075 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1076 hourly weather fields and 6 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 5 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

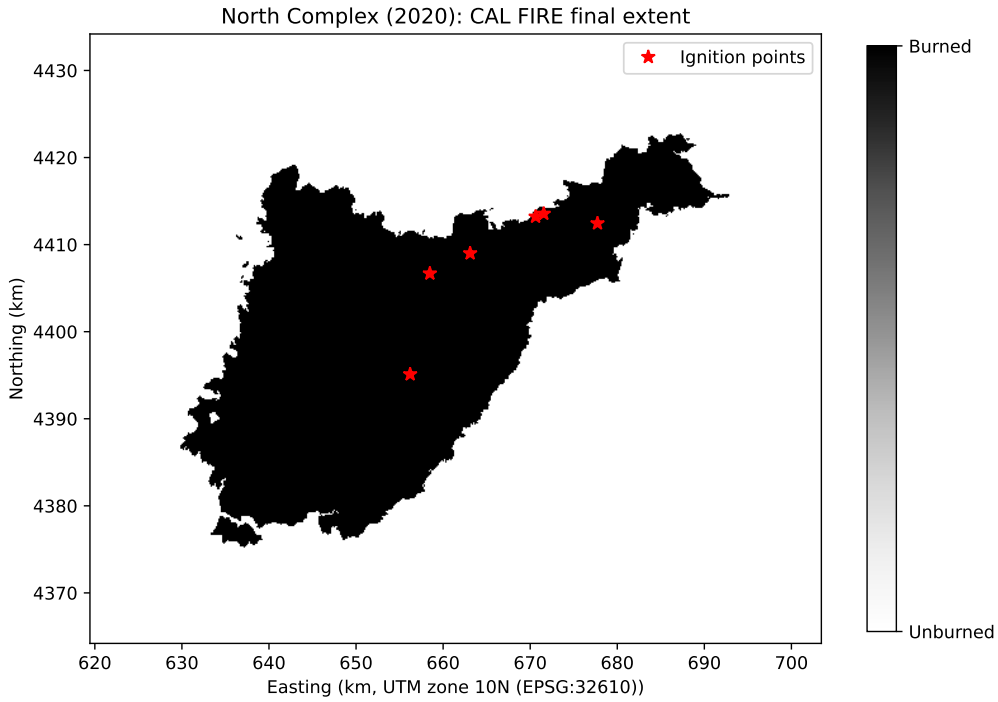


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 1,290 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the North Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 Red Salmon Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Red Salmon Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 27 July 2020 at 18:00 UTC to 10 October 2020 at 00:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1778 hourly perimeters, a final area of 575.114 km^2 and an official size of 144,698 ac. The configuration uses 2 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

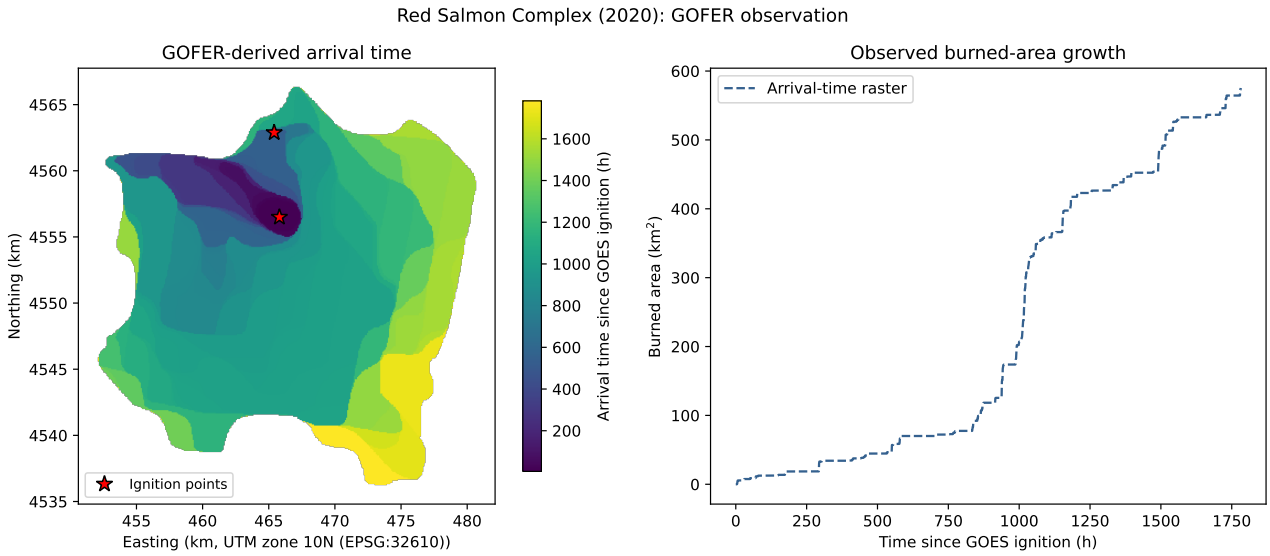


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,667 by 1,733 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 2,600 km^2. Elevation ranges from 60 to 2,303 m; median slope is 27°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 55%, and canopy base height has median 2.1 m.

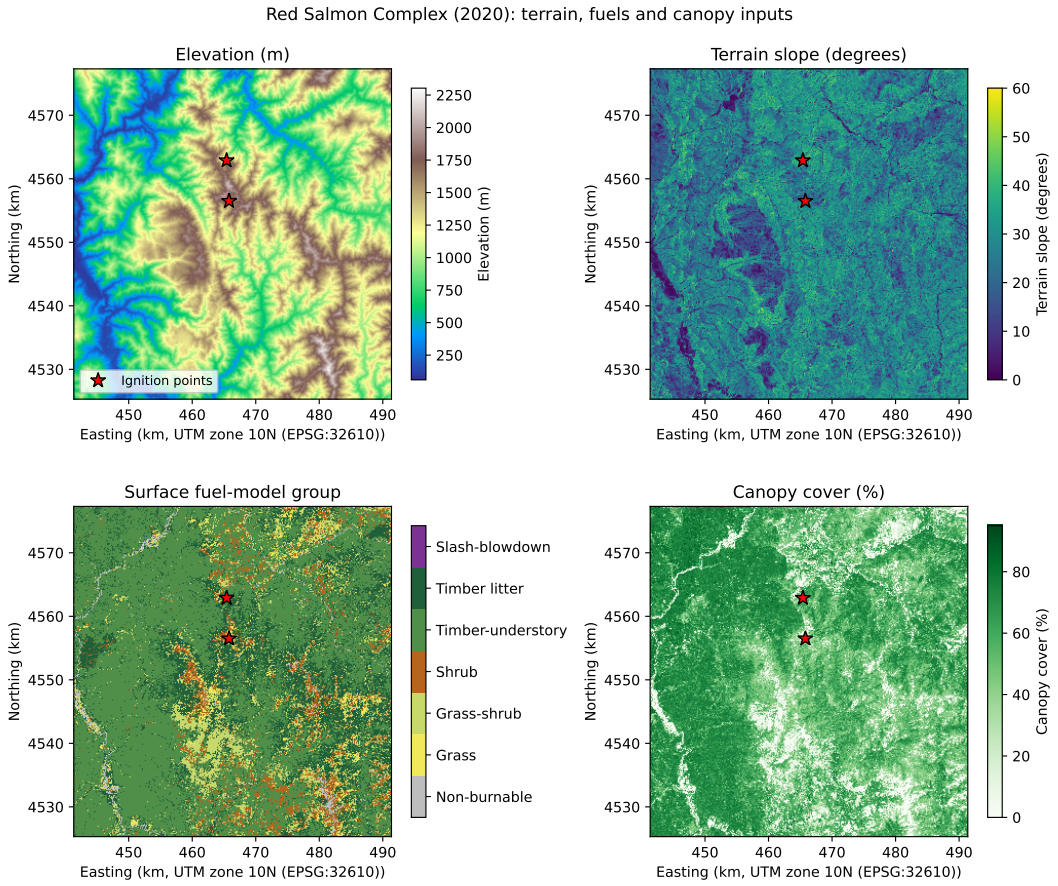


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

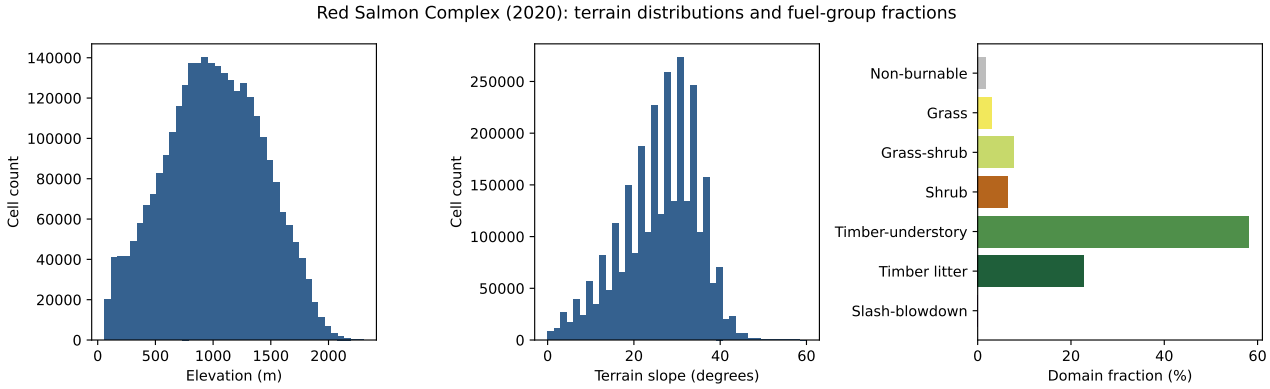


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,779 hourly fields. Median wind speed is 3.86 mph and its 95th percentile is 9.28 mph. Direction is summarized circularly: the mean direction is 272° and mean resultant length is 0.198. Median 1-h dead-fuel moisture is 7.55%. Spatial and temporal variability remain important when these domain summaries are interpreted.

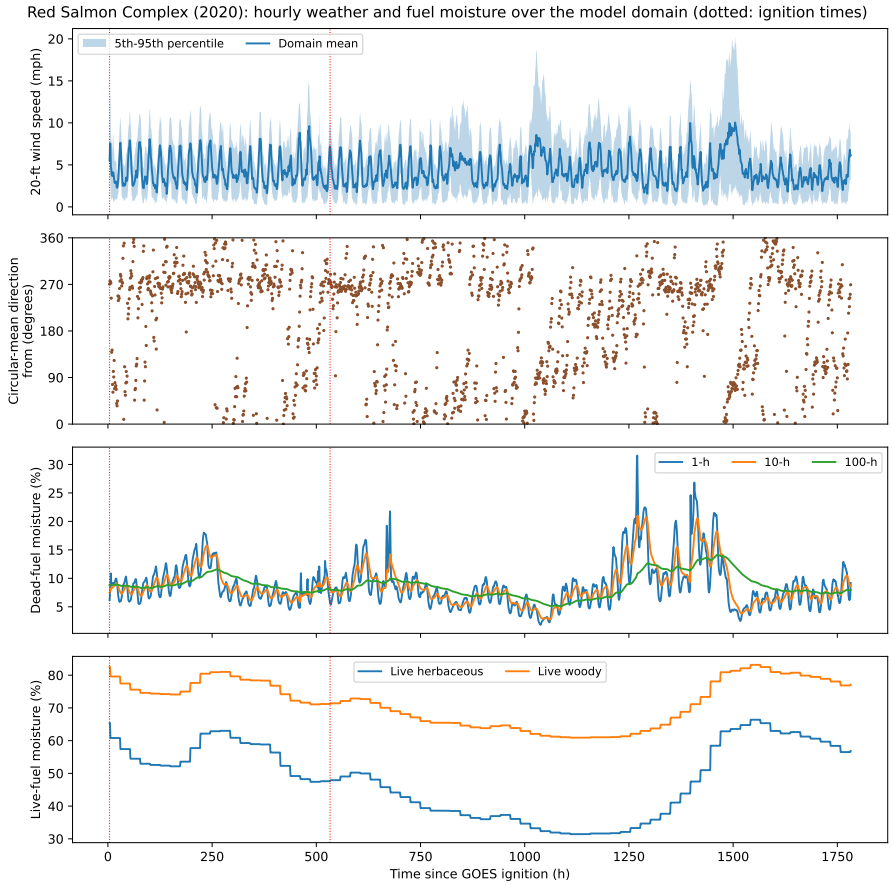


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

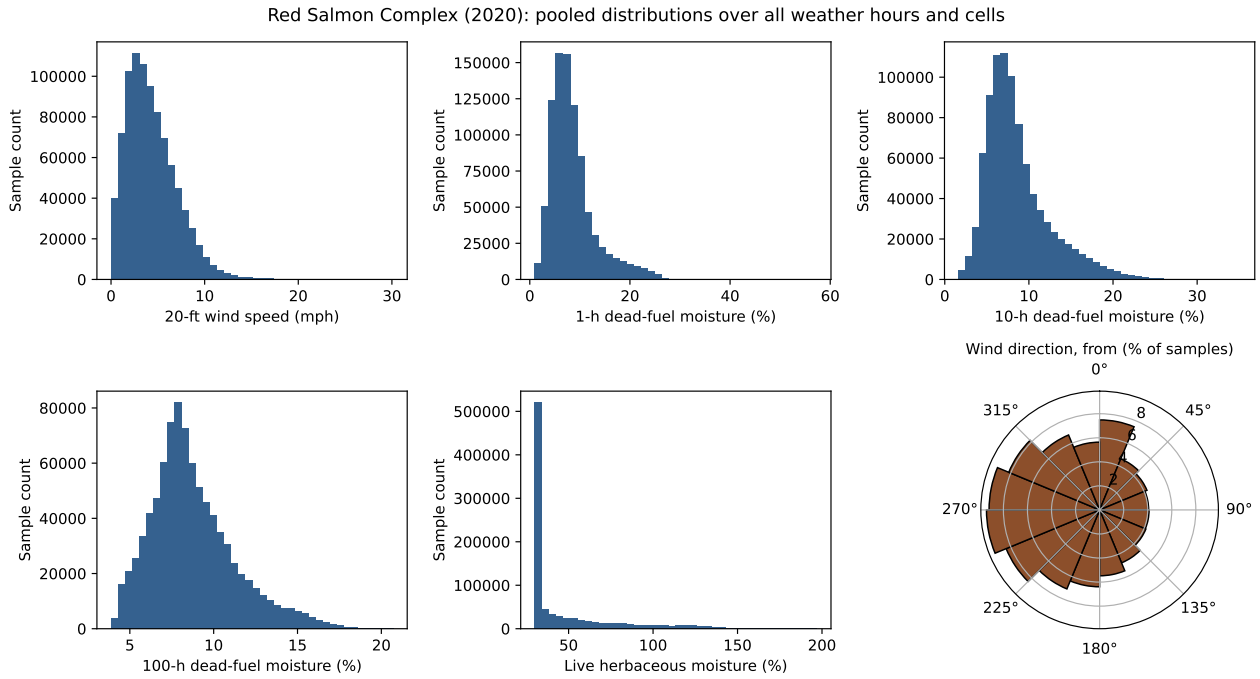


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1778 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1779 hourly weather fields and 2 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 4 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

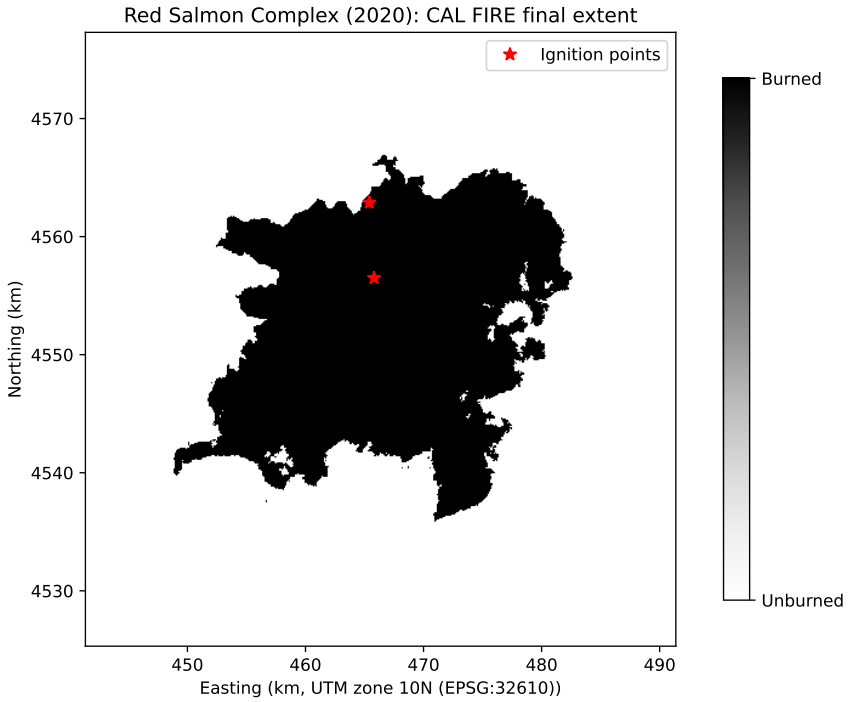


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 582 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Red Salmon Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 River Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the River Complex (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 30 July 2021 at 21:00 UTC to 26 September 2021 at 13:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1353 hourly perimeters, a final area of 931.308 km^2 and an official size of 199,359 ac. The configuration uses 5 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

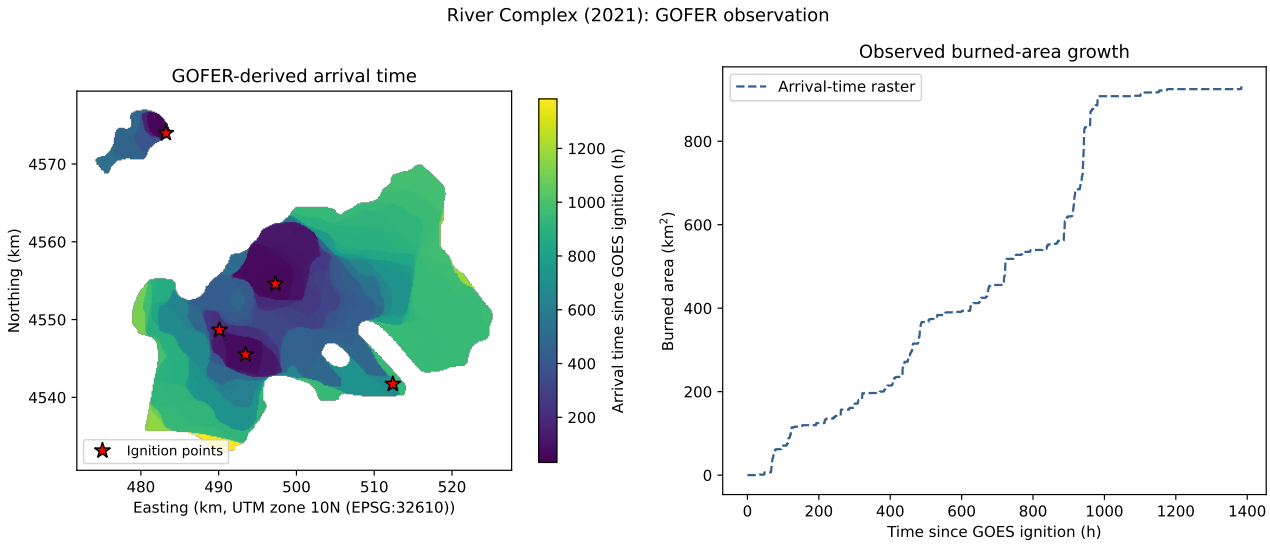


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,400 by 2,133 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 4,607 km^2. Elevation ranges from 154 to 2,726 m; median slope is 25°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 45%, and canopy base height has median 1.2 m.

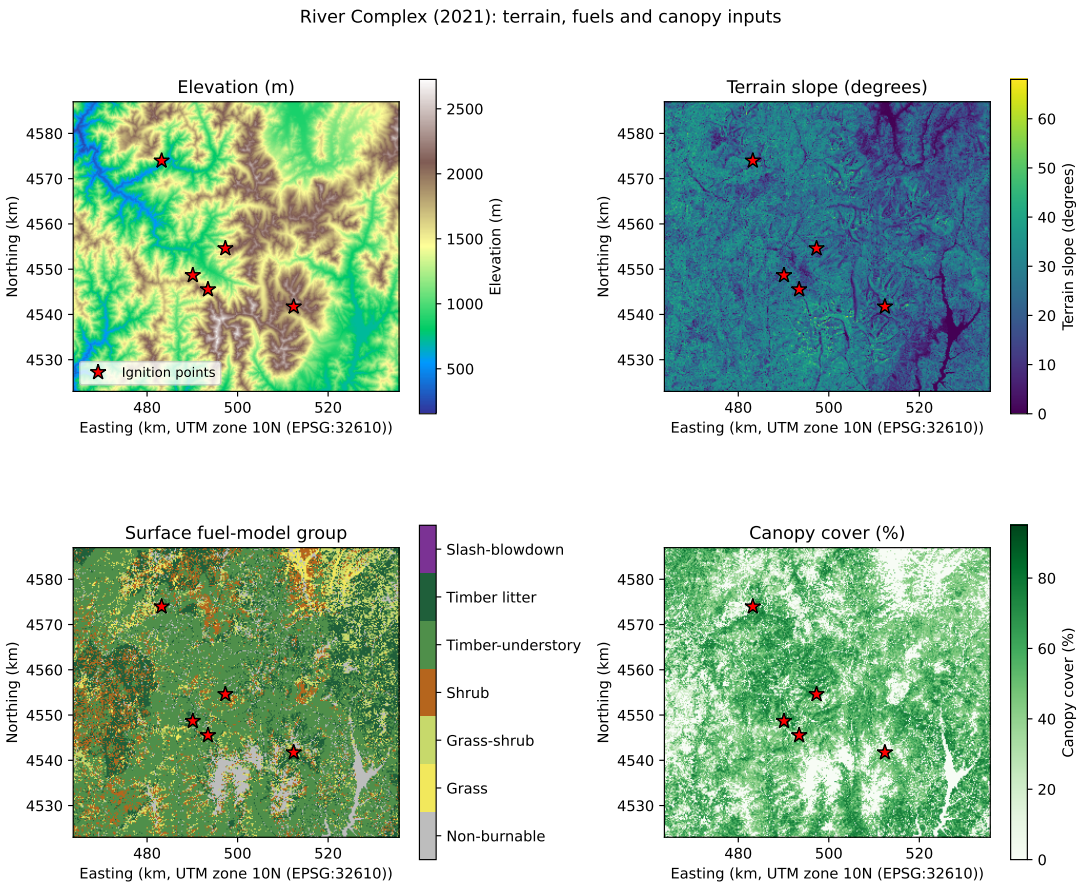


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

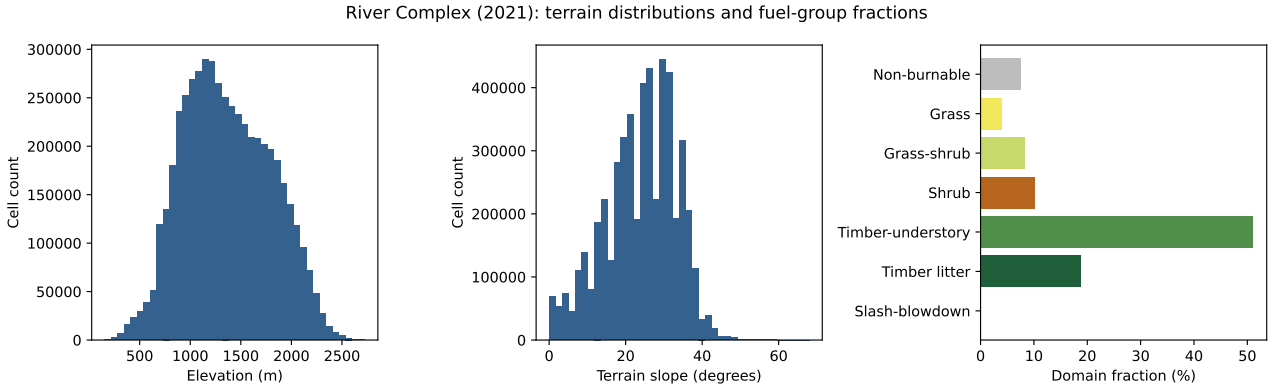


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,354 hourly fields. Median wind speed is 3.99 mph and its 95th percentile is 9.37 mph. Direction is summarized circularly: the mean direction is 214° and mean resultant length is 0.0792. Median 1-h dead-fuel moisture is 6.13%. Spatial and temporal variability remain important when these domain summaries are interpreted.

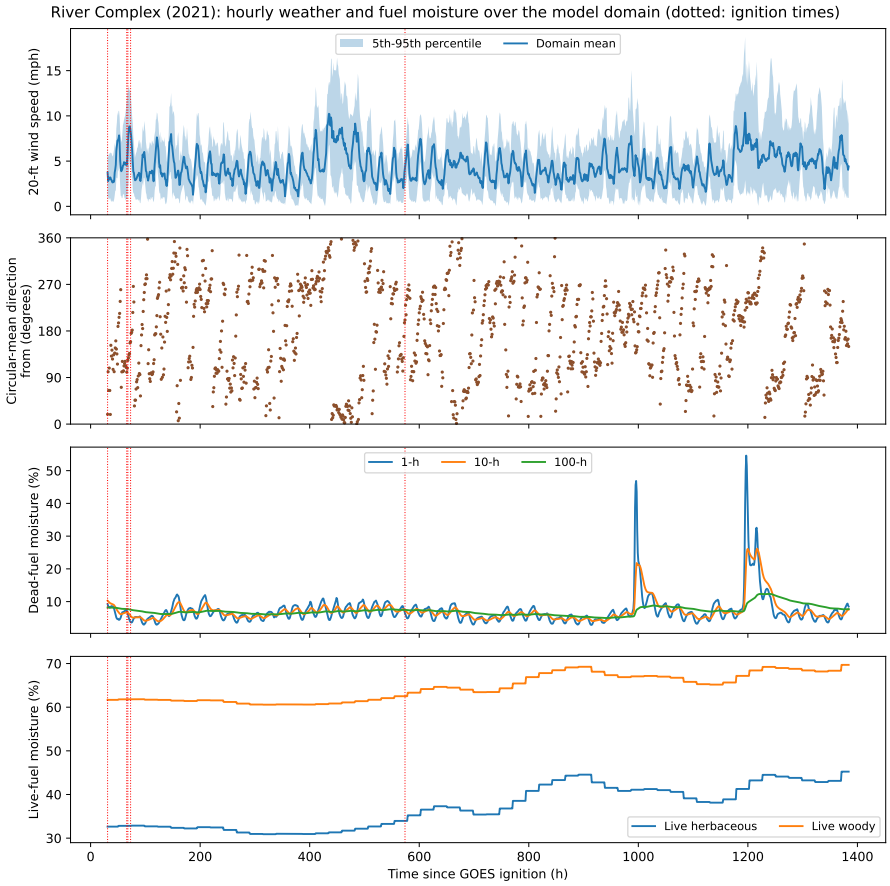


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

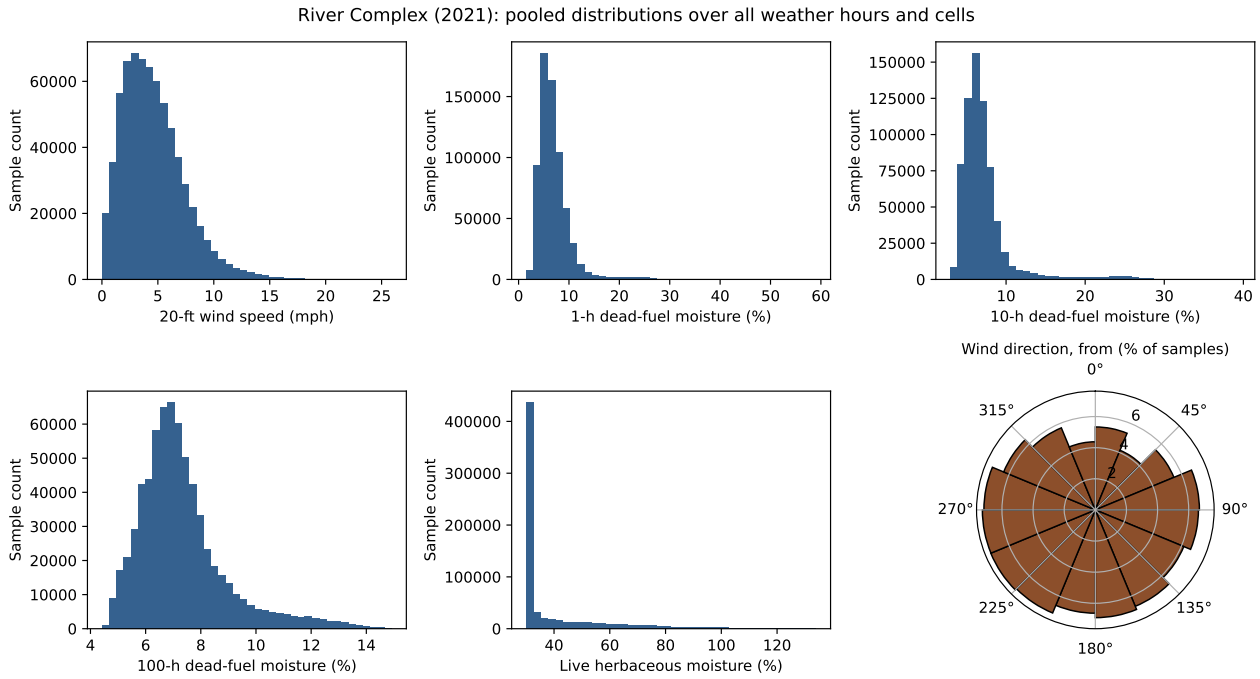


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1353 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1354 hourly weather fields and 5 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 31 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

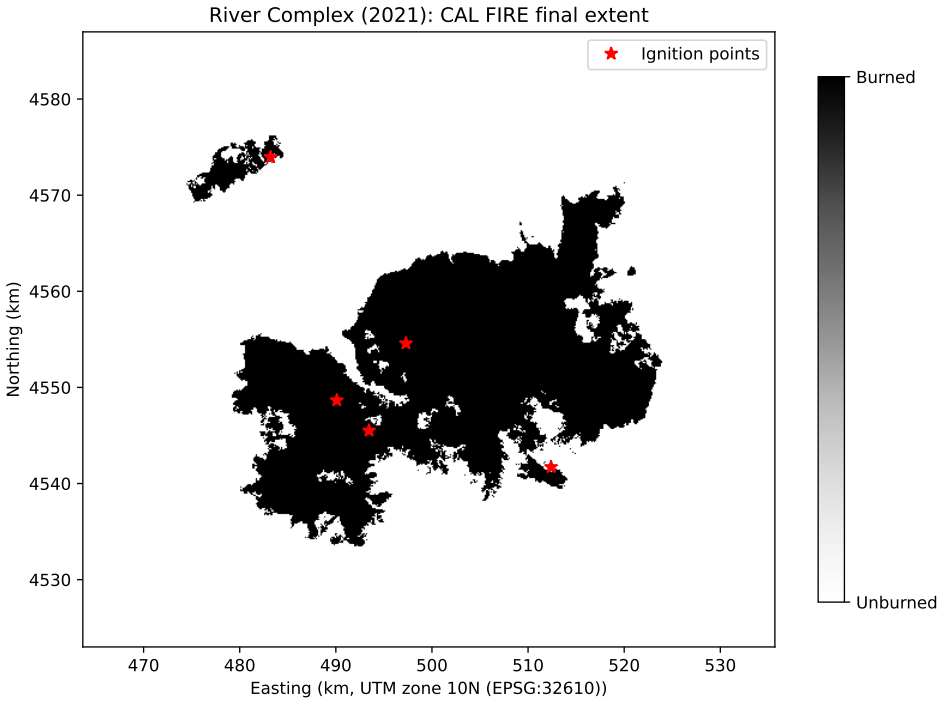


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 806 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the River Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 SCU Lightning Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the SCU Lightning Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 16 August 2020 at 11:00 UTC to 31 August 2020 at 03:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 347 hourly perimeters, a final area of 1525.8 km^2 and an official size of 396,625 ac. The configuration uses 5 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

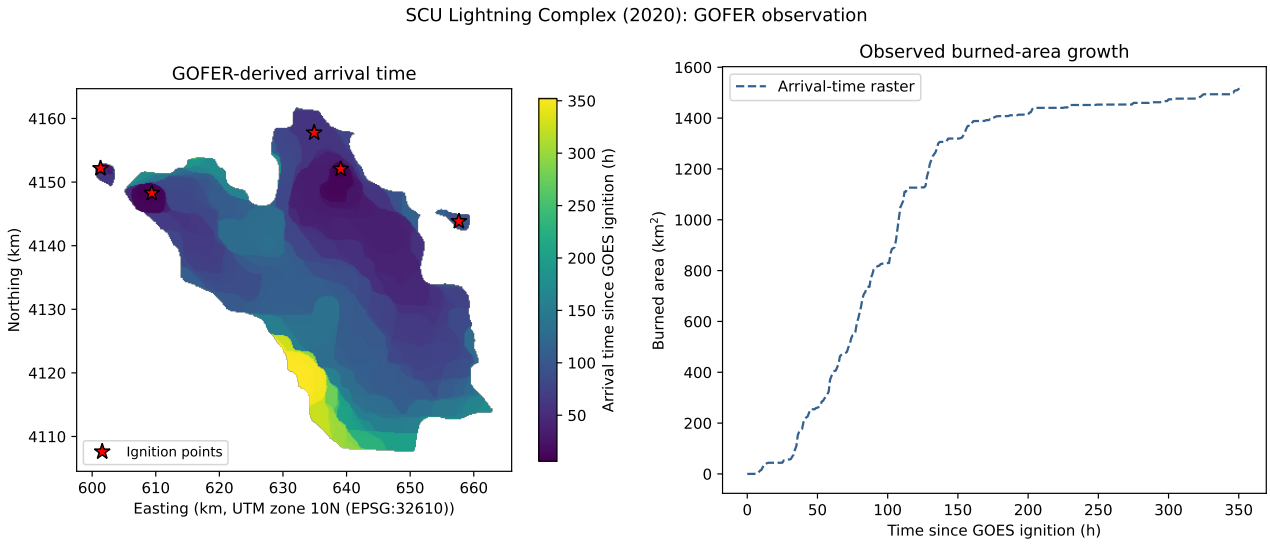


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,800 by 2,533 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 6,383 km^2. Elevation ranges from -4 to 1,323 m; median slope is 13°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

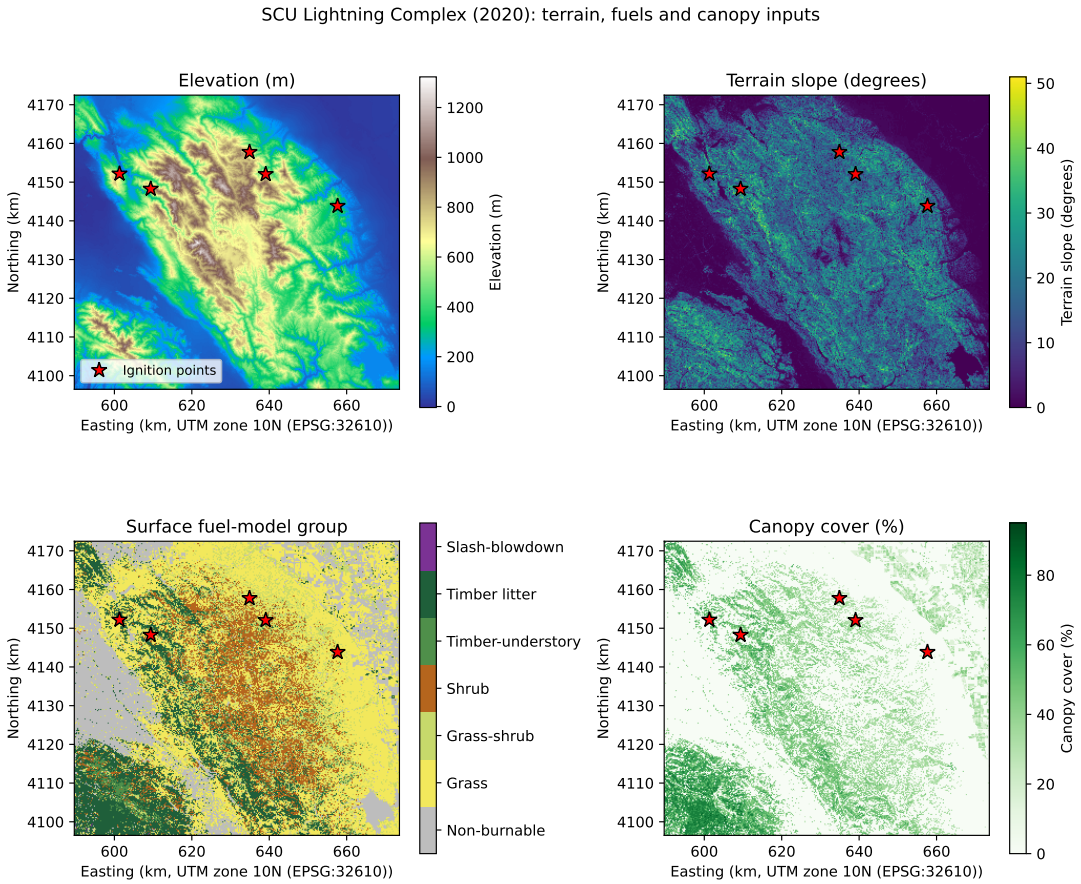


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

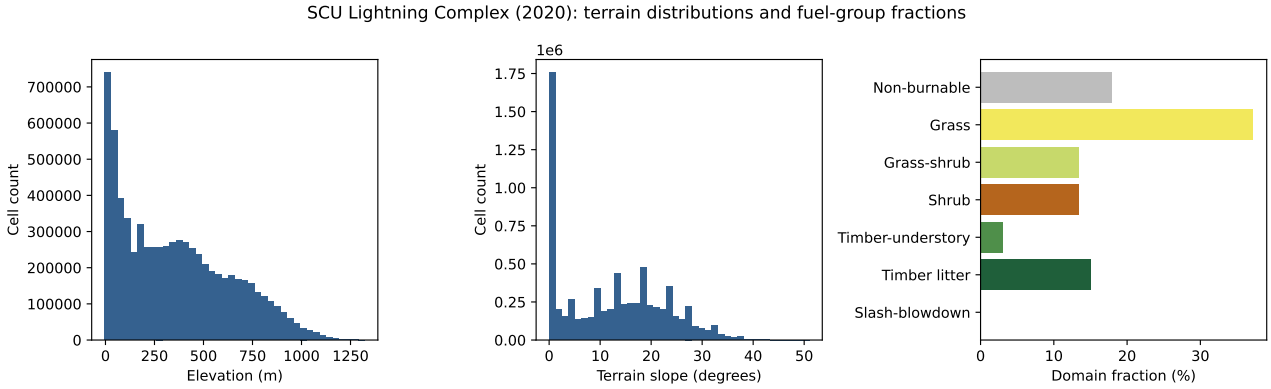


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 348 hourly fields. Median wind speed is 4.72 mph and its 95th percentile is 11.5 mph. Direction is summarized circularly: the mean direction is 293° and mean resultant length is 0.307. Median 1-h dead-fuel moisture is 8.73%. Spatial and temporal variability remain important when these domain summaries are interpreted.

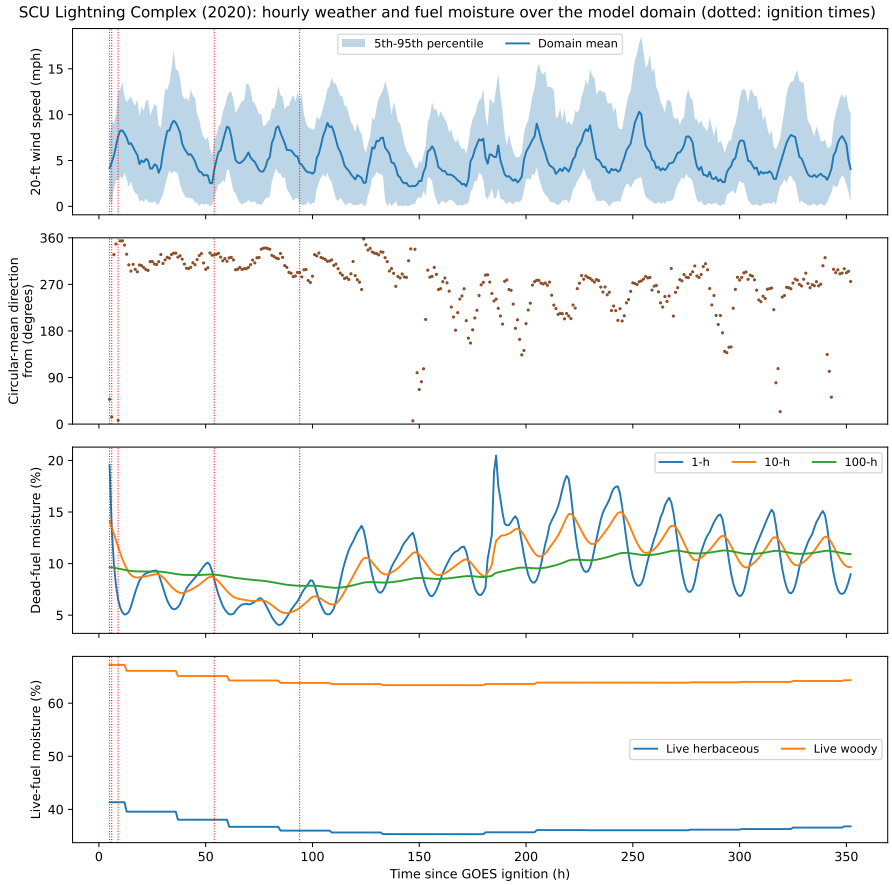


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

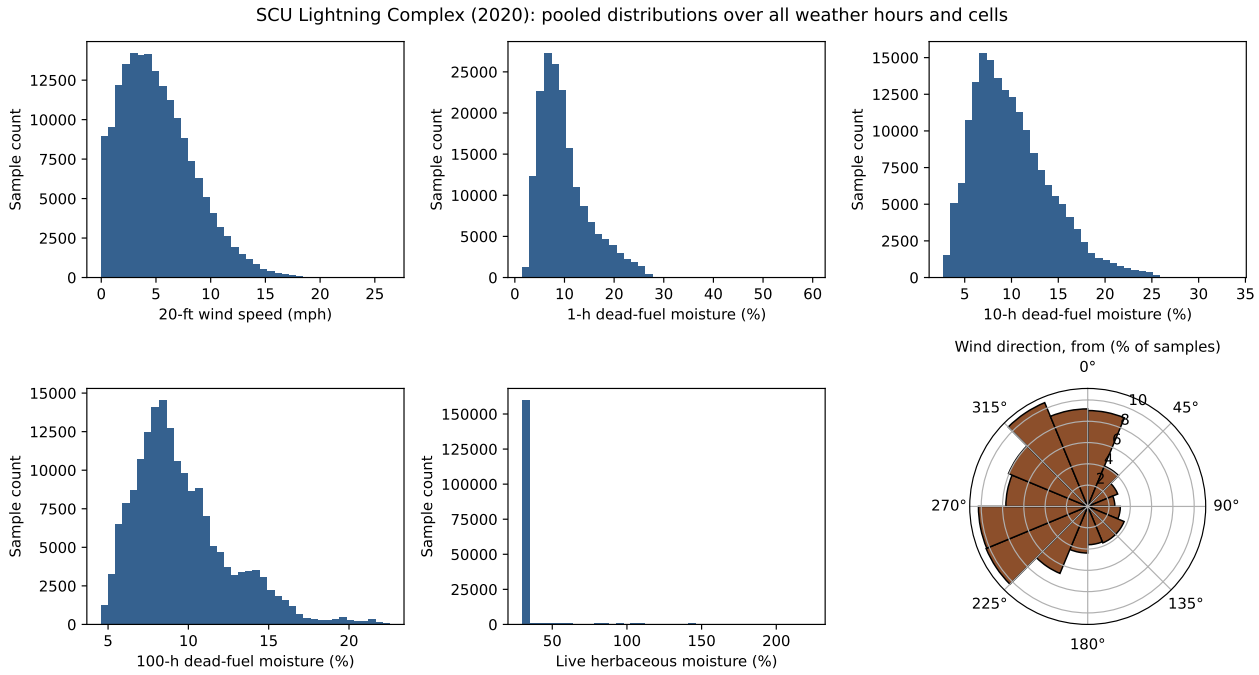


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 347 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 348 hourly weather fields and 5 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 5 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

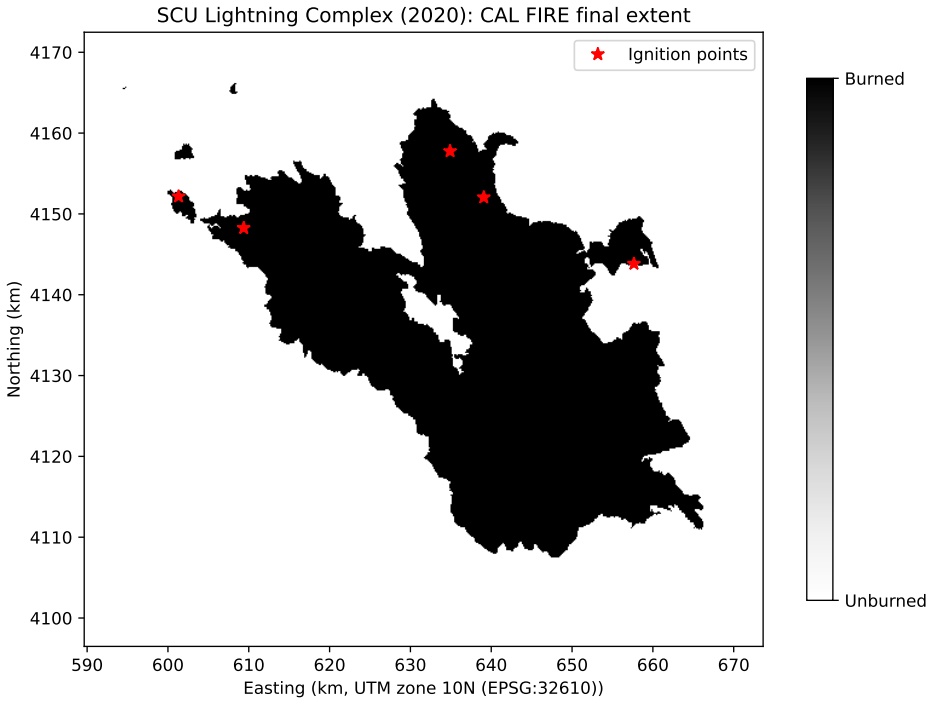


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 1,592 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the SCU Lightning Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 Slater and Devil fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Slater and Devil (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 8 September 2020 at 13:00 UTC to 22 September 2020 at 03:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 326 hourly perimeters, a final area of 697.049 km^2 and an official size of 166,127 ac. The configuration uses 4 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

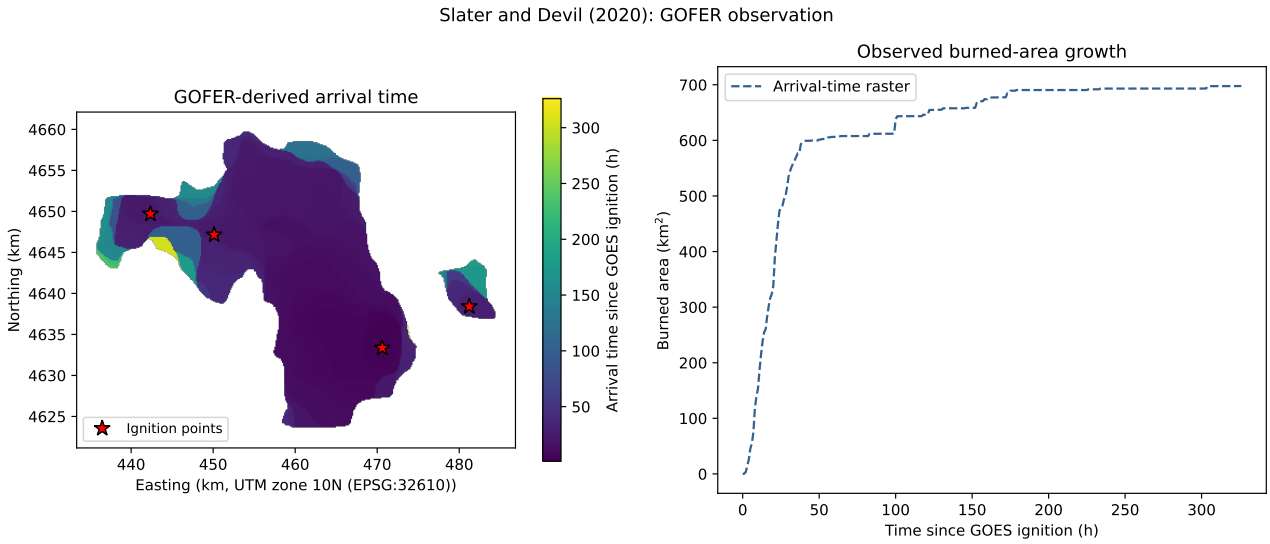


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,333 by 1,933 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 4,059 km^2. Elevation ranges from 164 to 2,203 m; median slope is 24°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 55%, and canopy base height has median 2.1 m.

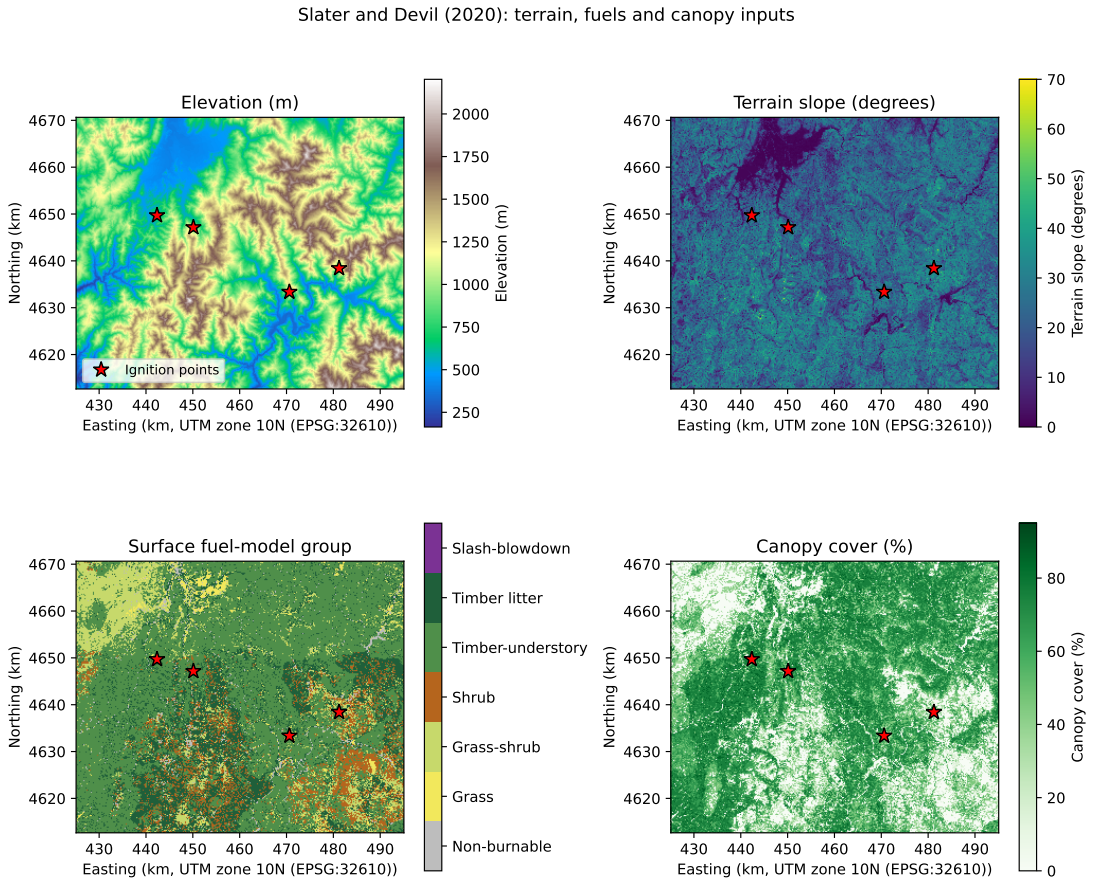


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

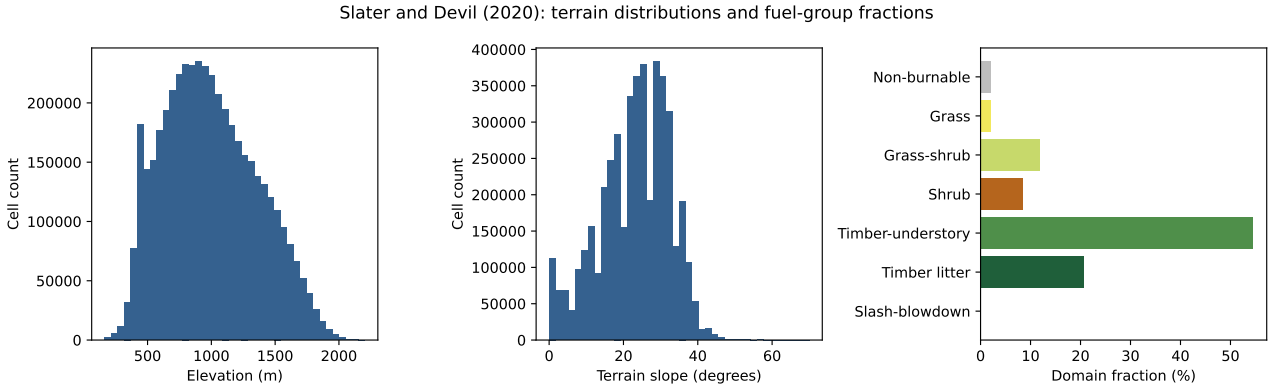


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 327 hourly fields. Median wind speed is 4.27 mph and its 95th percentile is 10.8 mph. Direction is summarized circularly: the mean direction is 201° and mean resultant length is 0.115. Median 1-h dead-fuel moisture is 9.13%. Spatial and temporal variability remain important when these domain summaries are interpreted.

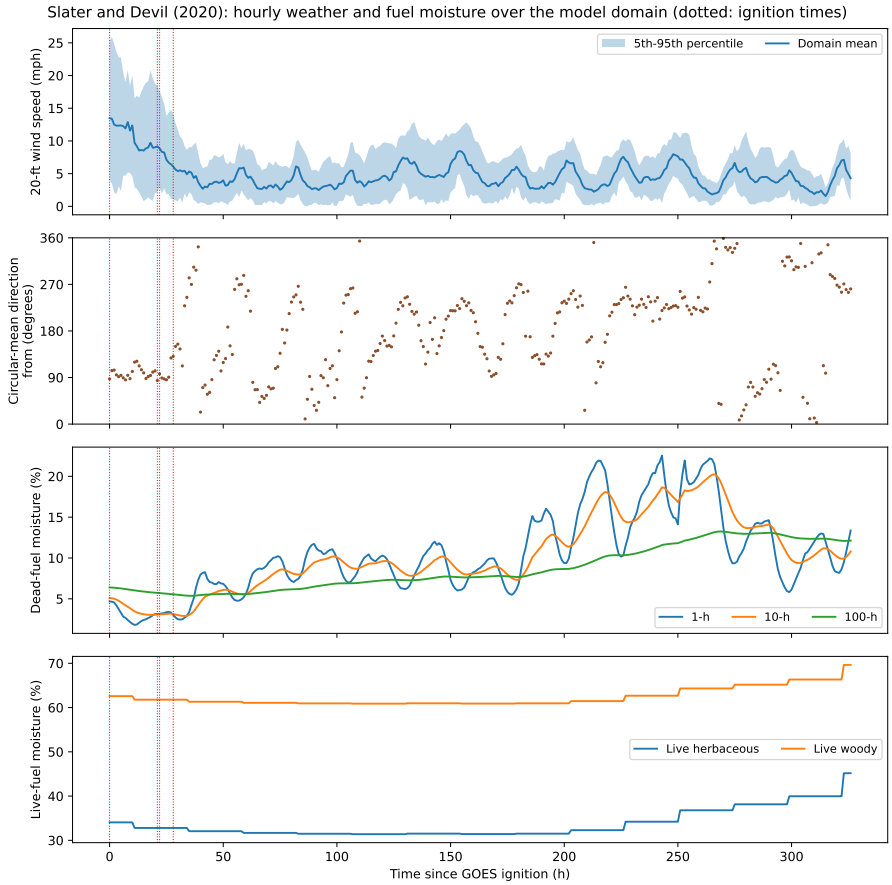


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

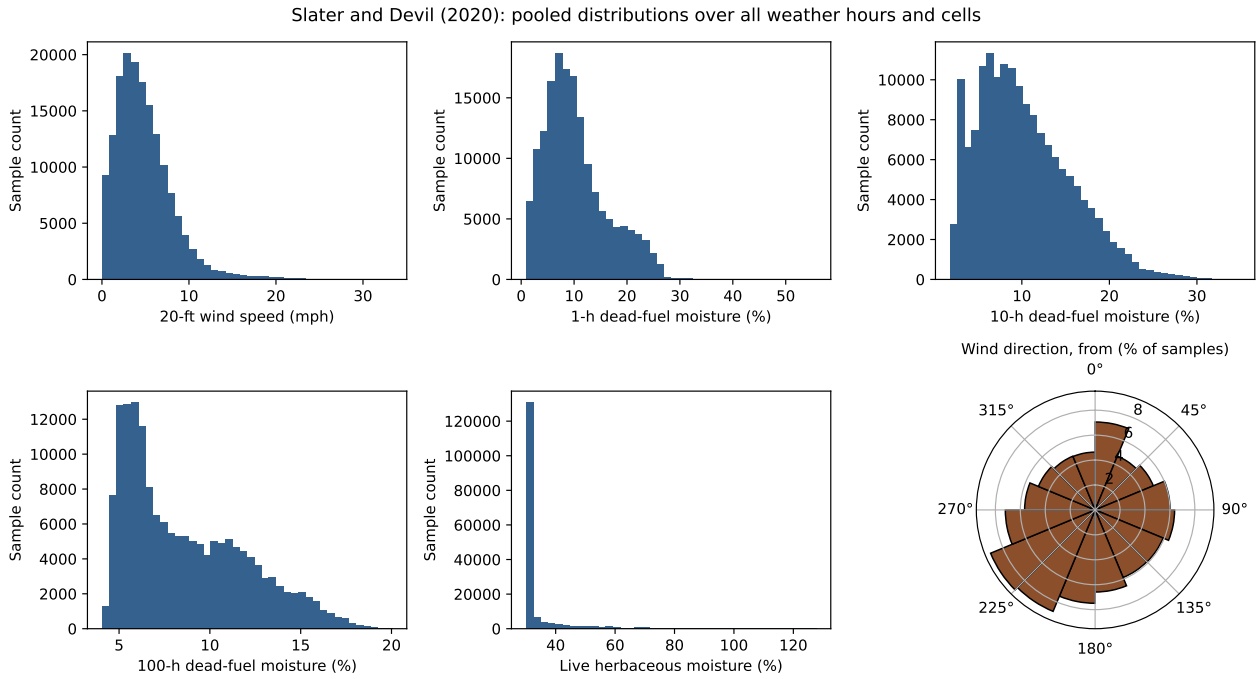


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 326 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 327 hourly weather fields and 4 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 0 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

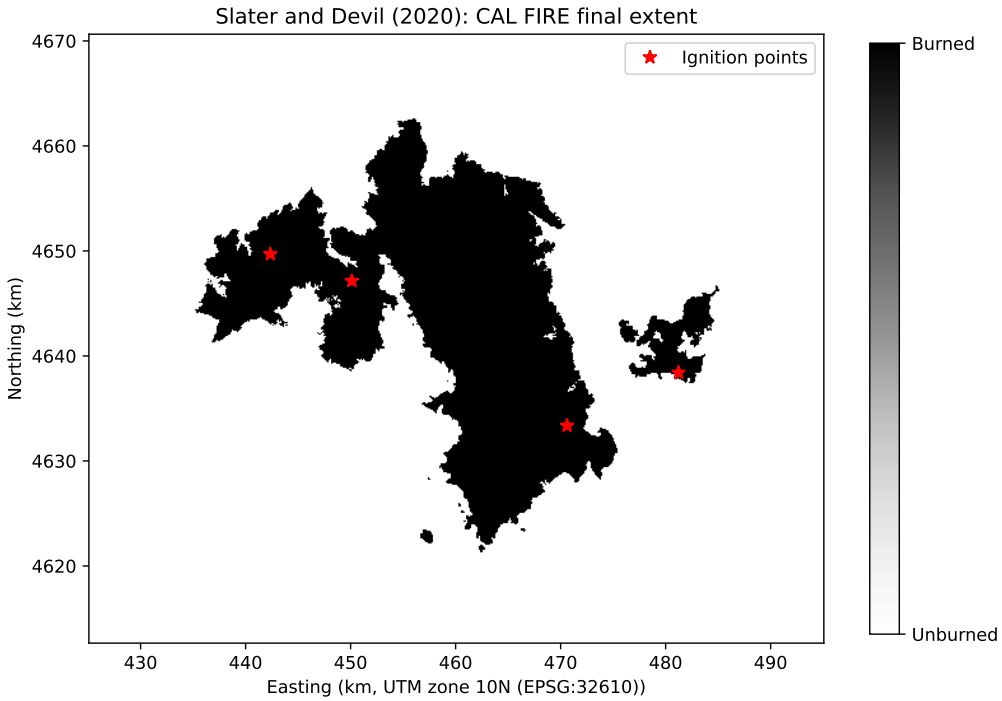


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 673 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Slater and Devil reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 SQF Complex fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the SQF Complex (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 19 August 2020 at 14:00 UTC to 7 October 2020 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 1152 hourly perimeters, a final area of 786.406 km^2 and an official size of 175,019 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

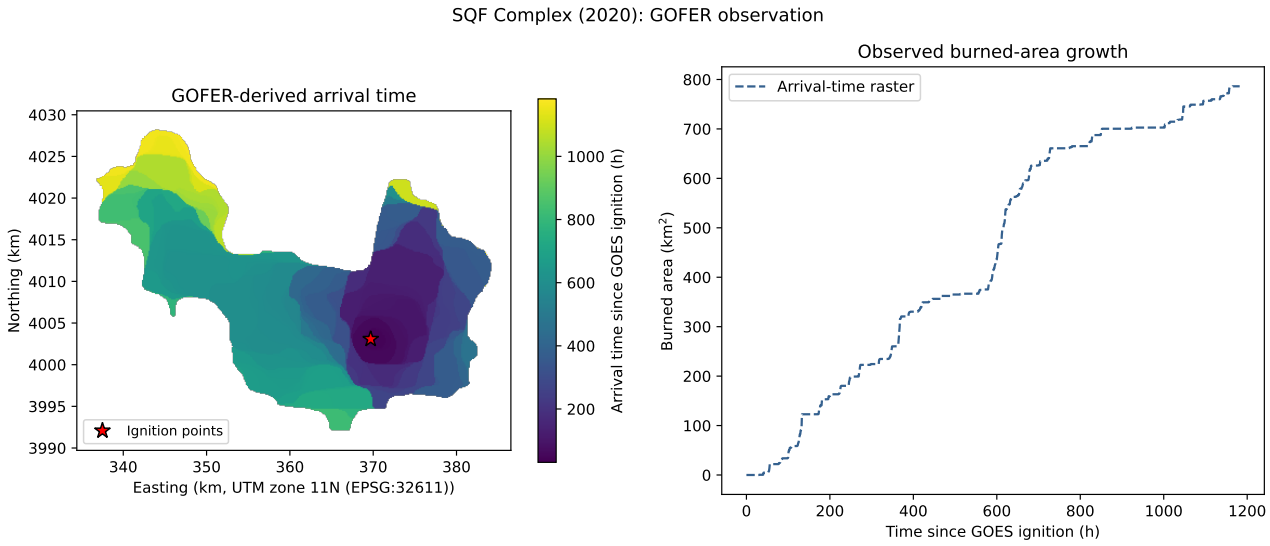


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 2,333 by 1,933 cells at 30 m spacing in UTM zone 11N (EPSG:32611), covering 4,059 km^2. Elevation ranges from 159 to 3,927 m; median slope is 18°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 25%, and canopy base height has median 0.5 m.

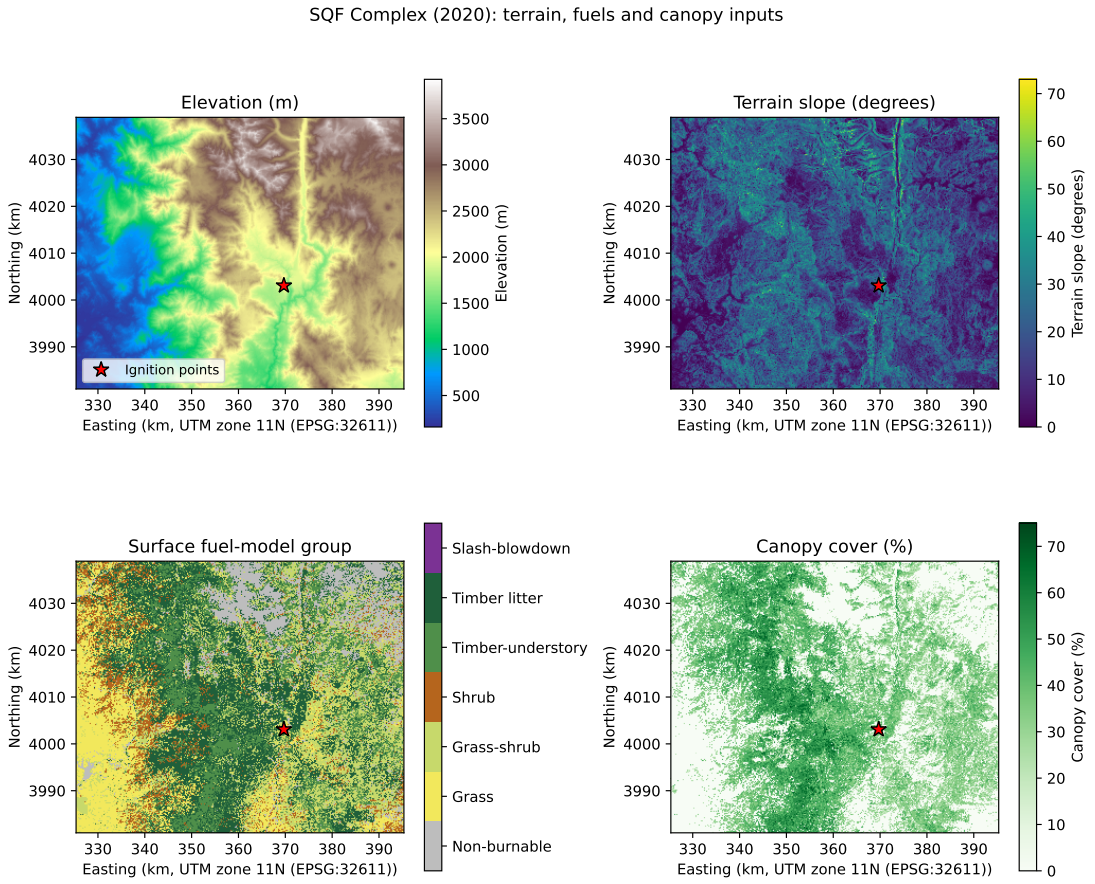


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 11N (EPSG:32611).

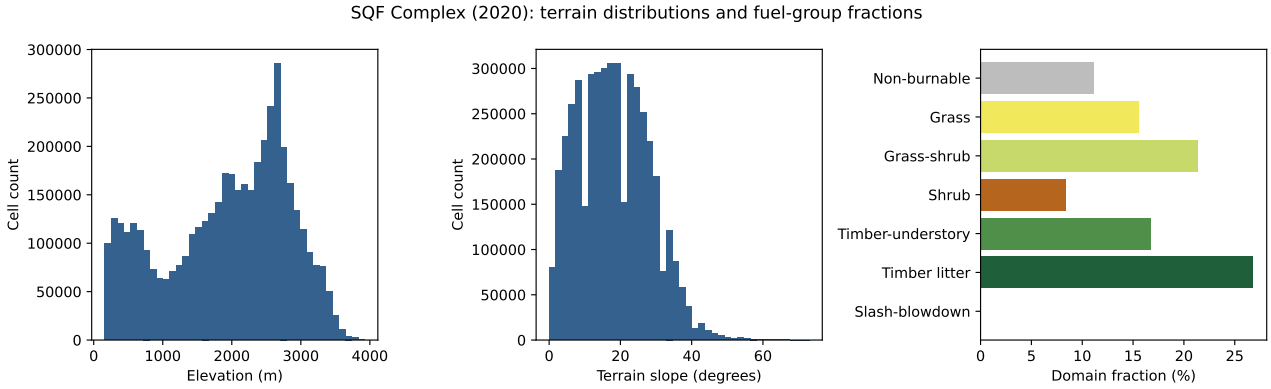


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 1,153 hourly fields. Median wind speed is 4.63 mph and its 95th percentile is 11.8 mph. Direction is summarized circularly: the mean direction is 254° and mean resultant length is 0.0991. Median 1-h dead-fuel moisture is 5.71%. Spatial and temporal variability remain important when these domain summaries are interpreted.

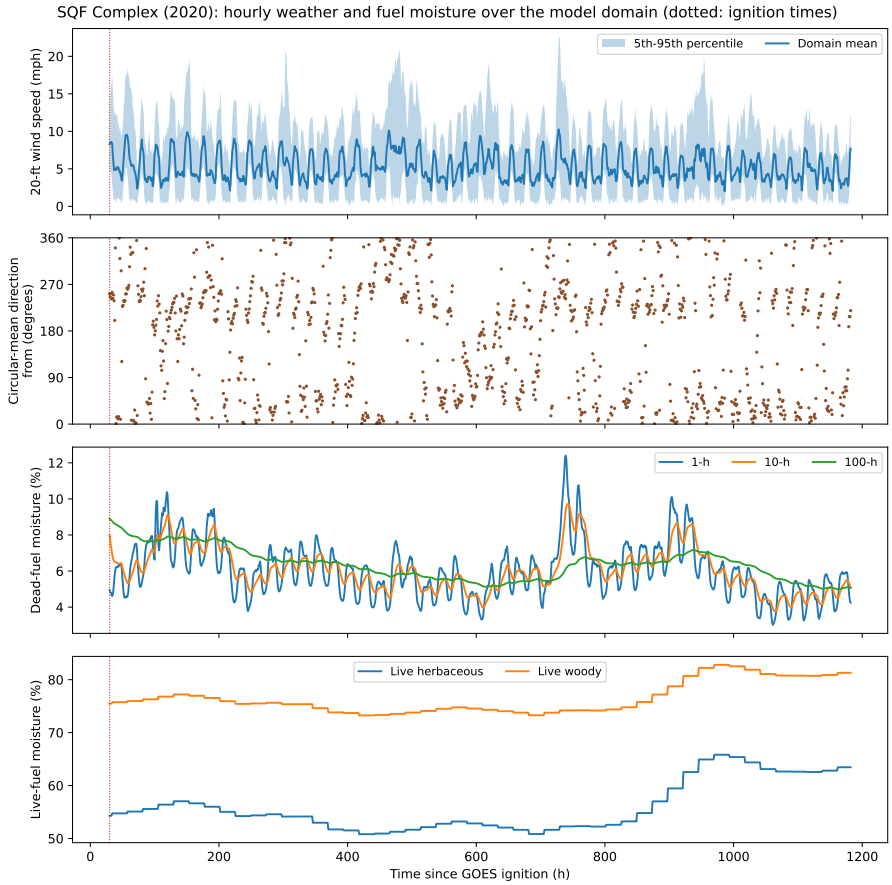


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

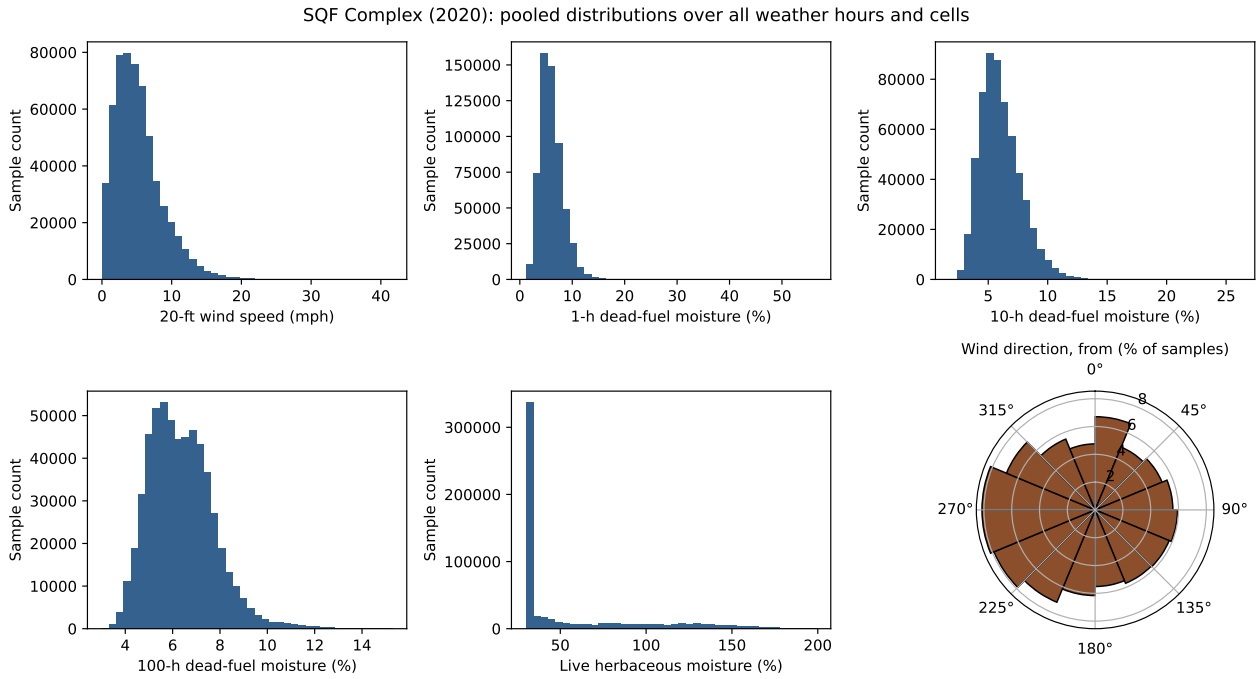


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 1152 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 1153 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 30 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

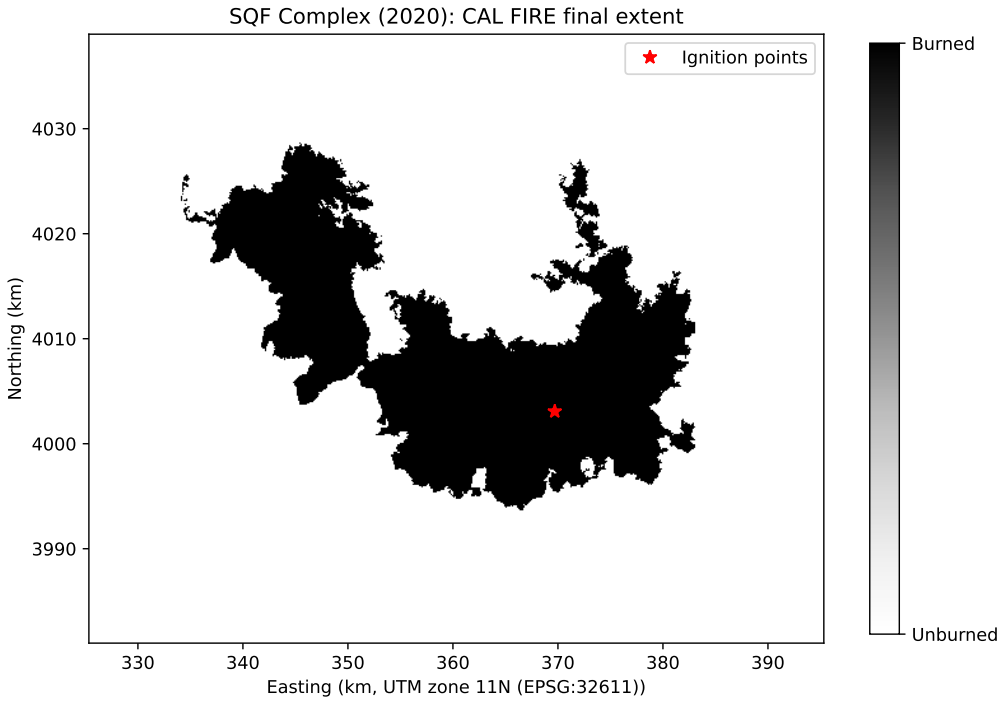


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 690 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the SQF Complex reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 Tamarack fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Tamarack (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 4 July 2021 at 18:00 UTC to 25 July 2021 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 504 hourly perimeters, a final area of 375.326 km^2 and an official size of 68,637 ac. The configuration uses 2 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

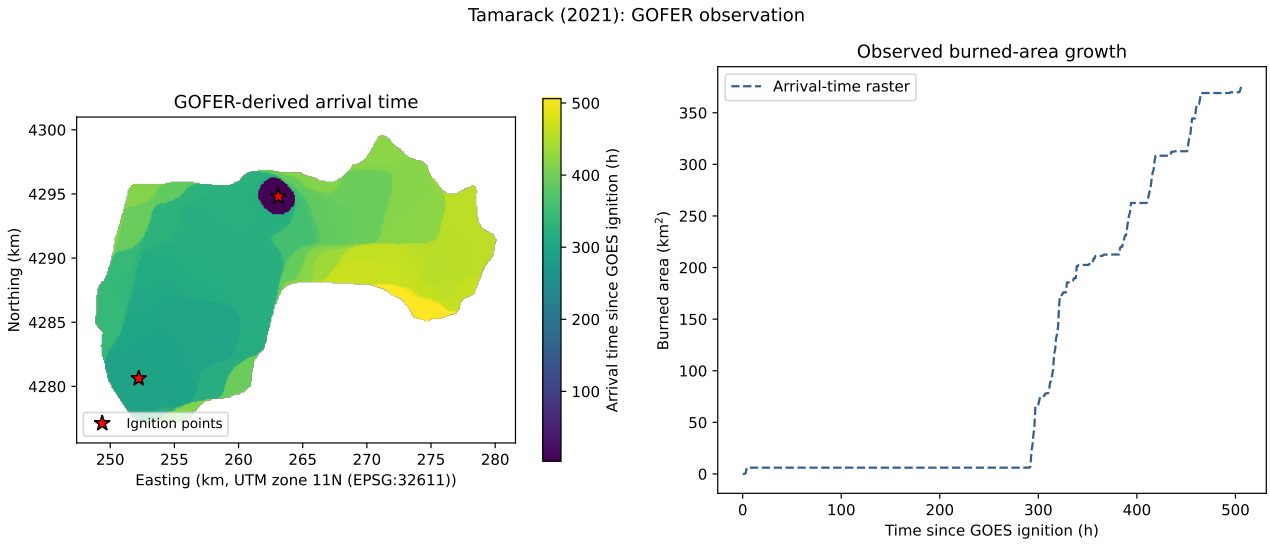


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,733 by 1,467 cells at 30 m spacing in UTM zone 11N (EPSG:32611), covering 2,288 km^2. Elevation ranges from 1,437 to 3,329 m; median slope is 13°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

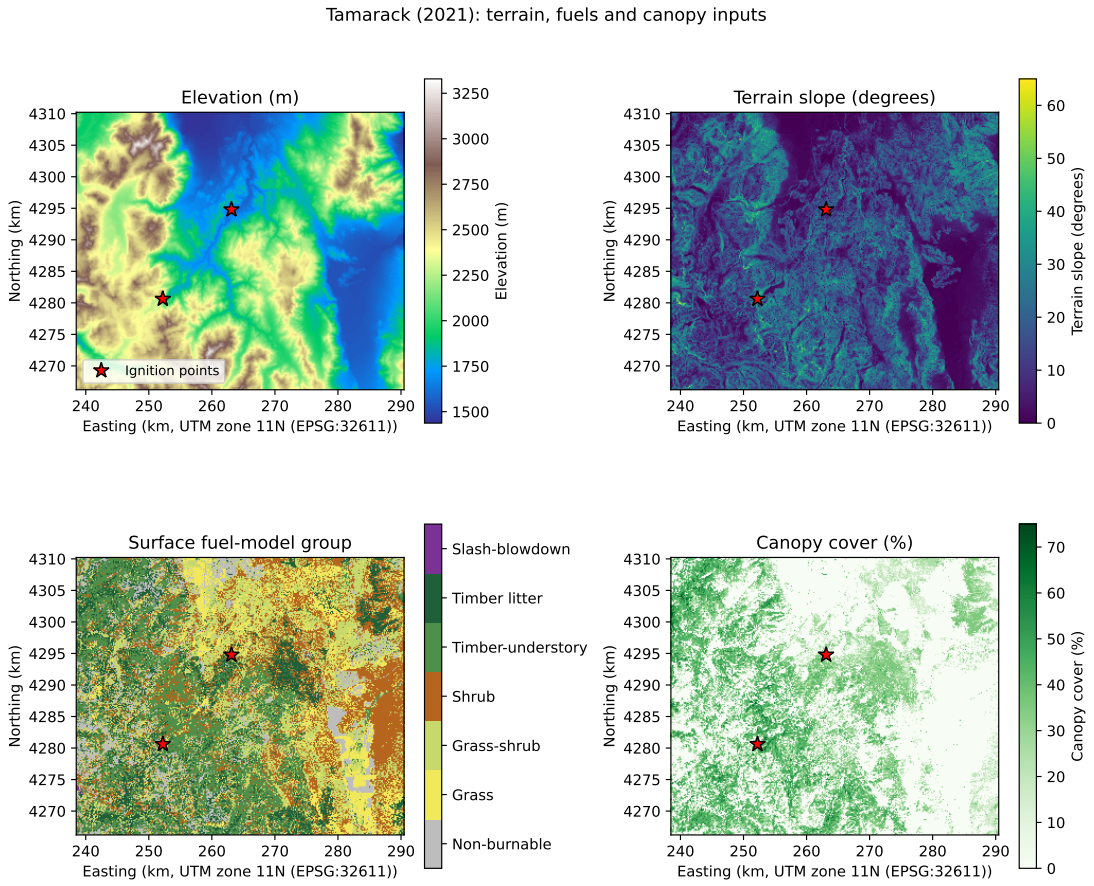


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 11N (EPSG:32611).

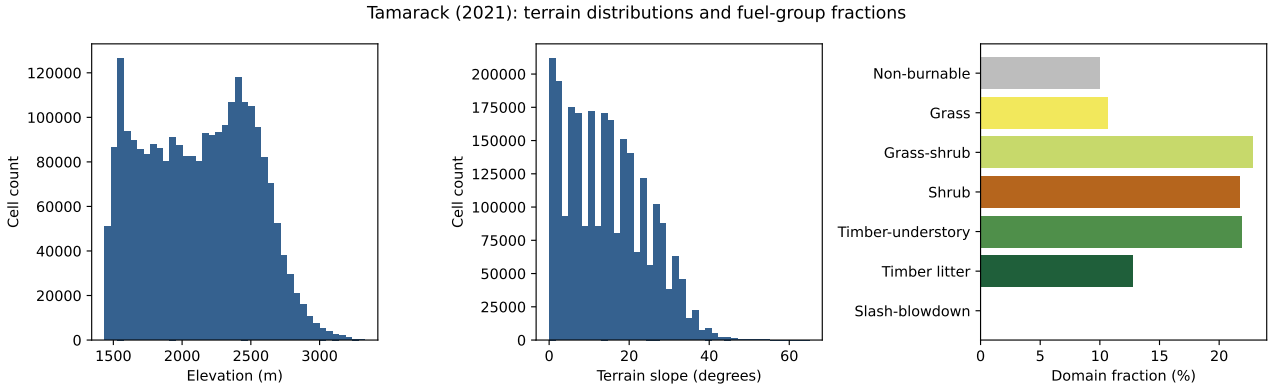


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 505 hourly fields. Median wind speed is 6.12 mph and its 95th percentile is 15.2 mph. Direction is summarized circularly: the mean direction is 226° and mean resultant length is 0.433. Median 1-h dead-fuel moisture is 5.76%. Spatial and temporal variability remain important when these domain summaries are interpreted.

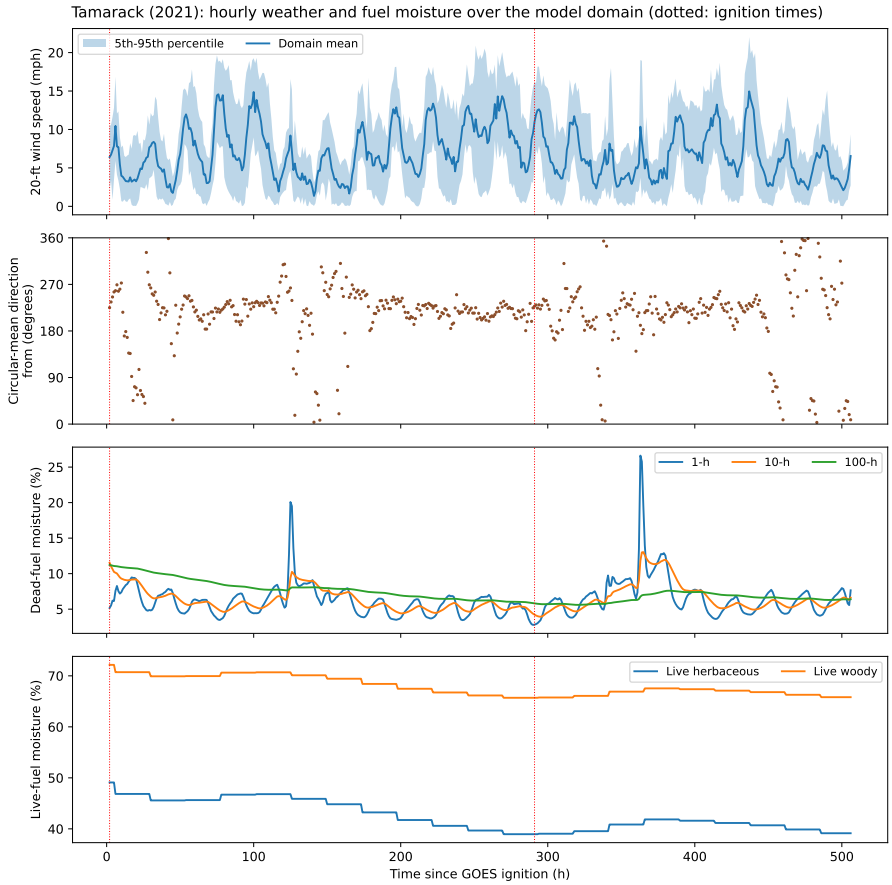


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

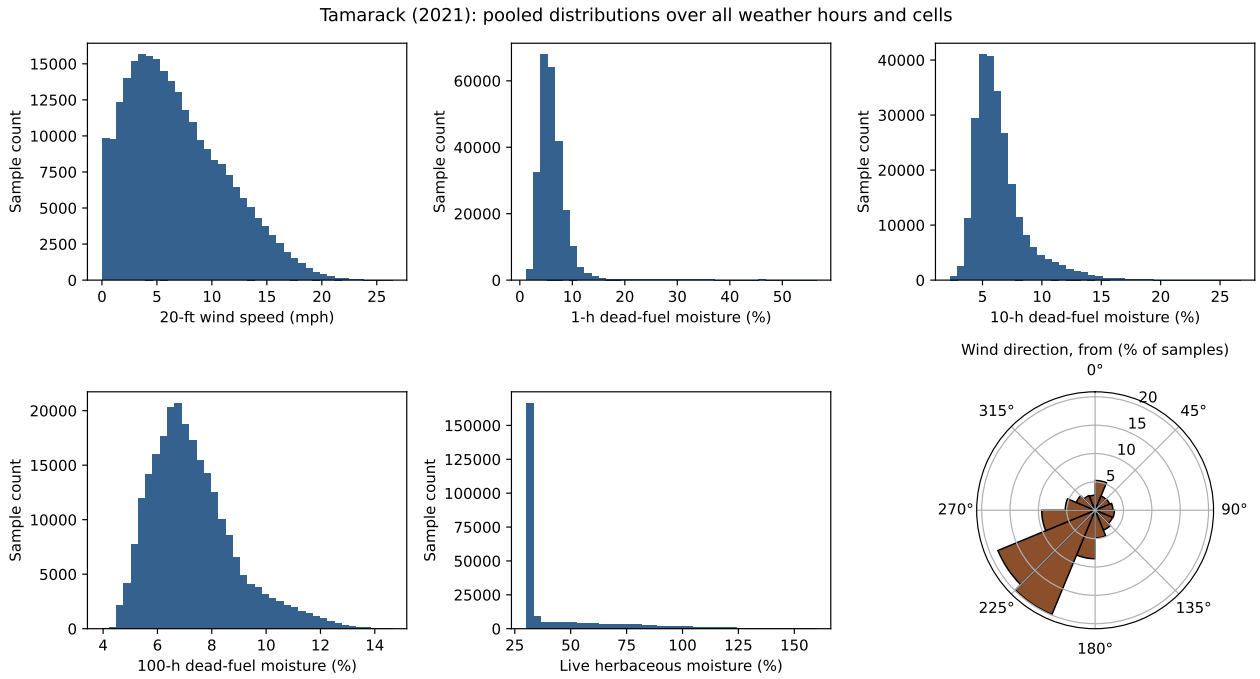


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 504 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 505 hourly weather fields and 2 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 2 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

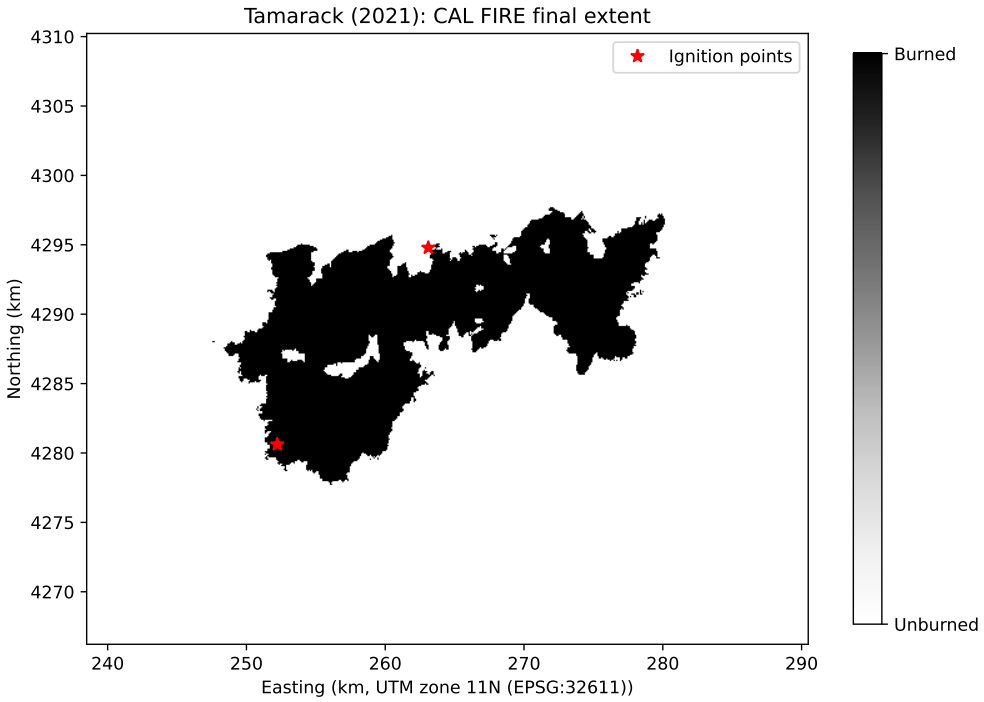


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 272 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Tamarack reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2017 Thomas Fire Landscape Reconstruction

1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2017 Thomas Fire Landscape Reconstruction through 17 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

2 Reference event, experiment, and observations

The configured event start, expressed in Pacific standard time (PST), is 4 December 2017 at 18:30 PST. The comparison instant is 5 December 2017 at 11:30 PST (19:30 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 6804 point records in the geographic coordinate system defined by EPSG:4326; 1446 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

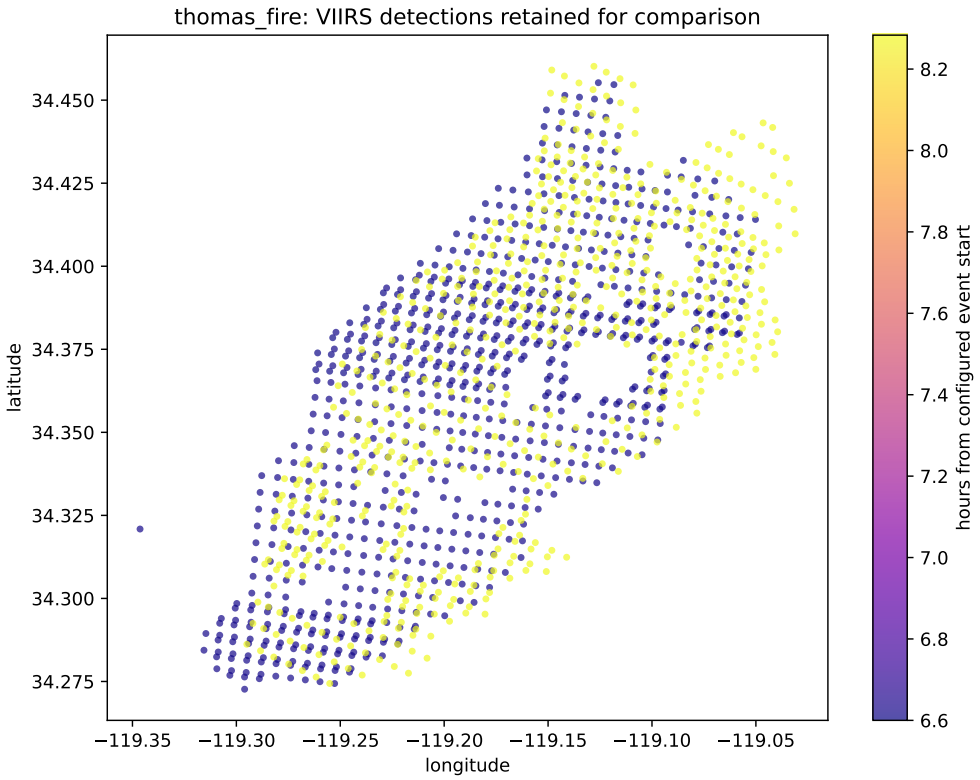


Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `642689c45406337886a8dea52e08ff413fd3e7c2a933dcc6e0f7f07954a3a044`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

Field	Minimum	Mean	Standard deviation	Maximum
Elevation	-1e+04	-1007	3832	2286
Slope	-1e+04	-1502	3591	68
Wind speed	0	26.46	12.74	62.97
1-h moisture	1.301	5.103	2.91	38.25
10-h moisture	2.696	6.503	2.641	15.76
100-h moisture	5.704	10.52	4.272	20.31

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

The active elevation raster contains values of -10000 even though its declared nodata value is -9999. The statistics retain those cells exactly as ELMFIRE sees them; the source owner must determine whether -10000 is a valid value or an undocumented sentinel before scientific interpretation.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 44.71° and 48.16°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most 512×512 cells per band, and are labeled accordingly.

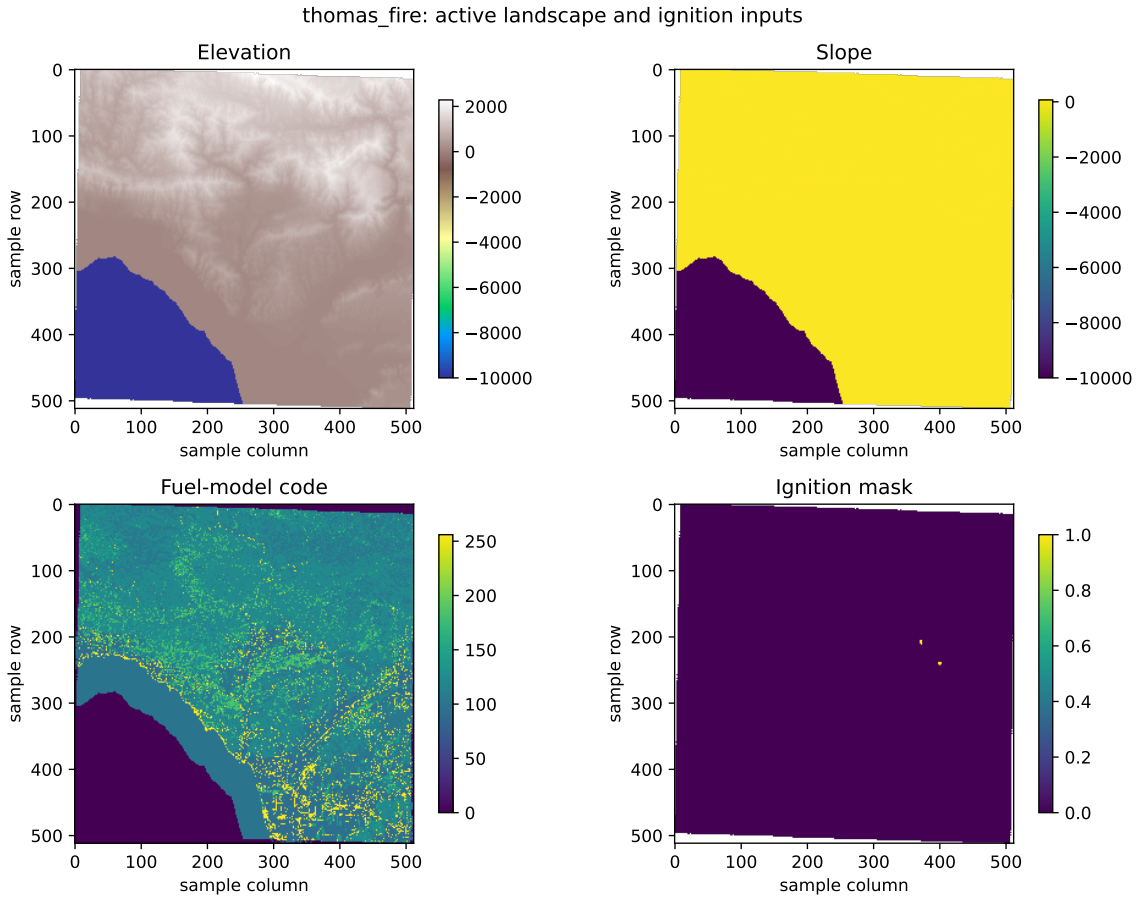


Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

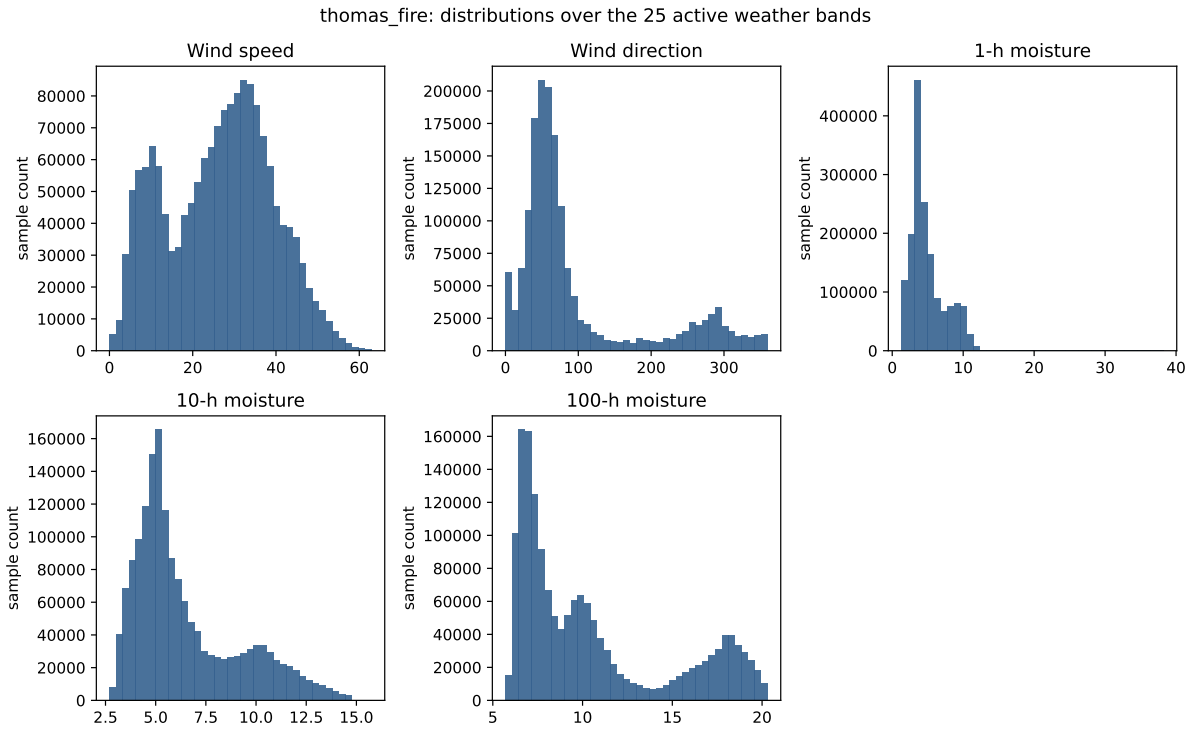


Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

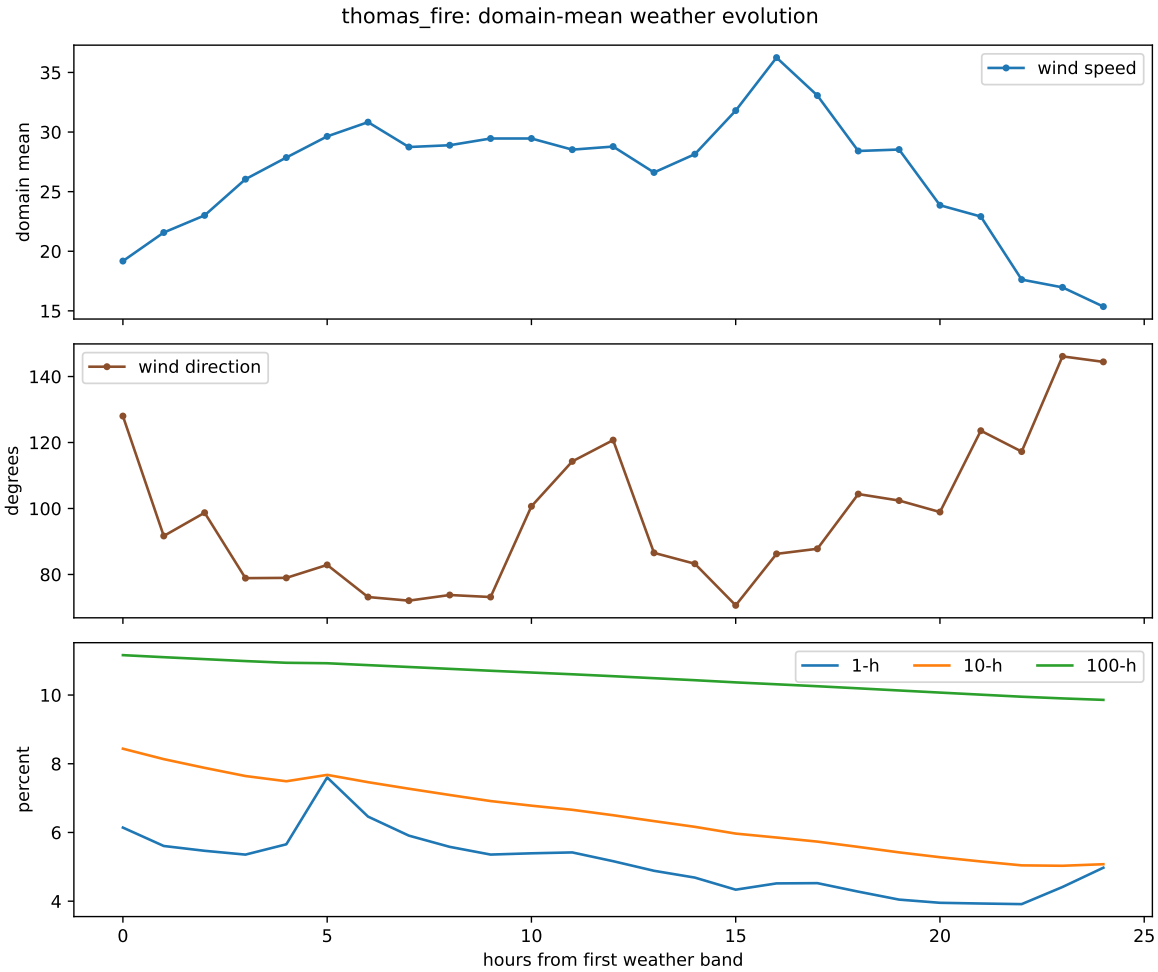


Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

4 Model configuration and rationale

The active domain is 2491 by 1779 cells at 30 m in EPSG:26911. Weather is 29-band weather rasters; bands 1–25 are selected at one-hour intervals, and the ignition treatment is two prescribed point ignitions at archived coordinates and times. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. Archived building spread type 3 was mapped to current University of California, Berkeley/University of Maryland building-spread formulation (type 2). The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

Choice	Evidence class	Basis and limitation
Landscape, weather, ignition, Monte Carlo, spotting, and WUI values	documented	Retained from the received archive; upstream derivation and event-specific tuning history are unknown.
Model-setting compatibility	derived	Required by the choices supported by the inspected model formulation; no physical values were tuned.
Comparison time	documented	Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent.
$p \geq 0.50$ footprint	selected for this comparison	Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use.
VIIRS hull ratio 0.2	selected for this comparison	Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter.

Table 2: Provenance of consequential configuration and comparison choices.

5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

6 Validation metrics and comparison method

For member i, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and t_c the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c] \,, \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support O, intersection $I = |P \cap O|$, and valid-cell count M, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and F_1 balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

7 Simulation results and field-level comparison

The current scientific status is **NOT EVALUABLE**. No current time_of_arrival GeoTIFFs were found for this case. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported and must not be inferred from WUI input rasters alone.

8 Integrated statistics and calculated metrics

Metric	Current value
Jaccard overlap	**MISSING**
Precision	**MISSING**
Recall	**MISSING**
F_1 score	**MISSING**
Brier score	**MISSING**
Predicted/observed area ratio	**MISSING**
Predicted area (km^2)	**MISSING**
Observation-support area (km^2)	**MISSING**
Ensemble members found	0

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **NOT EVALUABLE**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

2017 Tubbs Fire Landscape Reconstruction

1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2017 Tubbs Fire Landscape Reconstruction through 12 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

2 Reference event, experiment, and observations

The configured event start, expressed in Pacific daylight time (PDT), is 8 October 2017 at 21:45 PDT. The comparison instant is 9 October 2017 at 09:45 PDT (16:45 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 1168 point records in the geographic coordinate system defined by EPSG:4326; 597 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

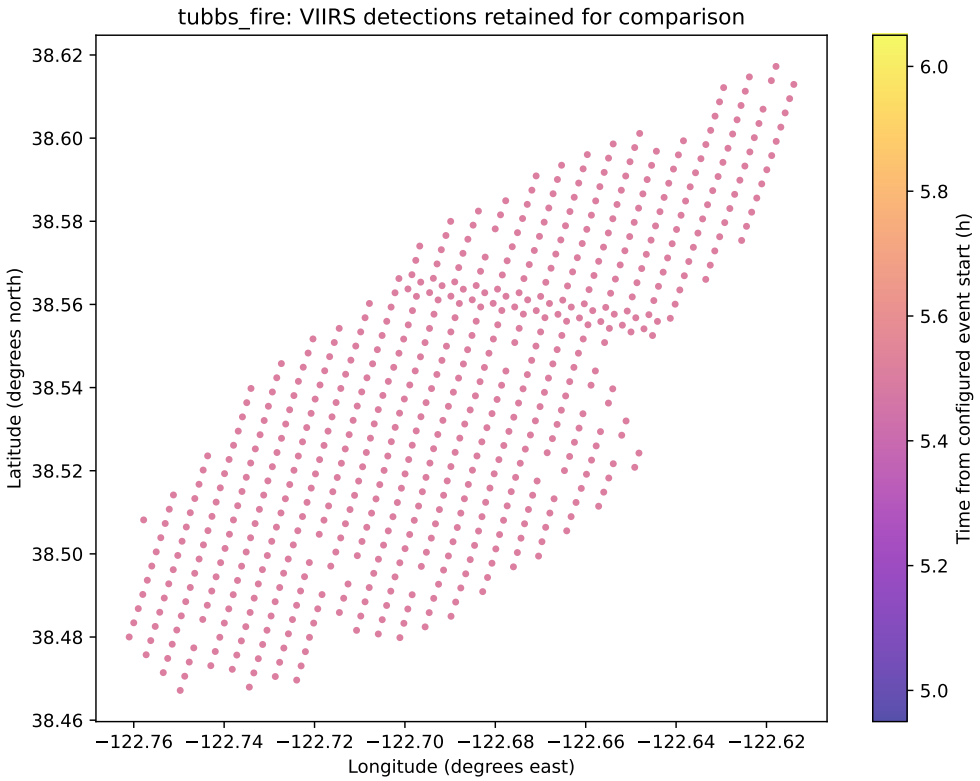


Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `e97bf430d251220 19211405f62f98979e929f80bc3f20d8c3b9f1f9f1a8a4955`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

Field	Minimum	Mean	Standard deviation	Maximum
Elevation	15	164.8	130.8	722
Slope	0	8.329	8.383	47
Wind speed	0.0608	24.32	13.19	54.81
1-h moisture	4.225	5.335	0.3679	6.793
10-h moisture	5.225	6.335	0.3679	7.793
100-h moisture	6.225	7.335	0.3679	8.793

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

The 300-m weather rasters cover the same extent and CRS as the 30-m landscape but intentionally do not share its pixel grid. The preprocessor checks weather extent, band count, and CRS separately rather than treating the resolution difference as an error.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 351.5° and 40.21°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most 512×512 cells per band, and are labeled accordingly.

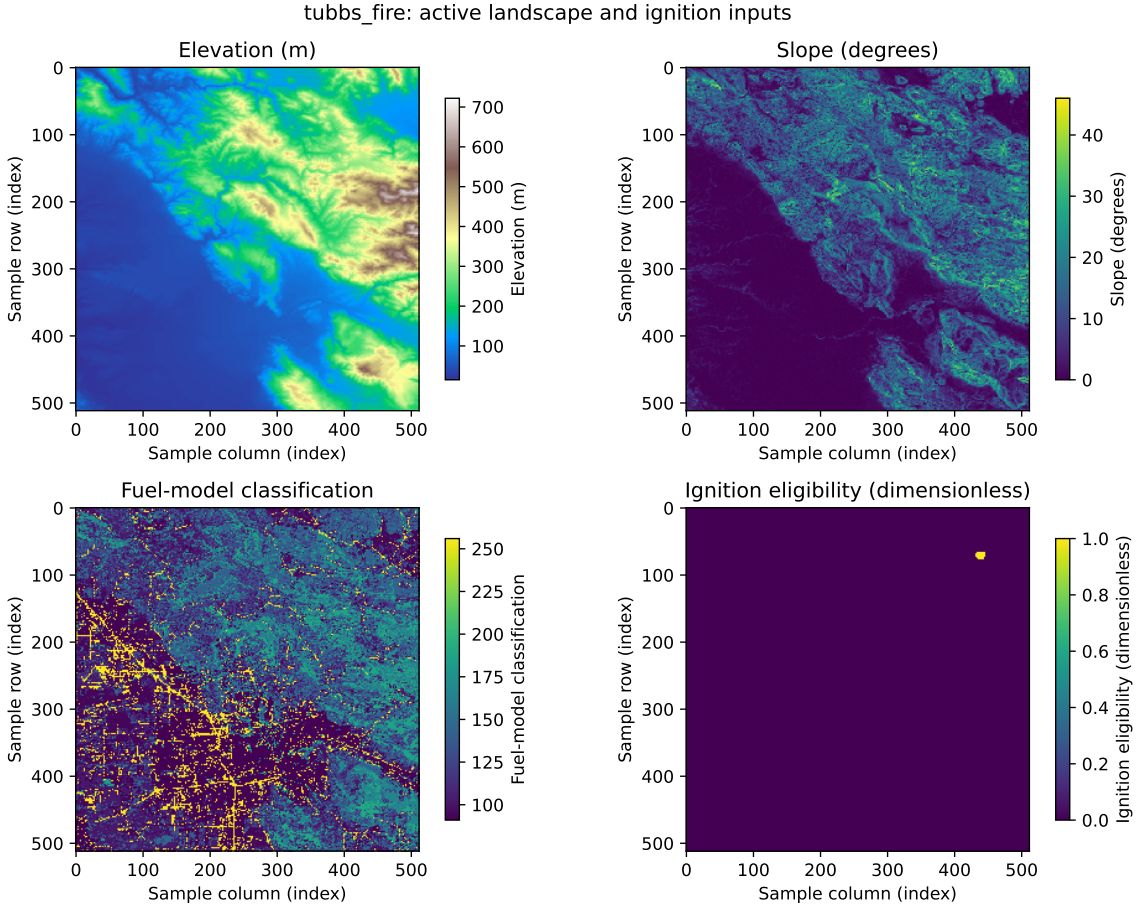


Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

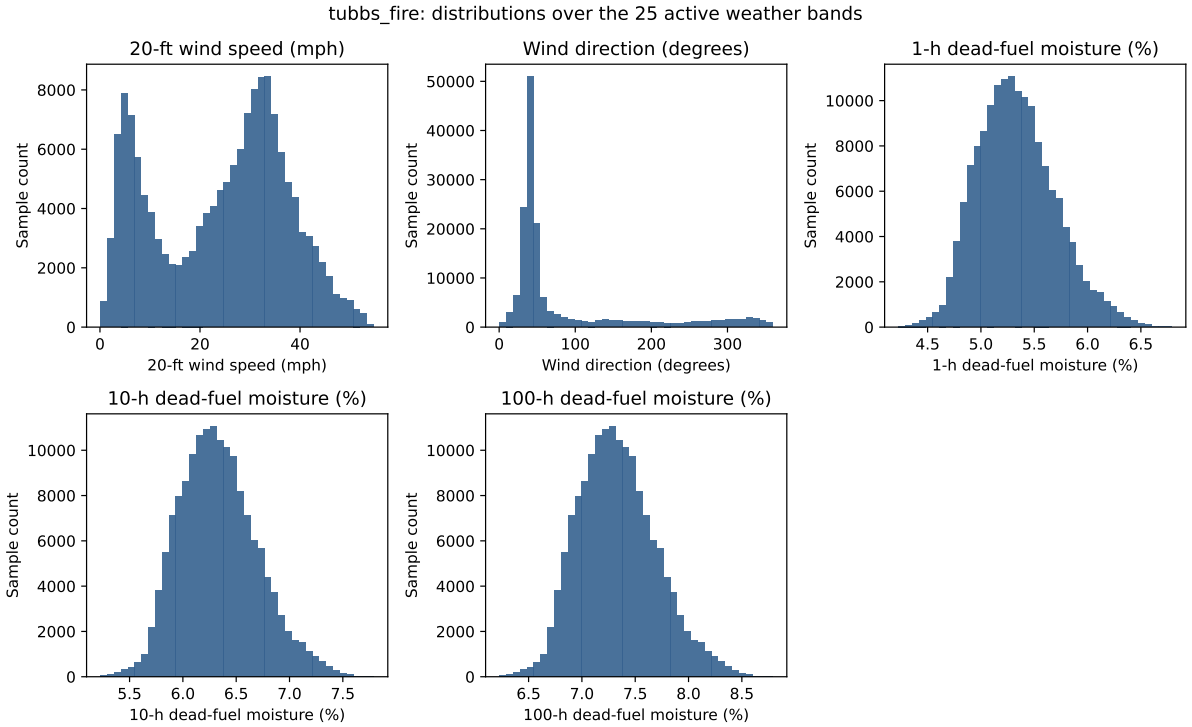


Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

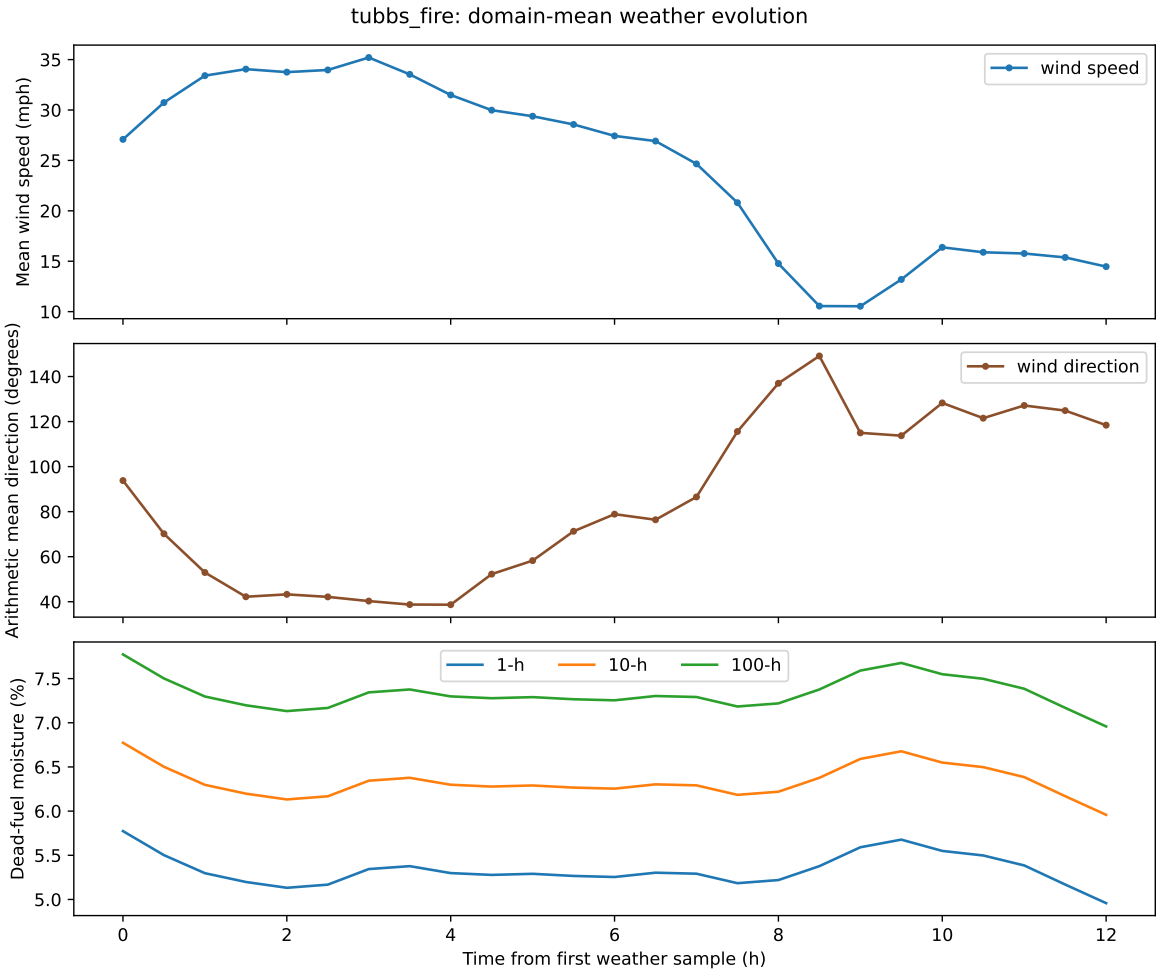


Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

4 Model configuration and rationale

The active domain is 700 by 900 cells at 30 m in the archived Lambert conformal conic CRS. Weather is 25-band, 300 m weather rasters at 30-minute intervals, and the ignition treatment is random ignition sampling from the archived ignition mask, with seed 2024. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive except for the documented removal of a contradictory fixed Tubbs ignition. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted, and the heat-release-rate ellipse adjustment was associated with the wildland–urban interface settings. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. Archived building spread type 3 was mapped to current University of California, Berkeley/University of Maryland building-

spread formulation (type 2). The contradictory archived fixed ignition was disabled because current ELMFIRE applies it in addition to the configured random ignition-mask sample, whereas this case specifies the latter alone. The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

Choice	Evidence class	Basis and limitation
Landscape, weather, ignition, Monte Carlo, spotting, and WUI values	documented	Retained from the received archive; upstream derivation and event-specific tuning history are unknown.
Model-setting compatibility	derived	Required by the choices supported by the inspected model formulation; no physical values were tuned.
Comparison time	documented	Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent.
$p \geq 0.50$ footprint	selected for this comparison	Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use.
VIIRS hull ratio 0.2	selected for this comparison	Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter.

Table 2: Provenance of consequential configuration and comparison choices.

5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

6 Validation metrics and comparison method

For member i, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and t_c the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c], \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support O, intersection $I = |P \cap O|$, and valid-cell count M, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and F_1 balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

7 Simulation results and field-level comparison

The current scientific status is **NOT EVALUABLE**. No current time_of_arrival GeoTIFFs were found for this case. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported and must not be inferred from WUI input rasters alone.

8 Integrated statistics and calculated metrics

Metric	Current value
Jaccard overlap	**MISSING**
Precision	**MISSING**
Recall	**MISSING**
F_1 score	**MISSING**
Brier score	**MISSING**
Predicted/observed area ratio	**MISSING**
Predicted area (km^2)	**MISSING**
Observation-support area (km^2)	**MISSING**
Ensemble members found	0

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **NOT EVALUABLE**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

2020 W-5 Cold Springs fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the W-5 Cold Springs (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 18 August 2020 at 18:00 UTC to 29 August 2020 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 264 hourly perimeters, a final area of 364.235 km^2 and an official size of 84,817 ac. The configuration uses 2 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

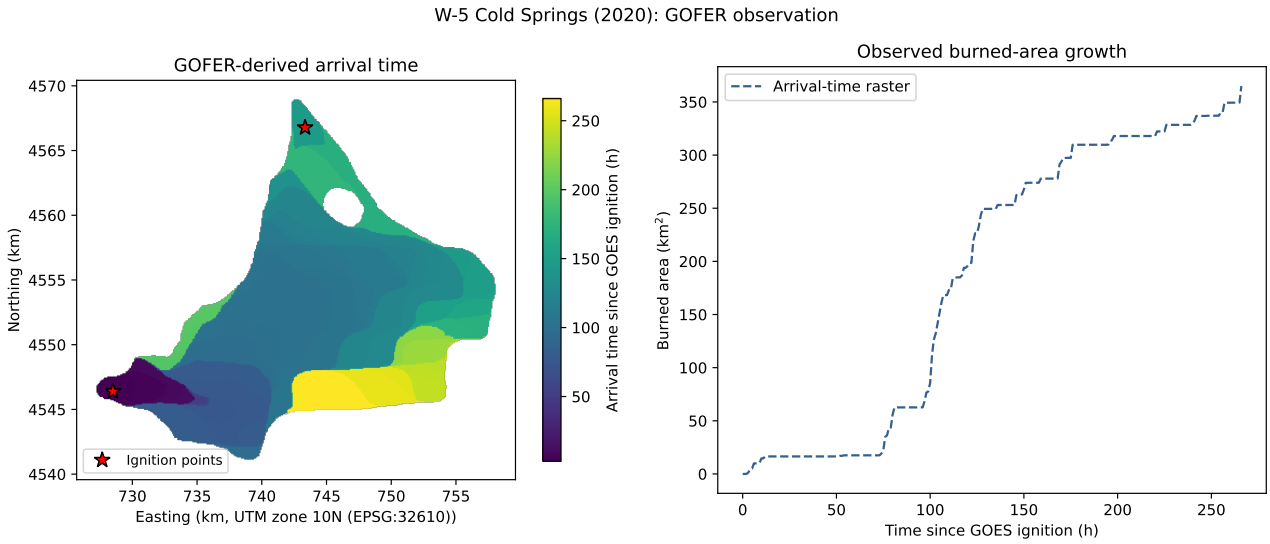


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,733 by 1,667 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 2,600 km^2. Elevation ranges from 1,362 to 3,014 m; median slope is 6°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

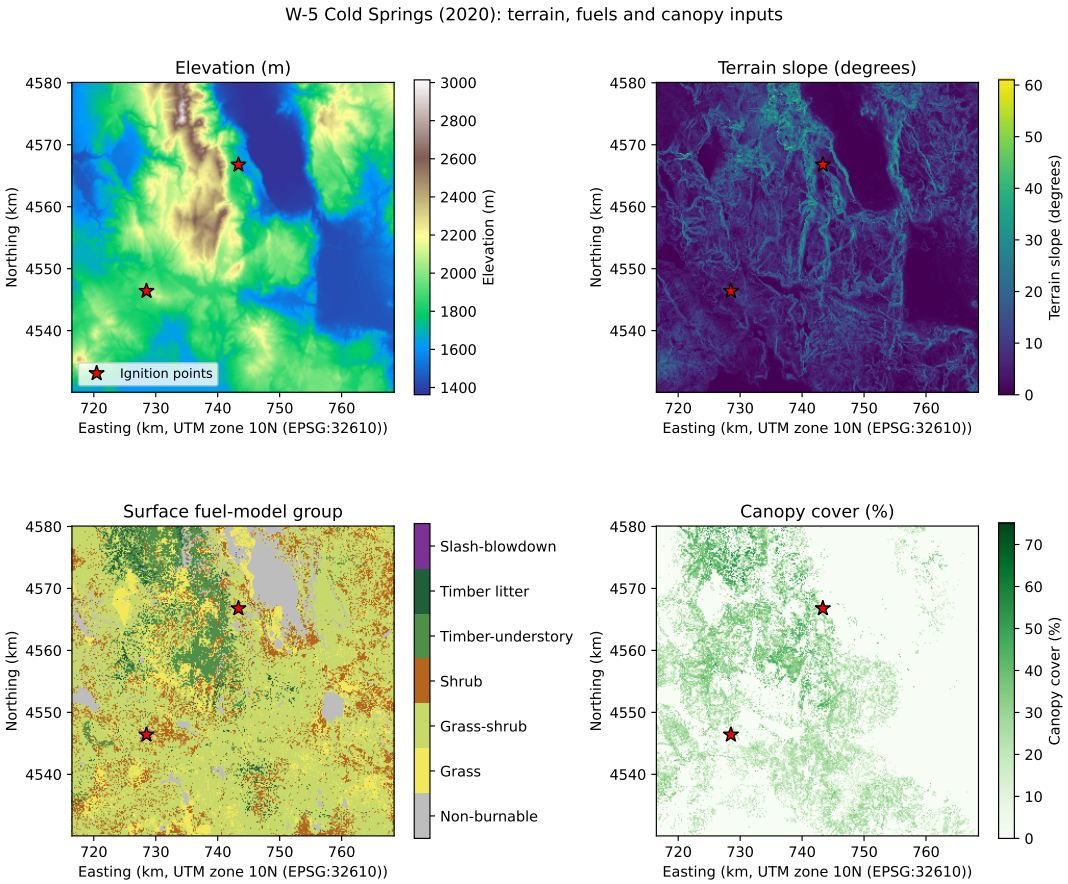


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

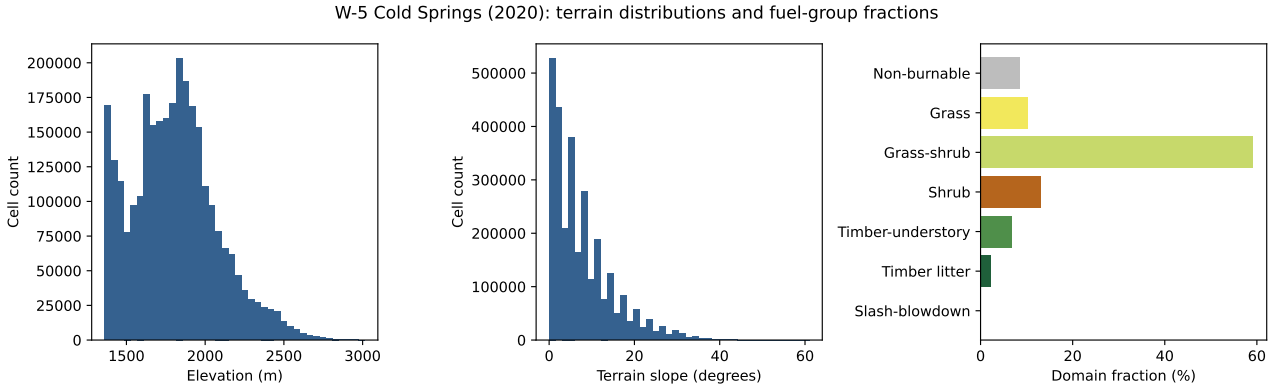


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 265 hourly fields. Median wind speed is 5.92 mph and its 95th percentile is 12.7 mph. Direction is summarized circularly: the mean direction is 260° and mean resultant length is 0.301. Median 1-h dead-fuel moisture is 6.43%. Spatial and temporal variability remain important when these domain summaries are interpreted.

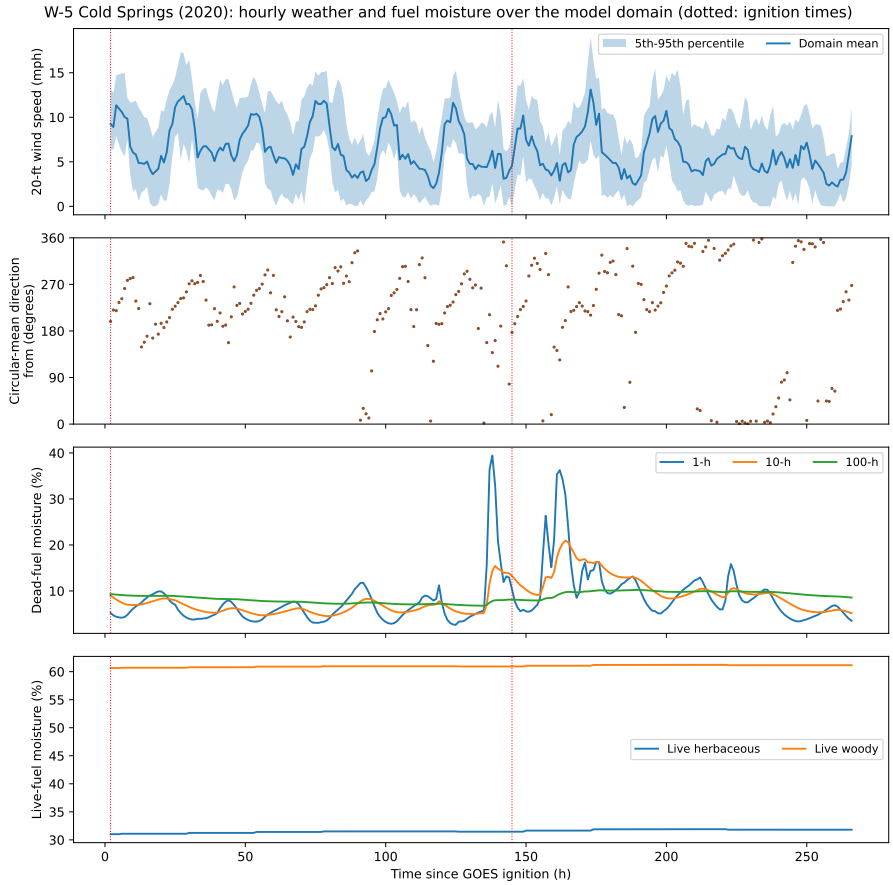


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

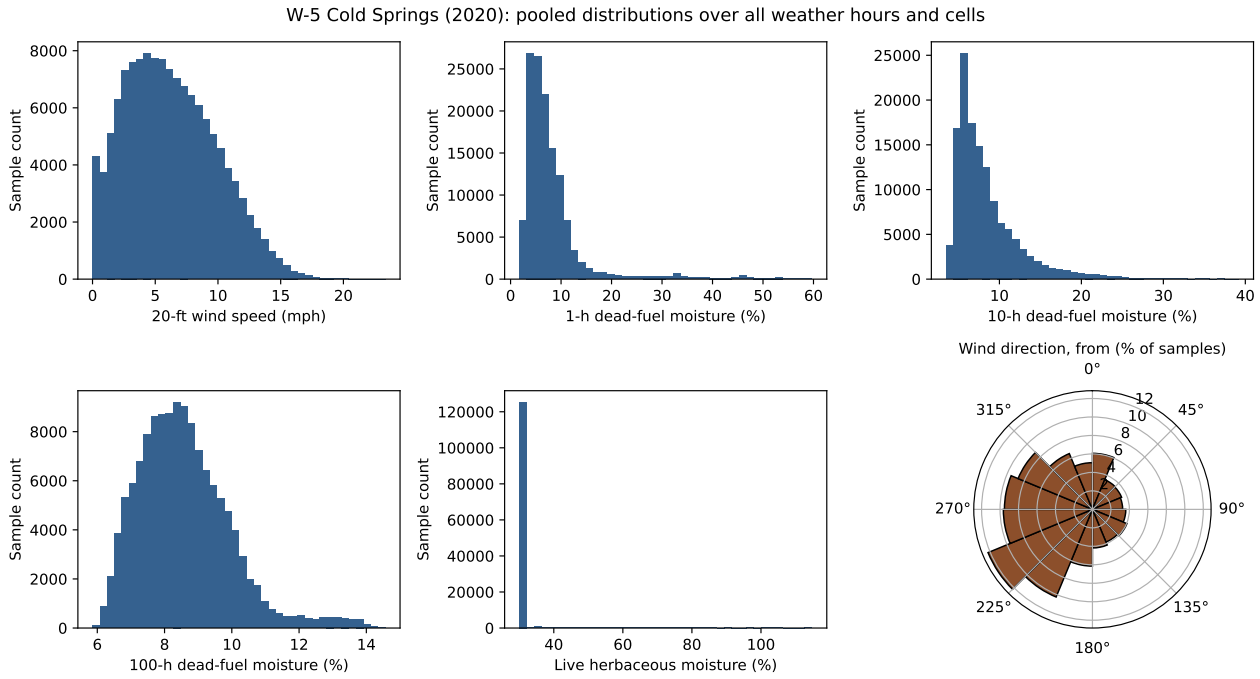


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 264 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 265 hourly weather fields and 2 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 2 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

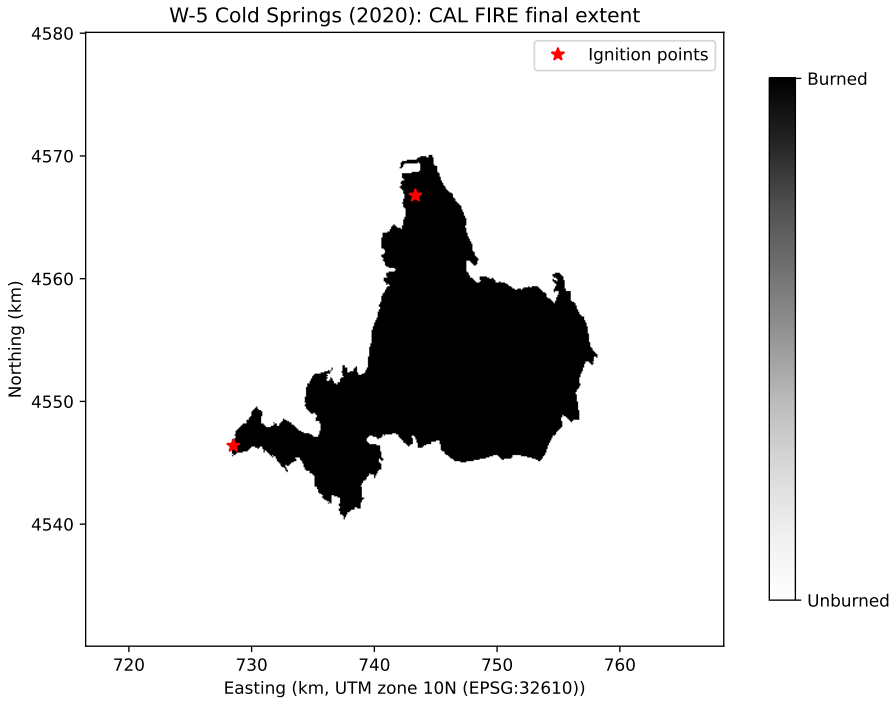


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 344 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the W-5 Cold Springs reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2019 Walker fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Walker (2019) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 4 September 2019 at 07:00 UTC to 12 September 2019 at 22:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 193 hourly perimeters, a final area of 267.693 km^2 and an official size of 54,612 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

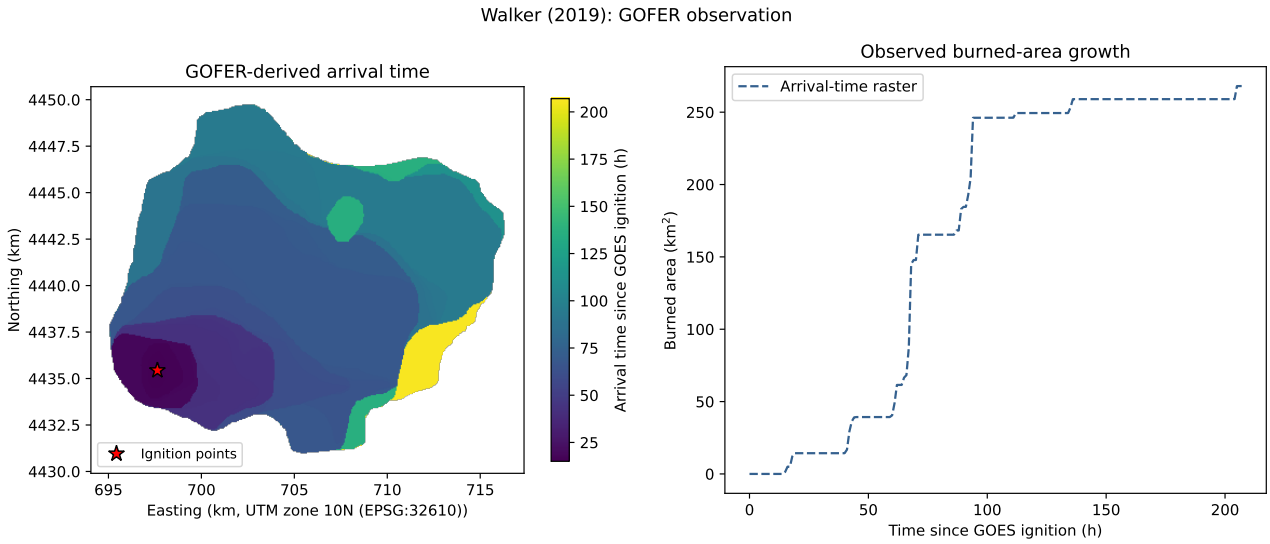


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.0.0_2019	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,400 by 1,333 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 1,680 km^2. Elevation ranges from 1,074 to 2,548 m; median slope is 11°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 35%, and canopy base height has median 1.1 m.

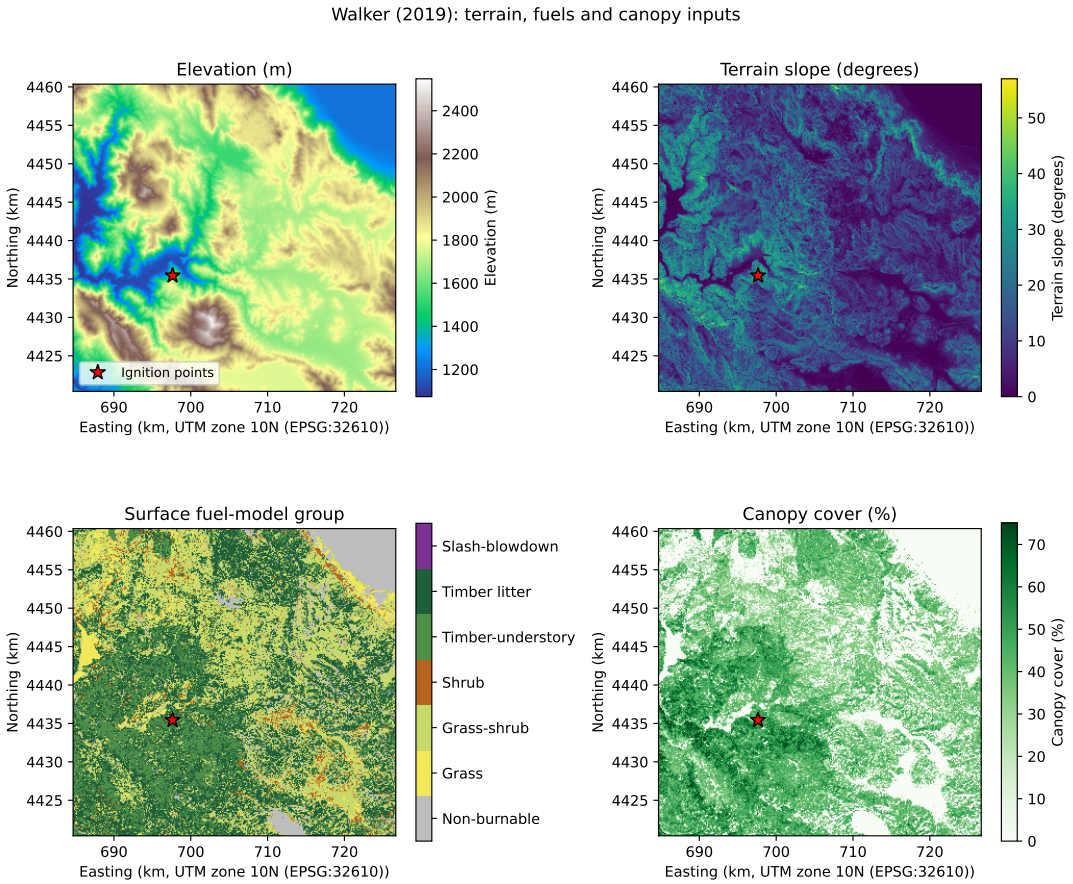


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

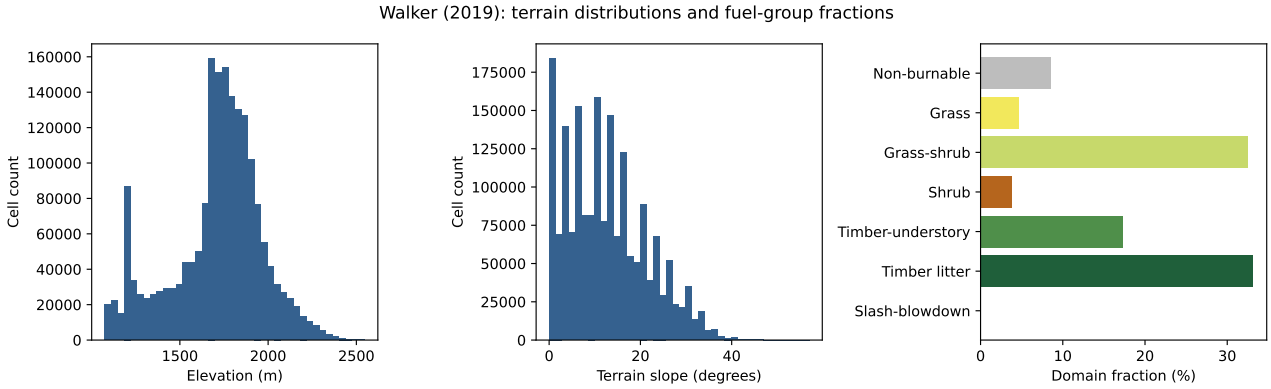


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 194 hourly fields. Median wind speed is 3.48 mph and its 95th percentile is 9.63 mph. Direction is summarized circularly: the mean direction is 262° and mean resultant length is 0.303. Median 1-h dead-fuel moisture is 9.47%. Spatial and temporal variability remain important when these domain summaries are interpreted.

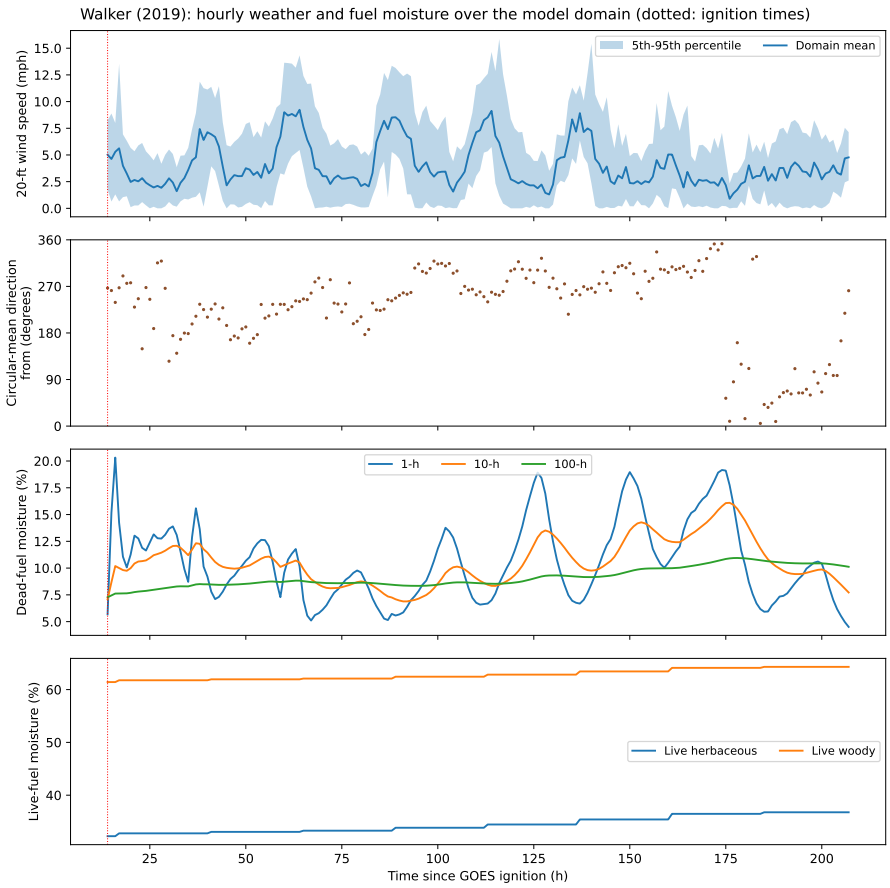


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

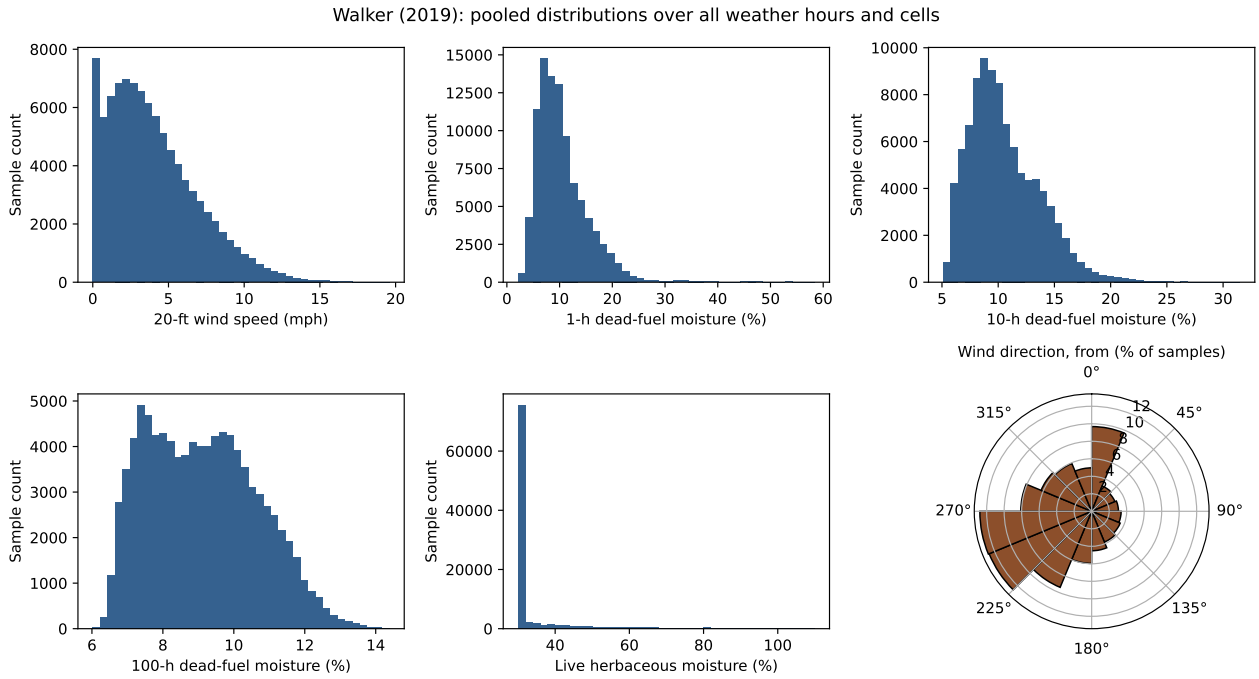


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 193 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 194 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 14 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

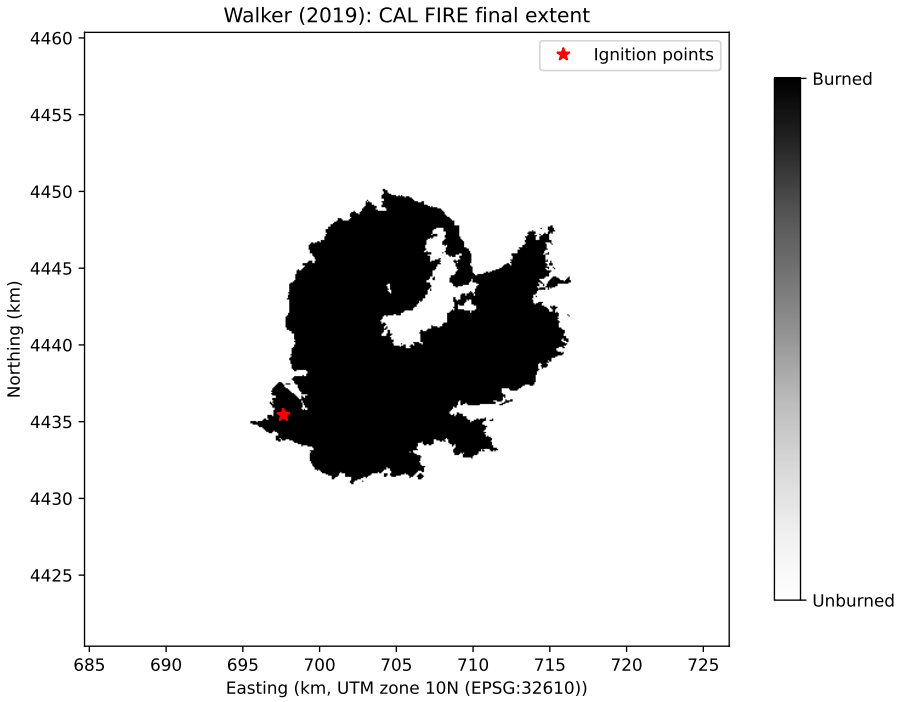


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 221 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Walker reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2021 Windy fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Windy (2021) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 10 September 2021 at 00:00 UTC to 6 October 2021 at 01:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 623 hourly perimeters, a final area of 427.142 km^2 and an official size of 97,528 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

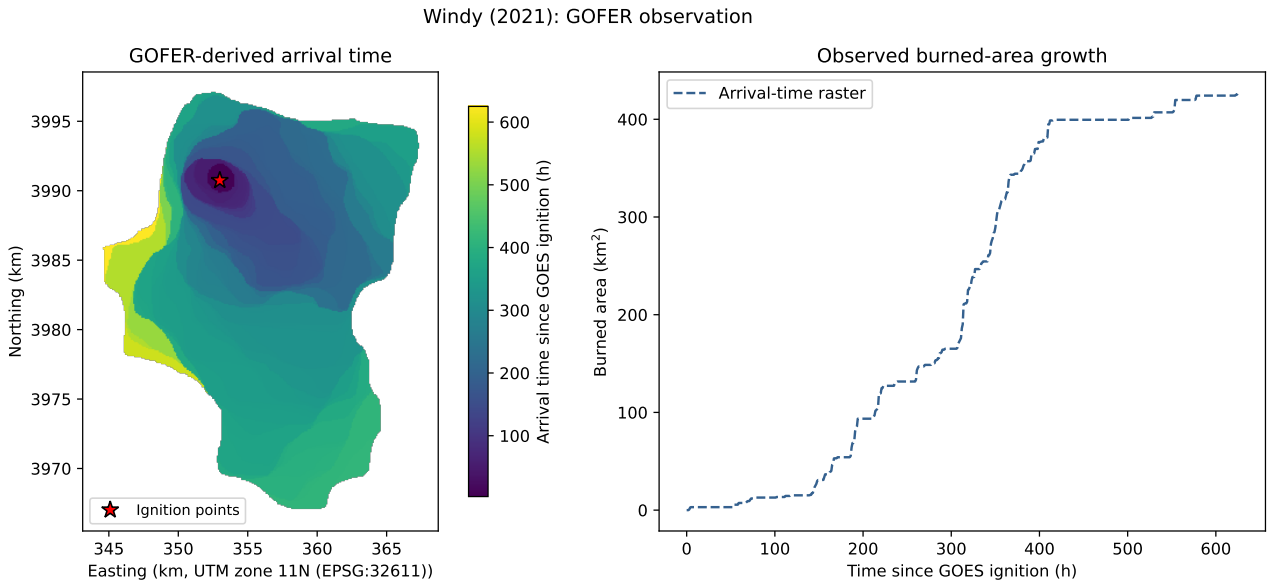


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.2.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,467 by 1,733 cells at 30 m spacing in UTM zone 11N (EPSG:32611), covering 2,288 km^2. Elevation ranges from 228 to 3,026 m; median slope is 19°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 25%, and canopy base height has median 0.7 m.

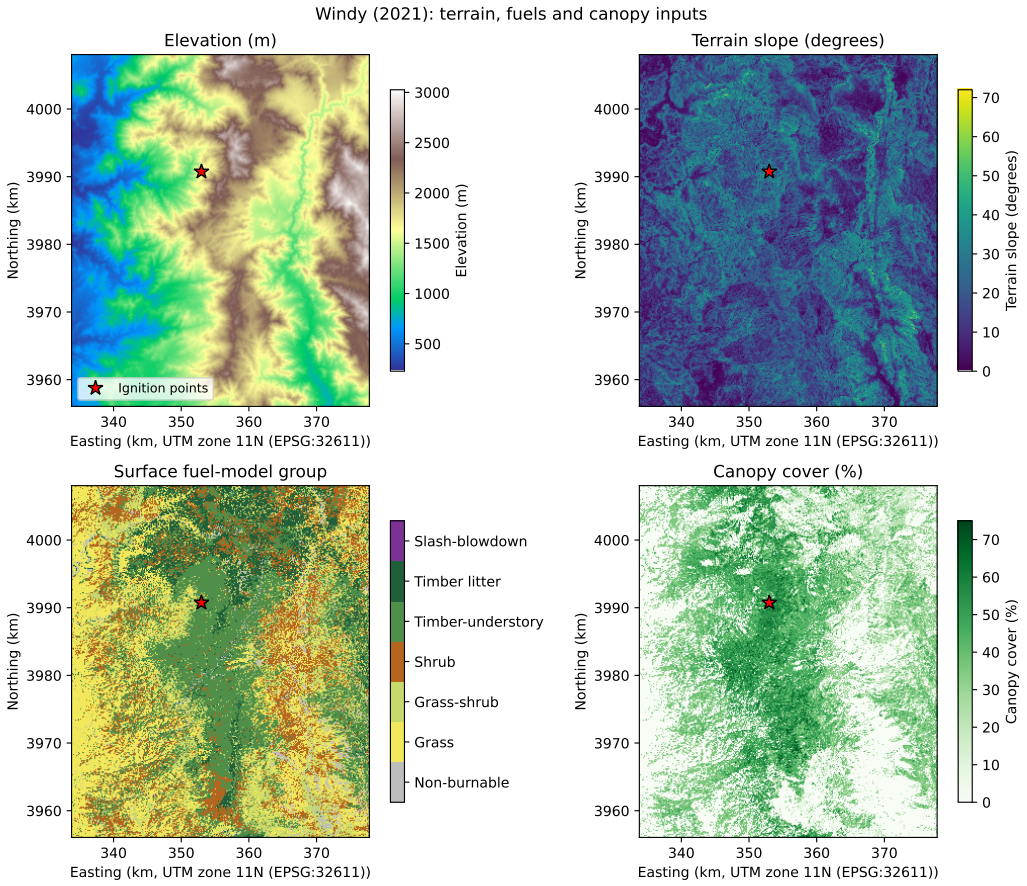


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 11N (EPSG:32611).

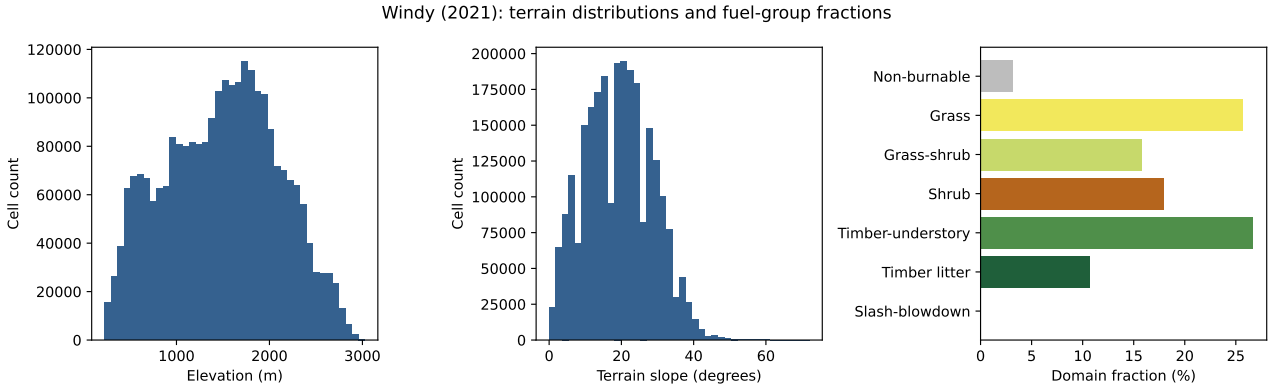


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 624 hourly fields. Median wind speed is 3.59 mph and its 95th percentile is 10.3 mph. Direction is summarized circularly: the mean direction is 284° and mean resultant length is 0.127. Median 1-h dead-fuel moisture is 6.17%. Spatial and temporal variability remain important when these domain summaries are interpreted.

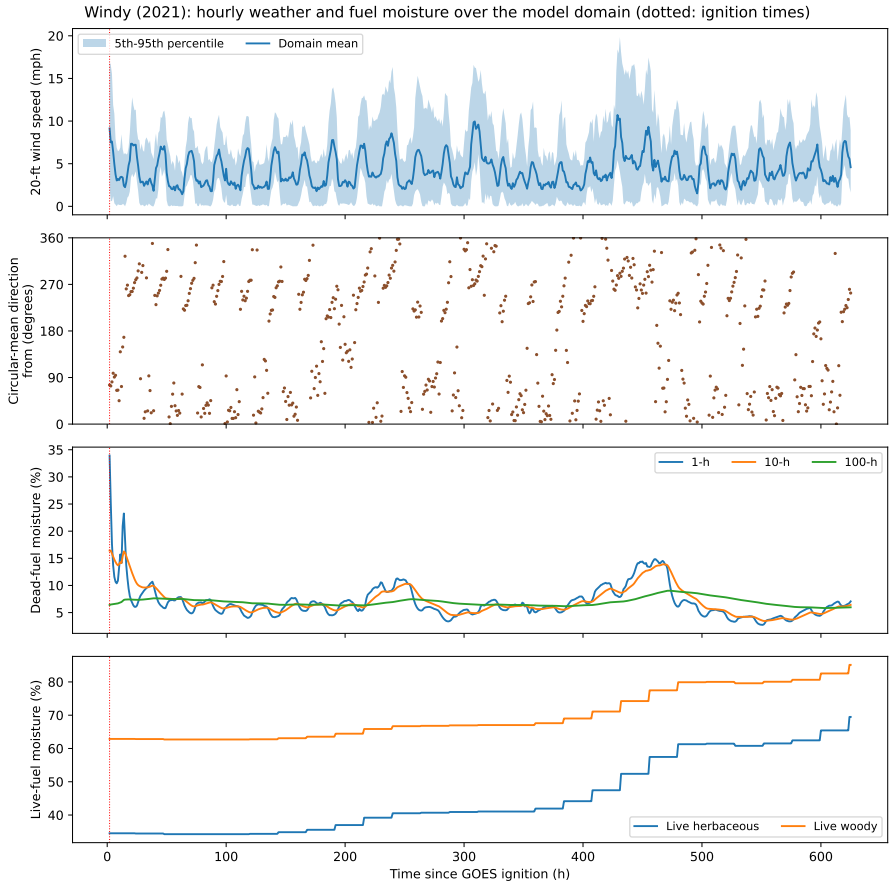


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

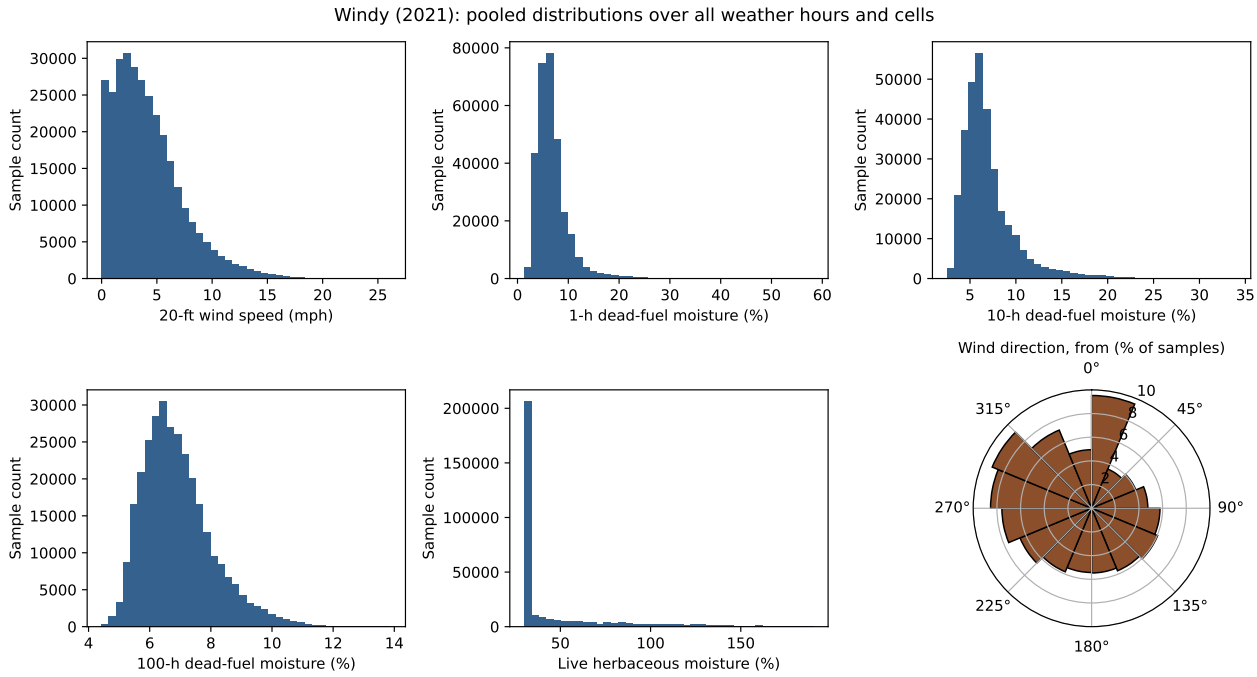


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 623 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 624 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

Landscape canopy quantities are decoded using the supplied LANDFIRE scale conventions. Aspect and wind direction use circular statistics. Observed growth is the cumulative area of finite arrival-time cells reached by each hour. Simulation and weather time zero is 2 h after the GOES ignition hour; comparisons apply this offset explicitly. A supplied PNG is a preview and is not used as quantitative evidence. The original data-generation sources are retained, but regeneration requires external GOFER and CAL FIRE source data and CloudFire access. The supplied materials do not include the GOFER arrival-time raster generator.

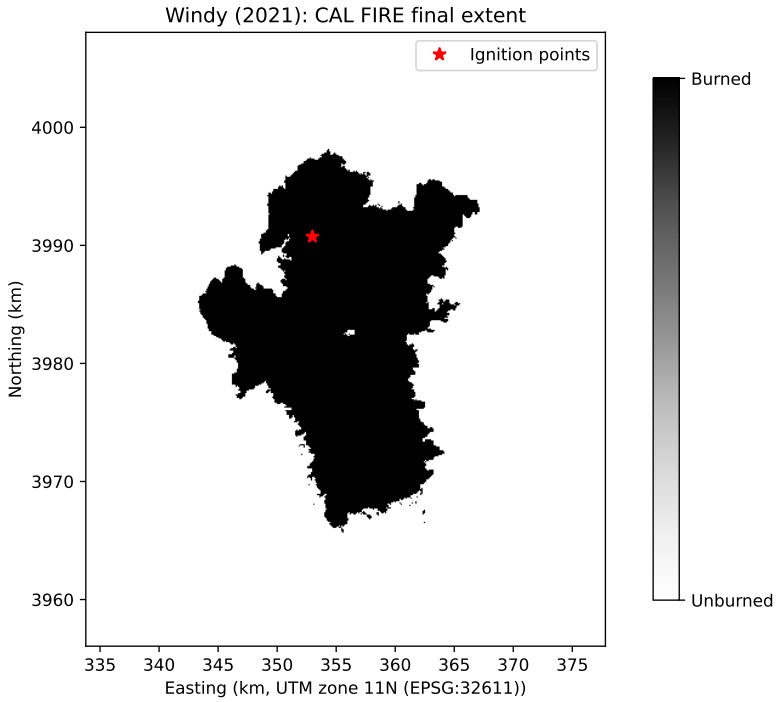


Figure 6: Supporting CAL FIRE final extent rasterized at model-cell centres, with used ignition points. The burned area is 395 km^2. This final extent is distinct from the GOFER timed-progression observation.

6 Validation metrics and comparison method

The unique latest arrival-time result of simulation member one is selected by its numeric output time. Negative arrivals are unburned; other values are converted from seconds since simulation start to hours since the GOES ignition. The comparison uses every valid landscape cell. Simulated grid coordinates must match the landscape or the explicitly reproduced single-precision header rounding of the inspected ELMFIRE source. GOFER times are transferred by nearest neighbour to these cells; times are not interpolated.

For simulated and observed footprints S_t and O_t, let $I_t = |S_t \cap O_t|$. At 24, 48 and 72 h, when earlier than the last observed arrival, and at the last observed arrival, report

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \tag{1}$$

These are Jaccard overlap, precision, recall, F1 and area ratio. Precision identifies commission and recall identifies omission; a correct area alone can hide displacement. Undefined ratios remain unavailable. Within cells burned in both footprints by the final comparison, report arrival-time bias, mean absolute error, root mean square error and median error in hours. Growth and hourly overlap describe the time history. Domain-edge contact and early termination are reported because they limit the observed simulation history.

The baseline is the supplied GOFER observation, rather than a tuned model run. Missing or malformed required evidence is NOT EVALUABLE. With valid results, metrics are CHARACTERIZED because no independent acceptance limits were supplied. CAL FIRE final extent supplies context and is not substituted for GOFER timing.

7 Simulation results and field-level comparison

The current status is **NOT EVALUABLE**. No simulation result is available for this case. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

8 Integrated statistics and calculated metrics

No simulation. Integrated growth and overlap histories require at least one simulation result. **No simulation.** No ELMFIRE result is available for this fire yet. Input and observation characterization is complete, but no footprint, arrival-time or growth metric can be calculated until the case configuration is simulated.

9 Interpretation and uncertainty

A low overlap with high recall and low precision indicates commission, while high precision and low recall indicate omission. Neither area agreement nor a visually similar boundary establishes predictive validity. Satellite sampling, coarse weather fields, fuel classification, undocumented moisture derivation, ignition uncertainty, stochastic spotting, omitted suppression and finite domain extent all affect interpretation. A single realization does not quantify uncertainty or separate these contributions. The CAL FIRE final perimeter and GOFER progression can legitimately differ in extent and timing.

10 Conclusions and limitations

This case defines a reproducible comparison of the Windy reconstruction with the supplied GOFER arrival times and characterizes the provided model inputs and supporting CAL FIRE extent. Its present status is **NOT EVALUABLE**, with 0 simulation result(s) evaluated. A defensible acceptance decision requires independently declared thresholds, documented source and calibration history, and an uncertainty analysis suitable for the intended application.

References

[1] T. Liu et al., "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, GOES-Observed Fire Event Representation product for 28 California wildfires, Zenodo, doi:10.5281/zenodo.8327264. Exact source version of the supplied arrival times is unverified.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] CAL FIRE FRAP, California Historic Fire Perimeters; the supplied preparation source identifies the release dated 26 May 2026.

2020 Zogg fire reconstruction against GOFER observations

1 Validation objective and scope

This case compares the ELMFIRE reconstruction of the Zogg (2020) with the GOES-Observed Fire Event Representation (GOFER) arrival-time observation [1, 2]. The event record extends from 27 September 2020 at 21:00 UTC to 30 September 2020 at 20:00 UTC. The simulated quantities are burned extent, fire arrival time, spread rate and fireline intensity. The comparison evaluates the combined surface-fire, crown-fire and firebrand-spotting response; it does not isolate one physical model. Suppression and undocumented interventions are not represented. No acceptance thresholds were supplied independently of these observations. An available result is therefore CHARACTERIZED, while missing or unusable results are NOT EVALUABLE. Prior calibration history is not documented in the supplied materials; no parameter tuning was performed during preparation of this case.

2 Reference event, experiment, and observations

The supplied GOFER event record contains 71 hourly perimeters, a final area of 222.818 km^2 and an official size of 56,338 ac. The configuration uses 1 timed ignition point(s). The GOFER-derived arrival time is measured in hours since the GOES ignition hour. Finite cells represent observed burning, and missing cells represent locations without an observed arrival. This distinction does not establish that missing cells were unburned at every later time. Satellite sampling, spatial detection limits and the conversion from hourly perimeters introduce uncertainty. The exact GOFER version and the complete raster-generation history are not documented in the available materials.

The observed growth curve below is calculated from the supplied arrival-time raster. The original hourly vector-area series was not supplied separately for this case, so the plotted curve must not be interpreted as an independent second observation. A CAL FIRE Fire and Resource Assessment Program (FRAP) final perimeter is also supplied as vector geometry and an aligned 30-m burned mask. The supplied generation source identifies the California Historic Fire Perimeters release dated 26 May 2026. Its three vector layers retain the dissolved final perimeter, contributing CAL FIRE records and final GOFER perimeter. CAL FIRE supplies supporting final-extent information; it does not supply the hourly arrival times used for the principal numerical comparison.

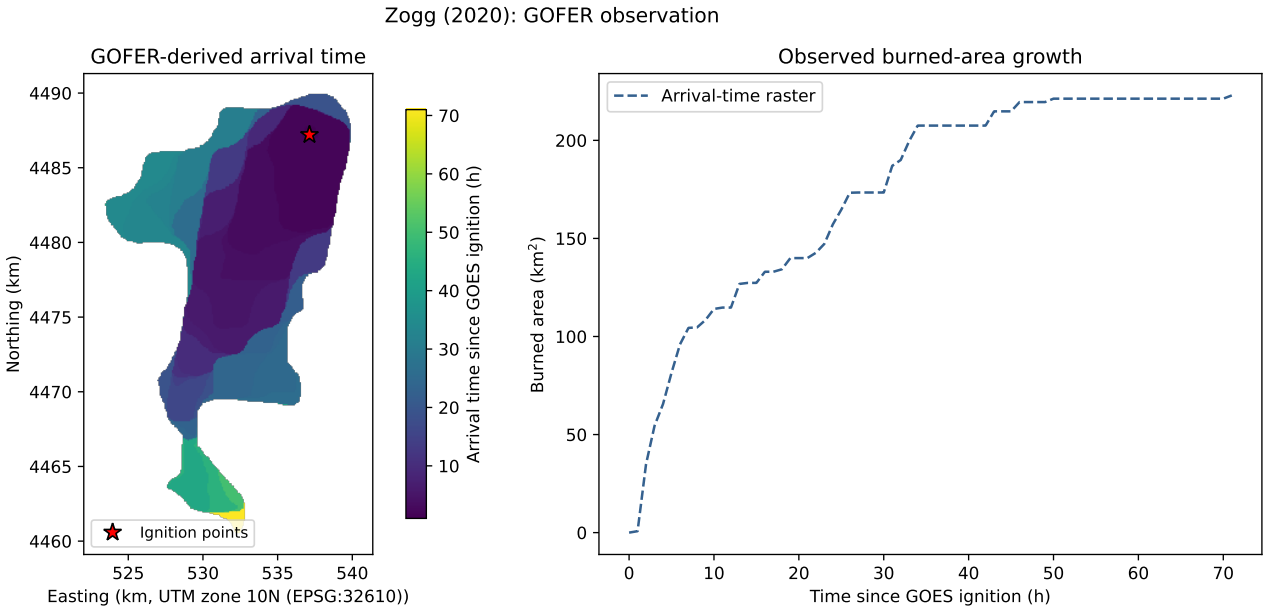


Figure 1: GOFER observation: spatial arrival time and burned-area growth derived from the same raster. Times are hours since the GOES ignition hour. Stars mark used ignition points.

3 Input data provenance and characterization

Quantity	Source	Interpretation and uncertainty
Terrain and surface fuels	LANDFIRE, LANDFIRE release 2.1.0	30-m elevation, slope, aspect and categorical Scott–Burgan fuel model; classification uncertainty is not quantified.
Canopy	LANDFIRE, same release	Cover (percent), height and base height (m), bulk density (kg m^{-3}), decoded from scaled integers.
Wind and moisture	Historical CloudFire service fields	Hourly wind at 20 ft (mph and direction from north), dead and live moisture (percent), following the conventions documented by the supplied preparation source. The underlying analysis product and moisture derivation are not identified there.
Ignition	Supplied GOFER ignition table	Projected positions and elapsed times; nonburnable-cell adjustments are recorded explicitly.
Comparison observations	GOFER and CAL FIRE FRAP	GOFER timed progression; CAL FIRE supporting final extent. Exact source-version and redistribution gaps remain recorded.

Table 1: Input sources and interpretation. Inputs were supplied on 22 October 2026; this receipt date does not establish their original acquisition dates.

The domain contains 1,267 by 1,667 cells at 30 m spacing in UTM zone 10N (EPSG:32610), covering 1,901 km^2. Elevation ranges from 136 to 2,118 m; median slope is 12°. Fuel categories are summarized by area rather than by numeric averages of model codes. Canopy cover has median 0%, and canopy base height has median 0 m.

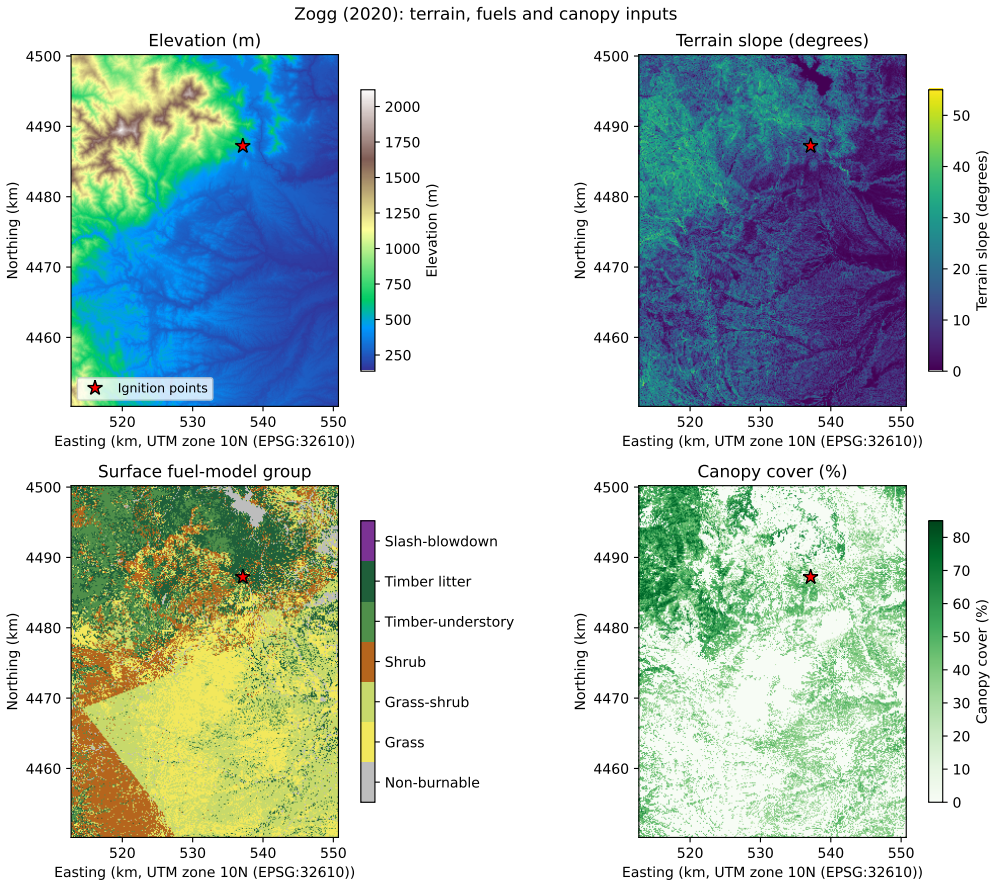


Figure 2: Whole-domain terrain, surface-fuel groups and canopy cover used as model inputs, with used ignition points. Coordinates are in UTM zone 10N (EPSG:32610).

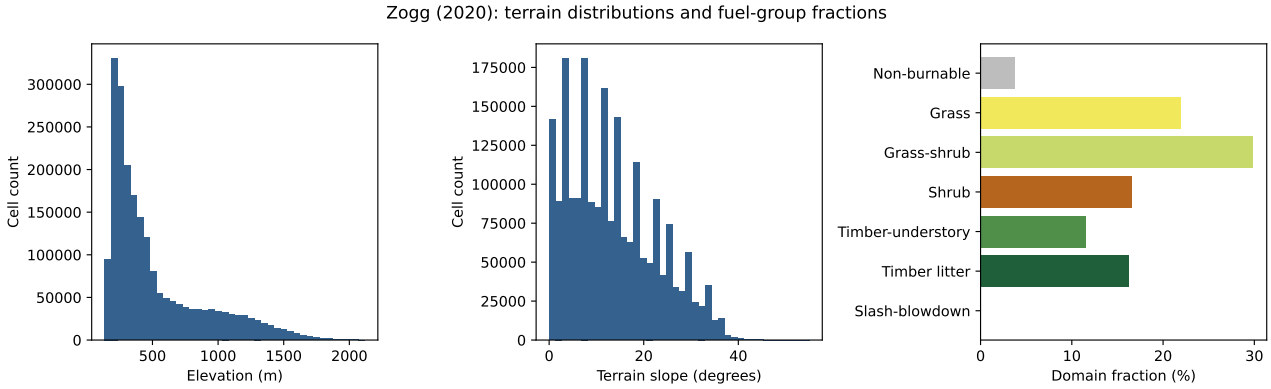


Figure 3: Input elevation and slope distributions and domain fractions occupied by surface-fuel groups.

The weather contains 72 hourly fields. Median wind speed is 4.43 mph and its 95th percentile is 12.9 mph. Direction is summarized circularly: the mean direction is 354° and mean resultant length is 0.386. Median 1-h dead-fuel moisture is 3.79%. Spatial and temporal variability remain important when these domain summaries are interpreted.

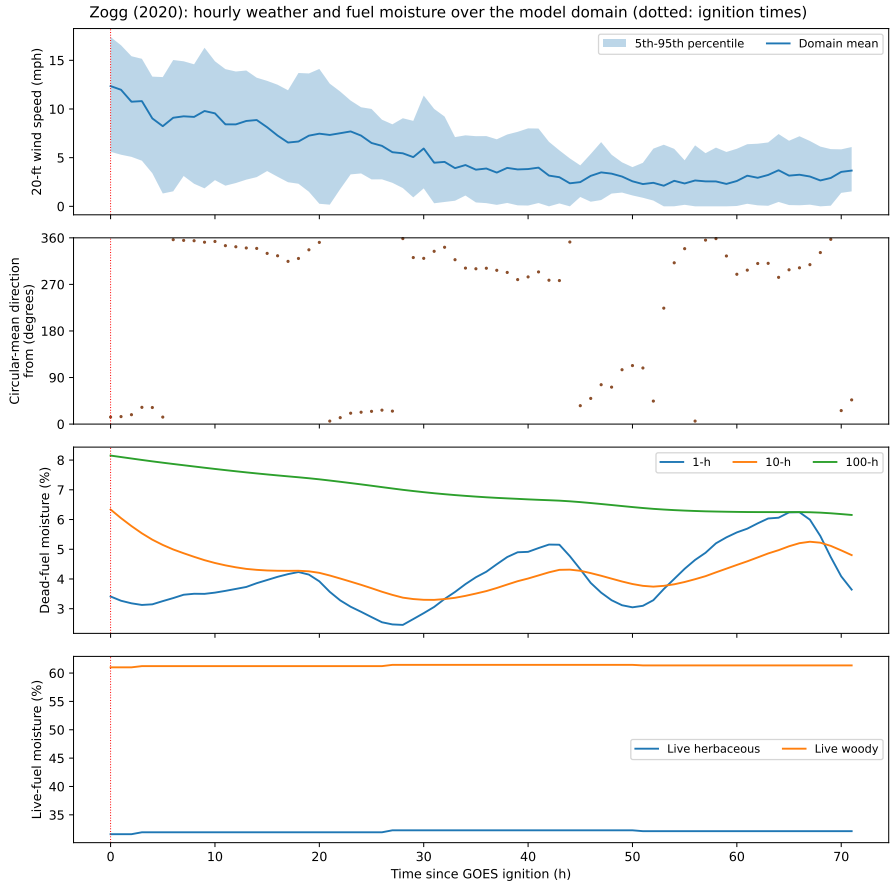


Figure 4: Hourly domain summaries of wind and fuel moisture. Wind-speed shading shows the 5th–95th percentile range; direction uses a circular mean. Dotted lines mark ignition times.

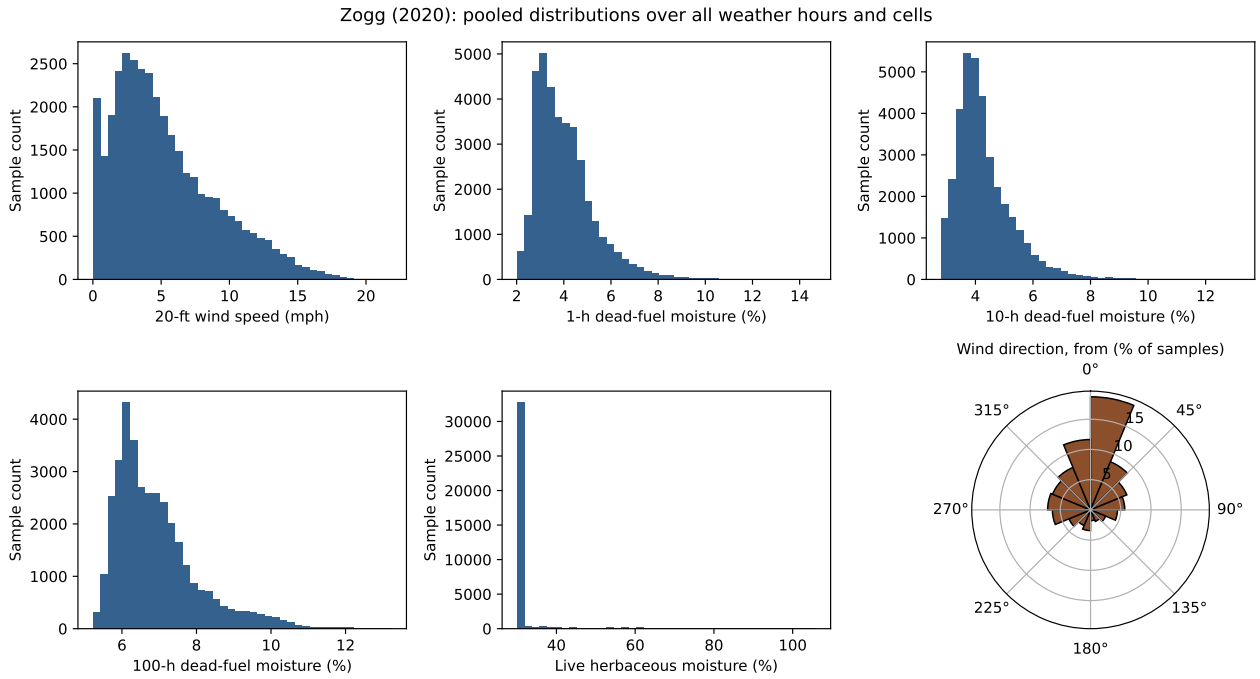


Figure 5: Weather and moisture distributions from a deterministic spatial sample at every weather hour, and a wind-direction frequency diagram.

4 Model configuration and rationale

The supplied simulation runs for 71 h, with an initial timestep of 1 s, target Courant–Friedrichs–Lewy number of 0.2, 72 hourly weather fields and 1 ignition point(s). Arrival time, spread rate, fireline intensity and fire-size statistics are requested at the supplied dump interval. Crown-fire model 1 and firebrand spotting are enabled. Unspecified settings retain the inspected source defaults, including fixed random seed 2024, foliar moisture 90% and maximum timestep 600 s. A single simulation member is configured. The target ELMFIRE source is branch yiren, revision 8981f14e0170. The source directory was inspected for accepted configuration variables, defaults and output conventions. Successful static checks establish configuration compatibility with that inspected source; they do not establish executable identity, successful execution or predictive accuracy.

Choice	Basis	Evidence and limitation
Ignitions and duration	documented	Supplied configuration and ignition table; all used points and times are retained.
Wind and moisture	documented	Supplied hourly rasters and preparation conventions; underlying analysis and moisture derivation remain incompletely identified.
Crown fire and spotting	documented	Supplied configuration retains crown-fire model 1 and enables spotting; other settings use the inspected source defaults.
Timestep and output cadence	documented	Supplied initial timestep, target CFL, stop time and dump interval are preserved.
Acceptance decision	unresolved	No independent numeric thresholds were supplied; only descriptive characterization is supported.

Table 2: Basis of consequential settings and limits on interpretation.

5 Preprocessing and reproducibility

Provided rasters, tables, previews and vector observations are retained unchanged. Sizes and cryptographic checksums are checked before characterization. The landscape must contain the eight documented bands. Weather must cover the landscape, share its coordinate reference system and contain the declared number of hourly bands. The CAL FIRE burned mask must match the model grid exactly, contain only burned, unburned or missing-data classes, and agree with the documented vector-layer coordinate system.

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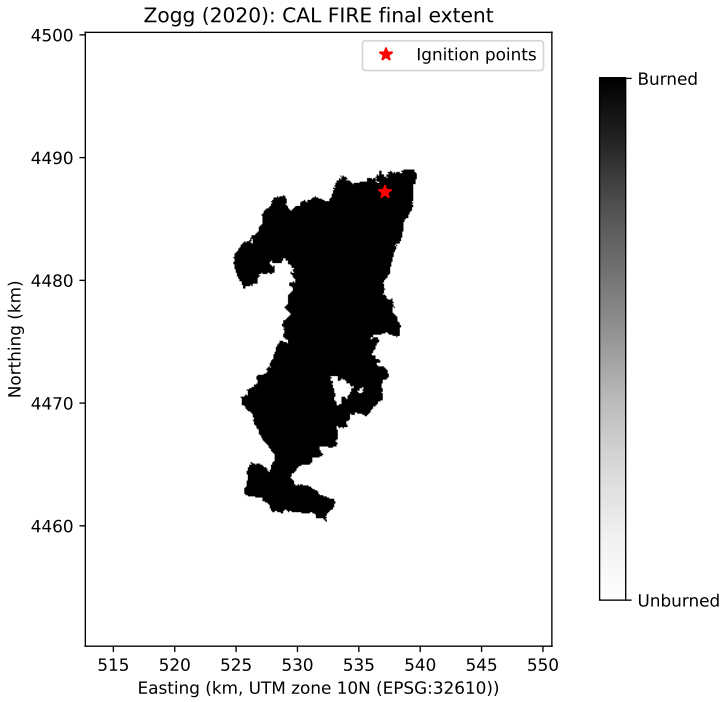


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